

Linear local state bounds, window certificates and first-order Wilson effects at the original coupling cap

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Round32 scientific addendum: AV1–AZ2

Abstract

This edition records the complete Round32 cycle: ten prospectively contracted investigations in five sub-rounds of two (AV1–AZ2), all accepted within their stated model scope. Each result is a fixed-spacing statement about a named SU(2) Kogut–Susskind model at the original coupling cap $|\tau| \leq 10^{-8}$, or about a named finite graph. In the zero-selected, nonuniform AQ family, AV1 replaces the Round31 square-root local state bound by one linear in τ : the trace distance on the 48-link Wilson cover is at most $D \approx 1.3612 \times 10^{-8}$ in every volume and for every AQ subsequential limit. AV2 replaces the Poisson kernel by a C^2 window, and the centered Euclidean Wilson correlation at $s = 1$ then lies within about 1.832×10^{-7} of an exact datum, meeting the 10^{-6} target that Round31 missed at the cap. AW1 proves Peter–Weyl parity and a link-flip lemma and finds the static Wilson mean $+\tau/144$ up to $K_2^+ \tau^2$, $K_2^+ \approx 3354.80$. AW2 encloses that mean at the rule-fixed coupling $\tau = 10^{-8}$ in about $[6.911, 6.978] \times 10^{-11}$, mirrored at -10^{-8} , so its sign is certified. AX1 re-derives the admitted chain by route B for the uniform Kogut–Susskind Hamiltonian at fixed spacing and strong bare coupling, $g^4 = 9.6 \times 10^9$ at the cap, and AX2 certifies its Euclidean node with radius about 1.912×10^{-7} . AY1 shows that any two subsequential limits of two named construction families lie within $2D \approx 2.72 \times 10^{-8}$ of each other on the cover, share one explicit first-order density and differ beyond it by at most $2K_2' \tau^2$, $K_2' \approx 13418$. AY2 lists six unproved obligations and certifies a static two-sided distance of every limit from the Haar product on the cover, about 4.38 to 4.41×10^{-10} . AZ1 states that along a trajectory with $g_n \rightarrow 0$ the family supplies one necessary estimate, the volume-uniform gap $g^2/(32a)$ at fixed a for $g^4 \geq 9.6 \times 10^9$, whose lattice-unit floor $a\Delta \geq 1250\sqrt{6} \approx 3062$ excludes a finite-mass limit inside the admitted regime, and none uniform along $a_n \rightarrow 0$. AZ2, on the Round11 two-plaquette graph with both faces coupled, a different finite model, finds the Wilson derivative $1/6$ exactly, a zero second-order coefficient and the third-order coefficient $-187/33696$, with sign-certified enclosures; under the dictionary $\tau_{\text{FG}} = \tau/24$ the value $1/6$ is consistent with $1/144$, a consistency check only. The free reference lies inside both Euclidean intervals, so no shift of the centered correlation is resolved. These are analytic certificates for fixed-spacing models and one finite graph. They establish no state uniqueness, no estimate uniform in the lattice spacing and no continuum construction. The four-dimensional continuum Yang–Mills problem remains open.

Status and reading guide

This is a separate scientific addendum; Draft03 and the Round31 addendum are unchanged. The Round32 cycle consists of ten prospectively contracted investigations in five sub-rounds of two. An expert panel of three methodological lenses (Newton/Tesla, Jung/Pauli and Penrose/Feynman), an advisor and a skeptic fixed the goals, directions and pre-registration rules in three deliberation loops before any production. After each sub-round one research assistant per lens tests assumptions and plans the next pair; assistant work counts as zero investigations. This is the complete edition. It records all five sub-rounds: ten investigations, all accepted within their stated model scope. As the AZ2 contract and gate state, scientific production stopped after the tenth review.

Three models occur, and a result proved in one is claimed for another only where it was re-derived there. The zero-selected, nonuniform AQ family carries AV1, AV2, AW1, AW2, AY1 and AY2; the uniform Kogut–Susskind model at fixed spacing and strong bare coupling carries AX1, AX2 and AZ1; and the Round11 two-plaquette graph carries AZ2. Five investigations are paired (AV1, AV2, AW1, AX1, AY1). Three are single-direction certificates admitted on one producer plus the skeptic’s independent pre-comparison replay (AW2, AX2, AZ2), and two are statements with a skeptic review (AY2, AZ1). AV2 and AX2 carry the sub-label `reference_unresolved`: the free reference lies inside both intervals. AW2 carries `static_not_dynamic`, and AW1 states the same restriction in words: the Wilson mean they treat is an equal-time quantity, not a dynamical correction. The sub-label `sign_certified_below_cap` does not apply to AW2, because the rule-fixed coupling is the cap itself. AY1 carries `uniform_local_closeness_not_uniqueness`, and AY2 carries it together with `static_not_dynamic`. AZ1 carries none. AZ2 carries `sign_certified_finite_graph` with `static_not_dynamic`; its sign certificate holds for the finite graph only. The Arb/mpmath previews can veto a run but supply no admitted value.

The headline results, each within the scope its gate records, are these.

- Cap-level Euclidean nodes at $s = 1$: radius about 1.832×10^{-7} in the zero-selected family (AV2) and about 1.912×10^{-7} in the uniform model (AX2), both against the unchanged target 10^{-6} and both with the free reference inside.
- A sign-certified static Wilson mean at the AW2 coupling $\tau = \pm 10^{-8}$, from the AW1 coefficient $+\tau/144$ and remainder constant $K_2^+ \approx 3354.80$ (AW1, AW2).
- The uniform Kogut–Susskind chain by route B: re-frozen contraction constant, unique ground state, gap $\alpha/16$ and a state bound of about 1.4446×10^{-8} (AX1).
- Local closeness of subsequential limits, $2D \approx 2.72 \times 10^{-8}$ on the cover, with a common first-order density and $K_2' \approx 13418$ (AY1), and a certified static two-sided distance from the Haar product of about 4.38×10^{-10} to 4.41×10^{-10} , with six obligations left unproved (AY2).
- The continuum-trajectory statement: one necessary estimate, the volume-uniform gap at fixed a with lattice-unit floor $a\Delta \geq 1250\sqrt{6}$, and none uniform along $a_n \rightarrow 0$ (AZ1).
- Exact finite-graph coefficients on the two-plaquette graph, a different finite model: $1/6$, 0 and $-187/33696$, consistent with $1/144$ under the dictionary and never transferred (AZ2).

The four-dimensional continuum Yang–Mills problem remains open, and no investigation count is a fraction of it.

Section 1 fixes the model dictionary, the Round31 checkpoint and the frozen plan. Section 2 derives the linear state bound and Section 3 the window certificate. Section 4 proves the parity theorem, the link-flip lemma and the first-order Wilson mean with its second-order remainder, and Section 5 instantiates the sign-certified enclosure. Section 6 re-derives the chain for the uniform model and closes with the AX2 certificate. Section 7 treats the local closeness of subsequential states (AY1, AY2), and Section 8 the continuum trajectory and the finite graph (AZ1, AZ2). Each investigation opens with its gate’s accepted statement, quoted verbatim in the gate’s ASCII notation as the reviewed scope, and closes with the limitations the gate recorded. Section 9 records the process, Section 10 what the record does and does not establish, and Appendix A the contribution and evidence ledger.

The human project author is Hruday N M (BUNZEEY). The advisor, the skeptic, the three lenses, the research assistants and the forward and reverse producers are separate model-agent roles with shared mathematical ancestry. Their agreement is correlated model-agent work, not external human peer review or independent physical observation. Historical figures supply documented methodological lenses; none participated in or endorsed this work, and no historical, occult or mystical text supplied a premise. HNM contribution aliases identify project records; scientific priority is unverified.

The AM2 creation-expansion gap and the AQ construction follow strategies present in Gauvin [arXiv:2503.15539](https://arxiv.org/abs/2503.15539) from v2 (2026-09-11; absent from v1 of 2025-03), which itself credits the creation-operator method to Bravyi, DiVincenzo and Loss (2008); the SU(2) selected-reference application

and its explicit constants are this project’s record; scientific priority is unverified and no independence is claimed. That sentence is fixed by the skeptic’s [version-diff record](#) and adopted in the [plan freeze](#).

1 Model dictionary, Round31 checkpoint and Round32 plan

1.1 Model dictionary

The theory is the SU(2) Kogut–Susskind form on the spatial lattice $\mathbb{Z}^3$ at fixed spacing a . A coarse site $b \in \mathbb{Z}^3$ owns the three positive links from each fine tail $(4b_x + r, 2b_y + q, b_z)$, $0 \leq r < 4$, $0 \leq q < 2$: exactly 24 untruncated $L^2(\text{SU}(2))$ link factors. Both original endpoint gauge actions are retained. Write $C_e = -\sum_a X_{ea}^2$ for the link Casimir, with generators $i\sigma_a/2$, and $W_f = \frac{1}{2} \text{Tr } U_f$ for the real fundamental Wilson plaquette, so $\|W_f\| = 1$.

An elementary face in directions $a < c$ at p has positive link tails at p , $p + e_a$, $p + e_c$ and p , so its coarse owner set is $\{\pi(p), \pi(p + e_a), \pi(p + e_c)\}$; there is no tail at $p + e_a + e_c$. Grouping by anchor gives 24 face classes per coarse site (Table 1), with relative supports inside the star $S = \{0, e_x, e_y, e_z\}$ (I1 dictionary). Three classes lie inside one factor. They form the ten-link selected strip, with coefficient triple $(\lambda_L, \mu_M, \lambda_R)$ constrained by $|\lambda_L|, |\lambda_R| \leq \alpha/2$ and $|\mu_M| \leq \alpha/8$; the other 14 links of the factor are free. The remaining 21 classes cross a block boundary. They are the *omitted* classes, with the common coefficient $\nu = \alpha\tau/24$.

Table 1: The I1 face classes per coarse site; q is the fine y phase.

Orientation and phase	Classes	Relative support	Role
$xy: r = 0, 1, 2; q = 0$	3	$\{0\}$	selected
$xy: r = 0, 1, 2; q = 1$	3	$\{0, e_y\}$	omitted
$xy: r = 3; q = 0$	1	$\{0, e_x\}$	omitted
$xy: r = 3; q = 1$	1	$\{0, e_x, e_y\}$	omitted
$xz: r = 0, 1, 2; q = 0, 1$	6	$\{0, e_z\}$	omitted
$xz: r = 3; q = 0, 1$	2	$\{0, e_x, e_z\}$	omitted
$yz: r = 0, \dots, 3; q = 0$	4	$\{0, e_z\}$	omitted
$yz: r = 0, \dots, 3; q = 1$	4	$\{0, e_y, e_z\}$	omitted

Throughout Sections 2–5 the selected triple is exactly $(0, 0, 0)$. This is an admissible choice inside the closed AQ coefficient box, not a replacement of a nonzero selected vacuum by Haar. It is also not the uniform model: equality with uniform Kogut–Susskind theory would require all 24 coefficients to equal $\alpha\tau/24$, which fails for $\tau \neq 0$. The original AL dictionary $\tau = 96/g^4$ is not assigned to this patterned subfamily; Section 6 assigns it to the uniform model.

Whole stars $b + S$ are retained only when $b + S \subset \Lambda_N = [-N, N]^3$. In normalized units $\delta = \alpha/8$ the I1.5 convention groups the omitted faces by anchor:

$$\phi_b = -\frac{\tau}{3} \sum_{f \in O_b} W_f, \quad |O_b| = 21, \quad \|\phi_b\| = 7|\tau|, \quad \hat{H}_N = \sum_{b \in \Lambda_N} h_b + \sum_{b+S \subset \Lambda_N} \phi_b, \quad h_b = 8 \sum_{e \text{ owned by } b} C_e. \quad (1)$$

At the zero triple the on-site operator is a pure Casimir sum with constant (Haar) vacuum projection P_b , zero ground scalar and $h_b \geq 6(I - P_b)$. The physical Hamiltonian is $H_N = \delta \hat{H}_N$. In the energy unit α each star is $V_b = \phi_b/8 = -(\tau/24) \sum_{f \in O_b} W_f$, with $\|V_b\| \leq 7|\tau|/8$. Mixing the two unit systems is an eightfold clock error, and every Round32 checker rejects it.

For the original xz Wilson plaquette W at the origin, the complete-factor cover is $R = \{0, e_z\}$: 48 stored links and 36 original endpoints (22 in each factor, eight shared). Its seven incident whole-star anchors are

$$R - S = \{0, -e_x, -e_y, -e_z, e_z, e_z - e_x, e_z - e_y\}. \quad (2)$$

All seven are retained for $N \geq 2$. The two-anchor orthant count, the four drawn links of W as a cover, and a one-star interaction budget are rejected controls.

The on-site product reference on R is the Haar vector Ω_R with projection P_R . Weyl integration gives $E_{\text{Haar}}[W^n] = 1, 0, \frac{1}{4}, 0, \frac{1}{8}$ for $n = 0, \dots, 4$, so the reference moments are $\omega_0(W) = 0$ and $\omega_0(W^2) = \frac{1}{4}$. The vector $W\Omega_R$ carries spin $\frac{1}{2}$ on four links with Casimir $\frac{3}{4}$ each; its energy is $8 \cdot 4 \cdot \frac{3}{4} \delta = 24\delta = 3\alpha$. The free centered spectral measure is therefore $\frac{1}{4}\delta_3$ in units of α , and

$$C_0(s) = \frac{1}{4}e^{-3s}, \quad c_0(\theta) = \frac{1}{4}e^{3i\theta}. \quad (3)$$

These formulas describe the reference, not the interacting spectrum.

AM2 expansion constants. AM2 (Round29) proves, for every finite complete-factor volume and both signs of $|\tau| \leq 10^{-8}$, a unique ground and a centered gap at least $\alpha/16$ (normalized $\frac{1}{2}$; [AM2 gate](#)). The proof uses a nilpotent commuting creation-operator expansion. For nonempty $I \subset \Lambda$ a creation vector $c_I \in \bigotimes_{x \in I} Q_x \mathcal{H}_x$, with $Q_x = I - P_x$, defines $\hat{c}_I = |c_I\rangle\langle\Omega_I| \otimes 1$. These operators commute, and $\hat{c}_I \hat{c}_J = 0$ whenever $I \cap J \neq \emptyset$. With $C = \sum_I \hat{c}_I$ and the anchored norm $\|c\|_a = \max_u \sum_{I \ni u} \|c_I\|$, the ground vector is $\psi = e^{-C}\Omega_0$, with vacuum coefficient one, and the unique fixed point satisfies

$$c = \sum_{k=0}^8 \frac{L_k(c, \dots, c)}{k!}, \quad (L_k)_M = H_M^{-1} P_M \text{ad}_{C_1} \cdots \text{ad}_{C_k}(V) \Omega_0, \quad \|c\|_a \leq R_a = \frac{1}{64}. \quad (4)$$

The commutator nest terminates at order eight. The per-site interaction sum counts all four stars that contain a site, including incoming stars: $J = \max_u \sum_{X \ni u} \|V_X\| \leq 4 \cdot 7|\tau| = 28|\tau| \leq J_0 = 7/25000000$. The all-order majorant and its evaluated constants are

$$\begin{aligned} G(t) &= 16e^{8t}(1 + 10t), & G(R_a) &< \frac{148}{7}, & G'(R_a) &< 352, \\ J_0 G(R_a) &< \frac{37}{6250000} < R_a, & 2J_0 G'(R_a) &< 1. \end{aligned} \quad (5)$$

The fixed-point equation gives the two majorant inequalities $\|c\|_a \leq JG(\|c\|_a)$ and $\|c - L_0\|_a \leq J(G(\|c\|_a) - 16)$, which AV1 uses. AM2 removes the on-site representation cutoff for the eigenvalues and the gap; AV1 removes it for the ground vector. We write R_a for AM2's anchored radius so that R always denotes the cover.

State and clock. AQ1 (Round29) supplies, by diagonal extraction along centered whole-star boxes, locally normal stationary subsequential states on the full coarse algebra, with trace-norm convergence of the local densities on every finite region. Let ω_τ be any such state for the declared coefficients, Ω_τ its GNS vacuum and $H_\tau \geq 0$ its physical energy generator. AQ2 gives a gap at least $\alpha/16$ in that representation; the Round32 window certificate uses only $H_\tau \geq 0$. No uniqueness, whole-sequence convergence or rate in N is assumed. Both the reference and the interacting clocks use

$$G_\tau = H_\tau/\alpha, \quad s = \frac{\alpha t_E}{\hbar}, \quad \theta = \frac{\alpha t}{\hbar}. \quad (6)$$

The positive energy α , action $\hbar$, spacing a and comparison energy $E_\star$ stay fixed, so s, θ and τ are dimensionless. The older normalized clock $u = s/8$, with free Wilson exponent 24, is forbidden in Round32 evidence packets. For the original plaquette,

$$m_\tau = \omega_\tau(W), \quad \chi_\tau = (\pi_\tau(W) - m_\tau)\Omega_\tau, \quad C_\tau(s) = \langle \chi_\tau, e^{-sG_\tau} \chi_\tau \rangle, \quad c_\tau(\theta) = \langle \chi_\tau, e^{i\theta G_\tau} \chi_\tau \rangle. \quad (7)$$

The correlation is dimensionless and centered as a vector in the actual state. Changing τ between investigations would be an explicit interaction change, not a change of units; in sub-rounds 1 and 2 the coupling is not changed.

1.2 Round31 checkpoint

Round31 closed three investigations in the same zero-selected family (Table 2; Round31 addendum). AT4 kept the original cap and missed its target. AT5 and AT6 met the target only after a prospectively declared change of coupling to 10^{-14} , which selects a different Hamiltonian and its own AQ state.

Table 2: Round31 results inherited as the starting point; decimals summarize outward rational bounds in the frozen outputs.

Loop	Design	Result	Verdict
AT4	$\tau = 10^{-8}$, $s = 1$, $L = 10^4$, target 10^{-6}	radius $\approx 8.41518704386267 \times 10^{-4}$; interval [0.0116052483875797, 0.0132882857963523]	limited
AT5	$\tau = 10^{-14}$, $s = 1$, $L = 10^9$	datum $\approx 0.0124467670919660$, radius $< 8.086109564653 \times 10^{-7}$	accepted
AT6	$\tau = 10^{-14}$, $0 \leq s \leq 128$, 4097 nodes	every node error $< 8.491932 \times 10^{-7}$; $I \in [0.083174562675757, 0.083535946641292]$, containing 1/12	accepted

The AT4 budget was dominated by the reset bound $D = 2\sqrt{49|\tau|/3} \approx 8.0829 \times 10^{-4}$, which scales as $|\tau|^{1/2}$. Before Round32 production, the advisor also observed that optimizing AT4’s Poisson certificate over the cutoff L leaves a dynamics-plus-tail cost of about 1.27×10^{-6} at $s = 1$ even if the state term vanished. That was a planning observation; AV2 later proves it as a retained failure (Section 3.7). The cap therefore had two separate obstacles: a square-root state term and a kernel whose absolute first moment diverges. Sub-round 1 addresses one each. All three Round31 intervals contain their free references, so no interaction shift was resolved there either. The ranked Round31 roadmap goals were cap-level precision, an interaction-induced effect, a uniform Hamiltonian, state identification and a continuum trajectory (Round31 roadmap).

1.3 Round32 plan

A user instruction of 23 September 2026 authorized Round32 as an adaptive expert-panel cycle with three deliberation loops and five sub-rounds of two investigations. In loop 1 each lens wrote a memo, source ledger and recommendation, and the skeptic wrote a prospective triage and control catalogue. In loop 2 the panel critiqued plan v1. It corrected the face count per factor from the bound 84 to the derived 49, set the AV1 target from AV2 feasibility, separated the AV1 and AV2 routes, and adopted a pre-registration block. Loop 3 was an objection pass. All four sign-offs arrived without a veto on the plan; the skeptic’s veto on freezing AV1 was cleared once the selection note matched the contract. Plan v2 was frozen before AV1 production (plan record). Table 3 lists it. All ten rows are gated.

Direction rules. Derivation-bearing investigations are paired, with genuinely different forward and reverse routes disclosed in the contract. Formula-instantiating certificates use one direction plus an independent skeptic replay written from the contract alone. State and continuum statements use a statement plus skeptical review. Same-author passes are never independent review. For paired investigations the reverse producer receives only `AGENTS.md`, the contract and its declared shared premises; the freezing tool enforces that inventory. After each sub-round the advisor may reorder, replace or retitle the next pair, and records any change in the next selection note and the plan history. After each of sub-rounds 1 to 4 the goals were kept (panel updates 1, 2, 3 and 4).

Pre-registration. Every contract freezes and hashes, before any outcome, the model identifier, the selected triple, the coupling and the signs evaluated, the state provenance, the observable with its centering convention, the nodes, an itemized list of error terms and the target. It also

Table 3: Frozen Round32 plan (v2). Direction: P = paired forward/reverse, S = single direction plus independent skeptic replay, T = statement plus skeptic.

Sub-round, goal	Investigations	Dir.	Status in this edition
1, cap	AV1: first-order local state lemma.	P	gated: accepted within scope
	AV2: window-kernel certificate for $C(1)$ at $\tau = 10^{-8}$.	P	gated: accepted within scope; reference unresolved
2, shift	AW1: Peter–Weyl parity, link-flip antisymmetry, first-order Wilson mean and second-order remainder K_2 , with the AW2 coupling rule frozen in advance.	P	gated: accepted within scope
	AW2: sign-certified enclosure of $\omega_\tau(W)$ at both signs, coupling fixed by the AW1 rule.	S	gated: accepted within scope
3, uniform	AX1: route-B re-derivation for uniform Kogut–Susskind $SU(2)$, J_0 re-frozen at $29/10^8$, dictionary $\tau = 96/g^4$.	P	gated: accepted within scope
	AX2: uniform-model datum $C(1)$ at the cap by the window route.	S	gated: accepted within scope; reference unresolved
4, state	AY1: uniform local closeness of all AQ-type subsequential states and the boundary-independent first-order density.	P	gated: accepted within scope; local closeness, not uniqueness
	AY2: two named construction families, topology, explicit constant, and what remains unproved.	T	gated: accepted within scope; local closeness, not uniqueness; static, not dynamic
5, continuum	AZ1: regulator/coupling trajectory and the one uniform estimate supplied or failed.	T	gated: accepted within scope; one uniform estimate at fixed a; nothing uniform in a
	AZ2: exact first- and second-order coefficients on the Round11 two-plaquette graph, labelled a different finite model.	S	gated: accepted within scope; sign-certified on the finite graph only; static, not dynamic; no transfer

fixes the outcome types (`accepted_within_scope`, `limited`, `insufficient`), the allowed sub-labels, the claim exclusions and the required control ids. Post-hoc node selection is forbidden and pilot runs are excluded from every count. An error term may be marked not applicable only with a stated reason. Each checker reads its target and reference from the hash-checked contract snapshot and records its own SHA-256 before the full-size evaluation. The allowed sub-labels are `reference_unresolved`, `sign_certified_finite_graph`, `sign_certified_below_cap`, `static_not_dynamic` and `uniform_local_closeness_not_uniqueness`. Four loop-2 vetoes stand: no AW2 coupling chosen after K_2 is known and no sign margin below 2, no uniform Wilson-mean sign rider on AX2, no coefficients fitted from finite-graph enclosures, and no radius target of 10^{-8} for AV. A fifth veto, on any priority or independence sentence about AM2, was resolved by the fixed attribution sentence of the status section. A limited or insufficient verdict is a retained outcome. Investigation counts are never a fraction of the continuum problem.

2 AV1: a local state bound linear in the coupling

Contribution HNM-C-AV1. The AV1 contract, recorded under the title “Hruday first-order local state lemma for the zero-selected AQ subfamily”, was frozen before production at the unchanged cap $\tau = \pm 10^{-8}$. It preregistered the target $D_{ii} \leq 4/10^7$ for the exact-first-order tier. That is the value AV2 needs, since $2(D + D^2) + 49 \cdot 10^{-8}/\pi \leq 10^{-6}$ requires $D \leq 4.22 \times 10^{-7}$; the contract kept 10^{-6} as a secondary comparison. Two routes were assigned. The forward producer derives the reduced density explicitly. The reverse producer, under premise isolation, bounds only the vacuum overlap. The product-ordering split and the location of the normalization trap are shared panel premises. Independence is claimed only for the inequality, the constants, the enumeration, the cutoff step and the passage step.

2.1 Reviewed scope

The [AV1 gate](#) records the verdict `accepted_within_scope`. Its accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section.

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In the AM2/AQ1 zero-selected patterned family (SU(2) Kogut-Susskind form on  $Z^3$  at
fixed spacing, coarse 24-link factors, selected triple exactly (0,0,0) with Haar
product reference, 21 omitted anchored faces per factor grouped into whole stars
 $\phi_b = -(\tau/3) \sum W_f$  in normalized units  $\delta = \alpha/8$ ), at either sign of  $\tau$  with
 $|\tau| \leq 10^{-8}$  (the cap values bound every smaller  $|\tau|$  because each bound is increasing
in  $|\tau|$ ), the unique untruncated ground state of every centered whole-star box
 $\Lambda_N$ ,  $N \geq 2$ , has reduced density on the complete cover  $R = \{0, e_z\}$  of the original  $xz$ 
Wilson loop satisfying  $||\rho_{\{N,R\}} - P_R||_1 \leq D$  uniformly in  $N$ . The bound follows from
the product-ordering split  $\psi = \psi_{out} + \delta$ ,  $(P_R \times 1)\psi = \psi_{out}$ ,
 $||\delta|| \leq \epsilon ||\psi_{out}||$  (straddling supports and the two-creation term included) by
two independently proved inequalities: the explicit-density bound  $2\epsilon(1+\epsilon)/(1+\epsilon^2)$ 
(forward) and the fidelity bound  $2\epsilon/\sqrt{1+\epsilon^2}$  (reverse). Tier (ii), with the exact
first-order face coefficient  $|\tau|/144$  and the self-consistent AM2 remainder
 $t \leq t_1/(1-352J)$ :
D_ii=585079838465912592144137406066050/42981220507576537932303142777593983768257
(~1.3612e-8; exact rational, certified by both inequalities), and the reverse's labelled
82-face refinement  $D_{ii} \leq 113902305553947264976096174312887/10^{40}$  (~1.1390e-8; directed
rational upper bound via the fidelity inequality); both meet  $4/10^7$  and  $10^{-6}$  at both
signs. Tier (i), crude majorant  $t \leq J_0G(R) = 37/6250000$ :
D_i=36133347268157653748322/1525878906463907516326874161 (~2.3680e-5; exact, forward)
and  $D_i \leq 236800700911401877571290493559933441/10^{40}$  (~2.3680e-5; directed, reverse),
failing both targets and retained. Both tiers scale linearly  $(D(\tau)/D(\tau/100))$  in
 $[99, 101]$ . Consequences:  $|\omega(W)| \leq D$ ,  $|\omega(W^2) - 1/4| \leq D/2$ , with  $m^2 \leq D^2$  charged.
The on-site cutoff is removed for the ground vector itself in each fixed box
(uniform-gap argument), and the bound passes by local trace-norm convergence to every
subsequential limit of AQ1's centered whole-star construction, including AQ1's chosen
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state: uniform local closeness, not uniqueness, whole-sequence convergence or a rate in N . No Euclidean node certificate, interaction-induced shift or its sign, first-order coefficient, uniform Wilson, weak-coupling or continuum claim follows.

The rest of this section derives that statement. The limitations recorded by the gate close the section (Section 2.10).

2.2 Product ordering and the exact split

Fix a centered whole-star box Λ_N , $N \geq 2$, and let $\psi = e^{-C}\Omega_0$ be the AM2 ground vector of (4). Because the creations commute and square to zero, e^{-C} factorizes, and the factors whose supports meet R can be placed last:

$$e^{-C} = \prod_I (1 - \hat{c}_I), \quad \psi = \prod_{I \cap R \neq \emptyset} (1 - \hat{c}_I) \psi_{\text{out}}, \quad \psi_{\text{out}} = \prod_{I \cap R = \emptyset} (1 - \hat{c}_I) \Omega_0 = \Omega_R \otimes \varphi_{\text{out}}. \quad (8)$$

The factorization of ψ_{out} holds because every creation disjoint from R acts as the identity on $\mathcal{H}_R$. In the remaining product, any two creations with overlapping supports multiply to zero. Pairwise disjoint supports that each meet the two-site set R number at most two: one contains 0 but not e_z , the other e_z but not 0. Hence

$$\delta := \psi - \psi_{\text{out}} = - \sum_{I \cap R \neq \emptyset} \hat{c}_I \psi_{\text{out}} + \sum_{\substack{I \ni 0, J \ni e_z \\ I \cap J = \emptyset}} \hat{c}_I \hat{c}_J \psi_{\text{out}}. \quad (9)$$

The second sum is the two-creation term. In the reverse form, with C_R the sum of creations meeting R , $C_R^3 = 0$ and $\delta = -C_R \psi_{\text{out}} + \frac{1}{2} C_R^2 \psi_{\text{out}}$.

Every term of δ contains a creation that is excited at some $x \in R$, and $P_x Q_x = 0$. Therefore

$$(P_R \otimes 1)\psi = \psi_{\text{out}}, \quad \langle \psi_{\text{out}}, \delta \rangle = 0, \quad \|\psi\|^2 = \|\psi_{\text{out}}\|^2 + \|\delta\|^2. \quad (10)$$

For a creation meeting R , $\hat{c}_I \psi_{\text{out}} = c_I \otimes \Omega_{R \setminus I} \otimes (\langle \Omega_{I \setminus R} | \otimes 1) \varphi_{\text{out}}$, and the partial vacuum bra is a contraction. The sum includes *straddling* supports, which meet both R and its complement. With $t = \|c\|_a$ and $\|\psi_{\text{out}}\| = \|\varphi_{\text{out}}\|$,

$$e := \frac{\|\delta\|}{\|\psi_{\text{out}}\|} \leq \varepsilon := \sum_{I \cap R \neq \emptyset} \|c_I\| + \sum_{\substack{I \ni 0, J \ni e_z \\ I \cap J = \emptyset}} \|c_I\| \|c_J\| \leq 2t + t^2. \quad (11)$$

The anchored maximum enters only at the two sites of R . A per-site maximum summed over the volume would be extensive; it is a rejected control.

2.3 Two inequalities

Forward: explicit reduced density. Put $n = \|\psi_{\text{out}}\|$, $d = \|\delta\|$, $\xi = (1_R \otimes \langle \varphi_{\text{out}} |) \delta$ and $\sigma = \text{Tr}_{\text{out}} |\delta\rangle\langle\delta|$. Direct partial traces and (10) give, exactly,

$$\rho_{N,R} = \frac{n^2 P_R + |\Omega_R\rangle\langle\xi| + |\xi\rangle\langle\Omega_R| + \sigma}{n^2 + d^2}, \quad \langle \Omega_R, \xi \rangle = 0, \quad P_R \sigma = 0, \quad \text{Tr } \sigma = d^2. \quad (12)$$

Cauchy–Schwarz gives $\|\xi\| \leq nd$, and on $\text{span}\{\Omega_R, \xi\}$ the cross operator has trace norm $2\|\xi\|$. Hence

$$\|\rho_{N,R} - P_R\|_1 \leq \frac{2nd + 2d^2}{n^2 + d^2} = \frac{2e(1+e)}{1+e^2} \leq f(\varepsilon) := \frac{2\varepsilon(1+\varepsilon)}{1+\varepsilon^2}. \quad (13)$$

The function f increases on $[0, 1 + \sqrt{2}]$, and $f(\varepsilon) \geq 2$ once $\varepsilon \geq 1$, so the last step holds for every ε .

Reverse: vacuum overlap and fidelity. By (10), $\text{Tr}(\rho_{N,R} P_R) = n^2/(n^2 + d^2) = 1/(1 + e^2)$ exactly. The pure-state mixture inequality admitted in AT4, $\|\rho - |\Omega\rangle\langle\Omega|\|_1 \leq 2\sqrt{1 - \langle\Omega, \rho\Omega\rangle}$, then gives

$$\|\rho_{N,R} - P_R\|_1 \leq \frac{2e}{\sqrt{1 + e^2}} \leq g(\varepsilon) := \frac{2\varepsilon}{\sqrt{1 + \varepsilon^2}} \leq f(\varepsilon). \quad (14)$$

The last inequality is $(1 + \varepsilon)^2 \geq 1 + \varepsilon^2$. Before comparing the producers, the skeptic re-derived g by a third route, purification plus partial-trace contractivity. Neither inequality estimates a norm of an outside product. The volume-dependent n and φ_{out} cancel in every ratio, and ε depends only on anchored sums at the two sites of R , which AM2 bounds independently of N and of the on-site cutoff.

2.4 The normalization trap

Both $\langle\psi, A\psi\rangle$ and $\|\psi\|^2 = n^2(1 + e^2)$ contain every creation in the box, and n^2 grows with the volume. Two shortcuts fail. The first keeps only creations meeting R in the numerator and normalizes by $1 + O(t^2)$ or by the full norm. This gives a volume-dependent value. Normalizing by the vacuum coefficient instead gives the global vacuum overlap, which decays with the volume. The second shortcut drops the outside creations everywhere. The value is then volume independent but wrong, because straddling creations contract their outside legs against φ_{out} . The only legitimate cancellation is inside the ratio e .

Exact three-site fixtures exhibit both failures. In the forward fixture, qubit sites r, o, o' with $R = \{r\}$ carry creations a on $\{r, o\}$ (straddling), b on $\{o\}$ and g on $\{o'\}$. Then

$$\rho_r = \frac{1}{1 + a^2 + b^2} \begin{pmatrix} 1 + b^2 & ab \\ ab & a^2 \end{pmatrix}, \quad \langle\psi, P_r\psi\rangle = (1 + g^2)(1 + b^2). \quad (15)$$

The decoupled creation g cancels from ρ_r but not from the unnormalized weight. At $a = \frac{1}{3}$, $b = \frac{1}{2}$ that weight runs through $\frac{5}{4}, \frac{29}{20}, \frac{5}{2}$ for $g = 0, \frac{2}{5}, 1$, while $\rho_r = \frac{1}{49} \begin{pmatrix} 45 & 6 \\ 6 & 4 \end{pmatrix}$. The outside creation b does not cancel: the off-diagonal vanishes when $b = 0$. The straddling creation enters e at first order but the off-diagonal only at second order. In the reverse producer's own fixture, naive restriction gives $e^2 = \frac{1}{2}$ instead of the actual $\frac{5}{8}$. There the outside creation *increases* the excitation of R , so naive restriction is neither exact nor a bound. With k decoupled outside sites the local overlap stays $\frac{8}{13}$, while the unnormalized weight is 2^k and the global vacuum overlap is $\frac{4}{13}2^{1-k}$. The excitedness premise $c_I \in \bigotimes_{x \in I} Q_x \mathcal{H}_x$ is also necessary: a creation with a vacuum component gives $\langle\psi_{\text{out}}, \delta\rangle = -\frac{5}{16} \neq 0$, and the split (10) fails.

2.5 Derived face counts and the first-order collection

The first-order collection is the $k = 0$ term of (4), $c_M^{(1)} = H_M^{-1} P_M V \Omega_0$. Five facts hold for every omitted face f . First, $\langle\Omega_0, W_f \Omega_0\rangle = 0$. Second, $W_f \Omega_0$ carries spin $\frac{1}{2}$ on the four distinct links of f and lies in the sector of its owner set M_f , with $|M_f| \geq 2$. Third, it is an eigenvector of H_0 with eigenvalue 24 in normalized units, that is 3 in units of α . Fourth, $\|W_f \Omega_0\|^2 = \frac{1}{4}$. Fifth, distinct faces share at most one link, so their vectors are orthogonal. Hence

$$c_M^{(1)} = -\frac{\tau}{72} \sum_{f: M_f=M} W_f \Omega_0, \quad \|c_M^{(1)}\| = \frac{|\tau|\sqrt{n_M}}{144}, \quad (16)$$

where n_M counts the faces with owner set M . The amplitude is $(\tau/3)/24$ in normalized units and $(\tau/24)/3$ in units of α . Mixing them would give $\tau/576$ or $\tau/9$; both are rejected controls.

The counts are derived from the I1 table (Table 1) by translation covariance. A face of class k anchored at b has owner set $b + K_k$, so it contains the factor u exactly when $b = u - d$ for some

$d \in K_k$. Therefore

$$\#\{f : u \in M_f\} = \sum_{d \in S} \#\{k : d \in K_k\} = 21 + 4 + 8 + 16 = 49. \quad (17)$$

Here 0 lies in all 21 omitted classes. The offset e_x lies in four: the two xy classes with $r = 3$ and the two xz classes with $r = 3$. The offset e_y lies in eight: the four xy classes with $q = 1$ and the four yz classes with $q = 1$. The offset e_z lies in all sixteen xz and yz classes. An owner set determines its anchor, its lexicographic minimum, and its class shape. The 49 faces therefore fall into the 15 owner sets of Table 4.

Table 4: Owner sets containing a factor $u = 0$, derived from the I1 table.

Shape	Owner sets containing 0	n_M each	Faces
$\{0, e_x\}$	$\{0, e_x\}, \{-e_x, 0\}$	1	2
$\{0, e_y\}$	$\{0, e_y\}, \{-e_y, 0\}$	3	6
$\{0, e_z\}$	$\{0, e_z\}, \{-e_z, 0\}$	10	20
$\{0, e_x, e_y\}$	three translates	1	3
$\{0, e_x, e_z\}$	three translates	2	6
$\{0, e_y, e_z\}$	three translates	4	12
Total	15		49

The faces meeting R number $49 + 49 - 16 = 82$. The 16 faces whose owner sets contain both 0 and e_z are the classes containing e_z , anchored at 0. Ten faces lie inside R , all with owner set $\{0, e_z\}$: the six xz classes with $r \leq 2$ and the four yz classes with $q = 0$, anchored at 0. The original W is one of them. The other 72 faces meeting R have straddling owner sets; of the 27 owner sets meeting R , 26 straddle. In a finite box only whole stars are retained, so every count is at most its bulk value. The explicit boxes Λ_2 and Λ_3 hold 64 and 216 stars (1344 and 4536 faces), with the maximum 49 reached at every bulk site. The product $4 \cdot 21 = 84$ is used only as a labelled bound. Both producers derived the counts before comparing them with the contract's candidates. A brute-force fine-lattice enumeration reproduced them without the class table, and source-mutation replays confirm that editing the table or copying a count aborts the run.

With orthogonality inside each owner set, the first-order anchored norm and the first-order sum over the cover obey

$$\|c^{(1)}\|_a \leq \frac{|\tau|}{144} (11 + 2\sqrt{3} + 3\sqrt{2} + 2\sqrt{10}) \leq \frac{49|\tau|}{144} =: t_1, \quad a_1 := \sum_{M \cap R \neq \emptyset} \|c_M^{(1)}\| \leq \frac{82|\tau|}{144}. \quad (18)$$

The admitted tier-(ii) values use the triangle forms t_1 and $82|\tau|/144$. The anchored norm remains an ℓ^1 sum over owner sets; replacing it by $\sqrt{49}$ or $\sqrt{82}$ is the rejected root- n misuse.

2.6 Two tiers at the cap

Tier (i): crude majorant. The fixed point gives $t \leq JG(t) \leq JG(R_a) \leq 28|\tau| \cdot \frac{148}{7} = 592|\tau|$, and only this t enters $\varepsilon_i = 2t + t^2$. At the cap, $t_i = 37/6250000$ and $\varepsilon_i = 462501369/39062500000000$.

Tier (ii): exact first order with a self-consistent remainder. Since G is convex and increasing and $t \leq R_a$,

$$\|c - c^{(1)}\|_a \leq J(G(t) - 16) \leq JtG'(R_a) \leq 352Jt, \quad t \leq t_1 + 352Jt \implies t \leq T := \frac{t_1}{1 - 352J}. \quad (19)$$

The remainder $\varrho := 352JT = T - t_1$ is charged at its positive majorant value and is never set to zero. Two tier mixtures are rejected: combining $c^{(1)}$ with the crude t , and using t_1 without

(19). The forward producer uses the contract’s global form $\varepsilon_{\text{fwd}} = 2T + T^2$. The reverse producer bounds the first-order sum over R directly by the 82 faces and charges the remainder at both sites, $\varepsilon_{\text{rev}} = a_1 + 2\rho + T^2$. Since $2T = 2t_1 + 2\rho$ and $a_1 < 2t_1$, $\varepsilon_{\text{rev}} < \varepsilon_{\text{fwd}}$, with ratio $\varepsilon_{\text{fwd}}/\varepsilon_{\text{rev}} \approx 1.19510$. The forward form double-counts the 16 faces that contain both sites.

At $\tau = \pm 10^{-8}$ the inputs are $J_0 = 7/25000000$, $352J_0 = 77/781250$, $t_1 = 49/14400000000$, $T = 49/14398580736$, $\rho = 3773/11248891200000000 \approx 3.35 \times 10^{-13}$ and $a_1 = 41/7200000000$. They give

$$\varepsilon_{\text{fwd}} = \frac{1411060914529}{207319127211110301696}, \quad \varepsilon_{\text{rev}} = \frac{461213409663624001}{80984034066839961600000000} \approx 5.6951153 \times 10^{-9}. \quad (20)$$

The resulting bounds follow. The forward values are exact rationals, the reverse values directed rational upper bounds, and the decimals readable summaries:

$$D_i^{\text{fwd}} = \frac{36133347268157653748322}{1525878906463907516326874161} \approx 2.36803504623 \times 10^{-5}, \quad (21)$$

$$D_i^{\text{rev}} \leq \frac{236800700911401877571290493559933441}{10^{40}} \approx 2.3680070 \times 10^{-5}, \quad (22)$$

$$D_{ii}^{\text{fwd}} = \frac{585079838465912592144137406066050}{42981220507576537932303142777593983768257} \approx 1.36124528703 \times 10^{-8}, \quad (23)$$

$$D_{ii}^{\text{rev}} \leq \frac{113902305553947264976096174312887}{10^{40}} \approx 1.1390231 \times 10^{-8}. \quad (24)$$

Every value is the same at both signs, because only $|\tau|$ enters. Table 5 compares them with the targets.

Table 5: AV1 state bounds at $\tau = \pm 10^{-8}$ against the preregistered target $4/10^7$ and the secondary comparison 10^{-6} .

Bound on $\ \rho_R - P_R\ _1$	Approximate value	$\leq 4/10^7$	$\leq 10^{-6}$
Round31 reset, $2\sqrt{49 \tau /3}$	$8.08290376865 \times 10^{-4}$	no	no
Tier (i), forward exact	$2.36803504623 \times 10^{-5}$	no	no
Tier (i), reverse directed	2.3680070×10^{-5}	no	no
Tier (ii), forward exact (both inequalities)	$1.36124528703 \times 10^{-8}$	yes	yes
Tier (ii), reverse directed (fidelity)	1.1390231×10^{-8}	yes	yes

Tier (i) fails both targets. It is retained as a limited-tier value and is not retuned. Iterating the fixed-point bound once (forward) or four times (reverse) lowers it only to about 1.79×10^{-5} . Tier (ii) meets the target with margins 29.38 (forward) and 35.12 (reverse). The two tier-(ii) numbers have different backing. D_{ii}^{fwd} is certified by both inequalities, because $g(\varepsilon_{\text{fwd}}) \leq f(\varepsilon_{\text{fwd}})$. D_{ii}^{rev} needs the fidelity inequality: the forward formula evaluated at ε_{rev} exceeds it. Both are valid upper bounds on the same quantity. The gate binds D_{ii}^{fwd} as the state premise for AV2.

2.7 From the cutoff box to a chosen subsequential limit

Cutoff-vector removal. AM2 removed the on-site cutoff for the eigenvalues and the gap only; a reduced density needs the ground vector. Fix N and let $Q_L = \bigotimes_{x \in \Lambda_N} 1_{[0,L]}(h_x)$. This is a finite-rank projection that commutes with H_0 and increases strongly to I . Let ψ_L be the normalized ground vector of $Q_L H Q_L$. AM2 applies in each cutoff space with rank-independent constants. For $L \geq 24$, $Q_L W_f \Omega_0 = W_f \Omega_0$ and $c_L^{(1)} = c^{(1)}$; for smaller L each face vector is kept or annihilated, so the counts only decrease. Hence $\|\rho_{L,R} - P_R\|_1 \leq D$ for every L . Monotone convergence of $\langle Q_L \psi, H_0 Q_L \psi \rangle$ and boundedness of V in the fixed box give $E_0 \leq E_{0,L} \rightarrow E_0$. The untruncated gap $E_1 - E_0 \geq \frac{1}{2}$ (AM2 §6) then gives the Eckart-type bound

$$1 - |\langle \psi, \psi_L \rangle|^2 \leq \frac{E_{0,L} - E_0}{E_1 - E_0} \leq 2(E_{0,L} - E_0) \rightarrow 0. \quad (25)$$

The pure-state trace distance is $2\sqrt{1 - |\langle \psi, \psi_L \rangle|^2}$, and partial trace is trace-norm contractive, so $\rho_{L,R} \rightarrow \rho_{N,R}$. The closed trace-norm ball keeps $\|\rho_{N,R} - P_R\|_1 \leq D$ for the untruncated finite-volume ground state. The reverse route reaches the same conclusion from a Rayleigh quotient with the uniform cutoff gap $\frac{1}{2}$. The gap is essential, because eigenvalue convergence alone does not control vectors. In $\text{diag}(0, 0, 1)$ the cutoff “ground” e_2 has the exact ground energy yet is orthogonal to e_1 ; the alternating family $\text{diag}(0, 1/n)$, $\text{diag}(1/n, 0)$ has converging eigenvalues and alternating ground vectors.

Passage to a chosen subsequential limit. AQ1 extracts subsequences N_k along which every local density converges in trace norm. For any such subsequence with limit density ρ_R ,

$$\|\rho_R - P_R\|_1 \leq \|\rho_R - \rho_{N_k,R}\|_1 + D \rightarrow D. \quad (26)$$

The quantifier is *every* subsequential limit of AQ1’s centered whole-star construction at either sign, AQ1’s chosen state included. Two such limits lie within $2D$ of each other on R , and the checker exhibits two distinct limits inside the same ball. This is uniform local closeness, not uniqueness, whole-sequence convergence or a rate in N . Trace-norm convergence is needed: a weak-* limit can lose mass, as $(1 - D/2)|e_0\rangle\langle e_0| + (D/2)|e_n\rangle\langle e_n|$ shows, and AQ1’s energy tightness excludes that loss.

2.8 Consequences, scaling and AV2 feasibility

Trace duality with $\|W\| \leq 1$ bounds the mean. For the second moment, a trace-zero difference has expectation at most half its trace norm against an effect, and $0 \leq W^2 \leq I$. Hence

$$|\omega_\tau(W)| \leq D, \quad |\omega_\tau(W^2) - \tfrac{1}{4}| \leq \tfrac{D}{2}, \quad m_\tau^2 \leq D^2. \quad (27)$$

At tier (ii) these bounds are about 1.3612×10^{-8} and 6.806×10^{-9} . The mean-square charge $m_\tau^2 \leq D^2$ is kept in every later budget. Neither the value nor the sign of $\omega_\tau(W)$ is claimed. A candidate first-order mean $|\tau|/144 \approx 6.94 \times 10^{-11}$ is consistent with the bound but is neither implied nor claimed; it belongs to AW1.

Every bound is increasing in $|\tau|$, so the cap values bound every smaller coupling. The preregistered scaling test compares the cap with 10^{-10} :

$$\frac{D(10^{-8})}{D(10^{-10})} \approx \begin{cases} 100.0098 & \text{tier (ii), forward,} \\ 100.0117 & \text{tier (ii), reverse,} \\ 100.0015 & \text{tier (i), forward,} \\ 100.0003 & \text{tier (i), reverse,} \end{cases} \quad \frac{2\sqrt{49 \cdot 10^{-8}/3}}{2\sqrt{49 \cdot 10^{-10}/3}} = 10. \quad (28)$$

All tier ratios lie in $[99, 101]$, so the bounds are linear in τ . The Round31 bound has ratio exactly 10, and relabelling it as linear is a rejected control. At the cap, tier (ii) is about 5.9×10^4 times smaller than the Round31 reset bound.

With a directed Machin bracket for π , the AV2 feasibility function $F(D) = 2(D + D^2) + 49 \cdot 10^{-8}/\pi$ satisfies $F(D_{\text{ii}}^{\text{fwd}}) \leq 1.8320 \times 10^{-7}$, $F(D_{\text{ii}}^{\text{rev}}) \leq 1.7876 \times 10^{-7}$ and $F(4/10^7) \leq 9.5597 \times 10^{-7}$. It meets 10^{-6} at 4.22×10^{-7} and fails at 4.23×10^{-7} and at D_{i} . This is arithmetic on the contract statement only; AV1 does not prove the window lemma.

2.9 Error ledger, controls and replays

The forward producer exports the preregistered error terms at tier (ii).

- **am2_remainder:** $\varrho \approx 3.35 \times 10^{-13}$; not applicable to tier (i), which bounds the whole collection directly.

- **two_creation**: $T^2 \approx 1.16 \times 10^{-17}$, inside ε .
- **straddling**: inside the $2t$ term; at first order the 72 straddling faces contribute at most $|\tau|/2$ to the sum over R .
- **density**: the σ and $d^2 P_R$ part of D , $2\varepsilon^2/(1+\varepsilon^2) \approx 9.3 \times 10^{-17}$.
- **onsite_cutoff_vector**: no numeric cost; it is an exact per-box limit.
- **arithmetic**: no numeric cost; every bound is an exact rational, and the only enclosures are directed upward ($e^{1/8} < 8/7$, $e^{8t} \leq 1/(1-8t)$, integer square-root brackets).

The reverse producer exports no six-name ledger. The skeptic's mapping supplies it: **am2_remainder** is 2θ , **density** is not applicable to the fidelity route, and **arithmetic** is outward rounding of at most 10^{-40} on D .

The contract listed 25 controls. The reverse checker, with 54 exact checks, implements all 25 as damaging mutations that must raise explicit exceptions. The forward checker, with 43 checks, implements 22 of them as mutations. The other three are positive exact checks that match their prospective semantics: **no_priority_or_continuum_claim**, **first_order_face_enumeration** and **av2_feasibility_threshold**. The forward report's sentence claiming that every control is a mutation is therefore not admitted. The skeptic's pre-comparison package (71 exact checks) was hash-recorded before either producer was opened. The post-comparison review adds 39 checks, including 12 source-mutation replays. Both producers reproduce their frozen outputs byte for byte under normal and optimized Python. The reverse input inventory is exactly **AGENTS.md**, the contract and its shared premises, as enforced by the freezing tool.

2.10 Limitations recorded by the gate

The AV1 gate records the following limitations verbatim.

- Only the AM2/AQ1 zero-selected patterned family, the cover $R=\{0,e_z\}$ and fixed spacing; nothing transfers to nonzero selected triples (non-Haar P_R), uniform Wilson theory, weak coupling or the continuum
- Upper certificates only: no lower bound on $||\rho_{R-P_R}||_1$; value and sign of $\omega(W)$, the first-order coefficient $1/144$ and any first-order vanishing are not claimed (AW1)
- Uniform local closeness of all AQ1 subsequential limits (pairwise within 2D on R), not uniqueness, whole-sequence convergence or a rate in N
- Inherited without re-proof: AM2 fixed point, multilinear majorant, cutoff-uniform and untruncated gap $1/2$ (AM2 section 6); AQ1 trace-norm compactness and diagonal extraction; the I1 dictionary
- Product-ordering split and normalization-trap location are shared panel premises; independence limited to the inequality, constants, enumeration, cutoff and passage steps; all agents are correlated model agents
- Sign-blind: the $\tau \approx 10^{-8}$ evaluation replays the same $|\tau|$ formula
- Tier (i) fails $4/10^{-7}$ and 10^{-6} and is retained as a limited-tier value; AT4 square-root bound not used
- $D_{ii}(\text{rev}) \sim 1.1390e-8$ rests on the fidelity inequality and the labelled 82-face refinement; $D_{ii}(\text{fwd}) \sim 1.3612e-8$ is certified by both inequalities
- Forward check.py implements 22 of 25 contract controls as exception-raising mutations; **no_priority_or_continuum_claim**, **first_order_face_enumeration** and **av2_feasibility_threshold** are positive exact checks; the forward sentence "Every control is a damaging mutation that must raise an explicit exception." is not admitted
- Reverse exports no six-name error ledger; the skeptic mapping in av1.md (N2) supplies it
- Reverse remark "no parity argument removes a first-order mean" is not admitted (AW1 exclusion)
- Forward attribution of a " $2(E_{\{0,L\}}-E_0)/\text{gap form}$ " to the contract is incorrect wording; the Eckart inequality itself is correct
- Both producers disclose undeclared protocol/code-style reads (tools README and freeze.py, AM2 and AT5 check.py); none carries premise weight
- Scientific priority unverified

Full derivations and reviewed implementations: [AV1 forward](#), [AV1 reverse](#), [independent model-agent review](#) and the [frozen contract](#). AV1 by itself certifies no Euclidean datum; Section 3 uses its forward tier-(ii) value.

3 AV2: a window certificate at the original cap

Contribution HNM-C-AV2. After the AV1 review, the AV2 contract fixed $\tau = +10^{-8}$, $s = 1$ and the unchanged absolute point-error target 10^{-6} , with $\tau = -10^{-8}$ as a control. Its recorded title is “Hruday window-kernel certificate for the centered Wilson correlation at the original cap”. The contract binds the AV1 forward tier-(ii) value (23) as the state premise and forbids tier (i) and the Round31 square-root bound. It freezes the C^2 window, its Fourier sign convention and its constants as shared panel premises, and it registers the C^1 window as a preview-only control. The forward producer proves the constants by half-line transforms. The reverse producer, under premise isolation, proves them by residues.

3.1 Reviewed scope

The [AV2 gate](#) records the verdict `accepted_within_scope`. Its sub-label is `reference_unresolved`, and its accepted statement, quoted verbatim below, is the scope of every claim in this section.

In the AM2/AQ1 zero-selected patterned family (SU(2) Kogut-Susskind form on Z^3 at fixed spacing, coarse 24-link factors, selected triple exactly (0,0,0) with Haar product reference, whole stars $\phi_b = -(\tau/3) \sum W_f$; original xz Wilson loop W with complete cover $R = \{0, e_z\}$ and seven incident stars; physical clock $s = \alpha \hbar / E$, $\theta = \alpha \hbar / G$, $G = H / \alpha$), at the original cap $\tau = +10^{-8}$ and, as a replay of the same $|\tau|$ formula at the mirrored coupling, $\tau = -10^{-8}$, every AQ1 subsequential state of the centered whole-star construction (each chosen state separately; no uniqueness, whole-sequence convergence or rate in N) has centered Euclidean Wilson correlation $C(s) = \langle \chi, e^{-sG} \chi \rangle$, $\chi = (\pi(W) - \Omega(W)) \Omega$, satisfying at the single preregistered node $s=1$: $|C(1) - d| \leq r$ with the exact rational datum $d = 497870683678639429793424156500617766317 / (4 \cdot 10^{40})$ ($\sim 0.012446767091965986$, the midpoint of the directed enclosure of $e^{-3/4}$) and $r \leq R = 1831967503411879425147166810021607 / 10^{40}$ ($\sim 1.83196750341e-7$, a directed upper bound of both producers' exact radii, which differ by $2.52e-41$ and exceed the analytic bound by less than 10^{-40}), so $C(1)$ in $[497863355808625782275723567833377679889 / (4 \cdot 10^{40}), 99575602309730615462224949033571570549 / (8 \cdot 10^{39})] \sim [0.0124465838952156, 0.0124469502887164]$, meeting the unchanged 10^{-6} target (margin 5.4586). The bound is $E = M_0(D + D^2) + k M_1 = 2(D + D^2) + 49|\tau|s/\pi$ plus the arithmetic half-width $1/(4 \cdot 10^{40})$: the frozen C^2 window $g(x) = e^{-sx}$ ($x \geq 0$), $e^{-sx}(1 - 2sx + 2s^2x^2)$ ($x < 0$) has $\hat{g} = 4s^3/(\pi(s - i\theta)^3(s + i\theta))$ in L^1 with $M_0 = \|\hat{g}\|_1 = 2$, $M_1 = 4s/\pi$, $M_2 = 2s^2$, proved by half-line transforms (forward) and by residues (reverse); the window lemma $C(s) = \int \hat{g} c$ uses L^1 inversion, Fubini and AQ1 nonnegativity only (no AQ2 gap); the seven-star relative-unitary Duhamel slope $k = 49|\tau|/4$ holds for every real θ uniformly in the box; the state term uses the AV1-admitted forward tier-(ii) bound $D = 585079838465912592144137406066050/42981220507576537932303142777593983768257$ ($\sim 1.3612e-8$) by trace duality with no $D/2$ effect refinement, and $m^2 \leq D^2$ is charged. The free reference $e^{-3/4}$ lies inside the interval (`reference_unresolved`): no interaction shift, sign or coefficient of $C(s)$ is claimed. Retained failures at unchanged τ and s : the AT4 Poisson certificate at $L = 10^4$ (radius $\sim 8.4152e-4$) and the Poisson floor, at least $1.2650882233e-6$ for every cutoff L even with $D \rightarrow 0$ (at least $1.2787e-6$ with the admitted D), whose truncated absolute first moment $(s/\pi) \log(1 + L^2/s^2)$ diverges against the window's finite $4s/\pi$, both insufficient; the tier-(i) window radius $\sim 4.7518e-5$ is a retained limited-tier value and the C^1 window (radius $\sim 1.7330e-7$) is a preview only. The analytic formula crosses 10^{-6} at s^* in $[6.236863446, 6.236863447]$ (~ 6.2368634460); only $s=1$ is certified. No grid or $[0, 128]$ claim, resolved interaction shift, uniform Wilson, weak-coupling, continuum, uniqueness or rate claim follows.

3.2 Spectral set-up

Let η be the spectral measure of G_τ in the vector χ_τ of (7). AQ1 gives $G_\tau \geq 0$, so η is a positive measure on $[0, \infty)$ with finite mass $\eta([0, \infty)) = \|\chi_\tau\|^2 = \omega_\tau(W^2) - m_\tau^2 \leq 1$, and

$$c_\tau(\theta) = \int_{[0, \infty)} e^{i\theta x} d\eta(x), \quad C_\tau(s) = \int_{[0, \infty)} e^{-sx} d\eta(x). \quad (29)$$

Only this nonnegativity enters. The AQ2 gap, which would restrict $\text{supp } \eta$ to $\{0\} \cup [1/16, \infty)$, is not used. The free reference has $\eta_0 = \frac{1}{4}\delta_3$ and the functions (3).

3.3 The window and its transform

With the frozen convention $\hat{g}(\theta) = (2\pi)^{-1} \int_{\mathbb{R}} g(x) e^{-i\theta x} dx$ and $s > 0$, the window and its transform are

$$g(x) = \begin{cases} e^{-sx}, & x \geq 0, \\ e^{sx}(1 - 2sx + 2s^2x^2), & x < 0, \end{cases} \quad \hat{g}(\theta) = \frac{4s^3}{\pi(s - i\theta)^3(s + i\theta)}, \quad |\hat{g}(\theta)| = \frac{4s^3}{\pi} (s^2 + \theta^2)^{-2}. \quad (30)$$

For $y = -x > 0$ the left branch is $e^{-sy}(1 + 2sy + 2s^2y^2) > 0$. So g is positive and continuous, lies in $L^1 \cap C_0$ with $\|g\|_1 = 8/s$, and is C^2 at the origin: both sides give the value, first and second derivatives $(1, -s, s^2)$. The third derivative jumps by $-8s^3$. That jump fixes the θ^{-4} decay $(2\pi)^{-1} 8s^3/\theta^4 = 4s^3/(\pi\theta^4)$. As a check, $\hat{g}(0) = 4/(\pi s) = \|g\|_1/(2\pi)$.

Forward route: half-line transforms. Put $a = s - i\theta$ and $b = s + i\theta$, so $a + b = 2s$ and $ab = s^2 + \theta^2$. Laplace transforms of the two branches give

$$2\pi\hat{g}(\theta) = \frac{1}{b} + \frac{1}{a} + \frac{2s}{a^2} + \frac{4s^2}{a^3} = \frac{8s^3}{a^3b}, \quad (31)$$

because $(2s - a)(a^2 + 2sa + 4s^2) = 8s^3 - a^3$. For a general left polynomial $1 + c_1y + c_2y^2$ the same calculation gives

$$2\pi\hat{g}(\theta) a^3b = (2s - c_1)a^2 + (2sc_1 - 2c_2)a + 4sc_2. \quad (32)$$

C^1 matching ($c_1 = 2s$) removes the θ^{-2} term, and C^2 matching ($c_2 = 2s^2$) removes the θ^{-3} term. The Poisson kernel is the kinked member $c_1 = c_2 = 0$, with $\hat{g} = s/(\pi(s^2 + \theta^2))$. Its θ^{-2} tail is exactly why its absolute first moment diverges. The C^1 window is $c_1 = 2s$, $c_2 = 0$, with $\hat{g} = 2s^2/(\pi a^2b)$.

Reverse route: residues. The reverse producer starts from the target. The only dynamical input is a real-time bound linear in $|\theta|$, so $C(s)$ must be written as $\int K(\theta)c(\theta) d\theta$ with $\int K(\theta)e^{i\theta x} d\theta = e^{-sx}$ on $x \geq 0$. Values on $x < 0$ are unconstrained. Exactness on $x \geq 0$ requires exactly one simple pole at $\theta = is$ in the upper half-plane. Placing every other pole at $\theta = -is$ gives the family

$$K_n(\theta) = \frac{(2s)^n}{2\pi(s + i\theta)(s - i\theta)^n}, \quad |K_n(\theta)| \sim |\theta|^{-(n+1)}, \quad (33)$$

whose normalization comes from the residue at is . Finite M_0 , M_1 and M_2 force $n = 3$: $n = 1$ is Poisson and $n = 2$ the C^1 window. The residue at the triple pole reproduces $e^{sx}(1 - 2sx + 2s^2x^2)$ on $x \leq 0$, and only then was the result compared with the frozen window. The skeptic added two further routes: the annihilator identity $(\frac{d}{dx} + s)(\frac{d}{dx} - s)^3g = -8s^3\delta_0$ with Wallis integrals, and a Leibniz expansion of the lower residues.

Constants. All four routes give

$$M_0 = \|\hat{g}\|_1 = 2, \quad M_1 = \int_{\mathbb{R}} |\theta| |\hat{g}(\theta)| d\theta = \frac{4s}{\pi}, \quad M_2 = \int_{\mathbb{R}} \theta^2 |\hat{g}(\theta)| d\theta = 2s^2. \quad (34)$$

The forward producer reads them from exact antiderivatives, each a rational function of θ plus $B \arctan(\theta/s)$, whose rational parts vanish at $\pm\infty$. The reverse producer uses the residue at is for M_0 and M_2 . For M_1 it uses a keyhole contour, $\int_0^\infty \theta(s^2 + \theta^2)^{-2} d\theta = -\sum_{\pm is} \text{Res}[z \log z / (z^2 + s^2)^2] = 1/(2s^2)$. Note that $\int \hat{g} = g(0) = 1$ while $\|\hat{g}\|_1 = 2$. The transform is complex, not a positive mass-one kernel, and charging the state term with mass one would under-charge it by half; that mutation is a rejected control.

3.4 Window lemma

L^1 inversion gives $g(x) = \int \hat{g}(\theta) e^{i\theta x} d\theta$ for almost every x . Both sides are continuous, so equality holds at every real x . Since $\iint |\hat{g}(\theta)| d\theta d\eta(x) = M_0 \eta([0, \infty)) < \infty$, Fubini gives

$$C_\tau(s) = \int_{[0, \infty)} g d\eta = \int_{\mathbb{R}} \hat{g}(\theta) c_\tau(\theta) d\theta, \quad C_0(s) = \int_{\mathbb{R}} \hat{g}(\theta) \frac{1}{4} e^{3i\theta} d\theta = \frac{1}{4} g(3) = \frac{1}{4} e^{-3s}. \quad (35)$$

Both hypotheses matter, and each is executed as a damaging control. A spectral atom at $x = -1$ would be read as $g(-1) = e^{-s}(1 + 2s + 2s^2)$, which is $5/e$ at $s = 1$, instead of the heat value e ; the window under-reads every negative atom. The mirrored convention $\theta \rightarrow -\theta$ evaluates $g(-3)/4 = 25e^{-3}/4$ on the free atom. A window matched to e^{-sx} only above an assumed gap is rejected, because only $G_\tau \geq 0$ is used. Fourier inversion and Fubini's theorem are cited, not machine-checked; the checkers verify the transform algebra, the modulus, the antiderivatives and inversion at $x = 0$ exactly.

3.5 Real-time comparison

The comparison restates AT4 with its domain argument. In each centered box Λ_N , $N \geq 2$, split $G_N = A_N + B_N$, where $B_N = \sum_{b \in R-S} V_b$ takes all seven incident stars whole. Then $\|B_N\| \leq 7 \cdot 7|\tau|/8 = 49|\tau|/8$, and A_N contains the free evolution on R and everything disjoint from R . On the common domain the relative unitary $U(\theta) = e^{i\theta G_N} e^{-i\theta A_N}$ has the bounded, strongly continuous derivative $i e^{i\theta G_N} B_N e^{-i\theta A_N}$. Vectorwise integration gives $\|U(\theta) - I\| \leq |\theta| \|B_N\|$. Only then is the bounded W inserted:

$$\left\| \alpha_\theta^N(W) - \alpha_\theta^0(W) \right\| \leq 2|\theta| \|B_N\| \leq k|\theta|, \quad k = \frac{49|\tau|}{4}. \quad (36)$$

The factor two is necessary. The exact fixture $U = \text{diag}(u, \bar{u})$ with $u = (3 + 4i)/5$ and $X = \sigma_x$ gives $\|U X U^* - X\|^2 / \|U - I\|^2 = 16/5$. The slope is local: an extensive star norm would grow like $(2N)^3$ and is rejected. AQ1's local norm convergence of the dynamics on compact intervals passes (36) to the actual AQ evolution at every real θ . Adding and subtracting the free evolution in the actual state, and then the local reference state, gives

$$|c_\tau(\theta) - c_0(\theta)| \leq k|\theta| + D + m_\tau^2 \leq k|\theta| + D + D^2 \quad (\theta \in \mathbb{R}). \quad (37)$$

The state term uses full trace duality. The operator $W \alpha_\theta^0(W)$ has free expectation $\frac{1}{4} e^{3i\theta}$, which is not real at $\theta = \pi/6$. It is therefore not an effect, and the $D/2$ refinement of AT4 is not applied; exact 2×2 fixtures show that the refinement fails for such operators.

3.6 Certificate at the cap

Subtract the two halves of (35) and integrate (37) against $|\hat{g}|$ over the whole real line:

$$|C_\tau(s) - C_0(s)| \leq M_0(D + D^2) + kM_1 = 2(D + D^2) + \frac{49|\tau|^s}{\pi} =: E(s). \quad (38)$$

There is no real-time cutoff and no tail. Enclose $e^{-3}/4$ outward as $[f^-, f^+]$ and export $d = (f^- + f^+)/2$ and $r = E^+(1) + (f^+ - f^-)/2$, where E^+ uses a lower bound π^- in the dynamics term. Then $|C_\tau(1) - d| \leq |d - e^{-3}/4| + |e^{-3}/4 - C_\tau(1)| \leq r$. At $\tau = +10^{-8}$, $s = 1$ and $k = 49/400000000$, the preregistered terms are those of Table 6. The state term is exactly

$$M_0D = \frac{1170159676931825184288274812132100}{42981220507576537932303142777593983768257}. \quad (39)$$

The dynamics term is about 85% of the radius.

Table 6: AV2 radius at $\tau = +10^{-8}$, $s = 1$. Decimals are rounded up; exact rationals are in the frozen outputs.

Preregistered term	Formula	Outward allowance
state	M_0D	$2.72249057406 \times 10^{-8}$
mean_square	M_0D^2	$3.70597746291 \times 10^{-16}$
kernel_dynamics	$kM_1 = 49 \tau /\pi^-$	$1.55971844231 \times 10^{-7}$
arithmetic	$(f^+ - f^-)/2$	2.5×10^{-41}
Complete radius	common outward bound R	$1.83196750342 \times 10^{-7}$

The exported datum, the admitted radius and the certified interval are

$$d = \frac{497870683678639429793424156500617766317}{4 \cdot 10^{40}} \approx 0.012446767091965986, \quad (40)$$

$$r \leq R = \frac{1831967503411879425147166810021607}{10^{40}} \approx 1.832 \times 10^{-7} < 10^{-6}, \quad (41)$$

$$C_{10^{-8}}(1) \geq \frac{497863355808625782275723567833377679889}{4 \cdot 10^{40}} > 0.0124465838952156, \quad (42)$$

$$C_{10^{-8}}(1) \leq \frac{99575602309730615462224949033571570549}{8 \cdot 10^{39}} < 0.0124469502887164. \quad (43)$$

The short decimals widen the exact endpoints outward. The target margin is $10^{-6}/R \approx 5.4586$. The numerator of d is $e^{-3} \cdot 10^{40}$ rounded to the nearest integer, m say. Both producers enclose e^{-3} in $[m - 1, m + 1]/10^{40}$, which gives the arithmetic half-width $1/(4 \cdot 10^{40})$. The two producers' exact radii differ by 2.52×10^{-41} , because they round the dynamics term differently. The forward producer divides by its Machin lower bound for π ; the reverse producer rounds $49|\tau|/\pi^-$ up to the 10^{-40} grid. Both are valid directed bounds, each exceeding the analytic $E(1)$ by less than 10^{-40} . The gate therefore quotes the common outward bound R , which also equals the skeptic's pre-comparison radius digit for digit.

At $\tau = -10^{-8}$ the same $|\tau|$ formula gives the identical datum and radius. This is a replay at the mirrored coupling, not a second confirmation, and it is not evidence for the planned link-flip lemma.

The reference is unresolved. The exact free enclosure $[f^-, f^+]$, of width $5 \cdot 10^{-41}$, lies inside $[d - r, d + r]$. The datum is the free midpoint, and the interaction enters only through the radius. Centering the interval on the computable free value does not make the interacting correlation equal to it. No interaction-induced shift, sign or coefficient of $C_\tau(s)$ is claimed, and AV2 is evidence neither for nor against first-order vanishing. For scale, the radius is about 2600 times the size $|\tau|/144$ of the candidate first-order Wilson mean.

3.7 Retained failures and the crossover

Table 7 lists the certificates evaluated at unchanged $\tau = 10^{-8}$ and $s = 1$ alongside the admitted one. The AT4 Poisson certificate at $L = 10^4$ is reproduced to 10^{-15} and remains insufficient. With $D \rightarrow 0$ and the tail $s/(\pi L)$, the Poisson dynamics-plus-tail cost obeys, for every $L > 0$,

$$\frac{ks}{\pi} \log\left(1 + \frac{L^2}{s^2}\right) + \frac{s}{\pi L} \geq \frac{s}{\pi} \left[2k \log \frac{L}{s} + \frac{1}{L}\right] \geq \frac{2ks}{\pi} \left(1 + \log \frac{1}{2ks}\right) \geq 1.2650882233 \times 10^{-6}, \quad (44)$$

with the minimum near $L = 1/(2k) \approx 4.08 \times 10^6$. The value at $L = 4081633$ is at most $1.265088224 \times 10^{-6}$. With the admitted D added, no cutoff brings the Poisson certificate below 1.2787×10^{-6} . No choice of L meets 10^{-6} at $s = 1$. This proves the Round31 planning observation of Section 1.2 and retains it as an insufficient outcome.

The reason is the kernel’s first moment:

$$\int_{-L}^L |\theta| \frac{s}{\pi(s^2 + \theta^2)} d\theta = \frac{s}{\pi} \log\left(1 + \frac{L^2}{s^2}\right) \xrightarrow{L \rightarrow \infty} \infty, \quad \int_{-L}^L |\theta| |\hat{g}(\theta)| d\theta = \frac{4s}{\pi} \frac{L^2}{s^2 + L^2} \uparrow \frac{4s}{\pi}. \quad (45)$$

At $s = 1$ the Poisson partial moment already exceeds $4/\pi$ at $L = 8$, since $\log 65 > 4$. At $L = 10^{32}$ the Poisson dynamics term alone exceeds 5.7×10^{-6} . The window removes both the cutoff and its tail.

Table 7: Certificates at $\tau = 10^{-8}$, $s = 1$, target 10^{-6} . Only the last row is admitted as a certificate.

Kernel	State bound	Radius or lower bound	Outcome
Poisson, $L = 10^4$ (AT4)	Round31 reset	$\approx 8.4152 \times 10^{-4}$	insufficient
Poisson, $L = 10^4$	AV1 tier (ii)	$\approx 3.2563 \times 10^{-5}$	insufficient
Poisson, every L	$D \rightarrow 0$	$\geq 1.2650882233 \times 10^{-6}$	insufficient
Poisson, every L	AV1 tier (ii)	$\geq 1.2787 \times 10^{-6}$	insufficient
C^2 window	AV1 tier (i)	$\approx 4.7518 \times 10^{-5}$	limited tier
C^1 window	AV1 tier (ii)	$\approx 1.7330 \times 10^{-7}$	preview only
C^2 window	AV1 tier (ii)	$\leq 1.832 \times 10^{-7}$	certified

The C^1 window has $M_0 = 4/\pi$, $M_1 = 4s/\pi$ and $M_2 = \infty$. Its preview radius is *smaller* than the admitted one, so the rejection of a kernel switch after D was known guards a real post-hoc gain. It is not vacuous. The frozen C^2 window also keeps the finite $M_2 = 2s^2$ that a later second-order refinement would need; that refinement is excluded here.

The bound (38) is linear in s and crosses 10^{-6} at

$$s^* = \frac{\pi(10^{-6} - 2(D + D^2))}{49|\tau|} \in [6.236863446, 6.236863447]. \quad (46)$$

This describes the analytic formula, not evaluated nodes. The preregistration forbids post-hoc node selection, so only $s = 1$ is certified; there is no grid and no $[0, 128]$ claim at the cap. At s^* the free value, below 1.9×10^{-9} , is about 500 times smaller than the radius.

3.8 Pre-registration defect, controls and replays

The gate records one clerical defect. The frozen contract lists 24 control ids, but its pre-registration mirror lists 23; `c1_window_preview_only` is missing. Both producers iterate the full 24-id list and execute the missing control as damaging mutations. The skeptic recorded the defect before production, and the Jung/Pauli research assistant found it separately in its audit. The frozen contract is not amended. The contract-freezing tool now refuses a contract whose mirror differs from its control list (commit `2e1bfc8`, made after this contract froze).

Each producer's checker runs 50 exact checks. The forward checker records 68 rejected mutations and the reverse checker 80, and each of the 24 controls has at least one. The skeptic's pre-comparison package (69 checks) was hash-recorded before either producer's commit. The post-comparison review adds 63 exact checks, including 24 source-edit mutations and two unmutated baselines. All four producer replays, forward and reverse under normal and optimized Python, reproduce the frozen outputs byte for byte. Each producer also exports a reusable calculator in the AT5 style. It accepts exact inputs only and computes D from the AV1 tier-(ii) formula rather than taking it as an input. It admits only the C^2 window. At $\tau = 0$ its radius reduces to the arithmetic term alone. In the skeptic's probe each calculator rejected 20 malformed or out-of-domain calls.

3.9 Limitations recorded by the gate

The AV2 gate records the following limitations verbatim.

- Only the AM2/AQ1 zero-selected patterned family, the cover $R=\{0,e_z\}$, fixed spacing and the single node $s=1$ at $\tau=\pm 10^{-8}$; nothing transfers to nonzero selected triples (non-Haar P_R), uniform Wilson theory, weak coupling or the continuum
- Upper certificate only: the free value $e^{-3}/4$ lies inside the interval (reference_unresolved); no interaction shift, sign or coefficient of $C(s)$ is claimed, and AV2 is not evidence for or against first-order vanishing (AW1)
- Each AQ1 subsequential state separately; no uniqueness, whole-sequence convergence or rate in N
- Sign-blind: the $\tau=10^{-8}$ value is a replay of the same $|\tau|$ formula at the mirrored coupling, not a second confirmation
- Inherited without re-proof: AQ1 construction, $G \geq 0$, GNS implementation and local norm dynamics on compact theta intervals; AT4 F10-F12 (restated); AV1's D and its passage to AQ limits; the I1 dictionary; Fourier inversion and Fubini are cited, not machine-checked
- The window, transform and constants are frozen shared premises; the forward read the skeptic's triage and loop-2 response (moment integrals) and the reverse read skeptic/av1.md (radius formula); independence is limited to the derivation routes (half-line transforms, residues, skeptic annihilator-jump plus Wallis and Leibniz), the code and the controls
- The crossover s^* is that of the analytic formula, not a node certificate; no grid or $[0,128]$ claim; at s^* the free value ($<1.9e-9$) is about 500 times smaller than the radius
- The two producers' exact radii differ by $2.52e-41$ (forward: $49|\tau|/\pi_{lo}$ with its Machin π_{lo} ; reverse: $49|\tau|/\text{floor}_{\{10^{-40}\}}(\pi)$ rounded up to 10^{-40}); both are valid directed bounds; the admitted headline is the common outward bound $R=1831967503411879425147166810021607/10^{40}$
- Pre-registration mirror: contract controls list 24 ids but preregistration.controls_required.ids lists 23 (missing `c1_window_preview_only`); both producers executed the missing id as damaging mutations and iterate the 24-id list; non-blocking clerical defect; the frozen contract is not amended; `freeze_contract.py` now enforces equality (commit 2e1bfc8, after this freeze)
- Tier (i) window radius $\sim 4.7518e-5$ retained as a limited-tier value; the contract preview 3.6-4.7e-5 is slightly low; the forward exports only a labelled lower value $2D_i+49|\tau|/\pi_{hi}$
- Forward sentence "The centered shift is expected to be $O(\tau^2)$ (triage (c)3)" is not admitted (AW1 obligation)
- Forward reverse_promise_isolation control mutates the contract-declared lists only; the actual isolation is enforced by the reverse checker, `freeze.py` and `replay_loop.py` (27 files, verified)
- Forward sign-convention control reads the free atom through the inversion identity $g(+3)$; the reverse evaluates it through its residue engine
- Reverse reference constants 3 and $1/4$ come from its calculator and are validated against the contract model string; target and D are read from the hash-checked snapshot in both
- Both producers disclose undeclared protocol/code-style reads (tools README and `freeze.py`, AT5 `check.py`, their own direction's AV1 `check.py`); none carries premise weight
- Scientific priority unverified

Full derivations and reviewed implementations: [AV2 forward](#), [AV2 reverse](#), [independent model-agent review](#) and the [frozen contract](#). Sub-round 1 is complete; Section 4 records AW1.

4 AW1: parity, link flip and the first-order Wilson mean

Contribution HNM-C-AW1. The AW1 contract, recorded under the title “Hruday parity theorem, link-flip antisymmetry and the first-order Wilson mean with an itemized second-order remainder”, was frozen after the AV2 gate and before production. It keeps the model of Sections 2 and 3: the zero selected triple, whole stars under I1.5, the cover $R = \{0, e_z\}$ and both signs of $|\tau| \leq 10^{-8}$. Its preregistered target is $K_2^+ \tau \leq 1/288$ at the cap, half the first-order coefficient. It also freezes the coupling rule for AW2 before any constant is known. Both directions are paired. Beyond the shared premises, the forward producer received the skeptic’s triage, the skeptic’s loop-2 response and the historical and modern loop-2 responses. These already state the parity rule, the flip set and the rough structure of K_2 , so they are disclosed as shared panel premises for that route. The reverse producer received only `AGENTS.md`, the contract and its 29 shared premises, 31 files in all. It reconstructs the parity rule and the flip set from the character structure.

Notation: $\mathbb{E} = E_{\text{Haar}}$ is the Haar expectation of the reference, E is the flip set, and E_N is the ground energy of the box Λ_N . In units of α the finite-box Hamiltonian is $G_N = H_N/\alpha$ and the omitted interaction is $V = \sum_b V_b$ with $V_b = -(\tau/24) \sum_{f \in O_b} W_f$; $\kappa = (\lambda_L, \mu_M, \lambda_R)$ is the selected triple. Boxes are open centered whole-star boxes, with no periodic identification.

4.1 Re-targeting roadmap goal 2

The second Round31 roadmap goal asks for “a certified quantity whose interacting enclosure excludes its matched free reference” ([Round31 roadmap](#)). AV2 did not meet it: the certified interval for $C_\tau(1)$ contains $e^{-3}/4$ (Section 3.6). Before AW1 was contracted, the panel observed in deliberation that $\text{SU}(2)$ Peter–Weyl parity removes every first-order term of the centered correlation in the zero-selected subfamily, so that its interaction shift is of second order in τ . AW1 proves this for finite boxes, with a constant not uniform in the volume (Section 4.3). A certificate whose error is linear in τ , such as AV2’s, cannot resolve a shift of second order.

The contract therefore re-targets goal 2 to the only first-order Wilson effect in this model, the uncentered equal-time mean $\omega_\tau(W)$. Its candidate expansion $+\tau/144 + O(\tau^2)$ was to be derived, not assumed. A sign certificate for the mean needs three ingredients: the exact coefficient, an exact antisymmetry control and a second-order remainder uniform in volume, taken from the same AM2 expansion admitted through AV1. AW1 supplies the three ingredients and fixes the coupling of the certificate. The certificate itself belongs to AW2 (Section 5). The mean is a static ground-state quantity. Neither its value nor its sign is a dynamical, spectral or mass-gap correction, and the shift of the centered correlation remains unresolved.

4.2 Reviewed scope

The [AW1 gate](#) records the verdict `accepted_within_scope`. The skeptic’s review carries the sub-label `static_not_dynamic`, and the gate’s statement and limitations carry the same restriction. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section. In it, $\mathbb{E}[\cdot]$ is a Haar expectation, the E in $G-E$ is the ground energy, and E alone is the flip set.

In the AM2/AQ1 zero-selected patterned family (SU(2) Kogut-Susskind form on $\mathbb{Z}^3$ at fixed spacing, coarse 24-link factors, selected triple exactly (0,0,0) with Haar product reference, whole stars $\phi_b = -(\tau/3) \sum W_f$ in normalized units $\delta\alpha = \alpha/8$; original xz Wilson loop W with complete cover $R = \{0, e_z\}$; clock $s = \alpha \tau E / \hbar$,

$G=H/\alpha$), at both signs of τ with $|\tau| \leq 10^{-8}$: (1) Parity theorem, finite boxes only: in every open centered whole-star box Λ_N , $N \geq 2$, and every on-site cutoff, the first-order τ -derivatives at $\tau=0$ of $\omega_N(W^2)$, of the real-time correlation $c_N(\theta) = \langle \chi, e^{i\theta(G-E)} \chi \rangle$ (every real θ) and of $C_N(s) = \langle \chi, e^{-s(G-E)} \chi \rangle$ (every $s \geq 0$) vanish, the state, Duhamel, energy and vector-centring terms separately, by the per-link Z_2 centre grading (SU(2) Peter-Weyl multiplicities): $E[W] = E[W^3] = 0$, $E[W^2] = 1/4$, $E[W^4] = 1/8$, $E[W^2 W_f] = 0$ for every omitted f including $f=W$, $P_{24} V W \Omega_0 = 0$ (only the identity component $(1/4)\Omega_0$ at energy 0 for $f=W$), and zero first-order splitting of the gauge-invariant energy-24 multiplet $\text{span}\{W_g \Omega_0\}$ (not claimed on the non-invariant eigenspace); hence $C_N(s) = e^{-3s}/4 + O_N(\tau^2)$ with the τ^2 constant explicitly unbounded (not uniform in N ; for AQ limits only evenness of $C(s)$, no AQ second-order statement), and $\omega(W^2) = 1/4 + O(\tau^2)$ with supplementary uniform constants (not targets; the free value lies inside, no shift resolved). (2) Flip lemma: $E = \{(p,x): p_y \text{ even}\} \cup \{(p,y): p_z \text{ even}\} \cup \{(p,z): p_x \text{ even}\}$ meets every plaquette of Z^3 in 1 or 3 links; U_E (central -1 on every link of E) commutes with every Casimir and every original endpoint gauge action, fixes Ω_0 and maps every W_f to $-W_f$, so $U_E H_N(\tau, \kappa)$ $U_E^* = H_N(-\tau, -\kappa)$ in every open centered whole-star box, including on-site cutoff compressions ($Q_L \rightarrow$ cutoff of $h_b(-\kappa)$); at $\kappa=0$ $\omega_{N,-\tau}(W) = -\omega_{N,\tau}(W)$ and $\omega_N(W^2)$, C_N , c_N are even in τ ; for AQ limits $S(-\tau) = S(\tau) \circ \alpha_E$ as whole sets of subsequential limits (pointwise only along a common subsequence); odd periodic sides and frozen or gauge-fixed links excluded; no τ -antisymmetry via U_E at $\kappa=0$; oddness gives no $O(\tau^3)$ remainder. (3) $\omega_{\tau}(W) = +\tau/144 + r(\tau)$ under I1.5 (AV1 creation convention $c^*(1) = L_0 = -(\tau/72)$ $\sum W_f \Omega_0$, $\omega = -2\text{Re} \langle W \Omega_0, c^*(1) \rangle + O(\tau^2)$); only $f=W$ contributes; first-order values $\pm 1/1440000000$ at $\tau = \pm 10^{-8}$. (4) $|r(\tau)| \leq K_2 \tau^2$ uniformly in N , cutoff and every AQ1 subsequential limit at the exact tier ($t \leq T = (49|\tau|/144)/(1-352J)$, $\rho = 352JT$, $J = 28|\tau|$): the bound value is $K_2 = 81108864767825329926713064490531229475390625/24176936535511801466930759024724079017984$ (~ 3354.80322946 , outward $3354803229461691/10^{12}$; the skeptic itemization, identical to the forward's labelled unpinned t -bounds variant, dominating the forward 3354.63832966 and reverse 3354.49942587 exact-tier values, all three valid). (5) $K_2 \tau^2 \leq 1/288$ at $\tau = 10^{-8}$, so the frozen decade-grid rule gives $\tau_{AW2} = 10^{-8}$ (the cap) with sign margin $1/(144 K_2 \tau^2) \sim 207.00005$ (directed floor $41400010489/200000000$). The crude tier ($t_c = 592|\tau|$, $K_2 \sim 7.9375e6$, margin 0.0875) fails at the cap and is retained as a limited-tier value. No enclosure or sign of $\omega(W)$, no dynamical or mass-gap statement, no third-order remainder, uniqueness, whole-sequence convergence, rate in N , uniform Wilson, weak-coupling, continuum or priority claim is admitted.

The rest of this section derives that statement. The limitations recorded by the gate close the section (Section 4.9).

4.3 The parity theorem

Centre grading. For a link e let $(\Gamma_e \psi)(\dots, U_e, \dots) = \psi(\dots, -U_e, \dots)$. Haar measure is invariant under the central element -1 , so Γ_e is unitary and fixes Ω_0 . On the Peter-Weyl block $V_j \otimes V_j^*$ of the link it acts as $(-1)^{2j}$, while the Casimir C_e acts as $j(j+1)$. The grading therefore commutes with every Casimir, with H_0 and with every function of the Casimirs: $H_0^{-1} P_\perp$, e^{-sH_0} , $e^{i\theta H_0}$ and every on-site spectral cutoff at the zero triple. Because -1 is central, Γ_e also commutes with left and right translations, hence with every original endpoint gauge action. Finally, $\Gamma_e W_f \Gamma_e = (-1)^{[e \in f]} W_f$. In character form, the Haar integral over one link of k spin- $\frac{1}{2}$ matrix elements projects onto the invariants of $V_{1/2}^{\otimes k}$. Their multiplicity is zero for odd k and the Catalan number of $k/2$ for even k . Hence, for centre-even A_i such as functions of the Casimirs,

$$\langle \Omega_0, A_0 W_{f_1} A_1 \cdots W_{f_k} A_k \Omega_0 \rangle = 0 \quad \text{unless every link lies in an even number of } f_1, \dots, f_k. \quad (47)$$

Haar moments and pairings. For one plaquette, $\mathbb{E}[W^n]$ is 2^{-n} times the multiplicity of spin 0 in $V_{1/2}^{\otimes n}$:

$$\mathbb{E}[W^n]_{n=0,\dots,8} = 1, 0, \frac{1}{4}, 0, \frac{1}{8}, 0, \frac{5}{64}, 0, \frac{7}{128}. \quad (48)$$

Three exact routes agree through $n = 8$: Clebsch–Gordan recursion, Weyl integration by Wallis ratios, and the contract’s named cross-check. For the cross-check, the Wilson face contains the free z link (e_x, z) . Under the reference its holonomy is therefore Haar, $W = q_0$ with q uniform on S^3 , and $\mathbb{E}[q_0^{2m}] = \prod_{i < m} (2i + 1)/(4 + 2i)$. This confirms $\mathbb{E}[W] = 0$ and $\mathbb{E}[W^2] = \frac{1}{4}$ without the character count. Distinct plaquettes share at most one link. So (47) gives $\mathbb{E}[W_f] = 0$ and $\mathbb{E}[WW_f] = \frac{1}{4}\delta_{f,W}$. It also gives $\mathbb{E}[W^2W_f] = 0$ for every omitted f , *including* $f = W$: for $f \neq W$ the links of f outside W carry one spin- $\frac{1}{2}$ factor each, and for $f = W$ every link of W carries three. Finally $\mathbb{E}[W_gW_fW_h] = 0$ for all plaquettes g, f, h . The reverse proof uses the flip set of Section 4.4: E meets each plaquette an odd number of times, so it meets the $\mathbb{Z}_2$ chain $g + f + h$ an odd number of times, and that chain is not empty. The forward proof notes that $g\Delta h$ is never a single face.

First-order terms. In each box the interaction is bounded and the ground energy is simple and isolated (AM2). Kato’s analytic perturbation theory (cited, not machine-checked) then makes $\omega_N(W^2)$, $c_N(\theta) = \langle \chi, e^{i\theta(G_N - E_N)} \chi \rangle$ and $C_N(s) = \langle \chi, e^{-s(G_N - E_N)} \chi \rangle$ real-analytic in τ near 0. Write $\psi^{(1)}$ for the first-order correction of the ground vector. AM2 writes $\psi = e^{-C}\Omega_0$, so with the first-order collection (16),

$$\psi^{(1)} = -c^{(1)} = +\frac{\tau}{72} \sum_f W_f \Omega_0. \quad (49)$$

The vector $W\Omega_0$ is an eigenvector of G_0 with eigenvalue 3. The first-order part of $C_N(s)$ is the sum of four terms, and each vanishes separately:

$$\begin{aligned} \text{state: } 2e^{-3s} \operatorname{Re} \langle W^2 \Omega_0, \psi^{(1)} \rangle &= \frac{\tau e^{-3s}}{36} \sum_f \mathbb{E}[W^2 W_f] = 0, \\ \text{Duhamel: } -s e^{-3s} \langle W \Omega_0, (V - E^{(1)}) W \Omega_0 \rangle &= \frac{\tau s e^{-3s}}{24} \sum_f \mathbb{E}[W^2 W_f] = 0, \\ \text{energy: } E^{(1)} = \langle \Omega_0, V \Omega_0 \rangle &= -\frac{\tau}{24} \sum_f \mathbb{E}[W_f] = 0. \end{aligned} \quad (50)$$

The fourth, vector-centring, term needs care. The uncentered mean has the nonzero first-order coefficient $1/144$ (Section 4.5). It drops out of the correlation only because it multiplies overlaps of $W\Omega_0$ with Ω_0 , which equal $\mathbb{E}[W] = 0$. Equivalently, since $e^{-s(G_N - E_N)}\psi_N = \psi_N$, one has $C_N(s) = \langle W\psi_N, e^{-s(G_N - E_N)} W\psi_N \rangle - m_N^2$ for the normalized ground vector, and m_N^2 is of second order. The real-time correlation has the same terms with $e^{3i\theta}$ and $i\theta e^{3i\theta}$ in place of e^{-3s} and $-se^{-3s}$. Its state term also pairs $c^{(1)}$ with $W\alpha_\theta^0(W)\Omega_0 = e^{3i\theta}W^2\Omega_0$, which is centre-even, so the pairing vanishes. For the second moment the first-order part is $2\operatorname{Re} \langle (W^2 - \frac{1}{4})\Omega_0, \psi^{(1)} \rangle = \frac{\tau}{36} \sum_f (\mathbb{E}[W^2 W_f] - \frac{1}{4}\mathbb{E}[W_f]) = 0$. Both checkers evaluate every sum exactly, over all 1344 retained faces of Λ_2 and over the 82 faces meeting R .

The energy-24 multiplet. In normalized units the per-link content of $W_f W\Omega_0$ fixes its energy. For $f = W$ every link carries $j \in \{0, 1\}$, so the energy lies in $\{0, 16, 32, 48, 64\}$. A face sharing one link with W gives $\{36, 52\}$, and a disjoint face gives 48. Hence

$$P_{24} V W\Omega_0 = 0. \quad (51)$$

The only component with the vacuum’s content is the identity component $\mathbb{E}[W^2]\Omega_0 = \frac{1}{4}\Omega_0$ of the $f = W$ term, at energy 0. On Λ_2 the odd-link sets $W\Delta f$ have sizes 0, 6 and 8, for 1, 10 and 1333 omitted faces; the size 4 of an energy-24 vector never occurs. For the multiplet itself, $8 \sum_e j_e(j_e + 1) = 24$ forces exactly four spin- $\frac{1}{2}$ links. Gauge invariance needs even degree at every vertex, so the four links form a 4-cycle, and the 4-cycles of $\mathbb{Z}^3$ are exactly the plaquettes. Each vertex of degree two carries a one-dimensional invariant. The gauge-invariant energy-24 multiplet is therefore $\operatorname{span}\{W_g\Omega_0\}$. On it $\langle W_g\Omega_0, V W_h\Omega_0 \rangle = -\frac{\tau}{24} \sum_f \mathbb{E}[W_g W_f W_h] = 0$, so its first-order

splitting is zero. The statement is *not* made for the full energy-24 eigenspace. There, four spin- $\frac{1}{2}$ links with two on a face f can be mapped back into the eigenspace by W_f . Both producers restrict the claim to the gauge-invariant multiplet. That is where χ and its dynamics live, because G_N commutes with the gauge actions.

Exact scope of the conclusion. For every open centered whole-star box Λ_N , $N \geq 2$, and every on-site cutoff,

$$C_N(s) = \frac{1}{4}e^{-3s} + O_N(\tau^2), \quad c_N(\theta) = \frac{1}{4}e^{3i\theta} + O_N(\tau^2), \quad \omega_N(W^2) = \frac{1}{4} + O(\tau^2). \quad (52)$$

Four qualifications are part of the admitted statement.

- *Finite boxes.* The first two expansions are Taylor statements in each box, from Kato analyticity per box.
- *Constant not uniform.* The τ^2 constant of C_N and c_N is explicitly unbounded. It is not proved uniform in N , so no second-order statement about $C(s)$ or $c(\theta)$ passes to AQ limits. A uniform constant would need a second-order dynamical estimate that neither route supplies.
- *Evenness in AQ limits.* For AQ limits only the evenness of $C(s)$ and $c(\theta)$ in τ passes, as a set-level statement from the flip lemma (Section 4.4).
- *Gauge-invariant multiplet.* Zero first-order splitting is claimed for $\text{span}\{W_g\Omega_0\}$ only.

For $\omega(W^2)$ the product split of Section 2.2 also gives supplementary constants that are uniform in N , the cutoff and every AQ subsequential limit: $|\omega(W^2) - \frac{1}{4}| \leq K_{W^2}\tau^2$ with K_{W^2} about 3354.567 (forward), 1677.331 (reverse) and 1677.518 (skeptical). The forward value is valid but conservative, because it charges 2ϱ and all 72 straddling faces. These constants are not targets. The free value $\frac{1}{4}$ lies inside each band, so no shift is resolved.

4.4 The link-flip lemma

Odd intersection. The flip set, read from the contract, is

$$E = \{(p, x) : p_y \text{ even}\} \cup \{(p, y) : p_z \text{ even}\} \cup \{(p, z) : p_x \text{ even}\}. \quad (53)$$

Each link direction is keyed to the parity of the next coordinate in cyclic order. In a face with directions $a < c$, the two a -links differ only in the c coordinate and the two c -links only in the a coordinate. In each of the three orientations exactly one of the two pairs is keyed to the coordinate in which its links differ. That pair contributes exactly one link of E , and the other pair contributes none or two. Hence $|f \cap E| \in \{1, 3\}$ for every plaquette f of $\mathbb{Z}^3$. Translation by an even vector preserves E , so $|f \cap E|$ depends only on the orientation and on $p \bmod 2$: 24 classes. Both producers enumerate them in full, as the contract requires. The forward producer also checks the 49 omitted faces of seven factors at both z -parities, their selected faces and every plaquette of the $N = 2$ box. The reverse producer checks every plaquette with base point in $[-9, 9]^3$: 20 577 plaquettes, 10 830 meeting E once, 9 747 three times and none evenly. It also checks all 49 omitted faces of the factors 0 and e_z . These two factors exhaust the z -parity classes, because coarse x and y translations are even fine shifts (4 and 2) while a coarse z translation is a unit shift. The Wilson face meets E in three links.

The reverse producer reconstructs E before parsing it. Consider the 27 rules “ $(p, d) \in E$ iff $p_{\sigma(d)}$ is even” for maps $\sigma : \{x, y, z\} \rightarrow \{x, y, z\}$. Exactly two of them meet every plaquette oddly: the cyclic set (53) and the anti-cyclic set $E' = \{(p, x) : p_z \text{ even}\} \cup \{(p, y) : p_x \text{ even}\} \cup \{(p, z) : p_y \text{ even}\}$. They differ by the coboundary of $p_x p_y + p_y p_z + p_z p_x \bmod 2$, which is a centre gauge transformation. A solution can exist only if the constant $\mathbb{Z}_2$ 2-cochain 1 is a cocycle. It is, because a cube has six faces. The set E is not itself a cocycle, since its coboundary is 1 on every plaquette. So U_E is not a gauge transformation, and it genuinely flips gauge-invariant observables; the solution is unique up to centre gauge transformations. Four mutations are rejected: E minus one link, which leaves

four plaquettes even; an odd periodic side, which leaves even plaquettes at the seam; a centre gauge transformation presented as the flip, which is even on every plaquette; and a spin-one face term claimed to flip, since $(-1)^{2j|f \cap E|} = +1$ for $j = 1$.

The operator identity. Let $U_E = \prod_{e \in E \cap \Lambda_N} \Gamma_e$. It is diagonal in the Peter–Weyl basis with entries $(-1)^{2j_e}$. It commutes with every Casimir and with every original endpoint gauge action, because $g(-U)h^{-1} = -(gUh^{-1})$. It fixes Ω_0 and maps W_f to $-W_f$ for every plaquette, selected or omitted. Therefore

$$U_E H_N(\tau, \kappa) U_E^* = H_N(-\tau, -\kappa), \quad U_E Q_L H_N(\tau, \kappa) Q_L U_E^* = Q'_L H_N(-\tau, -\kappa) Q'_L, \quad (54)$$

in every open centered whole-star box. Here Q'_L is the cutoff of $h_b(-\kappa)$, which equals Q_L at $\kappa = 0$. For $\kappa \neq 0$ the identity rests on the inherited form of the strip operator: Casimirs, the three selected face traces and scalars. A scalar shift cancels in the ground-subtracted on-site operator. The reverse checker verifies (54) term by term on Λ_2 : all on-site Casimirs, the 1344 omitted and 375 selected face terms, at $\kappa = 0$ and at $\kappa = (\frac{1}{7}, -\frac{1}{9}, \frac{1}{3})$.

The contract also demands a positive demonstration at $\kappa \neq 0$. The forward producer compresses $H_0 - (\tau/3)W - \kappa W_g$, with g a selected face, onto $\text{span}\{\Omega_0, W\Omega_0, W_g\Omega_0\}$. The flip maps it to $H(-\tau, -\kappa)$, not to $H(-\tau, \kappa)$. The reverse producer uses two truncated plaquettes, one omitted face carrying τ and one selected face carrying $\kappa = \frac{1}{7}$. There the selected-face mean at $(-\tau, -\kappa)$ is the exact negative of its value at (τ, κ) , and it is unchanged at $(-\tau, \kappa)$. Both are finite fixtures. Claiming τ -antisymmetry via U_E at a nonzero triple is the rejected control `flip_breaks_at_nonzero_kappa`.

Covariance at the zero triple. At $\kappa = 0$, AM2 gives a unique ground for both signs, in the full space and in every cutoff space. Since $U_E \psi_N(\tau)$ is a ground of $H_N(-\tau)$, $U_E \psi_N(\tau) \propto \psi_N(-\tau)$. With $U_E \Omega_0 = \Omega_0$ and the intermediate normalization $\langle \Omega_0, \psi \rangle = 1$ this gives $c_I(-\tau) = U_E c_I(\tau)$, so $c^{(1)}$ is odd and $\|c\|_a$ is even. Also $\chi_{-\tau} = -U_E \chi_\tau$ and $G_N(-\tau) - E_N = U_E (G_N(\tau) - E_N) U_E^*$. Hence, for every box and every cutoff,

$$\begin{aligned} \omega_{N,-\tau} &= \omega_{N,\tau} \circ \alpha_E, & \omega_{N,-\tau}(W) &= -\omega_{N,\tau}(W), \\ \omega_{N,-\tau}(W^2) &= \omega_{N,\tau}(W^2), & C_{N,-\tau} &= C_{N,\tau}, & c_{N,-\tau} &= c_{N,\tau}. \end{aligned} \quad (55)$$

AQ limits as whole sets. Let $\mathcal{S}(\tau)$ be the set of all local trace-norm limits of $\omega_{N_k,\tau}$ along subsequences of centered boxes. Conjugation by $U_{E \cap F}$ on each finite region F defines a consistent automorphism α_E of the quasi-local algebra. Along any subsequence on which $\omega_{N_k,\tau}$ converges, $\omega_{N_k,-\tau}$ converges to $\omega_\tau \circ \alpha_E$. Therefore

$$\mathcal{S}(-\tau) = \mathcal{S}(\tau) \circ \alpha_E \quad \text{as sets of subsequential limits.} \quad (56)$$

The automorphism α_E intertwines the finite-volume dynamics, and hence their norm limits in AQ1. So for each $\omega \in \mathcal{S}(\tau)$, $(\omega \circ \alpha_E)(W) = -\omega(W)$, while $\omega \circ \alpha_E$ has the same $\omega(W^2)$, $c(\theta)$ and $C(s)$ as ω . This is the AQ-level evenness used in Section 4.3. AQ1's chosen states at $+\tau$ and $-\tau$ come from τ -dependent diagonal extractions. They are related pointwise only along a common subsequence. In the reverse producer's toy sequence the finite-volume identity holds exactly and the two limit sets are negatives of each other, but independently chosen subsequences give the nonzero sum $\frac{1}{500}$; the pointwise claim without a common subsequence is rejected. Odd periodic sides and frozen or gauge-fixed links are excluded.

No third-order remainder from oddness. In each box, analyticity and oddness remove the τ^2 Taylor coefficient of $\omega_N(W)$, with a τ^3 constant that depends on N . They do not turn a uniform bound $|r| \leq K_2 \tau^2$ into $O(\tau^3)$: the function $r(\tau) = K\tau|\tau|$ is odd and saturates the bound. The

AQ limit, as a function of τ , is not even known to be continuous. Any third-order remainder is a separate obligation, and every claim flag records none. A loop-3 sign-off sentence that the direct remainder is $O(\tau^3)$ is used by neither producer and is not admitted.

The modified-flip-set remark (recorded, not admitted). Both producers record, without claiming it, a second cochain. Every unit cube contains 0 or 2 selected faces, so the selected-face indicator is a $\mathbb{Z}_2$ cocycle and hence a coboundary $\delta E''$ on every box. The forward producer's E'' consists of the x -links with $p_x \bmod 4 \in \{0, 1, 2\}$ and $p_y \bmod 4 \in \{1, 2\}$; the reverse producer's is a period-8 cochain on x - and y -links. The set $E^* = E \triangle E''$ then meets every omitted face oddly and every selected face evenly, so U_{E^*} would map $H(\tau, \kappa)$ to $H(-\tau, \kappa)$ and make $\omega(W)$ odd in τ at every selected triple. The skeptic confirmed the combinatorics independently: both E'' have the right coboundary on a $17 \times 17 \times 7$ fine box, they differ by a cocycle, and an exact 3×3 compression onto $\{\Omega_0, 2W\Omega_0, 2W_g\Omega_0\}$ shows both identities. The operator consequence at $\kappa \neq 0$ is not reviewed. It needs four premises: the on-site operator at a nonzero triple must contain only Casimirs, selected face traces and scalars (the strip form inherited from A1, which neither producer read and the skeptic did not review); the cutoffs must be spectral projections of $h_b(\kappa)$; the triple must be held fixed independently of τ ; and a unique ground and AM2 applicability must hold with the non-Haar reference. The gate records the remark and admits nothing at nonzero triples; a claim there would need its own contract. The remark does not contradict the contract or the control `flip_breaks_at_nonzero_kappa`, both of which concern U_E . The selection note's unqualified sentence “antisymmetry in tau holds only at the zero selected triple” is to be read as “via U_E ” (review, N3). The selection note is historical and unchanged.

4.5 The first-order Wilson mean

Exact decomposition. Use the product split (8)–(10): $\psi = \psi_{\text{out}} + \delta$ with $\psi_{\text{out}} = \Omega_R \otimes \varphi_{\text{out}}$, $n = \|\varphi_{\text{out}}\|$ and $e = \|\delta\|/n$. Since W acts on $\mathcal{H}_R$ and $\langle \Omega_R, W\Omega_R \rangle = 0$, and writing $w = W\Omega_R$,

$$\omega_N(W) = \frac{X + Y}{1 + e^2}, \quad X = \frac{2 \operatorname{Re} \langle w \otimes \varphi_{\text{out}}, \delta \rangle}{n^2}, \quad Y = \frac{\langle \delta, (W \otimes 1) \delta \rangle}{n^2}. \quad (57)$$

Single-component overlap. The four links of W have owners $0, 0, e_z, 0$, and $\|w\| = \frac{1}{2}$. Integrating over the link $(0, x)$, owned by 0, or over (e_z, x) , owned by e_z , kills a single spin- $\frac{1}{2}$ factor. So the partial vacuum contraction $\langle \Omega_x | w \rangle$ vanishes for each $x \in R$, and w is excited at both sites of R . A creation meeting only one site of R leaves the vacuum at the other site and is orthogonal to w . This includes the 66 first-order straddling faces that meet R once. At first order only the creation supported exactly on R overlaps w , with multiplier $2\|w\| = 1$:

$$-2 \langle W\Omega_0, c_R^{(1)} \rangle = \frac{\tau}{36} \sum_{M_f=R} \mathbb{E}[WW_f] = \frac{\tau}{144}. \quad (58)$$

Of the ten faces with owner set exactly R , only $f = W$ contributes. Hence

$$\boxed{\omega_\tau(W) = -2 \operatorname{Re} \langle W\Omega_0, c^{(1)} \rangle + O(\tau^2) = 2 \operatorname{Re} \langle W\Omega_0, \psi^{(1)} \rangle + O(\tau^2) = +\frac{\tau}{144} + r(\tau),} \quad (59)$$

$$\omega_{\pm 10^{-8}}^{(1)} = \pm \frac{1}{14400000000}.$$

The expansion holds for either sign of τ . Section 4.6 bounds $r(\tau)$.

The sign chain. The sign is fixed by four steps, each from a frozen convention.

- (i) I1.5 puts $-(\alpha\tau/24)W_f$ into H for every omitted face; in normalized units the star is $\phi_b = -(\tau/3) \sum W_f (1)$.

- (ii) Each $W_f\Omega_0$ is an eigenvector of H_0 with energy 24 (normalized) or 3 (units of α). The per-face coefficient of $c^{(1)} = L_0 = H_0^{-1}P_\perp V\Omega_0$ is therefore $(-\tau/3)/24 = (-\tau/24)/3 = -\tau/72$ in both unit systems.
- (iii) AM2 writes $\psi = e^{-C}\Omega_0 = \prod_I(1 - \hat{c}_I)\Omega_0$, so $\psi^{(1)} = -c^{(1)}$, which is (49).
- (iv) Only $f = W$ pairs with W , and $\mathbb{E}[W^2] = \frac{1}{4}$, so the mean is $2 \cdot \frac{\tau}{72} \cdot \frac{1}{4} = \frac{\tau}{144}$.

For $\tau > 0$ the interaction lowers the energy where $W > 0$, and the mean is positive. Mixing the unit systems gives $\tau/1152$ or $\tau/18$; both are rejected controls. Three further re-derivations agree. The forward producer minimizes over $\text{span}\{\Omega_0, W\Omega_0\}$ in normalized units, where $\langle W\Omega_0, V\Omega_0 \rangle = -\tau/12$ and $\langle W\Omega_0, HW\Omega_0 \rangle = 6$. The trial energy $6\varepsilon^2 - (\tau/6)\varepsilon$ is minimal at $\varepsilon = \tau/72$, and the mean is then $2\varepsilon \mathbb{E}[W^2] = \tau/144$. The skeptic repeats the minimization in units of α . Both producers also use a one-plaquette fixture, which is a finite graph with `transfers_to_aq` false. It is written in the character basis $\chi_j(U_f)$, with $W\chi_j = \frac{1}{2}(\chi_{j-1/2} + \chi_{j+1/2})$, free energy $4j(j+1)$ in units of α and coupling $-(\tau/24)W$. Its exact Rayleigh–Schrödinger series,

$$\begin{aligned} \langle W \rangle &= \frac{\tau}{144} + 0 \cdot \tau^2 - \frac{5\tau^3}{11943936} + 0 \cdot \tau^4 + \frac{289\tau^5}{6604518850560} + \cdots, \\ E_2 &= -\frac{\tau^2}{6912}, \quad \langle W^2 \rangle = \frac{1}{4} + \frac{7\tau^2}{331776} + \cdots, \end{aligned} \tag{60}$$

is stable under truncation. At $j \leq \frac{1}{2}$ the exact ground values of $\langle W \rangle$ at $\tau = \pm 10^{-8}$ are negatives of each other and carry the sign of τ .

The contract’s sign shorthand. Item 3 of the contract writes $\omega_\tau(W) = 2\langle W\Omega_0, c^{(1)} \rangle$. With AV1’s creation coefficient $c^{(1)} = L_0$, the literal display evaluates to $-\tau/144$. The correct identity is the one in (59); the display holds only if $c^{(1)}$ is read as the vector correction $\psi^{(1)}$. Both producers derive $+\tau/144$ with the full sign chain and record the shorthand as a convention finding. Both reject the literal reading as a damaging mutation: the forward as `plus_two_L0_convention_gives_minus_tau_over_144`, the reverse as `am2_sign_literal` and `contract_formula_with_am2_c1`. The skeptic recorded the same defect before production. The gate records it as a non-blocking wording defect. The frozen contract is not amended, and neither the gate nor this section uses the display.

Face controls. The pairing $\mathbb{E}[WW_f]$ vanishes for the 81 other faces meeting R ; counting all ten faces inside R would give $10\tau/144$. The Wilson face is the xz class with $r = 0, q = 0$ among the 21 omitted classes anchored at factor 0 (Table 1). The other twenty are kept apart throughout. Nine share the owner set R ; they lie in $c_R^{(1)}$, but their pairing with W is zero by parity. Six have owner sets strictly containing R , namely $\{0, e_x, e_z\}$ (two faces) and $\{0, e_y, e_z\}$ (four faces); they reach $\omega(W)$ only through the straddling term. Five have owner sets meeting R only at 0; they reach it only through the two-creation and density terms. Merging the Wilson face with the others, excluding it from the omitted set (coefficient 0) and counting it twice are rejected. For orientation invariance, every omitted class anchored at 0, used as the observable, has first-order mean $+\tau/144$, because only the class face pairs with itself. Reversal and a cyclic shift of the base point leave $\frac{1}{2} \text{Tr } U_f$ unchanged. A selected face used as the observable gives 0 at the zero triple.

4.6 The second-order remainder

Outside excitations. One lemma beyond AV1 is needed. The reverse producer identifies it as the missing premise, and the forward producer proves it separately. For a site u outside R , let C_u collect the creations of φ_{out} that contain u . They pairwise overlap, so $C_u^2 = 0$ and $\varphi_{\text{out}} = \varphi' - C_u\varphi'$ with $\varphi' = (P_u \otimes 1)\varphi_{\text{out}}$. Hence

$$\|(Q_u \otimes 1)\varphi_{\text{out}}\| = \|C_u\varphi'\| \leq t_u \|\varphi'\| \leq t \|\varphi_{\text{out}}\|, \quad t_u = \sum_{I \ni u, I \cap R = \emptyset} \|c_I\| \leq t = \|c\|_a. \tag{61}$$

Itemized bound. Let $a = |\tau|/144$ be the norm of one first-order face vector, and let T and $\varrho = 352JT$ be the exact-tier quantities of (19). Put $s_{\text{str}} = 6a + \varrho$, the sum over the six first-order faces whose owner sets strictly contain R ($6 = 16 - 10$), and $s_0 = s_{e_z} = 33a + \varrho$, the sum over the faces whose owner sets meet R at one site only ($33 = 49 - 16$). Inserting (9) into (57) and sorting the supports gives, with $\varepsilon \geq e$,

$$\left| \omega_N(W) - \frac{\tau}{144} \right| \leq \underbrace{2\|w\|\varrho}_{\text{am2_remainder}} + \underbrace{2\|w\|T s_{\text{str}}}_{\text{straddling}} + \underbrace{2\|w\|s_0 s_{e_z}}_{\text{two_creation}} + \underbrace{\|W\|\varepsilon^2}_{\text{density}} + \underbrace{\mathcal{N}}_{\text{normalization_order}} =: K_2 \tau^2. \quad (62)$$

The preregistered items are the following.

- **am2_remainder:** $|2\text{Re}\langle w, (c - c^{(1)})_R \rangle| \leq 2\|w\| \|(c - c^{(1)})_R\| \leq \varrho$, about $3354.108 \tau^2$. It dominates every exact-tier value.
- **overlap_multiplier:** $2\|W\Omega_R\| = 1$, with $\|W\Omega_R\| = \frac{1}{2}$ a positive exact control. The factor 4 ($2\text{Re}, \|W\| = 1$, two sites) is admitted only as a labelled conservative variant, about 13416.96 (forward) and 13416.82 (reverse).
- **straddling:** only supports strictly containing R pair with an outside excitation, of amplitude at most $t \leq T$ by (61). The 72 straddling faces split as $66 + 6$ by the size of $I \cap R$, and the 66 meeting R once contribute exactly zero.
- **two_creation:** disjoint pairs $I \ni 0, J \ni e_z$ contribute at most $2\|w\|A_0A_{e_z}$ with $A_x \leq 33a + \varrho$. This is conservative: every first-order support has at least two sites, so the pair term is actually of third order.
- **density:** $|Y| \leq \|W\|e^2 \leq \varepsilon^2$, from the traced density $\text{Tr}_{\text{out}} |\delta\rangle\langle\delta|$.
- **normalization_order:** the correction $-(X+Y)e^2/(1+e^2)$ is of third order and is still charged. The leading term is exactly $\tau/144$, so $a\varepsilon^2$ suffices; the reverse producer charges the looser $(\varepsilon + \varepsilon^2)\varepsilon^2$.
- **arithmetic:** no numeric cost. Every value is an exact rational, with triangle bounds instead of square roots.

The pins come from the contract enumeration: 15 owner sets with multiplicities $\{1 \times 5, 2 \times 3, 3 \times 2, 4 \times 3, 10 \times 2\}$, 49 faces per factor, 82 faces meeting R and 16 touching both of its factors (Table 4). Both producers derive them from the snapshotted I1 table, and the skeptic re-derived 49, 10, 6, $72 = 42 + 30$ and $33 + 33$.

Three valid values. The three itemizations differ in two places. For the density term, the forward producer and the skeptic use AV1's $\varepsilon = 2T + T^2$. The reverse producer pins ε to the 82 faces meeting R , $\varepsilon_{\text{rev}} = 82a + 2\varrho + s_0 s_{e_z}$, which is AV1's labelled 82-face refinement applied inside the density term; it is sharper because $2T \approx 98a$ exceeds $82a$. For the straddling and pair terms, the skeptic's pre-comparison headline charged $T \cdot T$ and T^2 without the counts. Table 8 lists the terms. The skeptic also computed a minimal accepted form that takes the pinned terms, the smaller ε and $a\varepsilon^2$. Every term of each itemization is at least the corresponding minimal term, so all three are valid upper bounds, ordered skeptic $>$ forward $>$ reverse. The skeptic's itemization coincides exactly with the forward producer's labelled variant **unpinned_t_bounds**.

The gate binds the largest of the three as the admitted constant:

$$K_2^+ = \frac{81108864767825329926713064490531229475390625}{24176936535511801466930759024724079017984} \approx 3354.80323, \quad (63)$$

$$K_2^+ \leq \frac{3354803229461691}{10^{12}}.$$

It uses only T , ϱ and $\varepsilon = 2T + T^2$, with no 6-, 33- or 82-face refinement. It therefore stays valid even if every refinement were disputed. The forward value is exact; the reverse value is a rational

Table 8: Exact-tier ledger of K_2 at $\tau = 10^{-8}$, each entry divided by τ^2 . Decimals are rounded previews of exact rationals in the frozen outputs and the review record. A formula is shown where it differs from the forward one.

Term	Order	Forward	Reverse	Skeptic
am2_remainder, ϱ	2	3354.10836	3354.10836	3354.10836
straddling	2	$T(6a + \varrho)$: 0.014191	0.014191	$T \cdot T$: 0.115812
two_creation	2	$(33a + \varrho)^2$: 0.052533	0.052533	T^2 : 0.115812
density	2	$(2T + T^2)^2$: 0.463247	$\varepsilon_{\text{rev}}^2$: 0.324343	0.463247
normalization_order	3	$a\varepsilon^2$: 3.2×10^{-11}	$(\varepsilon + \varepsilon^2)\varepsilon^2$: 1.8×10^{-9}	3.2×10^{-11}
overlap_multiplier	–	1	1	1
arithmetic	–	exact	exact	exact
K_2^+		3354.63832966	3354.49942587	3354.80322946

with a 114-digit numerator, quoted here by its outward 10^{-40} ceiling:

$$\begin{aligned}
K_{2,\text{fwd}}^+ &= \frac{81104877995836618199905135109761286809699489}{24176936535511801466930759024724079017984} \approx 3354.63833, \\
K_{2,\text{rev}}^+ &\leq \frac{33544994258669290845093439980837832702606211}{10^{40}} \approx 3354.49943.
\end{aligned} \tag{64}$$

The three values differ by at most about 0.304.

The bound holds uniformly in N . Every constant depends only on t , ϱ , ε and bulk counts, finite-box counts never exceed bulk counts, and the star of W is retained for $N \geq 2$. It holds at every on-site cutoff $L \geq 24$, where $c_L^{(1)} = c^{(1)}$. It passes to the untruncated ground vector by AV1’s cutoff removal (Section 2.7) and, because W is local on R , to every AQ1 subsequential limit. Each $K_2(\tau)$ is nondecreasing in $|\tau|$, so the cap value bounds every smaller coupling. The ratio $B(\tau)/B(\tau/100)$ of the total bounds is about 10000.98, which confirms quadratic scaling. The $-\tau$ ledger replays the same $|\tau|$ formula.

The AM2 generic majorant $352 JT$ is 99.98% of K_2^+ . Parity would remove the τ^2 coefficient of the direct overlap $\langle W\Omega_R, (c - c^{(1)})_R \rangle$, but using that needs a separate third-order bound, which is neither used nor claimed. Sharper directed forms of the AM2 remainder, of the type $G(t) - 16 \leq 288t/(1 - 8t)$, give labelled variants of about 2745 (forward 2744.88, reverse 2744.66). They are recorded, not admitted.

The crude tier fails and is retained. The crude tier uses only $t_c = J_0 G(R_a) = 592|\tau| = 37/6250000$ at the cap, with no face counts, so no tiers are mixed. Its terms divided by τ^2 are 5834752 (AM2 remainder $352 JT_c$), 350464 (straddling), 350464 (two-creation) and about 1401864.30 (density). Together they give

$$K_2^{\text{crude}} = \frac{1937877026146766159414097129}{244140625000000000000} \approx 7.9375443 \times 10^6 \tag{65}$$

for the forward producer and the skeptic, and about 7.9375609×10^6 for the reverse producer, which differs only in the normalization charge. Then $K_2^{\text{crude}}\tau \approx 0.0794$, which exceeds $1/288$ by a factor of about 22.9. The sign margin is 0.0875. The crude tier fails at the cap and is retained as a limited-tier value. Its rule value 10^{-10} is information only and is never substituted.

4.7 The frozen AW2 coupling rule

The contract freezes the rule before any constant is known: τ_{AW2} is “the largest element of the decade grid $\{10^{-8}, 10^{-9}, \dots\}$ with $K_2^+\tau \leq 1/288$ ”, where K_2^+ is the admitted exact-tier constant. Given $|r(\tau)| \leq K_2^+\tau^2$ and $\tau > 0$, the interval $\tau/144 \pm K_2^+\tau^2$ excludes zero with margin $(\tau/144)/(K_2^+\tau^2) =$

$1/(144K_2^+\tau)$. The threshold $1/288$ is exactly a margin of 2, the floor fixed by a loop-2 veto (Section 1.3). With the bound value (63), $K_2^+\tau \approx 3.35480 \times 10^{-5}$ at $\tau = 10^{-8}$, which is at most $1/288$. The rule therefore gives

$$\tau_{\text{AW2}} = 10^{-8} \text{ (the cap),}$$

$$\frac{1}{144K_2^+\tau} = \frac{42981220507576535941210238266176140476416}{207638693805632844612385445095759947457} \approx 207.00005 \geq \frac{41400010489}{200000000}. \quad (66)$$

The choice among the three valid values changes no decision. Any exact-tier constant up to $10^8/288 \approx 347222$ gives 10^{-8} , and the forward and reverse values give margins of about 207.010 and 207.019. The quotient of $1/288$ by $K_2^+\tau$ is about 103.5. The reverse headline “margin ≈ 103.5 ” is that quotient, not the contract’s sign margin.

The admitted numbers are compatible with AV1. The tier-(ii) bound (23) satisfies $D_{\text{ii}} \approx 1.3612 \times 10^{-8} \geq \tau/144 + K_2^+\tau^2 \approx 6.978 \times 10^{-11}$, about 196 times the first-order term. The feasibility Boolean is not a sign certificate. AW1 admits no enclosure and no sign of $\omega(W)$; instantiating the interval at τ_{AW2} is AW2’s task. The gate advises that AW2’s checker bind K_2^+ to the rational (63) from the gate rather than recompute it.

4.8 Controls, replays and review

The contract lists 30 controls. The forward checker runs 41 exact checks and rejects 68 damaging mutations; the reverse checker runs 57 exact checks and rejects 107. Each implements all 30 controls as damaging mutations, and neither uses a floating-point value or `sympy` in the admission path. Both record the same claim flags:

- true: `first_order_parity_claim` and `flip_lemma_claim`;
- false: `third_order_remainder_claim`, `continuum_claim`, `uniform_wilson_claim`, `resolved_interaction_shift` and `scientific_priority_verified`.

Both checkers verify the snapshot’s SHA-256 before parsing and read the target $1/288$, the rule and the reference values 0 and $\frac{1}{4}$ from it. After a coherently rehashed edit of the target to $1/144$, the reverse checker rejects the target and the forward checker records it. That edit requires changing `check.py`, which the freeze inventory binds, so the gate records the difference as non-blocking.

The skeptic’s pre-comparison package, with 79 exact checks, was committed before either producer’s commit. The post-comparison review adds 65 exact checks, including 30 source-edit mutation receipts. Twenty-five of the edits abort at the intended check, and one shows that a value is read from the contract. Four silent value edits run to completion in the producer checkers: forward density divided by 4, forward pair term divided by 9, reverse density pinned to 49 faces, and reverse pair term divided by 9. Only the skeptic’s term-by-term validator catches them, because neither producer pins individual term values. All four producer replays, forward and reverse under normal and optimized Python, reproduce the frozen outputs byte for byte. The closures hold 39 and 35 files, and the reverse inputs are exactly `AGENTS.md`, the contract and the 29 shared premises.

One isolation finding is recorded without being resolved. The shared session scratchpad root holds files written between 23:14Z and 23:20Z, before either producer commit. Their content matches the forward producer’s K_2 formulas, its E'' remark and its one-plaquette fixture, and the reverse producer’s α -unit fixture and four-site creation fixture. Neither producer disclosed scratch reads. The forward producer’s file overwrote the skeptic’s pre-comparison scratch file of the same name. Whether any agent read another’s scratch file cannot be determined. The routes and code differ materially: two different E'' constructions, two different density ε and different fixtures. Independence is therefore not refuted, but premise isolation is verified for repository inputs only.

4.9 Limitations recorded by the gate

The AW1 gate records the following limitations verbatim.

- Only the AM2/AQ1 zero-selected patterned family, the cover $R=\{0, e_z\}$, fixed spacing and $|\tau| \leq 10^{-8}$; nothing transfers to nonzero selected triples, uniform Wilson theory, weak coupling or the continuum
- Parity theorem for $C(s)$ and $c(\theta)$ is a finite-volume Taylor statement (Kato analyticity per box); the τ^2 constant is explicitly unbounded; AQ limits inherit only the set-level evenness
- Flip lemma needs open boxes and $\kappa=0$ for antisymmetry; the $\kappa \neq 0$ identity rests on the inherited strip-operator form; AQ statements are about whole sets of subsequential limits
- K_2^+ is an absolute second-order upper bound with no sign of $r(\tau)$ and no third-order remainder; the AM2 generic majorant 352JT is 99.98% of it; the $-\tau$ ledger is a replay
- No enclosure or sign of $\omega(W)$ is admitted (AW2); static equal-time mean, not a dynamical or mass-gap correction
- Inherited without re-proof: AM2 fixed point, majorant, gaps and ground uniqueness; AV1 product split, anchored-norm tier, cutoff removal and AQ passage; AQ1 construction and dynamics; I1 dictionary; Kato and Peter-Weyl cited, not machine-checked
- Fixtures (one plaquette, qubit/four-site creation algebras, κ compressions, 3x3 compression) are finite graphs or algebras, `transfers_to_aq` false
- Independence limited to routes and code: shared premises state the mechanism (loop-3 sign-offs, `selection-aw1.md`, `skeptic/av2.md`); the forward read the skeptic triage and loop-2 notes as declared premises
- N1: both producers used the shared scratchpad root (files matching the forward's K_2/K_W formulas, E' remark and one-plaquette fixture, and the reverse's alpha-unit fixture and four-site creation fixture, 23:14-23:20Z) without disclosing scratch reads; the forward's `k2.py` overwrote the skeptic's pre-comparison scratch `k2.py`; reads cannot be determined; isolation is verified for repository inputs only
- N4: the reverse's headline 'margin ~103.5' is the ratio to 1/288; the contract sign margin is $1/(144 K_2^+ \tau) \sim 207.02$
- N5: forward $K_W \sim 3354.57$ is conservative (2 ρ , all 72 straddling faces) against reverse 1677.33 and skeptic 1677.52; supplementary, not a target
- N6: reverse wording that U_E commutes with every on-site spectral cutoff and fixes P_R holds at $\kappa=0$; at $\kappa \neq 0$ these map to the $-\kappa$ objects (forward states Q_L)
- N7: the forward records a coherently rehashed 1/144 target instead of rejecting it (no semantic tie to half the coefficient; the reverse rejects); protected by the `check.py` hash in `freeze.json`
- N8: neither producer checker pins individual K_2 term values; four silent undercounts ran to completion and were caught only by the skeptic term validator; AW2 should bind K_2^+ from the gate
- N9: the modern loop-3 sign-off sentence that the direct remainder is $O(\tau^3)$ is used by neither producer and is not admitted
- Undeclared disclosed reads: forward tools `README`, `freeze.py`, AV1 forward `check.py` and two assistant scripts (named, not imported; the modern K_2 script holds only loop-2 tiers); reverse portions of `reverse/av1/check.py` and the line count of `reverse/av2/check.py`; the reverse's partial reads of the AV2 forward report and `skeptic/av2.md` are of declared shared premises
- Scientific priority unverified

Full derivations and reviewed implementations: [AW1 forward](#), [AW1 reverse](#), [independent model-agent review](#) and the [frozen contract](#). The gate selects AW2 next.

5 AW2: a sign-certified Wilson mean at the original cap

Contribution HNM-C-AW2. The AW2 contract, recorded under the title “Hruday sign-certified enclosure of the Wilson mean at the AW1-frozen coupling”, was frozen after the AW1 gate and

before production. It keeps the model of Sections 2–4: the zero selected triple with its Haar product reference, whole stars under I1.5, the original xz plaquette W with cover $R = \{0, e_z\}$, and fixed positive α , $\hbar$, E_* and spacing. The observable is the uncentered equal-time mean $\omega_\tau(W)$. Its free Haar reference is $\mathbb{E}[W] = 0$, which the checker reads from the contract. AW2 instantiates a formula that AW1 already admitted, so it runs in one direction (**single+skeptic**). There is one forward producer, and the skeptic wrote an independent replay from the contract alone and committed it before the producer’s package. The contract makes that replay a condition of admission. The producer received AGENTS.md, the contract and its 26 shared premises, 28 files in all. No reverse package exists, so reverse premise isolation is not applicable, and the packet says so.

5.1 Reviewed scope

The AW2 gate records the verdict **accepted_within_scope** with the sub-label **static_not_dynamic**. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section. In it, D is the common denominator of the endpoints, written Q below because D denotes AV1’s state bound; S is the set $\mathcal{S}$ of AQ1 subsequential limits; and $\circ$ denotes composition.

```
In the AM2/AQ1 zero-selected patterned family (SU(2) Kogut-Susskind form on  $Z^3$  at
fixed spacing, coarse 24-link factors, selected triple exactly (0,0,0) with Haar
product reference, whole stars  $\phi_b = -(\tau/3) \sum W_f$  under I1.5 in normalized units
 $\delta = \alpha/8$  with  $V_b = \phi_b/8$  in alpha units; original  $xz$  Wilson loop  $W$  with complete
cover  $R = \{0, e_z\}$ ; fixed positive  $\alpha$ ,  $\hbar$ ,  $E_*$  and spacing), with  $\tau_{AW2} = 10^{-8}$ 
(the cap) evaluated inside check.py by the frozen AW1 decade-grid rule (largest  $10^{-k}$ ,
 $k \geq 8$ , with  $K_2 + \tau \leq 1/288$ , half the first-order coefficient) from
 $K_2 = 81108864767825329926713064490531229475390625/$ 
 $24176936535511801466930759024724079017984$  ( $\sim 3354.80322946$ , outward
 $3354803229461691/10^{12}$ ) read from the hash-verified AW1 gate, never from a literal
coupling: for every state in the whole set of AQ1 subsequential limits at  $\tau = +10^{-8}$ ,
and in every centered whole-star box  $N \geq 2$  at every on-site cutoff (uniform in the box;
passed to the limits by local trace-norm convergence on  $R$ ,  $\|W\| \leq 1$ , closed interval),
the static equal-time Wilson mean  $\omega_\tau(W)$  lies in  $[\tau/144 - K_2 + \tau^2,$ 
 $\tau/144 + K_2 + \tau^2] = [42773581813770903096597852821080380528959/D,$ 
 $43188859201382168785822623711271900423873/D],$ 
 $D = 618929575309102117553427431032936422860390400000000$ , contained in  $[6.91089641214e-11,$ 
 $6.97799247674e-11]$  (decimals widened outward); at  $\tau = -10^{-8}$  the enclosure is the
mirror  $[-43188859201382168785822623711271900423873/D,$ 
 $-42773581813770903096597852821080380528959/D]$  (contained in  $[-6.97799247674e-11,$ 
 $-6.91089641214e-11]$ ), by the AW1 remainder bound at both signs and the AW1 flip lemma at
the zero triple ( $S(-\tau) = S(\tau) \circ \alpha_E$  as whole sets; a replay, not a second
confirmation). The free Haar reference 0 is strictly excluded at both signs and
 $\text{sign}(\omega_\tau(W)) = \text{sign}(\tau)$ : exclusion margin  $(|\tau|/144 - K_2 + \tau^2)/(K_2 + \tau^2) =$ 
 $42773581813770903096597852821080380528959/207638693805632844612385445095759947457$ 
( $\sim 206.00005 \geq 2$ , the contract target; directed floor  $41200010489/2000000000$ ) and sign
margin  $1/(144 K_2 + |\tau|) = 42981220507576535941210238266176140476416/$ 
 $207638693805632844612385445095759947457$  ( $\sim 207.00005$ ; directed floor
 $41400010489/2000000000$ ). Sub-label static_not_dynamic; sign_certified_below_cap does not
apply ( $\tau_{AW2}$  is the cap); resolved_interaction_shift refers to  $\omega_\tau(W)$  only. The
crude tier ( $t_c = 592|\tau|$ ,  $K_2 = 1937877026146766159414097129/2441406250000000000000$ 
 $\sim 7.9375e6$ ) gives an enclosure  $[-7.2431e-10, 8.6320e-10]$  containing 0 (sign margin
 $\sim 0.0875$ ) and is retained as a failed limited-tier control. No dynamical correction, mass
shift, mass-gap or susceptibility statement, no centered-correlation shift (AV2
reference_unresolved stands), no AQ uniqueness, whole-sequence convergence or rate in  $N$ ,
no third-order remainder, no statement at nonzero selected triples, and no uniform
Wilson, weak-coupling, continuum or priority claim follows. Admission rests on the
single forward producer plus the skeptic's independent pre-comparison replay from the
contract.
```

The rest of this section instantiates that statement. The limitations recorded by the gate close the section (Section 5.9).

5.2 The admitted premise

AW2 proves nothing new about the ground state. Its mathematical premises are items (3) and (4) of the AW1 gate (Sections 4.5 and 4.6), with the flip lemma of item (2) for the mirror: at the zero triple and for $|\tau| \leq 10^{-8}$,

$$\omega_\tau(W) = \frac{\tau}{144} + r(\tau), \quad |r(\tau)| \leq K_2^+ \tau^2, \quad (67)$$

at both signs, uniformly in the box Λ_N ($N \geq 2$), in the on-site cutoff and in every AQ1 subsequential limit, with the gate-bound constant $K_2^+ \approx 3354.80323$ of (63). The checker hard-codes the SHA-256 of the AW1 gate snapshot, verifies it before parsing, and reads K_2^+ from the gate’s decision text. It then checks exact equality with the gate’s accepted text, the contract parameter and the skeptic’s AW1 review. The forward AW1 report’s preview 3354.80322946 of its labelled `unpinned_t_bounds` variant is a truncation of the same number, and the outward ceiling $3354803229461691/10^{12}$ exceeds it by less than 10^{-12} .

As a consistency replay, not a source, the itemization named by the gate reproduces the constant exactly. With $J = 28|\tau|$, $T = (49|\tau|/144)/(1 - 352J)$, $\varrho = 352JT$ and $\varepsilon = 2T + T^2$ at $\tau = 10^{-8}$,

$$K_2^+ \tau^2 = \varrho + T \cdot T + T^2 + \varepsilon^2 + \frac{|\tau|}{144} \varepsilon^2. \quad (68)$$

The five terms are the AM2 remainder and the straddling, two-creation, density and normalization charges of the skeptic column of Table 8. The same T reproduces AV1’s D_{ii} exactly, and the AM2 term is 99.979% of the total. AW2 binds the gate value rather than recomputing it, as the AW1 gate advised after four silent term undercounts ran to completion in the AW1 producer checkers (Section 4.8). A recomputed or reduced constant, including the forward AW1 headline value, is a rejected mutation.

5.3 The coupling from the frozen rule

The rule was frozen in the AW1 contract before any constant was known (Section 4.7). The AW2 checker parses it from the AW1 contract snapshot, whose hash is a binding of the AW1 gate, and evaluates it on the parsed constant:

$$\tau_{\text{AW2}} = \max\{10^{-k} : k \geq 8, K_2^+ 10^{-k} \leq \frac{1}{288}\}. \quad (69)$$

The checker also requires the threshold to be half the first-order coefficient $1/144$. That tie answers AW1 review finding N7 (Section 4.8). The grid starts at the cap, and at its first element, exactly,

$$\begin{aligned} K_2^+ \cdot 10^{-8} &= \frac{207638693805632844612385445095759947457}{6189295753091021175534274310329364228603904} \leq \frac{1}{288}, \\ \frac{1}{288} - K_2^+ \cdot 10^{-8} &= \frac{21282971559982635125992733687992310290751}{6189295753091021175534274310329364228603904} > 0. \end{aligned} \quad (70)$$

The first line is about 3.35480×10^{-5} . No larger grid element exists, so the rule returns

$$\tau_{\text{AW2}} = 10^{-8}, \quad \text{the original cap.} \quad (71)$$

The contract and the AW1 gate both state this value. The checker uses those texts only as cross-checks of the rule’s output and never reads either as the coupling. Seven non-rule couplings are rejected: a literal coupling, the contract text used as the coupling, 10^{-9} (not the largest decade), the off-grid value $5 \cdot 10^{-9}$, the forward AW1 headline constant fed to the rule, the crude constant fed to the rule, and a threshold that is not half the coefficient. The skeptic confirmed that the rule really evaluates the constant: $1000K_2^+$ returns 10^{-9} and the crude constant returns 10^{-10} . The

ratio $(1/288)/(K_2^+\tau) \approx 103.50003$ measures the rule’s slack. It is not a margin of the enclosure, and any exact-tier constant up to $10^8/288 \approx 3.47 \times 10^5$ would also return the cap.

One gap is recorded as review finding N3. A literal coupling that happens to equal the rule value runs to completion, and only the recorded checker hash changes. The skeptic’s data-flow review of the source rejects that edit, and the hash of `check.py` in the freeze inventory protects the admitted source. This is the AW2 analogue of AW1 finding N7.

5.4 The enclosure at both signs

Write

$$I(\tau) = \left[\frac{\tau}{144} - K_2^+\tau^2, \frac{\tau}{144} + K_2^+\tau^2 \right], \quad I(-\tau) = -I(\tau). \quad (72)$$

By (67), $\omega_\tau(W) \in I(\tau)$ for every state in the whole set $\mathcal{S}(\tau)$ of AQ1 subsequential limits, and in every centered whole-star box $N \geq 2$ at every on-site cutoff. At $\tau = \tau_{\text{AW2}}$ the first-order value is $\tau/144 = 1/14400000000 \approx 6.9444 \times 10^{-11}$. With the common denominator

$$Q = 618929575309102117553427431032936422860390400000000 \quad (73)$$

the half-width is $K_2^+\tau^2 = 207638693805632844612385445095759947457/Q \approx 3.35480 \times 10^{-13}$, and the admitted enclosure is

$$\boxed{\begin{aligned} \omega_{10^{-8}}(W) &\geq \frac{42773581813770903096597852821080380528959}{Q} > 6.91089641214 \times 10^{-11}, \\ \omega_{10^{-8}}(W) &\leq \frac{43188859201382168785822623711271900423873}{Q} < 6.97799247674 \times 10^{-11}. \end{aligned}} \quad (74)$$

The short decimals widen the exact endpoints outward: the lower one is rounded down and the upper one up, at 12 significant digits. The checker re-parses each decimal exactly and verifies its direction and that it lies within one unit of its last digit.

Passage to the AQ limits. The interval does not depend on N . The checker verifies that the inputs of the formula are identical in the boxes $N = 2$ and $N = 3$: 82 retained faces meet R , four incoming stars act at each site of R , and each factor has 49 omitted faces. It rejects the box $N = 1$, which drops incoming stars. Along any subsequence on which the local densities converge in trace norm on R ,

$$|\omega(W) - \omega_{N_k}(W)| \leq \|W\| \|\rho_R - \rho_{N_k, R}\|_1 \rightarrow 0, \quad \|W\| \leq 1, \quad (75)$$

and $I(\tau)$ is closed, so every state in $\mathcal{S}(\tau)$ inherits the enclosure. No uniqueness of the AQ state, whole-sequence convergence or rate in N is used or claimed. The set $\mathcal{S}(\tau)$ may contain many states, and each of them obeys the enclosure.

The mirror at $\tau = -10^{-8}$. The AW1 bound holds at both signs, so (67) at $\tau = -10^{-8}$ gives

$$\omega_{-10^{-8}}(W) \in I(-10^{-8}) = -I(10^{-8}) \subset [-6.97799247674 \times 10^{-11}, -6.91089641214 \times 10^{-11}]. \quad (76)$$

The same interval is also the image of $I(10^{-8})$ under the AW1 flip lemma at the zero triple. There $\omega_{N, -\tau}(W) = -\omega_{N, \tau}(W)$ in every box and cutoff, and $\mathcal{S}(-\tau) = \mathcal{S}(\tau) \circ \alpha_E$ as whole sets (55)–(56). Both routes apply the same $|\tau|$ formula to the same constant, so the minus-sign enclosure is a replay, not a second confirmation. The packet records it as a replay. A sign-blind enclosure and a mirror counted as independent confirmation are both rejected mutations. Individual states at $\pm\tau$ are paired only along a common subsequence.

Table 9: Wilson-mean enclosures at $\tau = 10^{-8}$: the gate-bound exact tier and the crude tier with $t_c = 592|\tau|$. Decimals are outward previews of exact rationals in the frozen output; only the first row is admitted.

Tier	Constant	Enclosure of $\omega_{10^{-8}}(W)$	μ_{sign}	Outcome
exact	$K_2^+ \approx 3354.80323$	$[6.91089641214, 6.97799247674] \times 10^{-11}$	≈ 207.00005	accepted
crude	$K_2^{\text{crude}} \approx 7.9375443 \times 10^6$	$\approx [-7.2431, 8.6320] \times 10^{-10}$	≈ 0.0875	fails; retained

5.7 Exact scope

What is claimed. AW2 certifies a static equal-time effect. For every state in the whole set $\mathcal{S}(\pm 10^{-8})$ of AQ1 subsequential limits at the zero selected triple, and in every centered whole-star box $N \geq 2$ at every on-site cutoff, the uncentered Wilson mean lies in $I(\pm 10^{-8})$. That interval excludes the free reference 0 and has the sign of τ . The sub-label is `static_not_dynamic`. The flag `resolved_interaction_shift` is true only because the exclusion holds at the cap, and it refers to $\omega(W)$ alone. The sub-label `sign_certified_below_cap` is false. It would apply only if the rule had returned a coupling below the cap, and τ_{AW2} is the cap itself. Admission rests on the single forward producer plus the skeptic’s independent pre-comparison replay from the contract. The two share model-agent ancestry, and K_2^+ is the skeptic’s own AW1 itemization. Their exact agreement is therefore a replay of one admitted formula, not independent physical confirmation. The Arb/mpmath preview is a labelled comparison that can veto a run but supplies no admitted value (Section 5.8).

The [panel update](#) after sub-round 2 records roadmap goal 2 as met in the re-targeted form of Section 4.1. That is a planning record and admits nothing beyond the gate.

What is not claimed.

- No dynamical correction, mass shift, mass-gap or susceptibility statement. The mean is an equal-time expectation in the stationary state.
- No shift of the centered correlation $C_\tau(s)$ or $c_\tau(\theta)$. AV2’s `reference_unresolved` stands; by AW1 the first-order term of that shift vanishes in each finite box, and its second-order constant is not uniform in N .
- No uniqueness of the AQ state, whole-sequence convergence or rate in N , and no pairing of individual states at $\pm\tau$ without a common subsequence.
- No third-order remainder and no sign of $r(\tau)$.
- Nothing at nonzero selected triples, and no transfer from a finite graph.
- No uniform Wilson, weak-coupling or continuum statement.
- No priority. The forward report describes the result as a first for this model family. The review reads that as first in this workbench’s admitted record, not as a priority claim, and the gate does not adopt the wording. Scientific priority is unverified.

5.8 Controls, replays and review

The forward checker runs 37 exact checks. It implements all 23 contract controls as damaging mutations raised through explicit exceptions rather than assertions, so every control stays active under optimized Python. It rejects 116 mutations in all, 92 of them under the contract control ids. Its imports are standard-library only and every admission value is an exact rational; float, Boolean, NaN and division-by-zero inputs are rejected. The first-order chain is re-verified, not assumed. This covers the Haar moments $1, 0, \frac{1}{4}, 0, \frac{1}{8}, 0, \frac{5}{64}, 0, \frac{7}{128}$ by Clebsch–Gordan and by the free-link S^3 route, the face sum over all 82 faces meeting R and the sign chain of Section 4.5. The rejected alternatives include $\tau/72$, $\tau/288$, $\tau/1152$ and $\tau/18$; the literal contract display, which gives $-\tau/144$; another

face with owner set R paired with W ; and an unlabelled overlap factor 4. The checks of contract item 4 that have no id of their own are covered as well.

- *Sign convention.* The frozen I1.5 string is verified against the contract and the I1 snapshot. A flipped ϕ_b string is rejected, and so is a coherent flip in both the derivation and the fixture, which gives $-\tau/144$. The one-plaquette fixture of (60), a finite graph with `transfers_to_aq` false, reproduces the AW1 series at truncations 8 and 10. Its $j \leq \frac{1}{2}$ ground has $\det = -b^2 < 0$, which forces $E_0 < 0$ and $\text{sign}\langle W \rangle = \text{sign} \tau$. Its exact brackets lie inside $[6.94444444439, 6.94444444446] \times 10^{-11}$ and the mirror interval.
- *Boundary and state.* The formula inputs are identical at $N = 2$ and $N = 3$, and $N = 1$ is rejected.
- *Centring.* An exact 3×3 fixture separates vector from scalar centring. No $\hat{m}$ enters the $\omega(W)$ path, and the calculator has no such argument.
- *Coherent tampering.* Ten mutations with rebound hashes are rejected. Among them are a removed snapshot, a halved K_2^+ , a changed τ_{AW2} , and a contract target or gate constant edited with its hash rebound, which fails against the hard-coded pins.

The producer also exports a reusable exact calculator. It accepts exact inputs only and restricts the domain to the zero triple and $0 < |\tau| \leq 10^{-8}$. It reads K_2^+ , the coefficient $1/144$, the cap and the rule only from the hash-verified snapshots, and it takes no K or $\hat{m}$ argument; the checker records 13 domain rejections. Its fixed-design mode enforces $|\tau| = \tau_{\text{AW2}}$. Below the cap, its reusable mode attaches `sign_certified_below_cap` to any coupling whose enclosure excludes 0, with $\mu_{\text{sign}} = 207.00005 \cdot 10^{-8}/|\tau|$. Review finding N6 records those values as corollaries of the AW1 bound, not AW2 admissions. Any reuse needs its own contract.

The Arb/mpmath preview. A separate preview script recomputes the endpoints and margins with python-flint 0.9.0 (Arb) and mpmath 1.3.0 at 256 bits. It stores ball bounds in `arb-preview.json`, labelled as a preview and not used for admission. The checker reads only that stored file and verifies that every exact endpoint and margin lies inside both previews. A missing or inconsistent preview aborts the run, so the preview can veto an admission but cannot supply one. When the skeptic replaced every stored ball by the exact point value, the only output field that changed was the preview’s hash. Neither library is imported by the checker or the calculator, and an added `mpmath` import aborts. Regenerating the preview reproduces the stored file byte for byte. Finding N2 asked the gate to call the preview a labelled comparison, veto only, and the gate does.

Independent replay and review. The skeptic’s pre-comparison package holds a derivation, a contract review and a checker with 75 exact checks covering all 23 controls. It was hash-frozen and committed (86edd4d, 00:17:03Z) before the forward commit (4ef9352, 00:23:47Z). The two agree exactly on every exported rational: K_2^+ , τ_{AW2} , both enclosures, both margins, the rule arithmetic, the crude tier, the first-order values, the one-plaquette series, the counts 49, 82, 10, 16 and 21, and the Haar moments. The only difference is display precision. The producer prints 12 significant digits and the skeptic 16, and the skeptic’s outward decimals nest inside the producer’s. No admission Boolean reads a decimal. The post-comparison review adds 73 exact checks. All 33 of its must-abort source edits abort. One silent edit, the rule-valued literal of Section 5.3, is caught only by the static review, and two preview demonstrations support finding N2. Four edits weaken the producer’s own validators: non-strict exclusion, and disabled incoming-star, below-cap and claim-flag checks. Each lets a damaging mutation through, and the run then aborts with “damaging mutation accepted”. The controls are therefore live.

Replays and isolation. The skeptic’s replay tool always replays both directions, so it fails on AW2. It is not edited, because its hash is bound in the AV1, AV2 and AW1 reviews. A

wrapper imports it unchanged and replays only the declared direction. The forward closure holds 35 files, and its 28 input snapshots equal their repository sources byte for byte. Normal and optimized replays reproduce the frozen outputs byte for byte, and the freezing tool reports the closure verified. One isolation finding, N1, stands. The forward files carry times from 00:10Z to 00:23Z, and the package was committed after the skeptic's pre-comparison commit; both worked on a shared filesystem. The forward closure contains no `skeptic/aw2` file, the producer declares no such read, and no forward AW2 scratch file is in the shared scratchpad root. The codes differ in structure: 12- against 16-digit decimals, boxes $N = 2, 3$ against $N = 2, 7$, and different check ids. Reads cannot be determined, so isolation is verified for repository inputs only. Exact agreement is expected in any case, since both apply one admitted formula to one gate constant. The producer also disclosed reads of the tools documentation, the freezing tool and the AV2 and AW1 forward checkers for protocol and code style, and hash-only checks of the repository contract and gate (N8); none carries premise weight. The producer's flag `correlation_shift_resolved` and the skeptic's `centered_correlation_shift_resolved` are both false and mean the same (N4).

5.9 Limitations recorded by the gate

The AW2 gate records the following limitations verbatim.

- Single direction: one forward producer instantiates the admitted AW1 formula; admission rests on that producer plus the skeptic's pre-comparison replay from the contract (correlated model-agent ancestry; K_2^+ is the skeptic's own AW1 itemization); exact agreement is a replay of one admitted formula, not independent physical confirmation
- Only the AM2/AQ1 zero-selected patterned family, the cover $R=\{0,e_z\}$, fixed spacing and $\tau=\pm 10^{-8}$; nothing transfers to nonzero selected triples, uniform Wilson theory, weak coupling or the continuum; calculator values below the cap are corollaries of the AW1 bound, not AW2 admissions
- Inherited without re-proof: the AW1 first-order coefficient $+\tau/144$, remainder bound $|r|\leq K_2^+ \tau^2$ (uniform in N , cutoff and every AQ1 limit) and flip lemma; AV1 product split, anchored-norm tier, cutoff removal and AQ passage; AM2 fixed point, majorant, gaps and ground uniqueness; AQ1 construction; I1 dictionary
- K_2^+ is an absolute second-order bound with no sign of $r(\tau)$ and no third-order remainder; the AM2 majorant 352JT is 99.98% of it, so the enclosure is conservative
- Static equal-time mean in the AQ states only; `resolved_interaction_shift` refers to $\omega(W)$ alone; no dynamical, mass-gap, susceptibility or centered-correlation statement (AV2 `reference_unresolved` stands)
- Whole-set statement: no uniqueness of the AQ state, whole-sequence convergence or rate in N ; the $-\tau$ pairing of individual states holds only along a common subsequence; the $-\tau$ enclosure is a replay
- Fixtures (one plaquette, 3x3 centring, synthetic two-limit sequences) are finite exact audits (`transfers_to_aq` false); the Arb/mpmath preview is a labelled comparison that `check.py` reads as a veto only and that supplies no admitted value
- The rule $K_2^+ \tau \leq 1/288$ guarantees only $m \geq 1$; the target $m \geq 2$ is tested separately; literal-coupling protection for a literal equal to the rule value rests on source review and the `check.py` hash in `freeze.json`
- Read isolation verified for repository inputs only: the forward package was written after the skeptic pre-comparison commit on a shared filesystem; its closure excludes `skeptic/aw2*`, but reads cannot be determined
- Scientific priority unverified

Full derivation and reviewed implementation: [AW2 forward](#), [independent pre-comparison replay](#), [independent model-agent review](#) and the [frozen contract](#). Sub-round 2 is complete; the gate selects AX1 next.

6 AX1: the uniform Kogut–Susskind Hamiltonian by route B

Contribution HNM-C-AX1. The AX1 contract, recorded under the title “Hruday route-B re-derivation for the uniform Kogut–Susskind SU(2) Hamiltonian at fixed spacing”, was frozen after the AW2 gate and before production. It opens sub-round 3. Sections 2–5 hold for the zero-selected patterned subfamily only, which is not the uniform model (Section 1.1). The third Round31 roadmap goal asks for the uniform Kogut–Susskind SU(2) Hamiltonian at fixed spacing, in which every elementary face carries the same coefficient $\alpha\tau/24$. AX1 re-instantiates the admitted chain for that operator: the AM2 expansion, the AQ1 and AQ2 constructions, the incidence on the cover and the AV1 state lemma, each with explicit constants. Its preregistered target is $D'_{ii} \leq 4/10^7$ at both signs of the unchanged cap, below the threshold that the AX2 window budget needs. Both directions are paired. The forward producer received AGENTS.md, the contract, its 36 shared premises and three forward-additional premises (the skeptic’s triage, the skeptic’s loop-2 response and the modern loop-2 response), 41 files in all. The reverse producer received only AGENTS.md, the contract and the 36 shared premises, 38 files, and starts from the uniform Hamiltonian. The contract asked the reverse producer to find the single-factor grouping and the J_0 issue itself. The contract text and the selection note already state both, so reverse independence covers the proof route, the enumeration and the code only (Section 6.11).

Notation: primes mark route-B constants (J' , t'_1 , T' , ϱ' , D'). For a factor b , O_b is the set of its 21 omitted faces and Sel_b the set of its three selected xy faces, the classes with $r \leq 2$, $q = 0$ of Table 1. Other notation follows Sections 1–4.

6.1 Reviewed scope

The AX1 gate records the verdict `accepted_within_scope`. The skeptic’s review and the reverse packet carry the sub-label `uniform_local_closeness_not_uniqueness`, and the gate’s statement carries the same restriction. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section. In it, R inside $G(R)$ and $G'(R)$ is AM2’s anchored radius $R_a = 1/64$, and R elsewhere is the cover; E is the flip set, S is the set S of AQ1 subsequential limits and $\circ$ denotes composition.

For the uniform Kogut–Susskind SU(2) Hamiltonian on Z^3 at fixed spacing and strong bare coupling (every elementary face has $\text{nu}=\alpha\tau/24$; AL1 dictionary $\tau=96/g^4$, so $g^4=9.6\times 10^9$ at the cap; never weak coupling or continuum; $\tau<0$ has no real g and is the U_E image of $|\tau|$), both signs $|\tau|\leq 10^{-8}$ (the uniform triple $\tau/24$ lies in the admitted box), centered whole-star boxes Λ_N ($N=2$), the original xz Wilson loop W with cover $R=\{0,e_z\}$, clock $\theta=\alpha\tau/\hbar$, $G=H/\alpha$: (1) Route B, Haar reference: $h_b=8 \sum_e C_e=6Q_b$ is τ -independent; whole stars ϕ_b (21 omitted faces, $\text{norm } 7|\tau|$) plus one single-factor group $\psi_b=-(\tau/3) \sum$ of the three selected xy faces (support $\{b\}$, $||\psi_b||\leq |\tau|$); every face charged once; $J'=29|\tau|$; support 4 and termination order 8 unchanged. (2) J' exceeds AM2’s $J_0=7/25000000$ at the cap; with the re-frozen $J_0'=29/10^8$ (R1) the exact inequalities $J_0'G(R)<1073/175000000<1/64$ and $2J_0'G'(R)<319/1562500<1$ hold, so every finite complete-factor route-B volume has a unique gauge-invariant ground and full-space gap $>=1/2$ normalized ($\alpha/16$ physical), with a nonzero physical excited sector. (3) AQ1/AQ2 re-instantiation: $\text{reset } \omega(h_R)\leq 102|\tau|$; with the Haar gap six $\epsilon_R\leq 17|\tau|$ and $2\sqrt{17|\tau|}\leq 1/500$ (square-root control only; gap one fails), so the variance floor $61999/250000$ holds; $C_F=58|\tau||F|$ and $||\Phi'|||_F\leq 81J'=2349|\tau|$; compactness, stationarity, GNS, nonnegativity, gauge averaging, the gap $\alpha/16$, cutoff-vector removal and AQ passage re-apply verbatim. (4) Itemized incidence: exactly 7 whole stars (anchors R – S) and 2 single-factor groups meet R , charging 153 faces, 88 meeting R (82 omitted + 6 selected, the 6 selected being the two single groups) and 16 inside R , none twice; $||B_N||\leq 51|\tau|/8$, $k'=51|\tau|/4$ in G units. (5) Exact first-order count 52 faces per factor ($49+3$) and 88 for R ; 96 (4×24) and 168 (7×24) are bounds only. (6) State lemma at both signs: tier (ii) with $t_1'=52|\tau|/144$, $T'=t_1'/(1-352J')=13/3599632512$, $\rho'=352J'T'$ gives the bound value $D'_{ii}=2425369125199104794263242601250/167893028420061547330293754713793182097$

(~1.44459e-8; 2eps(1+eps)/(1+eps^2) with eps=2T'+T'^2; certified by both inequalities; equal to the skeptic pre-comparison and the reverse two-site variant), meeting 4/10^7 (margin ~27.7) and 10^-6; the reverse's labelled R-refinement D'_ii=30934916107401289/2530733246376451200000000 (~1.22237e-8; 2eps with eps=88|tau|/144+2rho'+T'^2, fidelity inequality) is also valid and differs by eps_fwd-eps_rev=16|tau|/144; tier (i) D'_i~2.4526e-5 fails and is retained; |omega(W)|<=D', |omega(W^2)-1/4|<=D'/2, m^2<=D'^2, passing to every AQ1 subsequential limit as uniform local closeness (not uniqueness, whole-sequence convergence or a rate in N). (7) Transfer, proved: every face coefficient is -(tau/3) (kappa tied to tau), h_b and its cutoffs commute with U_E, and E meets every plaquette, selected faces included, oddly; so U_E H_N(tau) U_E^*=H_N(-tau) in every box and cutoff, omega_{N,-tau}(W)=-omega_{N,tau}(W), omega_N(W^2), C_N, c_N even, and S(-tau)=S(tau) o alpha_E as whole sets; the AW1 link grading applies verbatim to selected faces and only f=W contributes at first order, so omega(W)^{1}={1}+=+tau/144 (+-1/14400000000) is unchanged. No Euclidean node, K_2 or uniform omega(W^2) constant (not transferred), uniform Wilson-mean sign certificate, weak-coupling, continuum, uniqueness, rate or priority statement is admitted; uniform_wilson_claim:true means only that the model is the uniform fixed-spacing Kogut-Susskind model as labelled.

The rest of this section derives that statement. The limitations recorded by the gate close the section (Section 6.12).

6.2 Dictionary, coefficient box and label

AL1 (Round29) fixes the uniform bulk dictionary in the convention it takes from Bauer et al. (arXiv:2307.11829), with $W_p = \frac{1}{2} \text{Tr } U_p$ and the raw magnetic term $\lambda \sum_p (1 - W_p)$ (AL1 gate):

$$H_{\text{KS}} = \alpha \sum_e C_e + \lambda \sum_p (1 - W_p), \quad \alpha = \frac{g^2}{2a}, \quad \lambda = \frac{2}{g^2 a}, \quad \frac{\lambda}{\alpha} = \frac{4}{g^4}, \quad (80)$$

$$\nu = \lambda = \frac{\alpha \tau}{24} \implies \tau = \frac{24\lambda}{\alpha} = \frac{96}{g^4}.$$

Dividing by $\delta = \alpha/8$ and dropping the scalar λN_p gives

$$\frac{H_{\text{KS}} - \lambda N_p}{\delta} = 8 \sum_e C_e - \frac{\tau}{3} \sum_f W_f, \quad (81)$$

so every face, selected or omitted, enters with the coefficient $-\tau/3$ of the I1.5 convention. The selected triple is therefore $(\lambda_L, \mu_M, \lambda_R)/\alpha = (\tau/24, \tau/24, \tau/24)$. The admitted box $|\lambda_L|, |\lambda_R| \leq \alpha/2$, $|\mu_M| \leq \alpha/8$ requires $|\tau|/24 \leq 1/8$, that is $|\tau| \leq 3$. This is AL1's bridge condition $g^4 \geq 32$. At the cap,

$$\frac{|\tau|}{24} = \frac{1}{2400000000}, \quad g^4 = \frac{96}{10^{-8}} = 9.6 \times 10^9 \quad (g^2 \approx 9.80 \times 10^4), \quad \frac{g^4}{32} = 3 \times 10^8. \quad (82)$$

The model is labelled *uniform Kogut–Susskind SU(2) at fixed spacing, strong bare coupling*. It is never described as weak coupling or as a continuum approach, and both checkers reject a label that uses either phrase. No path with $g \rightarrow 0$ stays inside the box, since the box needs $g^4 \geq 32$; the cap sits at the opposite, strong-coupling end. The physical gap $\alpha/16$ proved below equals $g^2/(32a)$.

For real g , $\tau = 96/g^4 > 0$, so $\tau = -10^{-8}$ has no real- g preimage. It is the U_E image of $+|\tau|$ (Section 6.9), and its evaluation replays the same $|\tau|$ formula at the mirrored coupling. It is neither a second Kogut–Susskind coupling nor a second confirmation. The checkers verify $\tau g^4 = 96$, $\alpha\tau/24 = \lambda$ and $(\alpha/8)(\tau/3) = \lambda$ on three exact (g^2, a) fixtures. Between them they reject a non-uniform or ends-only triple, the zero triple, $\tau = 4$ (bridge value $1/6 > 1/8$) and the normalized $\tau/3$ read in units of α .

6.3 Route B: Haar reference and single-factor groups

Why route B. The admitted chain needs a Haar reference in three places: the AV1 state lemma (face vectors $W_f \Omega_0$ at energy 24 and the Haar moments of P_R), the reset with on-site gap six, and the AW1 grading. Route A keeps the selected faces on site, $h_b^\Lambda = 8 \sum_e C_e - (\tau/3) \sum_{g \in \text{Sel}_b} W_g$. Its vacuum is not Haar. For one selected face with $k = \tau/3$, the trial vector $\Omega + (k/24)W_g \Omega$ has energy numerator $6t^2 - kt/2$ at $t = k/24$, that is $-k^2/96 = -\tau^2/864 < 0$, below the Haar value 0. Route A keeps the triple in the box and $J = 28|\tau|$, so the AM2 gate already covers the gap of this very operator. That coverage is an inherited cross-check only: route A does not supply the Haar vector $\psi = e^{-C} \Omega_0$, and claiming the Haar reference for it is the rejected control `selected_reference_not_haar`. Route B keeps the τ -independent Casimir sum on site and moves the selected faces into the interaction:

$$\hat{H}'_N = \sum_{b \in \Lambda_N} h_b + \sum_{b+S \subset \Lambda_N} \phi_b + \sum_{b \in \Lambda_N} \psi_b, \quad \phi_b = -\frac{\tau}{3} \sum_{f \in O_b} W_f, \quad \psi_b = -\frac{\tau}{3} \sum_{g \in \text{Sel}_b} W_g, \quad (83)$$

with $h_b = 8 \sum_{e \text{ owned by } b} C_e$ as in (1). All four links of a selected face are owned by b , so ψ_b has support $\{b\}$ and every single-factor group of the box is retained. Routes A and B split the same finite box operator up to on-site scalars. The route-B box therefore keeps AQ1's inventory: whole stars inside Λ_N and every selected face of every factor in Λ_N .

Norms and the per-site sum. Since $\|W_f\| \leq 1$,

$$\|\phi_b\| \leq 21 \cdot \frac{|\tau|}{3} = 7|\tau|, \quad \|\psi_b\| \leq 3 \cdot \frac{|\tau|}{3} = |\tau|, \quad J' = \max_u \sum_{X \ni u} \|V_X\| \leq 4 \cdot 7|\tau| + |\tau| = 29|\tau|. \quad (84)$$

Both norm bounds are attained at the identity configuration. A site lies in exactly four stars, anchored at $u - S$ with the incoming stars included, and in exactly one single-factor group. The reverse checker recomputes J' from the explicit group list of Λ_3 : the bulk maximum 29 is attained at the origin, and boundary sites are lower. Three other groupings are rejected: charging the selected faces in both a star and a single group (a duplicate face, per-site sum $33|\tau|$), merging them into the star ($32|\tau|$) and dropping them ($28|\tau|$, the zero-selected model). A per-face grouping has the smaller sum $52|\tau|/3$, but it changes the interaction decomposition and the boundary prescription; the reverse producer discloses it and does not use it.

No face is charged twice. A face has one anchor, the image under π of its base point, and one class among the 24 anchored classes. Stars hold only the 21 omitted classes and single groups only the 3 selected ones, so the two families are disjoint. The forward checker places each of 3600 plaquettes of a region of complete blocks in exactly one group. The reverse checker and the skeptic check the 5565 retained faces of Λ_3 , that is $216 \cdot 21 + 343 \cdot 3$.

Support and termination. Stars have four sites and single groups one. Two creations meeting the same site multiply to zero, so a nonzero nested word carries at most $|X|$ creations meeting X on each side of V_X , and

$$\text{ad}_{C_1} \cdots \text{ad}_{C_k}(V_X) = 0 \quad (k > 2|X|), \quad \max_X 2|X| = 8. \quad (85)$$

Exact qubit fixtures give $\text{ad}_C^8(V)\Omega = 8!|1111\rangle$ and $\text{ad}_C^9(V) = 0$ for a star, and a nonzero second order with $\text{ad}_C^3(V) = 0$ for a single site. AM2's multilinear constants $2^p(2p)^k(1+k(p+1)/p)$ increase in p , and every step of AM2's estimate uses $|X| \leq p$ only as an upper bound: at most $2^{|X|}$ output sets, $\sum_{I \cap X \neq \emptyset} \|c_I\| \leq |X| \|c\|_a$ and $|I_j| \leq |M| + |X|$. With $p = 4$, single-site groups satisfy AM2's majorant unchanged,

$$\|L_k(c_1, \dots, c_k)\|_a \leq J' 16 \cdot 8^k \left(1 + \frac{5k}{4}\right) \prod_j \|c_j\|_a, \quad (86)$$

and $G(t) = 16e^{8t}(1 + 10t)$ of (5) is unchanged. This monotonicity is the only new step in the majorant; the rest is inherited.

6.4 Re-frozen contraction and the finite-volume theorem

At the cap, $J' = 29/10^8$ exceeds AM2's frozen $J_0 = 7/25000000 = 28/10^8$. The AM2 gate therefore cannot simply be re-read, and citing it with its admitted constants is the rejected control `am2_gap_reuse_justified`. Resolution R1, frozen in the contract, re-freezes $J'_0 = 29/10^8$. With $e^{1/8} < 8/7$ one has $G(R_a) = \frac{37}{2}e^{1/8} < \frac{148}{7}$ and $G'(R_a) = 308e^{1/8} < 352$, as in (5). So

$$J'_0 G(R_a) < \frac{29}{10^8} \cdot \frac{148}{7} = \frac{1073}{175000000} < \frac{1}{64} = R_a, \quad 2J'_0 G'(R_a) < 2 \cdot \frac{29}{10^8} \cdot 352 = \frac{319}{1562500} < 1. \quad (87)$$

The Lipschitz constant is below $319/3125000$. The self-map inequality alone would allow $|\tau| \leq 7/274688$, about 2548 times the cap, and the exclusion inequality $|\tau| < 1/20416$. Resolution R2 keeps J_0 and lowers the cap to $|\tau| \leq J_0/29 = 7/725000000$. That is a changed coupling and is not selected; the controls `j0_resolution_declared` and `changed_model_relabelled` reject it under the cap label. Both packets pin J'_0 , $1073/175000000$ and $319/1562500$ together with the contract's SHA-256.

On-site theorem. By Peter–Weyl, C_e acts as $j(j+1)$ on the spin- j block, and its kernel is the constants. So $h_b = 8 \sum_e C_e \geq 0$ has the one-dimensional kernel spanned by the Haar vector Ω_b , and its next eigenvalue is $8 \cdot \frac{3}{4} = 6$:

$$h_b \geq 6 Q_b \geq Q_b, \quad P_b = |\Omega_b\rangle\langle\Omega_b|, \quad Q_b = I - P_b. \quad (88)$$

The Casimirs are bi-invariant, Ω_b is constant and every W_f is gauge invariant. The forward checker confirms the last point on exact rational-quaternion holonomies for six faces, among them W and selected faces.

Proposition 1 (route B, uniform model). *Take any nonempty finite complete-factor volume with route-B retention: whole stars inside the volume and every single-factor group. For either sign of $|\tau| \leq 10^{-8}$ the untruncated Hamiltonian has a unique ground $\psi = e^{-C}\Omega_0$ with $\|c\|_a \leq 1/64$, and*

$$E_1 - E_0 \geq \frac{1}{2} \quad (\text{normalized}), \quad \Delta_{\text{phys}} \geq \frac{\alpha}{8} \cdot \frac{1}{2} = \frac{\alpha}{16} = \frac{g^2}{32a}.$$

The ground is gauge invariant, and the physical subspace contains a nonzero excited sector.

The proof is AM2 Sections 2–6 with the new constants. It uses $h_x \geq Q_x$ (88); supports of at most four sites with termination at order eight (85); $J \leq J' \leq J'_0$ in every finite volume, because boundary per-site sums never exceed the bulk value (checked for $N = 1, 2, 3$); the self-map and the Banach contraction of (87); the shifted-resolvent exclusion $\|b\|_a \leq 2J'_0 G'(R_a) \|b\|_a < \|b\|_a$ for real $|z| < \frac{1}{2}$; and continuity of the isolated branch along the coupling from 0. AM2's cutoff removal applies verbatim, because h_b is a Casimir sum on a compact group with compact resolvent. The two producers prove the excited sector differently. The forward producer uses a selected face g : Ω_0 and $2W_g\Omega_0$ are orthonormal gauge-invariant vectors, since $\mathbb{E}[W_g] = 0$ and $\mathbb{E}[W_g^2] = \frac{1}{4}$, so the physical subspace has dimension at least two. The reverse producer uses the strict positivity of the ground, a Schrödinger ground of a Laplacian plus a bounded real potential on a connected compact manifold, so $(W_g - \langle W_g \rangle)\Omega_H$ is nonzero, gauge invariant and orthogonal to the ground. The skeptic accepts both arguments.

6.5 AQ1/AQ2 re-instantiation

Reset on the cover. Reset the finite ground density on R to the Haar product $P_R = P_0 \otimes P_{e_z}$ and keep the exterior marginal. Each group meeting R changes its expectation by at most twice its norm, and every other group is unchanged. Seven stars and two single groups meet R (Section 6.6), so the finite-ground variational principle gives

$$\omega_N(h_R) \leq 2(7 \cdot 7|\tau| + 2 \cdot |\tau|) = 102|\tau|. \quad (89)$$

The reverse producer rejects $100|\tau|$ as a miscount. The review notes that $2 \cdot 7 \cdot 7 + 2 = 100$ is also a valid labelled refinement, because the single groups lie inside R and have Haar mean 0. The admitted budget stays $102|\tau|$. The Haar product has the two-site gap six: $Q_0 \otimes I + I \otimes Q_{e_z} - (I - P_0 \otimes P_{e_z}) = Q_0 \otimes Q_{e_z} \geq 0$ gives $h_R \geq 6(I - P_R)$. Hence, at $|\tau| = 10^{-8}$,

$$\epsilon_R = 1 - \text{Tr}(\rho_R P_R) \leq \frac{102|\tau|}{6} = 17|\tau|, \quad \|\rho_R - P_R\|_1 \leq 2\sqrt{17|\tau|} \approx 8.2462 \times 10^{-4} \leq \frac{1}{500}. \quad (90)$$

The square-root bound is a control only. It scales by exactly 10 under $\tau \rightarrow \tau/100$, and the admitted state term is the linear D' of Section 6.8.

The Haar gap six is load-bearing. AQ2 needs $\|\rho_R - P_R\|_1 \leq 1/500$ for its Wilson variance floor. AQ2's own argument used only $h_R \geq I - P_R$, and its budget $98|\tau|$ passed with gap one only narrowly, $2\sqrt{98 \cdot 10^{-8}} \approx 1.980 \times 10^{-3}$. With the route-B budget, gap one fails and gap six holds:

$$(2\sqrt{102 \cdot 10^{-8}})^2 = \frac{51}{12500000} > \frac{1}{250000}, \quad (2\sqrt{17 \cdot 10^{-8}})^2 = \frac{17}{25000000} \leq \frac{1}{250000}. \quad (91)$$

The reference moments $\text{Tr}(P_R W) = 0$ and $\text{Tr}(P_R W^2) = \frac{1}{4}$ now hold because all 48 cover links are Haar. With $|\omega(W)| \leq 1/500$ and $\omega(W^2) \geq \frac{1}{4} - \frac{1}{500}$, AQ2's floor follows unchanged:

$$\text{Var}_\omega(W) \geq \frac{1}{4} - \frac{1}{500} - \frac{1}{250000} = \frac{61999}{250000} > \frac{1}{5}. \quad (92)$$

The forward checker retains the gap-one failure as a control. The producers classify this step differently. The reverse lists the floor as a verbatim re-application and puts the new constant in its overlap row; the forward lists the floor itself as needing the gap six. The substance is the same (review finding N9).

AQ1 constants. Resetting a finite region F changes at most $4|F|$ stars and exactly $|F|$ single groups. Every group has ℓ^1 -diameter at most two. With the Nachtergaele–Sims weight $\mathbf{F}(r) = (1+r)^{-4}$, whose norm (at most 7) and convolution constant (at most 224) are unchanged,

$$C'_F = 2(4 \cdot 7 + 1)|\tau| |F| = 58|\tau| |F|, \quad \|\Phi'\|_F \leq \mathbf{F}(2)^{-1} J' = 81 \cdot 29|\tau| = 2349|\tau|. \quad (93)$$

A single-site group contributes only at $x = y$, where $\mathbf{F}(0) = 1$; equivalently it is a bounded on-site perturbation of the unbounded Casimir sums. The skeptic also records a per-pair bound $567|\tau|$, labelled and not needed. Citing AQ1's $2268|\tau| = 81 \cdot 28|\tau|$ or $C_F = 56|\tau| |F|$ as unchanged would be a citation error. Table 10 lists every step. No named constant is left unproved, so the checklist does not trigger the contract's limited outcome.

6.6 Itemized incidence on the cover

A group meets $R = \{0, e_z\}$ exactly when its support does. A star $b + S$ meets R iff $b \in R - S$, the seven anchors of (2), all retained for $N \geq 2$; the farthest site needed is $2e_z$. A single group $\{b\}$ meets R iff $b \in R$: exactly two groups. In Λ_1 only four of the seven stars are retained, which is why

Table 10: AQ1/AQ2 re-instantiation for route B. “Verbatim” means the admitted argument re-applies unchanged; “new” means it needs the stated route-B constant. The last rows count J per $|\tau|$ and t_1, a_1 per $|\tau|/144$.

Step (source)	Status	Zero triple (admitted)	Route B
AM2 ground and gap	new	$J \leq 28 \tau \leq J_0$	$J' \leq 29 \tau \leq J'_0 = 29/10^8$
AM2 on-site hypothesis	verbatim	$h_x \geq Q_x$	$h_b \geq 6Q_b \geq Q_b$
AM2 majorant, termination	verbatim	support 4, order 8	unchanged; $ X = 1$ by monotonicity
AM2 gauge covariance, cutoff removal	verbatim		Casimir sum, compact resolvent
AQ1 reset energy on F	new	$56 \tau F $	$58 \tau F $
AQ1 compactness, diagonal extraction	verbatim	uses C_F	uses C'_F
AQ1 Nachtergaele–Sims placement	new	$2268 \tau $	$81J' = 2349 \tau $
AQ1 stationarity, GNS, $H \geq 0$	verbatim		
AQ2 gauge averaging, gap $\alpha/16$	verbatim	premise: AM2 full-space gap at J_0	premise: AX1 full-space gap at J'_0
AQ2 reset on R	new	$98 \tau $	$102 \tau $
AQ2 overlap, trace distance	new	$\epsilon \leq 98 \tau $, gap one	$\epsilon_R \leq 17 \tau $, gap six
AQ2 reference moments of W	verbatim	one free z link Haar	all 48 cover links Haar
AQ2 variance floor	verbatim	61999/250000	same; needs gap six
AT4 relative-unitary slope	new	$49 \tau /8$, $k = 49 \tau /4$	$51 \tau /8$, $k' = 51 \tau /4$
AV1 state-lemma tiers	new	$J, t_1, a_1 = 28, 49, 82$	$J', t'_1, a'_1 = 29, 52, 88$
AV1 cutoff removal, AQ passage	verbatim	gap $\frac{1}{2}$	gap $\frac{1}{2}$ at J'_0

Table 11: Groups meeting the cover $R = \{0, e_z\}$, with the faces each charges. Norms are in units of $G = H/\alpha$, where each group enters as $V_X/8$.

Group	Anchor	Sites of R met	Faces	Meeting R	Inside R	Norm
star	0	$0, e_z$	21	21	10	$7 \tau /8$
star	e_z	e_z	21	21	0	$7 \tau /8$
star	$-e_z$	0	21	16	0	$7 \tau /8$
star	$-e_x$	0	21	4	0	$7 \tau /8$
star	$-e_y$	0	21	8	0	$7 \tau /8$
star	$e_z - e_x$	e_z	21	4	0	$7 \tau /8$
star	$e_z - e_y$	e_z	21	8	0	$7 \tau /8$
single	0	0	3	3	3	$ \tau /8$
single	e_z	e_z	3	3	3	$ \tau /8$
Total	9 groups		153	88	16	$51 \tau /8$

the lemma needs $N \geq 2$. The forward checker enumerates $N = 2, 3, 4$. The reverse checker verifies the coordinate condition for $N = 2, \dots, 11$ and enumerates Λ_2 and Λ_3 . Both export every face with its class, fine base point and owner set, and recompute every total from that table. The three tables, forward, reverse and the skeptic's, agree row by row (Table 11).

As in AT4 and (36), every group meeting R enters B_N whole, so A_N evolves W by the free Casimir generator on R . In units of $G = H/\alpha$ and with the clock $\theta = \alpha t/\hbar$,

$$\|B_N\| \leq 7 \cdot \frac{7|\tau|}{8} + 2 \cdot \frac{|\tau|}{8} = \frac{51|\tau|}{8}, \quad k' = 2\|B_N\| = \frac{51|\tau|}{4}. \quad (94)$$

The 153 charged faces are $7 \cdot 21 + 2 \cdot 3$. Of them, 88 meet R (82 omitted and 6 selected) and 16 lie inside R (10 omitted and 6 selected); the other 65 are charged in full, as in AT4. The six selected faces inside R are exactly the two single groups: the xy faces with fine base points $(0, 0, 0)$, $(1, 0, 0)$ and $(2, 0, 0)$ of factor 0, and $(0, 0, 1)$, $(1, 0, 1)$ and $(2, 0, 1)$ of factor e_z . Two of them share one link with W , namely $(0, x)$ and (e_z, x) . No face appears in two rows. The faces anchored at 0 whose owner sets also contain e_z belong to star 0 only, and the selected faces belong to the single groups only. Rejected mutations include totals without the table, a missing single group, a group listed twice, the orthant two-anchor count, the outgoing-only sum $J = 8|\tau|$, a group count of 7 instead of 9 and the root-sum-of-squares $\sqrt{345}/8$ of the nine group norms (`root_n_misuse`). The contract registered $51|\tau|/8$; both producers derive it from the itemized table.

6.7 Face counts: 52 per factor and 88 for the cover

Every retained face f , omitted or selected, gives a vector $W_f\Omega_0$ with spin $\frac{1}{2}$ on its four links, normalized H_0 -energy $8 \cdot 4 \cdot \frac{3}{4} = 24$ and norm $\frac{1}{2}$, in the sector of its owner set M_f . Distinct faces share at most one link, so the vectors are orthogonal; the forward checker verifies this over all pairs of the 88 faces meeting R . A selected face has $M_g = \{b\}$. The first-order collection (16) therefore keeps its form in route B:

$$c_M^{(1)} = -\frac{\tau}{72} \sum_{f: M_f=M} W_f\Omega_0, \quad \|c_M^{(1)}\| = \frac{|\tau|\sqrt{n_M}}{144}, \quad \|c_{\{b\}}^{(1)}\| = \frac{\sqrt{3}|\tau|}{144} \neq 0. \quad (95)$$

The single-site creations are new. In the zero-selected model every first-order support had at least two sites. In route B the supports $\{0\}$ and $\{e_z\}$ are nonzero at first order, meet R and enter the two-creation term.

The counts are derived from the 24-class encoding as in (17). A face of class k anchored at b has u in its owner set iff $b = u - d$ with $d \in K_k$. The offset 0 lies in all 24 classes, and the selected classes contain only 0. Hence

$$\begin{aligned} \#\{f : u \in M_f\} &= \sum_{d \in S} \#\{k : d \in K_k\} = 24 + 4 + 8 + 16 = 52 = 49 + 3, \\ \#\{f : M_f \cap R \neq \emptyset\} &= 52 + 52 - 16 = 88. \end{aligned} \quad (96)$$

A site lies in 16 owner sets, with multiplicities $\{1 \times 5, 2 \times 3, 3 \times 3, 4 \times 3, 10 \times 2\}$; relative to Table 4 the new set is $\{u\}$ with $n = 3$. Sixteen faces lie inside R : the 10 omitted faces with owner set R and the 6 selected faces. Sixteen faces have owner sets containing both sites, and $16 + 16 - 3 = 29$ owner sets meet R . The omitted-only sub-table reproduces AV1's counts (49, 15, 82, 10). A brute-force fine-lattice enumeration without the class table reproduces 52, all 16 multiplicities and 88. Finite-box counts never exceed the bulk values (checked for $N = 1, 2, 3$).

The contract candidates are bounds. The contract registered “96 per factor” and “168 for R ” as candidates to be derived. They are the anchor-group bounds 4×24 and 7×24 : every face owning a link of u is anchored in $u - S$, but the four groups at those anchors also hold 44 faces

whose owner sets miss u . The candidates are admitted only as labelled bounds; the exact counts are 52 and 88. Using 96 as an exact count is a rejected mutation, and so are 49 (selected faces dropped), 61 (selected faces counted at all four anchors), a hard-coded 52 and a count restricted to 0 and e_z .

6.8 State-lemma tiers

Product ordering with the enlarged creation set. The split of Section 2.2 uses only that the creations commute, that overlapping creations multiply to zero and that $c_I \in \bigotimes_{x \in I} Q_x \mathcal{H}_x$ with Haar P_x . It holds verbatim for the route-B creations:

$$\psi = \psi_{\text{out}} + \delta, \quad (P_R \otimes 1)\psi = \psi_{\text{out}}, \quad \frac{\|\delta\|}{\|\psi_{\text{out}}\|} \leq \varepsilon = \sum_{I \cap R \neq \emptyset} \|c_I\| + \sum_{\substack{I \ni 0, J \ni e_z \\ I \cap J = \emptyset}} \|c_I\| \|c_J\| \leq 2t + t^2, \quad (97)$$

and both inequalities (13) and (14) apply. The pair of single-site creations $c_{\{0\}}$ and $c_{\{e_z\}}$ now enters the second sum with both factors of first order.

The two tiers. The crude tier uses only $t \leq J'G(R_a)$; at the cap $t_i = 1073/175000000$, the self-map rational. The exact tier uses the first-order collection and the self-consistent remainder of (19) with the route-B constants:

$$t'_1 = \frac{|\tau|}{144} \max_u \sum_{M \ni u} \sqrt{n_M} \leq \frac{52|\tau|}{144} = \frac{13|\tau|}{36}, \quad (98)$$

$$t \leq T' := \frac{t'_1}{1 - 352J'}, \quad \varrho' := 352J'T' = T' - t'_1.$$

At $\tau = \pm 10^{-8}$ the inputs are $J' = 29/10^8$, $352J' = 319/3125000$, $t'_1 = 13/3600000000$, $T' = 13/3599632512$ and $\varrho' = 4147/11248851600000000 \approx 3.687 \times 10^{-13}$. Three tier mixtures are rejected (`tier_mixing_rejected`): the exact $c'^{(1)}$ with the crude t , $t = t'_1$ without self-consistency, and a zero remainder.

Forward form, bound by the gate. The forward producer, and the skeptic before comparison, use AV1's global form $\varepsilon_{\text{fwd}} = 2T' + T'^2$ with the explicit-density bound f of (13):

$$\varepsilon_{\text{fwd}} = \frac{93590445481}{12957354221447430144}, \quad (99)$$

$$D'_{\text{ii}} = f(\varepsilon_{\text{fwd}}) = \frac{2425369125199104794263242601250}{167893028420061547330293754713793182097} \approx 1.4445919214 \times 10^{-8}.$$

Since $g(\varepsilon) \leq f(\varepsilon)$, both inequalities certify this value. It equals the skeptic's pre-comparison headline rational for rational, and the reverse reproduces it exactly as its labelled global two-site variant.

Reverse form, a labelled cover refinement. The reverse producer bounds the first-order sum over creations meeting R face by face, counting each of the 88 faces meeting R once, and charges the remainder at both sites:

$$a'_1 = \frac{88|\tau|}{144} = \frac{11}{1800000000}, \quad (100)$$

$$\varepsilon_{\text{rev}} = a'_1 + 2\varrho' + T'^2 = \frac{30934916107401289}{5061466492752902400000000} \approx 6.1118 \times 10^{-9},$$

$$D'_{\text{ii,rev}} = 2\varepsilon_{\text{rev}} = \frac{30934916107401289}{2530733246376451200000000} \approx 1.22237 \times 10^{-8}. \quad (101)$$

This is the route-B analogue of the AV1-reviewed 82-face refinement (24). Its headline 2ε dominates $g(\varepsilon)$ and therefore rests on the fidelity inequality. Its density form $f(\varepsilon_{\text{rev}}) \approx 1.22237 \times 10^{-8}$, certified by both inequalities, is also below the target.

The one-inequality difference. The two forms share t'_1 , T' , ϱ' and the pair term T'^2 . They differ at one inequality, the bound on the first-order sum $\sum_{I \cap R \neq \emptyset} \|c_I^{(1)}\|$. The forward bound $2\|c\|_a \leq 2T' = 2t'_1 + 2\varrho'$ charges $2 \times 52 = 104$ first-order faces, so the 16 faces whose owner sets contain both sites are counted twice. The reverse counts the 88 faces meeting R once. Exactly,

$$\varepsilon_{\text{fwd}} - \varepsilon_{\text{rev}} = \frac{(104 - 88)|\tau|}{144} = \frac{16|\tau|}{144} = \frac{|\tau|}{9} = \frac{1}{900000000}. \quad (102)$$

The final step also differs, f against 2ε , but that moves the bound by only about 1.04×10^{-16} . So $D'_{\text{ii}} - D'_{\text{ii,rev}} \approx 2|\tau|/9 + 1.04 \times 10^{-16} \approx 2.2222 \times 10^{-9}$, a ratio of about 1.1818, and

$$g(\varepsilon_{\text{rev}}) \leq D'_{\text{ii,rev}} \leq f(\varepsilon_{\text{rev}}) < 2\varepsilon_{\text{fwd}} \leq D'_{\text{ii}} \leq \frac{4}{10^7}. \quad (103)$$

The gate binds the forward D'_{ii} . It is the larger bound, both inequalities certify it, and it uses no cover refinement, so it survives any dispute of one. It is also the construction the AV1 gate bound: the skeptic’s code reproduces AV1’s D_{ii} (23) exactly from 49 faces and $J = 28|\tau|$. The reverse value is admitted as a labelled refinement.

Values against the targets. Every value is the same at both signs, because only $|\tau|$ enters. Table 12 compares them with the preregistered target $4/10^7$ and the secondary comparison 10^{-6} .

Table 12: Route-B state bounds at $\tau = \pm 10^{-8}$. Decimals are rounded previews of exact rationals in the frozen outputs; the AV1 row is the zero-selected value, shown for comparison.

Bound on $\ \rho_R - P_R\ _1$	Approximate value	$\leq 4/10^7$	$\leq 10^{-6}$
Square-root control, $2\sqrt{17 \tau }$	$8.246211251 \times 10^{-4}$	no	no
Tier (i), forward exact $f(\varepsilon_i)$	$2.452609023 \times 10^{-5}$	no	no
Tier (i), reverse $2\varepsilon_i$	$2.452578947 \times 10^{-5}$	no	no
Tier (ii), labelled 96-face bound (forward)	$2.666938952 \times 10^{-8}$	yes	yes
Tier (ii), forward exact, bound by the gate	$1.444591921 \times 10^{-8}$	yes	yes
Tier (ii), reverse cover refinement	$1.222369689 \times 10^{-8}$	yes	yes
AV1 tier (ii), zero triple	$1.361245287 \times 10^{-8}$	yes	yes

Tier (i), with $\varepsilon_i = 375551151329/3062500000000000$ and the exact forward value

$$D'_i = \frac{23002790096235779074916932482}{937890625141038667264537458466241},$$

exceeds $4/10^7$ by a factor of about 61. It fails both targets, is retained as a limited-tier value and is not retuned. Iterating the majorant lowers it only to about 1.856×10^{-5} . Tier (ii) meets the target with margins about 27.69 (forward) and 32.72 (reverse). Further labelled variants (grouped square roots, sharper remainder forms and the reverse’s 96/168 bounds) also lie below the target; they are recorded, not admitted. The contract’s “candidate about $2\text{--}3 \times 10^{-8}$ from 96 faces per factor” is the labelled 96-face row. The uniform value exceeds AV1’s zero-triple value mainly through the three selected faces per factor ($52/49 \approx 1.061$), and marginally through the extra $|\tau|$ in J' .

Consequences, passage and scaling. Trace duality and the effect bound give, at both tiers and both signs,

$$|\omega_\tau(W)| \leq D', \quad |\omega_\tau(W^2) - \tfrac{1}{4}| \leq \tfrac{D'}{2}, \quad m_\tau^2 \leq D'^2. \quad (104)$$

At the bound tier-(ii) value these are about 1.4446×10^{-8} and 7.223×10^{-9} , and $m_\tau^2 \leq D'^2$ is charged. The first-order mean $|\tau|/144 \approx 6.94 \times 10^{-11}$ is paid for, since D'_{ii} is about 208 times larger. AV1’s cutoff-vector removal (Section 2.7) re-applies with the route-B gap $\frac{1}{2}$; the selected face vectors also sit at energy 24, so $L \geq 24$ keeps $c'^{(1)}$. Passage by local trace-norm convergence to every

AQ1 subsequential limit re-applies verbatim. The result is uniform local closeness, not uniqueness, whole-sequence convergence or a rate in N . Every bound increases in $|\tau|$, so the cap values bound every smaller coupling. The exact ratios $D'(10^{-8})/D'(10^{-10})$ are about 100.0101 (forward tier (ii)), 100.0119 (reverse tier (ii)), 100.0015 and 100.0003 (tier (i)), all in $[99, 101]$, while the square-root control has ratio exactly 10.

AX2 feasibility, arithmetic only. With $k'M_1 = (51|\tau|/4)(4s/\pi)$ at $s = 1$, the AX2 window budget of (38) reads $F_u(D) = 2(D + D^2) + 51|\tau|/\pi \leq 10^{-6}$. With a directed Machin bracket for π , $F_u(D'_{ii}) \leq 1.9123 \times 10^{-7}$ (margin about 5.23) and $F_u(4/10^7) \leq 9.624 \times 10^{-7}$. The exact threshold lies in $[41883, 41884]/10^{11}$. The contract's note " $D' \leq 4.19 \times 10^{-7}$ " is therefore rounded the wrong way: $F_u(4.19 \times 10^{-7}) > 1.0003 \times 10^{-6}$, so 4.19×10^{-7} itself is infeasible. The tested target $4/10^7$ lies below the threshold, and the finding is non-blocking. The crude D'_i is infeasible. This is arithmetic on the contract statement: AX1 proves no window lemma and certifies no Euclidean node.

Error ledger. The preregistered items at tier (ii) are the following.

- **am2_remainder:** $\varrho' = 4147/11248851600000000$ (forward); the reverse charges $2\varrho'$. Not applicable at tier (i), which bounds the whole collection.
- **two_creation:** $T'^2 \approx 1.304 \times 10^{-17}$, now including the first-order pair $c_{\{0\}} \otimes c_{\{e_z\}}$.
- **straddling:** inside the $2t$ term (forward) or a'_1 (reverse); 72 of the 88 first-order faces meeting R have straddling owner sets.
- **density:** $2\varepsilon^2/(1 + \varepsilon^2)$, exported by the forward producer; not applicable to the reverse's fidelity route, with the reason stated. Review finding N6 notes that the exported value is an upper bound on $D' - 2\varepsilon = 2\varepsilon^2(1 - \varepsilon)/(1 + \varepsilon^2)$, not an identity; the excess is about 7.5×10^{-25} , and D' itself is exact.
- **onsite_cutoff_vector:** no numeric cost; it is an exact per-box limit.
- **arithmetic:** no numeric cost for the exact rationals; every enclosure is directed upward ($e^{1/8} < 8/7$, $e^{8t} \leq 1/(1 - 8t)$, square-root brackets, a Machin bracket for π).

6.9 Transfer of the flip identity and of $+\tau/144$

Every coefficient is tied to τ . In the uniform model every face term of (83) is $-(\tau/3)W_f$, selected faces included, and h_b does not depend on τ . In AW1's notation the uniform Hamiltonian is $H_N(\tau, \kappa(\tau))$ with $\kappa(\tau) = (\tau/24)(1, 1, 1)$ in units of α , an odd function of τ . A mixed convention is the rejected control **uniform_sign_convention**: selected faces with the opposite sign, with $\tau/24$ read in δ units, or with zero.

The flip identity, proved. Let $U_E = \prod_{e \in E \cap \Lambda_N} \Gamma_e$ with the flip set E of (53). Each Γ_e acts as $(-1)^{2j}$ on the spin- j block and, because -1 is central, commutes with left and right translations. So U_E commutes with every C_e , with h_b , P_b and every spectral cutoff $1_{[0, L]}(h_b)$, and with every endpoint gauge action, and it fixes Ω_0 . Every plaquette of $\mathbb{Z}^3$ meets E in one or three links (Section 4.4), the selected faces included, so $U_E W_f U_E^* = -W_f$ for every retained face. Hence, in every centered box and at every cutoff,

$$U_E \hat{H}'_N(\tau) U_E^* = \hat{H}'_N(-\tau), \quad U_E Q_L \hat{H}'_N(\tau) Q_L U_E^* = Q_L \hat{H}'_N(-\tau) Q_L. \quad (105)$$

Two differences from (54) follow. The patterned identity $(\tau, \kappa) \mapsto (-\tau, -\kappa)$ becomes $\tau \mapsto -\tau$, because κ is tied to τ . And neither the inherited strip-operator form nor a $Q_L \rightarrow Q'_L$ step is needed, because the reference and its cutoffs are Haar and Casimir objects that U_E fixes; AW1's modified flip set E^* is not needed either. The odd intersection is re-enumerated with the selected classes. The forward checker covers the 24 (orientation, base-parity) classes, the 24 anchored classes at 27 anchors (648 faces) and 3600 plaquettes. The reverse checker covers the 24 residue cases, the 6591

plaquettes with base point in $[-6, 6]^3$ (3042 meeting E once, 3549 three times, none evenly) and all 1719 faces of Λ_2 (1344 omitted and 375 selected). The Wilson face meets E in three links, and exact rational-quaternion holonomies show that the traces flip.

Consequences. Proposition 1 gives a unique ground at both signs, so $U_E \psi_N(\tau) \propto \psi_N(-\tau)$. With $U_E \Omega_0 = \Omega_0$ and the intermediate normalization $\langle \Omega_0, \psi \rangle = 1$, the creations obey $c_I(-\tau) = U_E c_I(\tau)$. In every box and at every cutoff,

$$\begin{aligned} \omega_{N,-\tau}(W) &= -\omega_{N,\tau}(W), & \omega_{N,-\tau}(W^2) &= \omega_{N,\tau}(W^2), \\ C_{N,-\tau} &= C_{N,\tau}, & c_{N,-\tau} &= c_{N,\tau}, & \rho_{N,R}(-\tau) &= U_E \rho_{N,R}(\tau) U_E^*. \end{aligned} \quad (106)$$

Since $U_E P_R U_E^* = P_R$, $\|\rho_{N,R} - P_R\|_1$ is exactly even in τ . For AQ limits, $\mathcal{S}(-\tau) = \mathcal{S}(\tau) \circ \alpha_E$ as whole sets of subsequential limits, as in (56); individual states at $\pm\tau$ are paired only along a common subsequence. Oddness gives no $O(\tau^3)$ remainder.

The tie is essential. The reverse checker applies U_E face by face on Λ_2 under five rules for the selected coefficient (Table 13). The forward producer compresses $H_0 - (\tau/3)W - (\kappa/3)W_g$, with g the selected face of factor 0 that shares the link $(0, x)$ with W , onto $\{\Omega_0, 2W\Omega_0, 2W_g\Omega_0\}$, where U_E acts as $D = \text{diag}(1, -1, -1)$:

$$H(\tau, \kappa) = \begin{pmatrix} 0 & -\tau/6 & -\kappa/6 \\ -\tau/6 & 24 & 0 \\ -\kappa/6 & 0 & 24 \end{pmatrix}, \quad D H(\tau, \kappa) D = H(-\tau, -\kappa). \quad (107)$$

With $\kappa = \tau$ this is $H(\tau) \mapsto H(-\tau)$, and the control `uniform_kappa_tied_to_tau` rejects the claimed uniform identity for $\kappa = \tau + 1/1000$, $\kappa = 1/100$ and $\kappa = 2\tau + 1/50$. The reverse producer's decoupled one-plaquette fixture gives the selected-face mean $k/(2\sqrt{576 + k^2})$ with $k = 8\kappa$. It is odd in τ for the tied rule and not for the mixed-sign and offset rules; for the uniform $k = \tau/3$ its first-order value is $k/48 = \tau/144$. These are finite compressions with `transfers_to_aq` false; they audit the algebra only.

Table 13: The flip identity under five selected-coefficient rules, applied face by face on Λ_2 by the reverse checker.

Selected coefficient over α	$U_E H(\tau) U_E^* = H(-\tau)$	Status
$\tau/24$ (tied)	yes	the uniform model
$ \tau /24$ (mixed sign)	no	identity breaks
$\tau/24 + 1/7$ (offset)	no	identity breaks
$\tau/12$	yes	odd, but not uniform
0	yes	the zero-selected model, not uniform

Parity on the selected faces. A selected face is a single-face character $W_g = \frac{1}{2} \text{Tr } U_g$, exactly like an omitted face, so the centre grading (47) applies verbatim. For every face f meeting R , selected or omitted,

$$\mathbb{E}[W^2 W_f] = 0, \quad \mathbb{E}[W W_f] = \frac{1}{4} \delta_{f,W}, \quad P_{24} V' W \Omega_0 = 0, \quad (108)$$

where V' is the route-B interaction. On Λ_2 the odd-link sets $W \triangle f$ have sizes 0, 6 and 8, for 1, 12 and 1706 faces. The twelve are the ten omitted faces that share a link with W and the two selected faces above. An energy-24 vector has exactly four odd links, so no face maps $W \Omega_0$ into the energy-24 multiplet. Also $\mathbb{E}[W_g W_f W_h] = 0$ for all 117 480 multisets of three faces meeting R , because each face meets E oddly; so the invariant multiplet has zero first-order splitting. The state, Duhamel, energy and vector-centring terms of Section 4.3 therefore vanish separately, and the finite-box parity theorem (52) transfers with its qualifications: finite boxes, a τ^2 constant for C_N and c_N that is explicitly unbounded, and only evenness for AQ limits.

The first-order Wilson mean. The first-order vector now sums over all retained faces, omitted and selected. By Kato analyticity in each box (V' bounded, a simple isolated ground with gap at least $\frac{1}{2}$),

$$\begin{aligned}\psi^{(1)} &= -c'^{(1)} = \frac{\tau}{72} \sum_f W_f \Omega_0, \\ \omega_{N,\tau}(W) &= 2 \operatorname{Re} \langle W \Omega_0, \psi^{(1)} \rangle + O_N(\tau^2) = \frac{\tau}{36} \sum_f \mathbb{E}[W W_f] + O_N(\tau^2) = +\frac{\tau}{144} + O_N(\tau^2).\end{aligned}\tag{109}$$

Only $f = W$ contributes. This is checked face by face over the 88 faces meeting R , and the two selected faces that share a link with W contribute exactly zero. In the product-split reading of (58), only c_R overlaps $W \Omega_R$. Its first-order part is unchanged, because the ten faces with owner set R are all omitted. The new single-site creations $c_{\{0\}}$ and $c_{\{e_z\}}$ are orthogonal to $W \Omega_R$, because W carries one spin- $\frac{1}{2}$ factor on a link of each site. The first-order values are $\pm 1/14400000000$ at $\tau = \pm 10^{-8}$, as in (59). Charging the selected faces at first order, and a zero first-order mean in the uniform model, are rejected mutations. The Rayleigh–Schrödinger series of the compression (107) reproduces $1/144$, with the W_g direction contributing zero.

What transfers and what does not. Table 14 lists the AW1 statements. The coefficient $+\tau/144$ transfers as a finite-box Taylor coefficient. Its uniform remainder does not. AW1’s step “the first-order parts of $c_{\{0\}}$ and $c_{\{e_z\}}$ vanish” is false in route B, and both producers say so. At the zero triple the pair term of K_2 was actually of third order, because every first-order support has at least two sites (Section 4.6); here it is of second order. A route-B K_2 would need its own contract, with T' , ϱ' , the single-site creations and new pins, for example 36 single-site faces per site of R instead of 33. Neither producer computes it. No uniform-in- N τ^2 constant, enclosure or sign of $\omega(W)$ is admitted for the uniform model.

Table 14: AW1 statements under uniform route B.

AW1 statement	Uniform route B	Needs
Per-link grading; commutes with Casimirs, gauge actions and cutoffs; fixes Ω_0	verbatim	none
E meets every plaquette in 1 or 3 links	verbatim; selected classes enumerated	none
$U_E H_N(\tau, \kappa) U_E^* = H_N(-\tau, -\kappa)$	becomes $H_N(\tau) \mapsto H_N(-\tau)$	κ tied to τ
Cutoff compressions $Q_L \rightarrow Q'_L$	$Q'_L = Q_L$	τ -independent h_b
Box covariance; $\omega_N(W)$ odd; $\omega_N(W^2)$, C_N , c_N even	transfers	route-B unique ground at J'_0
$\mathcal{S}(-\tau) = \mathcal{S}(\tau) \circ \alpha_E$ as whole sets	transfers	route-B AQ1 constants $58 \tau F $, $2349 \tau $
Parity theorem (finite boxes; τ^2 constant of C_N unbounded)	transfers, selected faces included	route-B box gap
$\omega_\tau(W) = +\tau/144 + r(\tau)$, only $f = W$	coefficient transfers	none for the coefficient
$ r(\tau) \leq K_2^+ \tau^2$ uniformly in N	not transferred	T' , ϱ' , ε' , single-site creations, new pins
Uniform $\omega(W^2)$ constant	not transferred	route-B constants
AW2 enclosure and sign	not transferred	claim exclusion

6.10 Controls, replays and review

The contract lists 28 controls. It defines new semantics for five of them.

- **j0_resolution_declared:** the R1 re-freeze with its two exact inequalities is present; the old J_0 at the cap is rejected.

- **uniform_sign_convention**: the selected faces enter with the same sign and magnitude $\nu = \alpha\tau/24$ as the omitted faces.
- **uniform_kappa_tied_to_tau**: an untied selected coefficient breaks the uniform flip identity.
- **route_b_no_double_count**: a face charged in both a star and a single group is rejected; the per-site sum stays $29|\tau|$.
- **incidence_table_itemized**: counts come from an itemized face table; asserted totals are rejected.

The forward checker runs 52 exact checks and the reverse checker 58. Each implements all 28 controls as damaging mutations raised through explicit exceptions, never assertions, so every control stays active under optimized Python. The forward producer reports 94 rejections; review finding N7 notes that 93 sit in the 28 control checks and one, the removal of a link from E , in **flip_set_odd_all_classes**. The reverse records 111. Both use only the standard library and exact rationals, and both reject float, Boolean, NaN and zero-denominator inputs. Both verify the contract snapshot’s SHA-256 before parsing and read the target $1/2500000$ and every candidate constant from it. After a coherently rehashed edit of the target to $1/3000000$, which is still met, the reverse records the new value and the forward aborts, because it pins the target to the acceptance text. A rehash requires editing `check.py`, which the freeze inventory binds. Both record the same claim flags:

- **true**: **uniform_wilson_claim**, which means only that the model is the uniform fixed-spacing Kogut–Susskind model as labelled (review finding N12);
- **false**: **continuum_claim**, **weak_coupling_claim**, **resolved_interaction_shift**, **scientific_priority_verified** and **euclidean_node_certified**.

The forward packet also sets **uniqueness_claimed**, **whole_sequence_convergence_claimed**, **rate_in_N_claimed**, **k2_uniform_claimed** and **wilson_mean_sign_certified** false.

Independent pre-comparison package. The skeptic’s package holds a derivation, a contract review and a checker with 80 exact checks implementing 26 of the 28 controls. The two deferred controls, **vector_versus_scalar_centering** (there is no centering here) and **reverse_premise_isolation** (a post-comparison check), are recorded with their reasons. The package was hash-frozen and committed (489e20e, 00:58:34Z) before the reverse commit (4eb7d8e, 01:08:09Z) and the forward commit (b2a1d86, 01:09:17Z). It already contains every producer headline exactly: the forward D'_{ii} , the reverse D'_{ii} and its density form, and both tier-(i) values. The skeptic records that its own triage seeded route B and its constants, so this agreement is correlated model-agent work, not an independent discovery.

Post-comparison review. The review adds 57 exact checks, including 29 source-edit mutation receipts. Unmutated copies of both closures reproduce the frozen outputs. Twenty-five source edits abort at the intended check. Among them are the selected row relabelled omitted, the coefficient $-1/4$, the single groups dropped ($J = 28|\tau|$), the remainder zeroed, the flip set re-keyed, a weak-coupling flag, a gap-one reset, a 49-face t_1 and forbidden files added to the reverse inputs. Three silent value edits run to completion in the producer checkers: the forward density form replaced by 2ε , and the pair term T'^2 dropped in either producer. Only the skeptic’s value validator and the cross-producer comparison catch them (N5). The AX2 checker must therefore read D'_{ii} from the AX1 gate as an exact rational and reject any recomputed or smaller value.

Replays and isolation. The declared-direction replay tool verifies both closures file by file (45 and 42 files), compares every premise snapshot with its repository source and replays under normal and optimized Python into fresh external directories. All four runs reproduce the frozen outputs byte for byte, and the freezing tool reports both closures verified. The reverse inputs are exactly **AGENTS.md**, the contract and the 36 shared premises; no triage, loop-2, deliberation, expert, forward-AX1 or

skeptic-AX1 file is among them. Isolation between the producers rests on disclosure and content (N11). The closures froze 74s apart (reverse 01:07:16Z, forward 01:08:30Z). The forward report was last written at 01:08:06Z, after the reverse freeze record; the forward checker was last written at 01:02:16Z, before the reverse checker. The forward shows no reverse-specific construct. Scratch was audited by names and times only: the private folders exist as disclosed, and the shared scratchpad root holds no AX1 file after the freeze. The disclosed protocol and code-style reads (the tools README, the freezing tool and the AV1 forward and reverse checkers) and two incidental name exposures (a forward listing of the scratch root and a reverse `git status`) carry no premise weight.

6.11 Non-blocking findings recorded by the gate

The gate’s decision records four non-blocking findings. Review findings N5–N9 and N11 appear above and in the limitations.

- *Mis-rounded feasibility note.* The contract’s `d_prime_note` writes “ $D' \leq 4.19 \times 10^{-7}$ ” for the AX2 window budget. The threshold lies in $[41883, 41884]/10^{11}$, so 4.19×10^{-7} is infeasible (Section 6.8). The skeptic’s contract review and both producers found this. The tested target $4/10^7$ lies below the threshold.
- *Bounds, not counts.* The contract’s candidates 96 per factor and 168 for R are the bounds 4×24 and 7×24 ; the exact counts are 52 and 88 (Section 6.7).
- *Limited reverse independence.* Contract item 8 asked the reverse producer to find the single-factor grouping and the J_0 issue itself. The contract and the [selection note](#) already state route B, $J' = 29|\tau|$ and J'_0 . The skeptic recorded the point before production, and both producers disclose it. Independence therefore covers the proof route, the enumeration and the code only, and the skeptic’s own triage seeded route B.
- *Pre-registration vocabulary.* The frozen preregistration lists the selected triple symbolically as `['tau/24', 'tau/24', 'tau/24']`. A Jung/Pauli research-assistant audit found three schema gaps in the sub-round 3–5 drafts. The vocabulary extension in the [plan record](#), recorded at 00:45:54Z on 24 September, about five minutes after the AX1 freeze, allows a selected triple given as an exact expression in τ and names AX1 as covered. The gate records it as an accepted extension. The contract is unamended, and both checkers parse and verify the triple.

6.12 Limitations recorded by the gate

The AX1 gate records the following limitations verbatim.

- Only the uniform Kogut-Susskind SU(2) model at fixed spacing and strong bare coupling ($g^4 \geq 9.6 \times 10^{-9}$, $|\tau| \leq 10^{-8}$), centered whole-star-plus-single-group boxes $N \geq 2$ and the cover $R = \{0, e_z\}$; never weak coupling or continuum; the box boundary is not identified with AL1’s all-contained-plaquette boundary; $\tau < 0$ is the U_E mirror, not a real- g coupling
- Upper certificates only: no lower bound on $||\rho_{R-P_R}||_1$; no value or sign of $\omega(W)$ beyond $|\omega(W)| \leq D'$ and the finite-box first-order coefficient; AQ statements are uniform local closeness of every subsequential limit, not uniqueness, whole-sequence convergence or a rate in N
- Inherited without re-proof, re-applied with the new constants: AM2 fixed point, multilinear majorant (the $|X| \leq 4$ monotonicity covering single-site groups is the only new step), exclusion and cutoff removal; AV1 product split, anchored-norm tier, cutoff-vector removal and AQ passage; AQ1 compactness and Nachtergaele-Sims dynamics; AQ2 gauge averaging and Fourier gap; AT4 reset and relative-unitary arguments; I1 dictionary; Kato, Peter-Weyl and Nachtergaele-Sims cited, not machine-checked
- D'_{ii} binding: the forward exact rational ($\sim 1.44459e-8$, both inequalities); the reverse $\sim 1.22237e-8$ is a valid labelled R-refinement (88 faces meeting R) whose headline rests on the fidelity inequality (its density form is also below target)
- Tier (i) $D'_i \sim 2.4526e-5$ fails $4/10^7$ and 10^{-6} and is retained as a limited-tier value

- Not transferred: K₂ (route-B T', rho', single-site first-order creations and new pins, e.g. 36 single-site faces per site of R), the uniform omega(W²) constant, the AW2 enclosure and every uniform-in-N tau² constant
- Contract d_prime_note mis-rounded (4.19x10⁻⁷ infeasible; threshold [41883,41884]/10⁻¹¹); contract candidates 96/168 are bounds, exact counts 52/88
- Reverse independence limited: the contract and selection-ax1.md state route B, J'=29|tau| and the J₀ issue; independence covers proof route, enumeration and code only; all agents are correlated model agents and the skeptic's triage seeded route B
- Pre-registration vocabulary: the symbolic selected triple ['tau/24','tau/24','tau/24'] in the frozen AX1 preregistration is covered by the plan.json vocabulary extension recorded after the freeze (2026-09-24T00:45:54Z, contract unamended); the gate should record it as an accepted extension
- N5: neither producer checker pins the pair term T'² (dropping it runs silently in both) and the forward does not pin its density form; caught only by the skeptic value validator and cross-producer comparison; AX2 must bind D'_ii from the gate
- N6: the forward's exported density_part_of_D=2eps²/(1+eps²) is an upper bound on D-2eps=2eps²(1-eps)/(1+eps²) (excess ~7.5e-25), not an identity; D is exact
- N7: the forward's '94 in total' counts 93 rejections in the 28 control checks plus 1 (E_minus_one_link) in flip_set_odd_all_classes
- N8: the reverse rejects 100|tau| as a miscount; 2*7*7+2=100|tau| is also a valid labelled refinement (single groups inside R, Haar mean 0); the admitted budget is 102|tau|
- N9: checklist classification of the variance floor differs (reverse verbatim, forward new constant); the substance is the same: the Haar gap six is load-bearing
- N11: producer isolation rests on disclosure and content: the closures froze 74 s apart and the forward report was last written after the reverse freeze.json; no reverse-specific construct appears in the forward; scratch audited by names and times only (private folders as disclosed; no AX1 file in the shared scratchpad root after the freeze); disclosed protocol/style reads (tools README, freeze.py, forward/av1/check.py, reverse/av1/check.py) and incidental name exposures (the forward's /tmp listing, the reverse's git status) carry no premise weight
- Fixtures (qubit termination, quaternion holonomies, 3x3 compressions, the route-A kappa=30 compression, the one-plaquette mean) are finite algebras, transfers_to_aq false
- Scientific priority unverified

Full derivations and reviewed implementations: [AX1 forward](#), [AX1 reverse](#), [independent model-agent review](#) and the [frozen contract](#). The gate binds the forward D'_{ii} as the uniform state bound for AX2 and selects AX2 next.

6.13 AX2: the uniform-model window certificate

Contribution HNM-C-AX2. The AX2 contract, recorded under the title “Hruday uniform-model window certificate for $C(1)$ at the original cap”, was frozen after the AX1 gate and before production. It closes sub-round 3. The model is the one of Sections 6.2–6.9: uniform Kogut–Susskind SU(2) at fixed spacing and strong bare coupling, with every elementary face, selected faces included, at $\nu = \alpha\tau/24$ and the dictionary $g^4 = 96/\tau = 9.6 \times 10^9$ at the cap $|\tau| = 10^{-8}$ (82). It is handled by route B (83): the Haar reference, the τ -independent on-site Casimir sum, whole stars and one single-factor group per factor, with $J' = 29|\tau|$ and the re-frozen contraction constant $J'_0 = 29/10^8$ (Section 6.4). The exact first-order count behind the state bound is 52 faces per factor, and 88 faces meet the cover $R = \{0, e_z\}$ (Section 6.7). The preregistered model identifier is `AQ_uniform_routeB`, with selected triple $(\tau/24, \tau/24, \tau/24)$ in units of α ; the states come from AQ1’s centered whole-star subsequences through a finite-volume uniform bound.

The observable is the centered Euclidean Wilson correlation $C_\tau(s)$ of (7) for the original xz plaquette W , with vector centering, in the physical clock (6), at the single preregistered node $s = 1$. The kernel is the frozen C^2 window of (30). The reference is the free correlation $C_0(s) = \frac{1}{4}e^{-3s}$ of (3), which the contract records as “exp(-3)/4 enclosed” on the Haar route. The coupling is $\tau = +10^{-8}$; $\tau = -10^{-8}$ has no real g and is evaluated only as the U_E mirror. AX2 applies a lemma that AV2 admitted with constants that AX1 admitted, so it runs in one direction (**single+skeptic**). There

is one forward producer, and the skeptic wrote an independent replay from the contract alone and committed it before the producer's package. The contract makes that replay a condition of admission. The producer received AGENTS.md, the contract and its 28 shared premises, 30 files in all. No reverse package exists, so reverse premise isolation is not applicable.

6.13.1 Reviewed scope

The [AX2 gate](#) records the verdict `accepted_within_scope`. Its sub-label is `reference_unresolved`, and its accepted statement, quoted verbatim below in the gate's ASCII notation, is the scope of every claim in this subsection. In it, R is the cover and R' the outward radius bound, written R' below; r is the exact radius; D' is the AX1 bound D'_{ii} of (99), written D' below; and π^- is a directed lower bound π^- for π .

In the uniform Kogut-Susskind SU(2) model on Z^3 at fixed spacing and strong bare coupling (every elementary face, selected faces included, $\nu=\alpha*\tau/24$; AL1 dictionary $\tau=96/g^4$, $g^4=9.6\times 10^9$ at the cap; label 'uniform Kogut-Susskind SU(2) at fixed spacing and strong bare coupling ($g^4=9.6\times 10^9$)'), handled by route B of AX1 (Haar reference, τ -independent on-site $h_b=8 \sum C_e$, whole stars plus one single-factor group per factor for its three selected xy faces), at $\tau=+10^{-8}$ ($\tau=-10^{-8}$ is the U_E mirror, no real g , evaluated only as a replay of the same $|\tau|$ formula, not a second confirmation), for the original xz Wilson loop W with cover $R=\{0, e_z\}$, clock $s=\alpha*\tau/h_b$ and $G=H/\alpha$, every AQ1 subsequential state of the centered whole-star-plus-single-group construction ($N\geq 2$; each state separately; no uniqueness, whole-sequence convergence or rate in N) has centered Euclidean Wilson correlation $C(1)$ satisfying $|C(1)-d|\leq r\leq R'$ at the single preregistered node $s=1$, with the exact datum $d=497870683678639429793424156500617766317/(4*10^{40})$ ($\sim 0.012446767091965986$; midpoint of the directed enclosure of $e^{-3}/4$; unchanged from AV2) and the exact radius bound $R'=1912298807996871790146581299723633/10^{40}$ ($\sim 1.91229880800e-7$; $r=2(D'+D'^2)+51|\tau|s/\pi^-$ plus the arithmetic half-width $1/(4*10^{40})$, $D'=2425369125199104794263242601250/167893028420061547330293754713793182097$ ($\sim 1.44459e-8$) the AX1 gate forward tier-(ii) value, $k'=51|\tau|/4$ from 7 whole stars and 2 single-factor groups, $M_0=2$, $M_1=4s/\pi$), so $C(1)$ lies in $[0.012446575862085186, 0.012446958321846786]$, inside $[99572606896681488461252714035083774357/(8*10^{39}), 497878332873871417280584742825816660849/(4*10^{40})]$, meeting the 10^{-6} target (margin ≥ 5.22930). The free reference $e^{-3}/4$ lies inside the interval (reference_unresolved). Retained failures at unchanged τ and s : the AT4-type Poisson certificate at $L=10^4$ with $D=2\sqrt{17|\tau|}$ ($\sim 8.5790596e-4$), the Poisson floor $\geq 1.313477283e-6$ for every L even with $D>0$ ($\geq 1.3279232e-6$ with D'), and the tier-(i) window ($\geq 4.921572155e-5$), all insufficient. The analytic formula $E'(s)=2(D'+D'^2)+51|\tau|s/\pi$ crosses 10^{-6} at s^* in $[5.982012284161, 5.982012284163]$ (~ 5.98201228416), a crossover of the formula only; only $s=1$ is certified. No interaction shift, sign or coefficient of $C(s)$, no uniform Wilson-mean sign rider or uniform K_2 , no grid, no weak-coupling or continuum statement, and no priority claim follows. Admission rests on the single forward producer plus the skeptic's independent pre-comparison replay from the contract.

The gate's decision also binds the hash of the AX1 gate, which the contract named without a hash. The rest of this subsection instantiates the statement. The limitations recorded by the gate close it (Section 6.13.10).

6.13.2 The lemma transfer

AX2 proves no new lemma. The AV2 window lemma (35) and its bound (38) need four things: $G_\tau \geq 0$, so that the centered spectral measure η of (29) lives on $[0, \infty)$; a finite mass of η ; $g \in L^1 \cap C_0$ with $\hat{g} \in L^1$, which gives inversion at every x ; and Fubini. The last two depend on the window only. The first two hold for every uniform route-B AQ1 state, because AQ1's nonnegativity and GNS construction re-apply with the route-B constants (Section 6.5, AX1 gate item (3)). As in AV2, the AQ2 gap is not used. The frozen window, its Fourier sign convention and the constants (34),

$$M_0 = \|\hat{g}\|_1 = 2, \quad M_1 = \frac{4s}{\pi}, \quad M_2 = 2s^2,$$

are therefore unchanged. The checker parses them from the hash-bound AV2 contract and gate and re-audits them in exact arithmetic: the C^2 matching with third-derivative jump $-8s^3$, the half-line identity $2\pi\hat{g}a^3b = 8s^3$ of (31) in Gaussian rationals, the modulus, and exact antiderivatives giving M_0, M_1, M_2 and $\int \hat{g} = 1$. Only two inputs change, the real-time slope and the state bound (Table 15). The contract’s insufficient outcome, “The transfer of the window lemma fails”, is not triggered.

Table 15: Transfer of the AV2 window certificate to uniform route B, as exported by the forward checker.

Hypothesis or ingredient	Uniform route B
Window, conventions, M_0, M_1, M_2	verbatim (AV2)
$G \geq 0$, finite η , inversion, Fubini	verbatim; AQ1 nonnegativity re-applies (AX1 gate item 3)
Reference $c_0(\theta) = \frac{1}{4}e^{3i\theta}$	verbatim value (Section 6.13.3)
Real-time slope	new constant $k' = 51 \tau /4$ (Section 6.13.4)
State term, mean square	new constant D' bound by the AX1 gate (Section 6.13.5)
Passage to every AQ1 subsequential limit	verbatim with the route-B constants (AX1 gate items 3 and 6)
AQ2 gap	not used

6.13.3 The reference is unchanged in route B

Route B keeps the Haar product P_R on all 48 cover links and the τ -independent on-site generator $h_b = 8 \sum_e C_e$. Both single-factor groups that meet R , ψ_0 and ψ_{e_z} , lie inside R and are placed in B_N together with the seven stars. So A_N evolves W by the free Casimir generator on R , with Casimir $\frac{3}{4}$ on each of the four links of W , and

$$G_{0,R}W = 3W, \quad \omega_0(W) = 0, \quad \omega_0(W^2) = \frac{1}{4}, \quad c_0(\theta) = \frac{1}{4}e^{3i\theta}, \quad C_0(s) = \frac{1}{4}e^{-3s}, \quad (110)$$

exactly as in (3). The selected faces could correlate with W only through shared links. W has the two z links $(0, z)$ and (e_x, z) , and every selected face is an xy face with no z link. In W times any product of the six selected faces of R each z link of W therefore occurs once, and the Haar mean vanishes by the AW1 grading (47); the checker verifies all 64 subsets. The two selected faces that share an x link with W , at $(0, x)$ and (e_z, x) , enter only through B_N , never through the reference. The skeptic’s contract review adds that the free z links give only the two moments. The free energy 3 also needs the Haar reference on the whole cover and the τ -independent h_b , with the selected faces in the interaction only, and the forward report states both. The exported datum equals the AV2 gate datum rational for rational.

6.13.4 The uniform slope

The forward checker re-derives the route-B groups by a fine-lattice brute force over 80 anchors. Each anchored face falls in exactly one group, a 21-face star or a 3-face single group, with support inside the group’s support. The groups meeting R are the seven stars with anchors $R - S$ (2), which meet R in 21, 21, 16, 8, 8, 4 and 4 faces, and the two single-factor groups $\{0\}$ and $\{e_z\}$ with three selected faces each. That makes 153 faces charged, 88 meeting R (82 omitted and 6 selected) and 16 inside R : the itemized incidence of Table 11, row for row. As in (36) and (94), in units of $G = H/\alpha$ and for every real θ ,

$$\|B_N\| \leq 7 \cdot \frac{7|\tau|}{8} + 2 \cdot \frac{|\tau|}{8} = \frac{51|\tau|}{8}, \quad (111)$$

$$\left\| \alpha_\theta^N(W) - \alpha_\theta^0(W) \right\| \leq 2|\theta| \|B_N\| \leq k'|\theta|, \quad k' = \frac{51|\tau|}{4},$$

so $k' = 51/400000000$ at the cap. The estimate is local: the boxes $N = 2, 3, 4$ each have exactly seven stars and two single groups meeting R , and $N = 1$ has only four stars. Three slopes are rejected. The extensive norm grows with the box. The seven-star slope $49|\tau|/4$ is AV2’s zero-selected slope and drops the two groups, a changed model. The third drops the conjugation factor 2. AQ1’s norm dynamics on compact θ intervals re-applies with $\|\Phi'\|_F \leq 2349|\tau|$ (93), so (111) holds for the actual AQ evolution at every real θ .

6.13.5 The state premise and its tiers

The state term uses the AX1 gate’s forward tier-(ii) bound (99),

$$\|\rho_R - P_R\|_1 \leq D' = D'_{\text{ii}} = \frac{2425369125199104794263242601250}{167893028420061547330293754713793182097} \quad (112)$$

$$\approx 1.44459192143 \times 10^{-8},$$

valid at both signs and for every AQ1 subsequential limit (Section 6.8). The checker hard-codes the SHA-256 of the AX1 gate snapshot, verifies it before parsing and reads D' from the gate’s decision. It then requires exact equality with the contract parameter, the gate’s accepted text, the calculator’s pinned constant and the gate formula recomputed at the cap: $J' = 29|\tau|$, $t'_1 = 52|\tau|/144$, $T' = t'_1/(1 - 352J') = 13/3599632512$, $\varepsilon = 2T' + T'^2$ and $D' = 2\varepsilon(1 + \varepsilon)/(1 + \varepsilon^2)$. This answers AX1 review finding N5 (Section 6.10). A recomputation with the pair term T'^2 dropped ($\approx 1.4445919188 \times 10^{-8}$), or with the density form replaced by 2ε ($\approx 1.4445919110 \times 10^{-8}$), is a rejected mutation, not a silent run. So are the AX1 reverse cover refinement (101), the AX1 target $4/10^7$, the mis-rounded contract note 4.19×10^{-7} , AV1’s zero-selected D (23), tier (i) and a D supplied to the calculator. With two unadmitted calculator tiers these make the ten rejections of `av1_tier_bound`, whose contract semantics read: “ D' equals the AX1 gate’s admitted forward tier-(ii) value; any other D is rejected”. Tier mixing is rejected separately (`tier_mixing_rejected`): a tier-(i) mean square with the tier-(ii) state term, a reverse state term with a forward mean square, an exact $c^{(1)}$ without the AM2 remainder, and a zero-selected state term with the uniform slope.

Table 16 shows why the value, not the Boolean, decides. Every substitute except tier (i) still meets 10^{-6} ; the AX1 target itself passes with margin about 1.04. The contract’s limited outcome, “Only the crude tier admitted”, is moot, because the AX1 gate admitted tier (ii).

Table 16: State bounds and the window radius each would give at $\tau = 10^{-8}$, $s = 1$. Decimals are previews of exact rationals; the radii of the three substitute rows are the skeptic’s pre-comparison values. Only the first row is admitted.

State premise	Bound on $\ \rho_R - P_R\ _1$	Window radius	Status
AX1 tier (ii), forward exact	$1.44459192143 \times 10^{-8}$	$\leq 1.91229880800 \times 10^{-7}$	admitted
AX1 tier (ii), reverse refinement	$1.22236968878 \times 10^{-8}$	$\approx 1.8679 \times 10^{-7}$	not substituted
AX1 target	4×10^{-7}	$\approx 9.6234 \times 10^{-7}$	not substituted
AV1 tier (ii), zero triple	$1.36124528703 \times 10^{-8}$	$\approx 1.8956 \times 10^{-7}$	other model
AX1 tier (i)	$2.45260902281 \times 10^{-5}$	$\geq 4.921572155 \times 10^{-5}$	fails; retained

The comparison. Trace duality against the operator $W\alpha_\theta^0(W)$ on R , of norm at most one, together with $m_\tau^2 \leq D'^2$ (104), gives the AV2 comparison (37) with the route-B constants:

$$|c_\tau(\theta) - c_0(\theta)| \leq k'|\theta| + D' + m_\tau^2 \leq k'|\theta| + D' + D'^2 \quad (\theta \in \mathbb{R}). \quad (113)$$

The operator $W\alpha_\theta^0(W)$ is not an effect, so the $D'/2$ refinement is not applied; the control `state_term_not_effect` rejects it. The mean-square term is substantive here. In the uniform model the first-order Wilson mean is $+\tau/144 \neq 0$ in every finite box (109), but no uniform K_2 is admitted (Table 14), so only $|m_\tau| \leq D'$ is uniform. Charging $(\tau/144)^2$ in place of D'^2 , or assuming a zero mean, is rejected under `first_order_mean_charged`.

The mirror at $\tau = -10^{-8}$. For real g , $\tau = 96/g^4 > 0$, so $\tau = -10^{-8}$ has no real- g preimage. It is the U_E image of $+10^{-8}$: $U_E \hat{H}'_N(\tau) U_E^* = \hat{H}'_N(-\tau)$ in every box and at every cutoff (105), C_N is even, and $\mathcal{S}(-\tau) = \mathcal{S}(\tau) \circ \alpha_E$ as whole sets (106). Every term of (114) depends on $|\tau|$ only, and the calculator returns the identical datum and radius at $\tau = -10^{-8}$. This is a replay at the mirrored coupling, not a second confirmation, and the packet sets `mirror_is_second_confirmation` false.

The reference is unresolved. The exact free enclosure $[f^-, f^+]$ of $e^{-3}/4$, of width $5 \cdot 10^{-41}$, lies inside $[d - r, d + r]$. The datum is the free midpoint, and the interaction enters only through the radius. No interaction-induced shift, sign or coefficient of $C_\tau(s)$ is claimed, and the sub-label is `reference_unresolved`, as for AV2.

6.13.7 Retained failures and the crossover

Table 18 lists the certificates evaluated at unchanged $\tau = 10^{-8}$ and $s = 1$, with directed π , logarithms and square roots. The AT4 formula

$$D + D^2 + \frac{ks}{\pi} \log\left(1 + \frac{L^2}{s^2}\right) + \left(\frac{1}{2} + \frac{D}{2}\right) \frac{2s}{\pi L} \quad (120)$$

reproduces the frozen AT4 decimal 0.000841518704386267 to 10^{-15} with AT4's own inputs. With the uniform inputs at $L = 10^4$, the route-B square-root control $D = 2\sqrt{17|\tau|}$ of (90) and k' , it lies in $[8.5790595, 8.5790596] \times 10^{-4}$. The Poisson floor (44) at the uniform slope,

$$\frac{k's}{\pi} \log\left(1 + \frac{L^2}{s^2}\right) + \frac{s}{\pi L} \geq \frac{2k's}{\pi} \left(1 + \log \frac{1}{2k's}\right) \quad (L > 0), \quad (121)$$

and its minimum over L lies in $[1.313477283, 1.313477284] \times 10^{-6}$: the right-hand side is at least the lower value, and the cost at $L = 3921569$, near $1/(2k')$, is at most the upper one. No L meets 10^{-6} , even with $D \rightarrow 0$. With D' added, no cutoff brings it below 1.3279232×10^{-6} . The truncated absolute first moment of the Poisson kernel is at least 46.907878 at $L = 10^{32}$, so its product with k' alone exceeds 10^{-6} , while the window's $M_1 = 4s/\pi$ is finite (45). The tier-(i) window uses the AX1 value

$$D'_i = \frac{23002790096235779074916932482}{937890625141038667264537458466241} \approx 2.45260902281 \times 10^{-5}$$

and gives a radius of at least $4.921572155 \times 10^{-5}$. The gate records all three as insufficient. They are retained failures, and the tier-(i) radius is a limited-tier value, never the bound premise. At $\tau = 3739/5000000000000$ the Poisson floor with $D \rightarrow 0$ would meet the target (upper value 9.9985841×10^{-7}); retuning the coupling to it after the outcome is a rejected mutation under `insufficient_verdict_retained`.

Table 18: Certificates for uniform route B at $\tau = 10^{-8}$, $s = 1$, target 10^{-6} . Only the last row is admitted as a certificate.

Kernel	State bound	Radius or lower bound	Outcome
Poisson, $L = 10^4$ (AT4 form)	$2\sqrt{17 \tau }$	$\approx 8.5790596 \times 10^{-4}$	insufficient
Poisson, every L	$D \rightarrow 0$	$\geq 1.313477283 \times 10^{-6}$	insufficient
Poisson, every L	AX1 tier (ii)	$\geq 1.3279232 \times 10^{-6}$	insufficient
C^2 window	AX1 tier (i)	$\geq 4.921572155 \times 10^{-5}$	insufficient
C^2 window	AX1 tier (ii)	$\leq 1.91229880800 \times 10^{-7}$	certified

Crossover. The bound (114) is linear in s (`window_linear_in_s`) and crosses 10^{-6} at

$$s^* = \frac{\pi(10^{-6} - 2(D' + D'^2))}{51|\tau|} \in [5.982012284161, 5.982012284163], \quad (122)$$

about 5.98201228416, below AV2’s $[6.236863446, 6.236863447]$ (46). The gate quotes the skeptic’s pre-comparison bracket. The producer’s exact bracket, with ten-digit previews $[5.9820122841, 5.9820122842]$, is built with $\pi^\mp$ so that $E'^+(s^{*-}) = E'^-(s^{*+}) = 10^{-6}$ exactly, and the review records the brackets as nested. This describes the analytic formula, not evaluated nodes. The preregistration fixes the single node $s = 1$ and forbids post-hoc node selection. The mutation “node $s = 5$ with the $s = 1$ radius” is rejected, `grid_claim` is false and there is no $[0, 128]$ claim. At s^* the free value $e^{-3s^*}/4$ is below 10^{-8} .

6.13.8 Exact scope

What is claimed. For the uniform Kogut–Susskind $SU(2)$ model at fixed spacing and strong bare coupling, route B, $\tau = +10^{-8}$ and the node $s = 1$, every AQ1 subsequential state of the centered whole-star-plus-single-group construction ($N \geq 2$), each state separately, satisfies $|C_\tau(1) - d| \leq r \leq R' < 10^{-6}$. The exported label is exactly the contract’s: *uniform Kogut–Susskind $SU(2)$ at fixed spacing and strong bare coupling ($g^4 = 9.6 \times 10^9$)*. The flag `euclidean_node_certified` is true and means only the $s = 1$ enclosure within 10^{-6} . The flag `uniform_wilson_claim` is true and means only that the model is the uniform fixed-spacing model at strong bare coupling as labelled; its exported scope string adds “no uniform Wilson-mean sign certificate”. The sub-label is `reference_unresolved`. Admission rests on the single forward producer plus the skeptic’s independent pre-comparison replay from the contract. The two share model-agent ancestry, so their exact agreement is a replay of one admitted formula, not independent physical confirmation.

The [panel update](#) after sub-round 3 records roadmap goal 3 as met within its stated scope. That is a planning record and admits nothing beyond the AX1 and AX2 gates.

What is not claimed. The contract excludes a resolved interaction shift, a uniform Wilson-mean sign certificate, weak coupling or the continuum, AQ uniqueness and scientific priority. The preregistration adds that a free reference inside the enclosure means no interaction claim, and excludes whole-sequence convergence or a rate in N , transfer from a finite graph and relabelling a static shift as dynamical. In detail:

- No interaction shift, sign or coefficient of $C_\tau(s)$: the free value lies inside the interval.
- No uniform Wilson-mean sign certificate, no uniform K_2 and no uniform $\omega(W^2)$ constant. The loop-2 veto forbids the sign rider (Section 1.3), and the contract’s `K_2_note` repeats it.
- No node other than $s = 1$, no grid and no $[0, 128]$ claim.
- No uniqueness of the AQ state, whole-sequence convergence or rate in N .
- No transfer from a finite graph; the fixtures are finite algebra.
- No weak-coupling or continuum statement. No path with $g \rightarrow 0$ stays in the admitted box (Section 6.2).
- No priority. Scientific priority is unverified.

The packet records the following claim flags.

- true: `euclidean_node_certified` (the $s = 1$ enclosure only) and `uniform_wilson_claim` (the model label only);
- false: `continuum_claim`, `weak_coupling_claim`, `resolved_interaction_shift`, `scientific_priority_verified`, `grid_claim`, `wilson_mean_sign_certified`, `k2_uniform_claimed`, `state_uniqueness_claim`, `independent_review_claimed` and `mirror_is_second_confirmation`.

6.13.9 Controls, replays and review

The contract lists 25 controls, and its preregistration mirror lists the same 25, so AV2’s clerical defect (Section 3.8) does not recur. Two carry new semantics: `single_producer_declared`, under

which the producers are [forward] and admission requires the skeptic’s independent replay, and `av1_tier_bound` (Section 6.13.5). The forward checker runs 52 exact checks. It implements all 25 controls as damaging mutations raised through explicit exceptions, never assertions, so every control stays active under optimized Python. It records 98 rejections in the control checks and 31 calculator-domain rejections, 129 in all. Its imports are standard-library only, every admission value is an exact rational, and float, Boolean, NaN and zero-denominator inputs are rejected. It verifies the SHA-256 of the contract, AX1 gate and selection-note snapshots, all pinned, and checks every other snapshot against the AX1 or AV2 gate bindings. It reads τ , s , k' , D' , the target, the reference and the control list from the contract, and it records its own SHA-256 and the calculator’s before the evaluation. Controls not yet described include the following.

- `selected_incidence_count`: forgotten groups (49/8), faces counted as groups (55/8), single groups placed at the seven star anchors (56/8) and the selected coefficient read in δ units are rejected.
- `uniform_label_strong_coupling`: a weak-coupling label, continuum wording in the verdict, a label missing “fixed spacing” or “strong bare coupling”, a negative τ read as a real g and τ read in δ units are rejected.
- `changed_model_relabelled`: twelve relabellings are rejected, among them $\tau = 10^{-14}$, the R2 cap, $s = 1/2$, the zero triple, route A, a finite-graph model, the C^1 and Poisson kernels and the slope 49/4.
- `root_n_misuse`: root sums of squares of the four terms or of the nine group norms ($\sqrt{345}/8$), and division by $\sqrt{4}$ or $\sqrt{9}$, are rejected.
- `coherent_evidence_tampering`: nine edits with the packet hash rebound are rejected, among them a removed AX1 gate snapshot, a halved radius, the reverse D' , the seven-star slope and a claimed reverse producer.
- `kernel_negative_atom_misread` and `window_fourier_sign_convention`: as in AV2 (Section 3.4), $g(-1) = 5/e < e$ and $g(-3)/4 = 25e^{-3}/4$ at $s = 1$.
- `wrong_delta_alpha_hbar_clock`: the exponent 24 with s and normalized group norms in the G clock are rejected; the non-unit fixture $\alpha = 5$, $\hbar = 7$ gives physical time 7/5.

Calculator. The producer exports a reusable exact calculator. It accepts exact inputs and the model `uniform_KS_routeB` only, with the uniform triple, $|\tau| \leq 10^{-8}$ at either sign, $s > 0$ and positive physical scales. D' and k' are not inputs; only the tier-(ii) state tier and the C^2 window are accepted. A fixed-design mode enforces $|\tau| = 10^{-8}$, $s = 1$ and the target 10^{-6} , and it flags the negative sign as the mirrored replay. At the cap D' is the pinned gate rational. Below the cap it is the gate’s admitted tier-(ii) $|\tau|$ -form, which increases in $|\tau|$ and never exceeds the gate value. At $\tau = 0$ the calculator accepts the free Casimir limit with $D = 0$, where the uniform triple coincides with the zero triple and the radius is the arithmetic half-width. Review finding N2 and the gate record the below-cap and $\tau = 0$ values as corollaries, not AX2 admissions; any reuse needs its own contract. The skeptic’s probe of a hash-verified copy recorded 24 domain refusals.

Independent replay and review. The skeptic’s pre-comparison package holds a derivation, a contract review and a checker with 72 exact checks covering all 25 controls. It was hash-frozen and committed (85df74d, 01:48:07Z) before the forward commit (0ae59e6, 01:50:59Z). Its routes differ from the producer’s: Hutton’s formula for π and a power series with geometric remainder for e^{-3} , both to 10^{-60} , against Machin and an alternating Taylor bracket with halving to 10^{-40} ; the window constants by the annihilator–jump route and Wallis integrals; and its own fine-lattice enumeration. The two agree exactly on D' , k' , the datum and the outward radius R' . The producer’s exact interval $d \pm r$ lies inside the pre-comparison $d \pm R'$ and contains the skeptic’s 10^{-60} interval, the s^* brackets are nested, and the retained failures and tier (i) agree. The only difference is display precision. The report text truncates D' to $1.44459192142 \times 10^{-8}$, while the packet’s outward preview is $1.44459192143 \times 10^{-8}$ (N4). No admission Boolean reads a decimal.

The skeptic’s contract review records that the contract, the selection note and the skeptic’s own AX1 review already state every constant and the value 1.9123×10^{-7} (Section 6.8), so the single producer could have copied the headline. The replay is therefore independent in code and route only. Its producer-error checklist shows why the formula has to be read: the seven-star slope gives about 1.8486×10^{-7} , the $D'/2$ effect refinement about 1.7678×10^{-7} , and the substitute state values of Table 16 likewise pass 10^{-6} ; only tier (i) fails.

The post-comparison review adds 30 exact checks. All 42 must-abort source edits abort at the intended check. Twenty-two are in the calculator, among them the single groups set to 0, the pair term dropped, the density form 2ε , the state term $D/2$, π^+ in the kernel term and $M_0 = 1$. Ten are in `check.py`: six validator weakenings, each of which ends in “damaging mutation accepted”, so the controls are live, and a changed selected predicate, dropped incoming-star anchors, tier (i) used as D' and a `decimal` import. Ten are in the inputs. A contract target of 10^{-7} with its hash rebound aborts at `target_boolean_1e-6`, and a halved contract τ with its hash rebound also aborts, which shows that both are read from the contract. Two edits run silently by design (N3). A literal D' equal to the gate value in `check.py` runs to completion and changes only the recorded checker hash. The skeptic’s static data-flow validator rejects it, and the hash of `check.py` in the freeze inventory protects the admitted source; this is the AX2 analogue of AW2 finding N3 (Section 5.3). Removing only the box-dependence test of the Duhamel validator also runs to completion, as the producer disclosed. The slope-value test is the redundant guard, and removing both aborts. The skeptic’s own validator rejects 11 damaged packets.

Replays and isolation. The declared-direction replay tool, which imports the unchanged two-direction tool as in AW2 (Section 5.8), verifies the forward closure of 35 files. It compares the 30 input snapshots with their repository sources byte for byte and replays `check.py` under normal and optimized Python into fresh external directories. Both runs reproduce the frozen outputs byte for byte, and the freezing tool reports the closure verified. The inputs are exactly `AGENTS.md`, the contract and the 28 shared premises. There is no `skeptic/ax2` or `reverse/ax2` input, and no reverse package exists. One isolation finding, N1, stands. The forward calculator (01:38:07Z) and checker (01:45:43Z) were written at the same time as the skeptic’s package (checker 01:41:35Z, commit 01:48:07Z), on a shared filesystem; the forward report (01:50:06Z) and freeze (01:50:26Z) followed the skeptic’s commit. The producer declares no `skeptic/ax2` read, and its closure holds none. Its disclosed scratch matches the listings, which were checked by name and time only. The shared scratchpad root holds an AX2 budget preview, `ax2feas.py`, also checked by name only; the same budget appears in committed assistant tools and in the skeptic’s AX1 review. Reads cannot be determined, so isolation is verified for repository inputs only. The identical R' is expected by construction, since both round one admitted formula to the same 10^{-40} grid, and the skeptic’s pre-comparison R' used the admitted AV2 calculator’s primitives, which the producer’s calculator reuses. The producer also disclosed reads of the tools documentation, the freezing tool and the AV2 forward checker for protocol and code style (N5); partially read and unopened premises are hash-verified, and none of these reads carries premise weight.

6.13.10 Limitations recorded by the gate

The AX2 gate records the following limitations verbatim.

- Single direction: one forward producer applies the admitted AV2 window lemma with the admitted AX1 route-B constants; admission rests on that producer plus the skeptic’s pre-comparison replay from the contract (correlated model-agent ancestry; the $1.9123e-7$ value was already in shared premises); exact agreement is a replay of one admitted formula, not independent physical confirmation
- Only the uniform Kogut-Susskind SU(2) model at fixed spacing and strong bare coupling ($g^4=9.6 \times 10^{-9}$ at $|\tau|=10^{-8}$), route B, Haar reference, centered whole-star-plus-single-group boxes $N \geq 2$, cover $R=\{0,e_z\}$, node $s=1$; $\tau < 0$ is the U_E

- mirror (no real g); calculator values below the cap and at $\tau=0$ (free Casimir limit) are corollaries, not AX2 admissions
- Inherited without re-proof: the AV2 window lemma and constants (Fourier inversion, Fubini); AT4 relative-unitary Duhamel and trace duality; AX1 route-B constants (J_0' , gap $\alpha/16$, reset, incidence, D' and its passage to every AQ1 limit, flip identity); AQ1 construction, nonnegativity and norm dynamics; AL1/I1 dictionary
 - Upper certificate only: r bounds $|C(1)-d|$; the free value $e^{-3}/4$ is inside (reference_unresolved); no interaction shift, sign or coefficient of $C(s)$; no uniform K_2 , $\omega(W^2)$ constant or uniform Wilson-mean sign rider (loop-2 veto)
 - Each AQ1 subsequential state separately; no uniqueness, whole-sequence convergence or rate in N
 - $s \sim 5.98201228416$ is the crossover of the linear formula only; no grid and no node other than $s=1$
 - Retained failures remain failures at this τ and s : the AT4-type Poisson certificate, the Poisson floor for every L and the tier-(i) window
 - Read isolation verified for repository inputs only: the forward calculator and check.py were written concurrently with the skeptic pre-comparison package; the forward report and freeze followed the skeptic commit; scratch and shared-filesystem reads cannot be determined
 - No weak-coupling or continuum statement follows
 - Scientific priority unverified

Full derivation and reviewed implementation: [AX2 forward](#), [independent pre-comparison replay](#), [independent model-agent review](#) and the [frozen contract](#). Sub-round 3 is complete; the gate selects AY1 next.

7 Sub-round 4: local closeness of subsequential states

The fourth Round31 roadmap goal, “Identify limiting states and boundary effects”, asks either for state identification or for a quantitative boundary comparison between two frozen construction families, with a local observable class and a common physical clock ([Round31 roadmap](#)). AQ1 supplies its states by diagonal extraction along centered whole-star boxes (Section 1.1) and does not prove uniqueness, whole-sequence convergence or a rate in N . Sub-round 4 does not identify states either. AY1 compares every subsequential limit of two named finite-volume constructions on the cover R , with explicit constants. AY2 states, with those constants, what the comparison does and does not establish, lists the obligations that remain unproved and certifies a static two-sided distance from the Haar product on R . The model returns to the zero-selected patterned family of Sections 2–5. It is not the uniform model of Section 6.

7.1 AY1: uniform local closeness of subsequential limits

Contribution HNM-C-AY1. The AY1 contract, recorded under the title “Hruday uniform local closeness of all AQ-type subsequential states and the boundary-independent first-order density”, was frozen after the AX2 gate and before production. It opens sub-round 4. The preregistered model identifier is `AQ_patterned_zero_selected`: the selected triple is exactly $(0,0,0)$ with the Haar product reference P_R ; the 21 omitted faces per anchor enter with coefficient $-\tau/3$ in normalized units (1); both signs of $|\tau| \leq 10^{-8}$ are covered; and the cover is $R = \{0, e_z\}$, with its 48 links, 36 endpoints and seven incident anchors (2). The observable class is $B(\mathcal{H}_R)$, and the clock is the common one of (6). There is no centering and no Euclidean node. The preregistered target is $2D \leq 1/1250000$ at the cap, twice the AV1 target $4/10^7$, with D the AV1 admitted tier. The itemized error terms are `state_boundary`, `second_order_difference` and `arithmetic`. In the title, “boundary-independent” refers to the first-order density on R , which is the same for both families. No boundary independence of the dynamics is claimed.

Both directions are paired. The contract records the historical lens’s sub-round-3 preference for one direction plus a skeptic replay as a dissent, because the padded-family verification, the explicit

first-order density and the R -local constant K'_2 are new derivations. The forward producer received AGENTS.md, the contract, its 28 shared premises and two forward-additional premises, the skeptic's triage and the Jung loop-2 response: 32 files in all. It derives the reduced density on R explicitly. The reverse producer received only AGENTS.md, the contract and the 28 shared premises, 30 files, and works from the vacuum overlap and the fidelity. The triage's goal-4 note and the Jung note already state the mechanism of the closeness and of the common first-order density, so on the forward route they are shared panel premises. The contract and the [selection note](#) state it too. Independence therefore covers the proof routes, the enumeration and the code only (Section 7.1.9).

Notation: F1 and F2 are the two construction families of Section 7.1.2. ρ_R and ρ'_R are the reduced densities on R of two subsequential limits, and $\rho_{N,R}$ is that of a finite-volume ground state. F_R is the set of the ten faces with owner set exactly R , and $a = |\tau|/144$ is the norm of one first-order face vector. T , $\varrho = 352JT$ and $\varepsilon = 2T + T^2$ are the exact-tier quantities of (19) and (20), and K_2^+ is the AW1 constant (63). Other notation follows Sections 1–4.

7.1.1 Reviewed scope

The [AY1 gate](#) records the verdict `accepted_within_scope`. Its accepted statement, quoted verbatim below in the gate's ASCII notation, is the scope of every claim in this subsection. The sub-label, carried by the skeptic's review, both producer packets and the gate, is `uniform_local_closeness_not_uniqueness`: the statement is local closeness, not uniqueness. In it, R inside $G(R)$ and $G'(R)$ is AM2's anchored radius $R_a = 1/64$, and R elsewhere is the cover. ρ alone is the AM2 remainder ϱ , while ρ_R , ρ'_R and $\rho^{(1)}_R$ are densities on R . a inside the items of K_2' is $|\tau|/144$, while "lattice spacing a " is the spacing. s inside the face labels is the fine y phase q of Table 1, and s elsewhere is the Euclidean clock.

```
In the AM2/AQ1 zero-selected patterned family (model AQ_patterned_zero_selected: SU(2)
Kogut-Susskind form on  $Z^3$  at fixed spacing, coarse 24-link factors, selected triple
exactly (0,0,0) with Haar product reference  $P_R$ , 21 omitted faces per anchor entering
as  $-(\tau/3)W_f$  in normalized units  $\delta = \alpha/8$ ; both signs  $|\tau| \leq 10^{-8}$ ; cover
 $R = \{0, e_z\}$  of the original  $xz$  Wilson loop, 48 links, 36 endpoints, 7 incident anchors;
observable class  $B(H_R)$ ; common clock  $s = \alpha \cdot t_E / \hbar$ ,  $\theta = \alpha \cdot t / \hbar$ , with the
two named construction families F1 = centered whole-star boxes  $\Lambda_N$  (AQ1) and F2 =
all-contained-face boxes with padding (I1 section 6) on the same  $\Lambda_N = [-N, N]^3$ ,
 $N \geq 2$ : (1) both families satisfy the AV1 premises: F1 by the admitted chain  $I1 \rightarrow AM2 \rightarrow$ 
 $AQ1/AQ2 \rightarrow AV1 \rightarrow AW1$ ; F2 item by item (anchor groups of the retained faces supported
in  $(b+S)$  cap  $\Lambda_N$ ,  $|X| \leq 4$ , termination order 8, per-site sum
 $J \leq 28|\tau| = J_0 = 7/25000000$  at the cap with  $J_{OG}(R) < 37/6250000 < 1/64$  and
 $2J_{OG'}(R) < 77/390625 < 1$ , at most 49 first-order faces per site, cutoff-vector removal,
reset  $\omega(h_R) \leq 98|\tau|$  and  $C_F = 56|\tau||F|$ , local trace-norm compactness and a
diagonal extraction of its own); F2 retains  $28N(5N+1)$  boundary faces more than F1, and
both retain the same 82 faces meeting  $R$  for  $N \geq 2$ ; states are compared in the trace norm
on  $B(H_R)$ , and dynamics is named in the norm on compact time windows with no dynamical
statement. (2)–(3) For every pair of subsequential limits of the named construction
families F1 and F2 at the same coupling, on the fixed cover  $R$  and for the frozen
observable class, the reduced densities satisfy  $\|\rho_R - \rho'_R\|_1 \leq 2D$  and agree to
first order in  $\tau$ ; this does not assert equality of the states, whole-sequence
convergence, translation invariance, boundary independence of the dynamics, or a rate
in  $N$ . Constants:  $2D = 1170159676931825184288274812132100/$ 
 $42981220507576537932303142777593983768257$  ( $\sim 2.72249057405e-8$ ;  $D =$ 
 $585079838465912592144137406066050/42981220507576537932303142777593983768257$  the AV1
gate forward tier (ii)), meeting the preregistered  $1/1250000$  with margin  $\sim 29.38$  (tier
(i)  $2D_i \sim 4.736e-5$  fails and is retained); the common first-order density is
 $\rho^{(1)}_R = (\tau/72) \sum_{f \in F_R} (|W_f \Omega_R| < \Omega_R| + |\Omega_R| < W_f \Omega_R|)$  over
the 10 faces  $F_R$  with owner set exactly  $R$  ( $xz$   $r=0,1,2$ ,  $s=0,1$  and  $yz$   $r=0..3$ ,  $s=0$ ,
anchored at 0; the 72 straddling faces among the 82 meeting  $R$  have zero  $R$ -marginal and
the family has no single-site creations), with  $\|\rho^{(1)}_R\|_1 = \sqrt{10}|\tau|/72$ 
( $\sim 4.39205e-10$  at the cap) and  $\text{Tr}(\rho^{(1)}_R W) = +\tau/144$  (AW1); for every subsequential
limit of either family  $\|\rho_R - \rho^{(1)}_R\|_1 \leq K_2' \tau^2$  with the  $R$ -local
trace-norm constant  $K_2' = 966771578474926086618624139557778885954947547760216752246561/$ 
```

The gate’s decision binds the triangle values common to the forward producer, the reverse producer and the skeptic’s pre-comparison package. It names the Bures pairwise constant and the $\sqrt{2}$, 288-majorant and combined variants of K'_2 as labelled refinements only, and it records the gate fields of Section 7.1.8. The rest of this subsection derives the statement. The limitations recorded by the gate close it (Section 7.1.11).

Table 19: Retained omitted faces of the two families, enumerated identically by the forward producer, the reverse producer and the skeptic. B_+ is the padded set of I1 §6.

Box	F1 faces	F2 faces	F2 – F1	Partial groups in F2	Sites of B	Sites of B_+
Λ_2	1344	1960	616	60	125	200
Λ_3	4536	5880	1344	126	343	490

Other prescriptions are not covered. The literal vertex boxes of I1 §7 clip the on-site operator, so their reference is not Haar and they form another model; relabelling them as F2 is a rejected mutation. Periodic and orthant boxes are not among the named families either.

Two topologies and one clock. The contract requires two topologies, named separately.

- *States*: the trace norm on $B(\mathcal{H}_R)$, $\|\rho - \rho'\|_1 = \sup\{|\text{Tr}((\rho - \rho')A)| : A \in B(\mathcal{H}_R), \|A\| \leq 1\}$. Convergence on fixed observables is not enough. The densities $(1 - D/2)|e_0\rangle\langle e_0| + (D/2)|e_n\rangle\langle e_n|$ converge weak-* to a limit of mass $1 - D/2$ while staying a trace distance D apart. The reset energy supplies the tightness that excludes this, as in (26).
- *Dynamics*: the operator norm on each fixed bounded local observable, uniformly on compact time windows (Nachtergaele–Sims, as placed in AQ1). It is not uniform over the unit ball of a local algebra, and it is named only. F1 inherits the star-indexed interaction. For F2 the native owner-set-indexed interaction has supports of ℓ^1 -diameter at most two and per-site sum at most $49|\tau|/3$, so its weighted norm is at most $81 \cdot 49|\tau|/3 = 1323|\tau|$ (reverse). No dynamics, stationarity or GNS generator is constructed for F2 limits, and neither producer claims that the dynamics of the two families coincide.
- *Clock*: $s = \alpha t_E/\hbar$, $\theta = \alpha t/\hbar$ and $G = H/\alpha$, as in (6); the normalized clock $u = s/8$ and the exponent 24 are forbidden in packets. The comparison is static. Both families are evaluated at the same coupling and in the same unit convention, with face coefficient $-\tau/3$ in units of $\delta = \alpha/8$.

7.1.3 The AV1 premises for both families

F1 is covered by the admitted chain $\text{I1} \rightarrow \text{AM2} \rightarrow \text{AQ1/AQ2} \rightarrow \text{AV1} \rightarrow \text{AW1}$, which both producers cite step by step; nothing is re-proved for F1. The forward checker reads every admitted constant from the hash-bound gate and report snapshots, and the reverse checker recomputes the AV1 and AW1 gate values exactly. The forward checker also verifies, for $N = 2, 3$, that F1 retains all seven incident stars and that each retained group has 21 faces. F2 is verified item by item, not by analogy. Table 20 lists the premises, and the paragraphs below give the arguments.

The AM2 contraction for F2. AM2’s proof uses the interaction only through a short list of hypotheses. Both producers verify each of them for F2 at every N , and the checkers audit them on the actual boxes $N = 2, 3$.

- *On-site operator*. $h_x = 8 \sum_e C_e$ is unchanged: $h_x \geq Q_x$, with the Haar vacuum and compact resolvent. The padded sites carry the same operator.
- *Grouping and supports*. Each retained face is charged exactly once, at its anchor, and $\phi_b^{(\Lambda_N)}$ acts on $X_b = \bigcup\{M_f : f \in O_b, M_f \subset \Lambda_N\} \subset (b + S) \cap \Lambda_N$. AM2’s counting uses $|X| \leq p = 4$ only as an upper bound: at most $2^{|X|}$ output sets, $\sum_{I \cap X \neq \emptyset} \|c_I\| \leq |X| \|c\|_a$ and $|I_l| \leq |M| + |X|$. Smaller supports are therefore covered, as in (86). A nested word with more than eight creations puts at least five of them on one side of V_X , and two of those overlap, so the commutator nest terminates at order $2|X| \leq 8$. The exact four-qubit fixture gives $\text{ad}_C^8(V)\Omega = 8!|1111\rangle$ and $\text{ad}_C^9(V) = 0$.

Table 20: The AV1 premises for the two families at $\tau = \pm 10^{-8}$. F1 values are admitted; F2 values are proved in AY1 and enumerated on Λ_2 and Λ_3 .

Premise	F1 (admitted chain)	F2 (verified in AY1)
On-site operator	$h_b = 8 \sum_e C_e \geq 6Q_b$, Haar P_b	unchanged; padded sites the same
Grouping	whole stars ϕ_b , 21 faces	anchor groups $\phi_b^{(\Lambda_N)}$; each retained face once
Supports	$b + S$, $ X = 4$	inside $(b + S) \cap \Lambda_N$, $ X \leq 4$
Termination	order 8	order $2 X \leq 8$
Per-site sum	$J \leq 28 \tau \leq J_0$	$J \leq 28 \tau = J_0$ at the cap; enumerated maximum 28
Contraction	$J_0 G(R_a) < 37/6250000 < 1/64$, $2J_0 G'(R_a) < 77/390625 < 1$	the same, at the same $J_0 = 7/25000000$
First-order faces per site	49	at most 49; enumerated maximum 49
Tier (ii) state bound	D of (23)	the same rational
Cutoff-vector removal	AV1, gap $\frac{1}{2}$	verbatim, $L \geq 24$
Reset on R ; on F	$98 \tau $; $56 \tau F $	$98 \tau $ (seven full groups); $56 \tau F $
Compactness, extraction	AQ1	trace-norm precompact; own extraction
Dynamics	AQ1, Nachtergaele–Sims	named only

- *Per-site sum.* If $u \in X_b$ then $b \in u - S$: at most four anchors, incoming anchors included. Each group has at most 21 faces of norm $|\tau|/3$, so

$$J^{(2)} = \max_u \sum_{b: u \in X_b} \|\phi_b^{(\Lambda_N)}\| \leq 4 \cdot 21 \cdot \frac{|\tau|}{3} = 28|\tau| \leq J_0 = \frac{7}{25000000} \quad (|\tau| \leq 10^{-8}), \quad (124)$$

with equality at bulk sites; the enumerated maximum on Λ_2 and Λ_3 is exactly 28. At the cap $J = J_0$, which AM2 allows. The constants of (5) depend only on p and J_0 , so $J_0 G(R_a) < (148/7)(7/25000000) = 37/6250000 < 1/64$ and $2J_0 G'(R_a) < 2 \cdot 352 \cdot 7/25000000 = 77/390625 < 1$ hold unchanged. The forward checker evaluates the series enclosure $e^{1/8} < 53823253862885575661009/47498854821020880076800 < 8/7$ exactly. Grouping by owner set instead gives $J = 49|\tau|/3$ with supports of at most three sites. It is a labelled alternative and is not used.

- *Cutoffs.* Product spectral cutoffs commute with H_0 , keep supports and do not raise $J^{(2)}$, and each h_x has compact resolvent, so AM2’s cutoff removal applies verbatim.

Every F2 box at every on-site cutoff therefore has the fixed point $\psi = e^{-C}\Omega_0$ with excited c_I and $\|c\|_a \leq 1/64$, the two majorant inequalities of Section 1.1, and a simple ground with full-space gap at least $\frac{1}{2}$. The ground is gauge invariant, because every retained term is a Wilson loop. The rejected mutations include a merged two-anchor group with support five, faces charged with norm one ($J = 84|\tau| > J_0$), a face charged in two groups, a per-site sum from the outgoing star alone ($7|\tau|$), a padded site given an interaction, termination at order six and the literal vertex boxes relabelled as F2.

Two valid padding conventions. The producers index the F2 groups differently. The forward producer, like the skeptic, uses the actual supports $X_b \subset (b + S) \cap \Lambda_N$. With that convention AM2 needs no padding, and the padded sites only fit II’s source empty-boundary form. The reverse producer indexes each clipped group by its declared support $X_b = b + S$ inside the AM2 volume B_+ . It follows the AM2 reverse report’s scope statement, under which each group has “its declared support of size at most four”. With that convention padding is required, and the reverse checker rejects its removal, because an indexed support would then leave the AM2 volume. The fixed-point map preserves the collections supported in B , so $\psi^{\text{pad}} = \psi_B^{(2)} \otimes \Omega_{B_+ \setminus B}$ in either reading. Both conventions are valid, and the density on R is the same (review finding N1).

Tier-(ii) inputs and the state bound. The first-order collection for F2 is $c^{(1)} = L_0 = -(\tau/72) \sum_f W_f \Omega_0$ over the retained faces, as in (16). At most 49 retained faces have an owner set containing a given site, the enumerated maximum is exactly 49, and no retained face meets a padded site. So $\|c^{(1)}\|_a \leq t_1 = 49|\tau|/144$. With $J = 28|\tau|$ the self-consistent tier (19) is unchanged: $T = 49/14398580736$, $\varrho = 3773/11248891200000000$ and $\varepsilon = 1411060914529/207319127211110301696$ at $\tau = \pm 10^{-8}$. The product split (8)–(11), the explicit density (12), both inequalities (13) and (14), and the cutoff-vector removal (25) use only commuting excited creations, anchored sums at the two sites of R and the untruncated gap $\frac{1}{2}$; for $L \geq 24$, $c_L^{(1)} = c^{(1)}$. So every F2 box Λ_N , $N \geq 2$, satisfies $\|\rho_{N,R}^{(2)} - P_R\|_1 \leq D$ with the AV1 rational (23). Both checkers recompute D and require equality with the AV1 gate value.

Reset and compactness. Reset the finite ground density on R to P_R and keep its exterior marginal, as in AQ1. The on-site terms on R drop to zero, groups not meeting R keep their expectations, and unbounded exterior energies cancel after bounded spectral truncation. Each group meeting R changes by at most twice its norm. The groups meeting R are exactly the seven anchors $R - S$, each of norm at most $7|\tau|$, and on any finite F at most $4|F|$ groups meet F . Hence

$$\omega_N^{(2)}(h_R) \leq 2 \cdot 7 \cdot 7|\tau| = 98|\tau|, \quad \omega_N^{(2)}(h_F) \leq 2 \cdot 4|F| \cdot 7|\tau| = 56|\tau||F| =: C_F. \quad (125)$$

Enumeration on Λ_2 and Λ_3 gives exactly $98|\tau|$ for both families. Charging face by face gives the labelled refinement $2 \cdot 82|\tau|/3 = 164|\tau|/3$, which is not used, and the two-anchor orthant reset $28|\tau|$ is rejected. Since h_F has compact resolvent, $\text{Tr}(\rho_{N,F}(1 - Q_{F,L})) \leq C_F/L$ for $Q_{F,L} = 1_{[0,L]}(h_F)$, and purification with partial-trace contractivity gives $\|\rho - Q\rho Q\|_1 \leq 2\sqrt{C_F/L}$. The F2 densities on every finite region are therefore trace-norm precompact. Diagonal extraction over the nested cubes gives subsequences along which every local density of F2 converges in trace norm, and the compatible limits define locally normal states. F2 needs this extraction of its own, because its limits do not come from AQ1's subsequences; the preregistered state provenance names only the F1 construction (wording defect W4). A subsequential limit of F2 means any such limit, and a chosen subsequential limit is one member of that set.

7.1.4 The closeness $2D$

Let ω and ω' be subsequential limits at the same τ , ω along boxes of family F_i and ω' along boxes of family F_j , where $i = j$ is allowed. Here D is the AV1 gate's forward tier-(ii) bound (23), $D = 585079838465912592144137406066050/42981220507576537932303142777593983768257$, about 1.3612×10^{-8} . Every finite-volume density on R lies in the closed trace-norm ball of radius D about P_R : by AV1 for F1, and by Section 7.1.3 for F2. Trace-norm limits stay in closed balls (26). Hence

$$\begin{aligned} \|\rho_R - \rho'_R\|_1 &\leq \|\rho_R - P_R\|_1 + \|P_R - \rho'_R\|_1 \leq 2D \leq \frac{1}{1250000}, \\ 2D &= \frac{1170159676931825184288274812132100}{42981220507576537932303142777593983768257} \approx 2.72249057405 \times 10^{-8}. \end{aligned} \quad (126)$$

The target margin is about 29.38. The target is read from the contract: the forward checker pins it to $1/1250000$, and the reverse checker ties it to twice the AV1 target. For effects $0 \leq A \leq 1$ the trace-zero difference gives $|\omega(A) - \omega'(A)| \leq D$. Since D increases in $|\tau|$, the cap values bound every smaller coupling at either sign, and the ratio $D(\tau)/D(\tau/100)$ is about 100.0098, so the bound is linear. The $-\tau$ evaluation replays the same $|\tau|$ formula.

Reverse: the vacuum fidelity passed to every limit. In every box of either family, at every cutoff $L \geq 24$ and for the untruncated ground, the AV1 split gives exactly $\text{Tr}(\rho_{N,R} P_R) = 1/(1 + e_N^2)$

with $e_N \leq \varepsilon$. Because $|\text{Tr}((\rho - \sigma)P_R)| \leq \|\rho - \sigma\|_1$, the overlap is trace-norm continuous, so every subsequential limit of F1 or F2 satisfies

$$F_{\text{vac}} := \text{Tr}(\rho_R P_R) \geq \frac{1}{1 + \varepsilon^2} = \frac{42981220507576535941210238266176140476416}{42981220507576537932303142777593983768257}. \quad (127)$$

The mixture inequality used in (14) then gives $\|\rho_R - P_R\|_1 \leq 2\varepsilon/\sqrt{1 + \varepsilon^2} \leq D$. So the reverse inequality also certifies D , and the triangle inequality gives $2D$.

Labelled variant: the Bures angle. Let $A(\rho, \sigma) = \arccos \sqrt{F_U(\rho, \sigma)}$, with F_U the Uhlmann fidelity; it is a metric on states. For the pure P_R , (127) gives $A(\rho_R, P_R) \leq \arctan \varepsilon$, so $A(\rho_R, \rho'_R) \leq 2 \arctan \varepsilon \leq \pi/2$. The Fuchs–van de Graaf inequality then gives $\frac{1}{2}\|\rho_R - \rho'_R\|_1 \leq \sin(2 \arctan \varepsilon) = 2\varepsilon/(1 + \varepsilon^2)$, that is,

$$\begin{aligned} \|\rho_R - \rho'_R\|_1 &\leq \frac{4\varepsilon}{1 + \varepsilon^2} = \frac{1170159668967453566242603438964736}{42981220507576537932303142777593983768257} \\ &\approx 2.7224905552 \times 10^{-8}. \end{aligned} \quad (128)$$

This bound is genuinely pairwise and slightly below $2D$, and the skeptic re-derived it. It shows that $2D$ is not an artefact of the triangle through P_R . The contract names $2D$, so the Bures value is a labelled refinement, not the headline.

Tiers and retained failures. Table 21 lists the pair bounds at the cap. Tier (i), $2D_i = 72266694536315307496644/1525878906463907516326874161$, fails the target and is retained as a limited-tier value; it is not retuned. The doubled Round31 square-root bound fails as well.

Table 21: Bounds on $\|\rho_R - \rho'_R\|_1$ for any two subsequential limits of F1 or F2 at the same $\tau = \pm 10^{-8}$, against the preregistered target $1/1250000$. Decimals are previews of exact rationals in the frozen outputs.

Bound	Approximate value	$\leq 1/1250000$
Round31 reset, doubled, $4\sqrt{49 \tau /3}$	1.61658×10^{-3}	no
Tier (i), $2D_i$	4.73607×10^{-5}	no
Tier (ii), $2D$, bound by the gate	$2.72249057405 \times 10^{-8}$	yes
Bures pairwise (reverse, labelled)	$2.7224905552 \times 10^{-8}$	yes
Second order after $\rho_R^{(1)}$, $2K'_2\tau^2$	$2.68350550879 \times 10^{-12}$	yes

Quantifiers. The statement covers every pair of subsequential limits of F1 or F2 at the same τ , from the same family or from different families. It holds for a chosen subsequential limit of either family, every one separately, and it does not identify limits with one another. It is uniform local closeness, not uniqueness, whole-sequence convergence, translation invariance or a rate in N . The proof uses only a bound that holds in every box, so no rate can come from it. The triangle route is sharp in form. The forward checker takes pure states $v_{\pm} = (\cos \phi, \pm \sin \phi)$ with rational $\cos \phi = (1 - m^2)/(1 + m^2)$ and $m = 1/1000$. Each lies at distance $D_{\text{ex}} = 2 \sin \phi$ from P , yet they are $2D_{\text{ex}} \cos \phi$ apart. So two states that both satisfy the one-state bound may differ by nearly $2D$.

The same coupling is essential. P_R does not depend on τ , so $2D$ also bounds two states at different couplings. The limits at $+\tau$ and $-\tau$ are certifiably different on R , by at least $2\sqrt{10}|\tau|/72 - 2K'_2\tau^2 \approx 8.757 \times 10^{-10}$ (Section 7.1.6), yet both lie within $2D$ of each other. The bound $2D$ alone is therefore a common enclosing ball, not a boundary comparison. The comparison content is the matching first-order term of Section 7.1.5, and every statement keeps “at the same coupling” (review finding N5).

7.1.5 The common first-order density

Forward: the R -marginal of the first-order vector. AV1's creation convention is $\psi = e^{-C}\Omega_0$ with $\langle\Omega_0, \psi\rangle = 1$ and $c^{(1)} = L_0 = -(\tau/72)\sum_f W_f\Omega_0$ ((16), (49)). Because $\langle\Omega_0, c_I\rangle = 0$,

$$|\psi\rangle\langle\psi| = |\Omega_0\rangle\langle\Omega_0| - \sum_I (|\hat{c}_I^{(1)}\Omega_0\rangle\langle\Omega_0| + \text{h.c.}) + O(\tau^2), \quad \|\psi\|^2 = 1 + O(\tau^2).$$

Contract item 3 asks for the R -marginal of $-(c^{(1)}\Omega_0^* + \text{h.c.})$, restricted to the creations meeting R . With the AV1 convention that form gives $+\tau/144$ for the Wilson mean, unlike AW1's display (Section 4.5); wording finding W7 records it as verified. The supports sort into three kinds.

- *Disjoint from R .* The marginal is $\langle\Omega_{\text{out}}, c_I\rangle|\Omega_R\rangle\langle\Omega_R| = 0$.
- *Straddling.* Here $\text{Tr}_{\text{out}}|W_f\Omega_0\rangle\langle\Omega_0| = |\langle\Omega_{\text{out}}|W_f\Omega_0\rangle\rangle\langle\Omega_R|$. Every straddling face has a link owned outside R that occurs once, and its Haar integral vanishes; the forward checker verifies this for all 72.
- *Inside R .* The supports are R , $\{0\}$ and $\{e_z\}$. No omitted face has a one-site owner set, so $c_{\{0\}}^{(1)} = c_{\{e_z\}}^{(1)} = 0$. The zero-selected family has no single-site creations, unlike route B (Section 6.7).

Hence

$$\rho_R^{(1)} = -(|c_R^{(1)}\rangle\langle\Omega_R| + |\Omega_R\rangle\langle c_R^{(1)}|) = \frac{\tau}{72} \sum_{f \in F_R} (|W_f\Omega_R\rangle\langle\Omega_R| + |\Omega_R\rangle\langle W_f\Omega_R|). \quad (129)$$

The ten faces of F_R are anchored at 0: the six xz classes with $r \leq 2$ and the four yz classes with $q = 0$ (Section 2.5). The Wilson face W is the xz class with $r = 0$, $q = 0$. The coefficient $-\tau/72$ of $c^{(1)}$ is the same in both unit systems, and the mixed values $\tau/576$ and $\tau/9$ are rejected. Table 22 recounts the creations meeting R by anchor. The 82 faces meeting R have 27 owner sets. The 72 straddling faces split as $66 + 6$ by the size of $I \cap R$, with 33 single-contact faces per site, and every one of them drops out of the marginal.

Table 22: First-order faces meeting R by incident anchor, and those that survive the partial trace over the complement of R .

Anchor	0	e_z	$-e_z$	$-e_x$	$-e_y$	$e_z - e_x$	$e_z - e_y$	Total
Faces meeting R	21	21	16	4	8	4	8	82
Owner set exactly R	10	0	0	0	0	0	0	10

Reverse: the vector overlap identity. Decompose any density on R in blocks relative to P_R : $\rho = F_{\text{vac}}P_R + |\rho_{\perp 0}\rangle\langle\Omega_R| + |\Omega_R\rangle\langle\rho_{\perp 0}| + \rho_{\perp\perp}$, with $\rho_{\perp 0} = (1 - P_R)\rho\Omega_R$ and $\rho_{\perp\perp} = (1 - P_R)\rho(1 - P_R) \geq 0$.

- The scalar fidelity identity $F_{\text{vac}} = 1/(1 + e^2)$ with $e \leq \varepsilon$ gives $0 \leq 1 - F_{\text{vac}} \leq \varepsilon^2$. The diagonal block has no first-order part, because the τ -derivative of the scalar identity vanishes.
- Positivity gives $\text{Tr}\rho_{\perp\perp} = 1 - F_{\text{vac}} \leq \varepsilon^2$, so the excited block has no first-order part either. Exactly, $\|(F_{\text{vac}} - 1)P_R + \rho_{\perp\perp}\|_1 = 2(1 - F_{\text{vac}}) = 2e^2/(1 + e^2)$.

A first-order term is therefore off-diagonal and fixed by the single vector $\rho_{\perp 0}$. The scalar identity cannot supply it; its vector form can. For every box of either family, every cutoff $L \geq 24$ and the untruncated ground,

$$\rho_R\Omega_R = \frac{(1_R \otimes \langle\varphi_{\text{out}}|)\psi}{\|\psi\|^2}, \quad \varphi_{\text{out}} = (\langle\Omega_R| \otimes 1)\psi. \quad (130)$$

Indeed $\langle v, \text{Tr}_{\text{out}}(|\psi\rangle\langle\psi|)\Omega_R \rangle = \langle v \otimes \chi, \psi \rangle$ with $\chi = (\langle\Omega_R| \otimes 1)\psi$, and the AV1 split $(P_R \otimes 1)\psi = \Omega_R \otimes \varphi_{\text{out}}$ gives $\chi = \varphi_{\text{out}}$. The Ω_R component of (130) is the scalar fidelity identity. With $n = \|\varphi_{\text{out}}\|$, $(1_R \otimes \langle\varphi_{\text{out}}|)\psi = n^2(\Omega_R + \hat{\xi})$ with $\hat{\xi} \perp \Omega_R$, and $\rho_{\perp 0} = \hat{\xi}/(1 + e^2)$. Sort the supports of δ in (9).

The creations inside R contribute $-c_R - c_{\{0\}} \otimes \Omega_{e_z} - \Omega_0 \otimes c_{\{e_z\}}$ exactly, with n^2 cancelled. Each straddling creation contributes at most $T\|c_I\|$, by the outside-excitation lemma (61), and each pair at most $\|c_I\|\|c_J\|$. The first-order part of $\hat{\xi}$ is therefore $-c_R^{(1)}$, and the reverse route yields the same operator (129). The contract’s phrase “derivative of the vacuum-overlap/fidelity identity” thus holds in vector form only; the scalar derivative vanishes, and the gate records the point as non-blocking. Derivatives are taken in finite boxes only.

Properties. The vectors $W_f\Omega_R$, $f \in F_R$, satisfy $\langle W_f\Omega_R, W_g\Omega_R \rangle = \frac{1}{4}\delta_{fg}$ and $\langle \Omega_R, W_f\Omega_R \rangle = 0$, because $\mathbb{E}[W_f] = 0$, $\mathbb{E}[W_f^2] = \frac{1}{4}$ and distinct faces share at most one link. The skeptic checks all $82 \cdot 81$ ordered pairs of faces meeting R . With the orthonormal vectors $e_f = 2W_f\Omega_R$,

$$\rho_R^{(1)} = \frac{\tau}{144} \sum_{f \in F_R} (|e_f\rangle\langle\Omega_R| + |\Omega_R\rangle\langle e_f|), \quad (\rho_R^{(1)})^3 = \frac{10\tau^2}{144^2} \rho_R^{(1)}, \quad \text{Tr } \rho_R^{(1)} = 0. \quad (131)$$

So $\rho_R^{(1)}$ has rank two, eigenvalues $\pm\sqrt{10}|\tau|/144$, a zero diagonal and a zero excited block, and

$$\begin{aligned} \|\rho_R^{(1)}\|_1 &= \frac{\sqrt{10}|\tau|}{72} \approx 4.39205230578 \times 10^{-10}, \\ \|\rho_R^{(1)}\|_1^2 &= \frac{1}{5184000000000000000}, \quad \|\rho_R^{(1)}\|_1 \leq \frac{52704627669473}{12000000000000000000000} \end{aligned} \quad (132)$$

at $\tau = \pm 10^{-8}$; the last bound is the directed upper bracket. The labelled triangle bound is $5|\tau|/36$. The operator is self-adjoint, traceless and gauge invariant. The consistency checks with AW1 and AV1 are exact.

- $\text{Tr}(\rho_R^{(1)}W) = 2 \cdot \frac{\tau}{144} \langle \Omega_R, W e_W \rangle = +\tau/144$, the AW1 coefficient (59), that is $\pm 1/14400000000$ at $\tau = \pm 10^{-8}$. Only $f = W$ contributes.
- $\text{Tr}(\rho_R^{(1)}(W^2 - \frac{1}{4})) = 0$, by the parity rule (47).
- $\|\rho_R^{(1)}\|_1 \leq D$.
- Under the AW1 flip every face term changes sign, so $\rho_R^{(1)}$ is odd in τ .

The checkers reject the literal AM2-sign display, which gives $-\tau/144$; a dropped adjoint; a scalar or diagonal first-order ansatz; a first-order excited block; a zero density; and straddling faces treated as operators on $\mathcal{H}_R$.

The same operator for every limit. For every $N \geq 2$ both families retain exactly the same ten faces with owner set R . This is checked face by face for $N = 2, 3$ and follows in general from the geometry of Section 7.1.2. So $c_R^{(1)}$, and hence $\rho_R^{(1)}$, is the same operator for every box of both families and every cutoff $L \geq 24$. With the uniform remainder of Section 7.1.6, every subsequential limit ω_τ of either family satisfies, for every $|\tau| \leq 10^{-8}$,

$$\|\rho_R(\omega_\tau) - P_R - \rho_R^{(1)}\|_1 \leq K'_2\tau^2. \quad (133)$$

This is a statement about a coefficient, not about states. Subsequential limits are chosen separately at each τ , so a limit has no τ -derivative. If operators X and X' both satisfy the expansion along a sequence $\tau_j \rightarrow 0$, then $\|X - X'\|_1 \leq (K + K')|\tau_j| \rightarrow 0$. The first-order coefficient therefore does not depend on the family, the subsequence or the selection $\tau \mapsto \omega_\tau$, even one that switches families with τ . It does not identify the states: they may still differ at order τ^2 on R , and without bound elsewhere.

7.1.6 The R -local constant K'_2

Exact decomposition. Use the notation of (12): $n = \|\varphi_{\text{out}}\|$, $e = \|\delta\|/n$, $\xi = (1_R \otimes \langle \varphi_{\text{out}} |) \delta$ and $\sigma = \text{Tr}_{\text{out}} |\delta\rangle\langle \delta|$. Sorting δ by support gives $\xi/n^2 = -c_R^{(1)} + \eta$ with $\eta = \eta_{\text{am2}} + \eta_{\text{str}} + \eta_{\text{pair}}$, where

$$\eta_{\text{am2}} = -(c - c^{(1)})_R - c_{\{0\}} \otimes \Omega_{e_z} - \Omega_0 \otimes c_{\{e_z\}},$$

$$\eta_{\text{str}} = - \sum_{I \text{ straddling}} \frac{(1 \otimes \langle \varphi_{\text{out}} |) \hat{c}_I \psi_{\text{out}}}{n^2}, \quad \eta_{\text{pair}} = \sum_{\text{pairs}} \frac{(1 \otimes \langle \varphi_{\text{out}} |) \hat{c}_I \hat{c}_J \psi_{\text{out}}}{n^2}. \quad (134)$$

Exact algebra then gives

$$r_R := \rho_R - P_R - \rho_R^{(1)} = \frac{(|\eta\rangle\langle\Omega_R| + \text{h.c.}) + \sigma/n^2 - e^2 P_R}{1 + e^2} + \frac{e^2}{1 + e^2}(|c_R^{(1)}\rangle\langle\Omega_R| + \text{h.c.}). \quad (135)$$

Each piece has an exact trace norm. Since $\eta \perp \Omega_R$, the first operator has trace norm $2\|\eta\|$. Since $\sigma \geq 0$, $P_R\sigma = 0$ and $\text{Tr } \sigma = d^2$, one has $\|\sigma/n^2 - e^2 P_R\|_1 = 2e^2$. The last operator has trace norm $2\|c_R^{(1)}\| = 2\sqrt{10}a$, which is at most $20a$.

Itemized bound. Each component is bounded with the AW1 ingredients, restricted to supports meeting R .

- **am2_remainder:** $\varrho = 352JT$ bounds the anchored remainder at each site of R . So $\|(c - c^{(1)})_R\| + \|c_{\{0\}}\| \leq \varrho$ at 0, $\|c_{\{e_z\}}\| \leq \varrho$ at e_z , and $\|\eta_{\text{am2}}\| \leq 2\varrho$.
- **straddling:** for a straddling I and a site $u \in I \setminus R$, c_I is excited at u , so the outside-excitation lemma (61) gives $\|\eta_{\text{str}}\| \leq T \sum_{I \text{ straddling}} \|c_I\| \leq T(72a + 2\varrho)$, where 72 counts the straddling faces meeting R .
- **two_creation:** $\|\eta_{\text{pair}}\| \leq A_0 A_{e_z}$ with $A_x \leq 33a + \varrho$, where 33 faces contain one site of R but not the other.
- **density:** $e \leq \varepsilon_R := 82a + 2\varrho + (33a + \varrho)^2$, the ε pinned to the 82 faces meeting R , admitted as valid in the AV1 and AW1 gates. It is about 5.695×10^{-9} at the cap.
- **normalization:** the last term of (135) is of third order and is still charged, as $20a\varepsilon_R^2$.

Collecting the pieces,

$$\begin{aligned} \|r_R\|_1 \leq & \underbrace{4\varrho}_{\text{am_remainder}} + \underbrace{2T(72a + 2\varrho)}_{\text{straddling}} + \underbrace{2(33a + \varrho)^2}_{\text{two_creation}} \\ & + \underbrace{2\varepsilon_R^2}_{\text{density}} + \underbrace{20a\varepsilon_R^2}_{\text{normalization}} =: K'_2\tau^2. \end{aligned} \quad (136)$$

Table 23 lists the items at the cap. The gate binds the sum,

$$K'_2 = \frac{966771578474926086618624139557778885954947547760216752246561}{72052885697817210754545804931891200000000000000000000} \quad (137)$$
$$\approx 13417.5275439,$$

and the pair constant at $\tau = \pm 10^{-8}$,

[illegible]

Each is the same exact rational in the forward output, the reverse output and the skeptic's pre-comparison package. The items add linearly; a root-sum-of-squares assembly, a zero item and a crude-tier item are rejected.

Table 23: The items of K'_2 at $\tau = \pm 10^{-8}$, each divided by τ^2 . Decimals are previews of exact rationals in the frozen outputs.

Item	Formula	Value/ τ^2	Order
am2_remainder	4ϱ	$94325000000/7030557 \approx 13416.4334347$	2
straddling	$2T(72a + 2\varrho)$	$538349237890625/1581719415367968 \approx 0.340356976502$	2
two_creation	$2(33a + \varrho)^2$	$166184094520081/1581719415367968 \approx 0.105065470465$	2
density	$2\varepsilon_R^2$	≈ 0.648686759083	2
normalization	$20a\varepsilon_R^2$	$\approx 4.50476916030 \times 10^{-10}$	3
arithmetic	exact rationals	0	–
K'_2	sum	≈ 13417.5275439	

Uniformity and scaling. The bound holds uniformly in N , in the cutoff $L \geq 24$ and in both families. Finite-box face counts never exceed the bulk counts 82, 72 and 33, and T and ϱ are the same for both families. It passes to the untruncated ground by AV1’s cutoff removal and to every subsequential limit because the trace-norm ball is closed; that is (133). Every item is tier (ii). The only inputs that are not R -local are T and ϱ , which are anchored maxima of the AM2 fixed point and are uniform in N , as in AW1. The forward checker has a second code path that recomputes each item from the pins and requires equality, which guards against the silent undercounts of AW1 review finding N8. The ratio $B(\tau)/B(\tau/100)$ of the total bounds is about 10000.976, so the bound is quadratic. The values of K'_2 at τ , $\tau/10$, $\tau/100$ and $\tau/1000$ are about 13417.528, 13416.337, 13416.218 and 13416.206, so K'_2 is nondecreasing in $|\tau|$ and the cap value bounds every smaller coupling.

Since $\rho_R^{(1)}$ is common to all limits, (133) gives $\|\rho_R - \rho'_R\|_1 \leq 2K'_2\tau^2$ for any two subsequential limits of either family at the same τ (137). That is a factor of about 1.0145×10^4 below $2D$. It is the “agree to first order in tau” of the template (closeness order 2). It is still not equality, and it says nothing outside R .

Comparison with K_2^+ . Contract item 3 asks for K'_2 “built from the AW1 remainder items restricted to faces meeting R (not the whole-box K_2^+)” and for a comparison. K'_2 is larger:

$$\frac{K'_2}{K_2^+} = \frac{3867086313899704346474496558231115543819790191040867008986244}{966893014524284957965768152362480514948256313800811767578125} \approx 3.99949762363. \quad (138)$$

The two constants bound different quantities. K_2^+ bounds the single observable W . Since $W\Omega_R$ is excited at both sites of R , W pairs only with c_R , with multiplier $2\|W\Omega_R\| = 1$, and it annihilates the single-site remainders $c_{\{0\}}$ and $c_{\{e_z\}}$ and the 66 one-site straddling faces. The trace norm over $B(\mathcal{H}_R)$ has no such overlap structure. It pays the multiplier 2, and it pays all three excited sectors, $\{0\}$, $\{e_z\}$ and $\{0, e_z\}$, at both sites. The dominant item is the AM2 generic remainder $\varrho = 352JT$, a per-site constant that is not R -local. It is 99.979% of K_2^+ and 99.99% of K'_2 . Every trace-norm constant built from the AW1 items is therefore at least $2\varrho/\tau^2 \approx 6708.2$, and at least about 5488.7 with the sharper majorant $288t/(1-8t)$; both floors exceed K_2^+ . The restriction to faces meeting R helps where it can. ε_R , about $0.5695|\tau|$, lies below $2T + T^2$, about $0.6806|\tau|$, and the pins 72 and 33 replace the unpinned $T \cdot T$ and T^2 . Per unit trace multiplier the count-dependent items total 0.547, against 0.695 in K_2^+ (Table 24).

Only the W -projected R -local analogue, $\varrho + T(6a + \varrho) + (33a + \varrho)^2 + \varepsilon_R^2 + a\varepsilon_R^2$, about 3354.49943, lies below K_2^+ , by 0.30380. It is AW1’s minimal form, not a density bound. The gate therefore reads the contract’s parenthetical as “built from R -local face pins” (82, 72, 33 and 10), unlike the unpinned $T \cdot T$, T^2 and $2T + T^2$ of K_2^+ . It does not read it as a claim that K'_2 is smaller. Nor is K_2^+ a “whole-box” constant: it is uniform in N (wording defect W1). The modern lens’s update-3 expectation that K'_2 would be strictly smaller than K_2^+ does not hold for a trace-norm constant.

Table 24: Items of K_2^+ and K_2' at the cap, each divided by τ^2 (reverse comparison ledger). The last column divides the K_2' items by the trace-norm multiplier 2.

Item	K_2^+ (W only)	K_2' (trace norm on $B(\mathcal{H}_R)$)	K_2' per unit multiplier
am2_remainder	3354.108358 (ϱ)	13416.433435 (4ϱ)	6708.216717
straddling	0.115812 ($T \cdot T$)	0.340357	0.170178
two_creation	0.115812 (T^2)	0.105065	0.052533
density	0.463247 ($(2T + T^2)^2$)	0.648687 ($2\varepsilon_R^2$)	0.324343
normalization	3.2×10^{-11}	4.5×10^{-10}	2.3×10^{-10}
Total	3354.80322946	13417.5275439	

Labelled variants. Four further values of K_2' are valid upper bounds and are recorded as previews only (Table 25). The $\sqrt{2}$ variant uses that η_{am2} is a sum of three mutually orthogonal vectors, so $\|\eta_{\text{am2}}\| \leq \sqrt{2} \varrho$, with a directed bracket for $\sqrt{2}$. The 288-majorant variant replaces ϱ by $288JT/(1 - 8T)$, from $G(t) - 16 \leq 288t/(1 - 8t)$. Each rests on an extra labelled step, and each still exceeds K_2^+ . The gate binds the triangle value, which is purely rational and identical in all three computations.

Table 25: Labelled variants of K_2' (previews only, not headline).

Variant	K_2'	Ratio to K_2^+
Orthogonal in- R sectors ($\sqrt{2}$)	≈ 9487.945	≈ 2.8281674
Directed AM2 remainder (288-majorant)	≈ 10978.18	≈ 3.2723756
Both refinements combined	≈ 7763.06	≈ 2.3140145
Admitted $\varepsilon = 2T + T^2$ in the density item	≈ 13417.81	≈ 3.9995804
Triangle headline, bound by the gate	≈ 13417.5275439	≈ 3.99949762363

7.1.7 A recorded observation

The reverse triangle inequality applied to (133) gives, for every subsequential limit of either family at either sign,

$$\frac{\sqrt{10}|\tau|}{72} - K_2'\tau^2 \leq \|\rho_R - P_R\|_1 \leq \frac{\sqrt{10}|\tau|}{72} + K_2'\tau^2. \quad (139)$$

At the cap the interval lies in $[4.3786 \times 10^{-10}, 4.4055 \times 10^{-10}]$. The forward previews of the two ends are $4.37863477824 \times 10^{-10}$ and $4.40546983333 \times 10^{-10}$. The skeptic verified each end exactly as a directed bound, $(\text{lower} + K_2'\tau^2)^2 \leq 10\tau^2/5184 \leq (\text{upper} - K_2'\tau^2)^2$. The review asks that the observation be recorded from these exact values, because the forward checker does not pin its lower end (Section 7.1.9). The upper end is about 30.9 times below D . AV1's ε charges the 72 straddling faces at first order, but they reach the off-diagonal block only at second order. The lower end would be the first lower bound on $\|\rho_R - P_R\|_1$ in Round32; the AV1 gate provides none (Section 2.10).

The gate records (139) as a labelled observation only. It is not a preregistered target or comparator and not an admitted tier, and it does not replace D anywhere, including in $2D$. It is not a resolved interaction shift: `resolved_interaction_shift` stays false, and the static Wilson mean belongs to AW2 (Section 5). AY2 may certify it under its own contract.

7.1.8 The mandatory sentence, gate fields and definitions

The template. The preregistration freezes a mandatory sentence template from the Jung lens, and contract item 4 requires it. Items (2)–(3) of the gate statement contain it verbatim:

For every pair of subsequential limits of the named construction families F1 and F2 at the same coupling, on the fixed cover R and for the frozen observable class, the reduced densities satisfy $\|\rho_R - \rho'_R\|_1 \leq 2D$ and agree to first order in τ ; this does not assert equality of the states, whole-sequence convergence, translation invariance, boundary independence of the dynamics, or a rate in N .

The gate then fills its slots with the constants of this subsection (Table 26).

Table 26: The slots of the mandatory sentence template and the constants with which the gate fills them.

Slot	Filled with
named construction families F1 and F2	F1: centered whole-star boxes Λ_N (AQ1); F2: all-contained-face boxes with padding (I1 §6), on the same Λ_N , $N \geq 2$
the same coupling	$\tau = \pm 10^{-8}$, and every $ \tau \leq 10^{-8}$
the fixed cover R	$R = \{0, e_z\}$: 48 links, 36 endpoints, seven incident anchors
the frozen observable class	$B(\mathcal{H}_R)$, in the trace norm
$2D$	the exact rational of (126), about $2.72249057405 \times 10^{-8}$
agree to first order in τ	the common $\rho_R^{(1)}$ of (129), with $\ \rho_R - \rho'_R\ _1 \leq 2K'_2\tau^2$, about $2.68350550879 \times 10^{-12}$ (137)

Both producers also fill the Jung loop-2 form. It quantifies over $A \in B(\mathcal{H}_R)$ with $\|A\| \leq 1$, which is equivalent by trace duality. It adds that the statement does not assert boundary independence beyond R , and that two states satisfying it “may differ by the stated order on R and without bound elsewhere”. The frozen template and the Jung text differ in wording (review reading R5), and both are filled.

Gate fields. The preregistration requires five gate fields, which record what is not claimed, and the gate exports all five as false. It also exports the Jung field `rate_in_N_claimed` as false (wording defect W3).

- not claimed, exported as false: `uniqueness_claimed`, `whole_sequence_claimed`, `rate_claimed`, `translation_invariance_claimed`, `boundary_independence_of_dynamics_claimed` and `rate_in_N_claimed`.

The descriptive fields read, verbatim:

- `states_compared`: all subsequential limits of F1 and F2 (every pair, same or different family, same τ)
- `region`: $R=\{0, e_z\}$ fixed before production (complete cover of the original xz Wilson loop)
- `topology`: trace norm on $B(\mathcal{H}_R)$; dynamics named in the norm on compact time windows (no dynamical statement)
- `closeness_order`: $[1, 2]$

The closeness order is 1 for $2D$ and 2 after subtracting $\rho_R^{(1)}$. The forward producer exported 2 and the reverse $[1, 2]$; the gate adopts $[1, 2]$ (review finding N4).

Quantitative boundary comparison. Contract item 6 requires a definition, and both producers give it in substance. In AY1 a quantitative boundary comparison is the pair of two statements: (i) a closeness bound, $\|\rho_R - \rho'_R\|_1 \leq 2D$ for every pair of subsequential limits of the two named prescriptions at the same τ ; and (ii) a matching first-order term, the same $\rho_R^{(1)}$ for every such limit, with second-order difference at most $2K'_2\tau^2$. It is explicitly not a variational statement about which boundary condition the infinite-volume theory selects. It compares no energies or free energies of the prescriptions and ranks neither. It does not say that the limits of either family are canonical, and it says nothing outside R or about the dynamics. A definition that claims a variational selection, or drops the first-order term, is rejected by both checkers.

Uniform in N is not uniform in the spacing. Every constant here is uniform in the box size N , the on-site cutoff, the family and the subsequence. It holds at fixed lattice spacing, fixed α , $\hbar$ and E_* , and fixed τ . Nothing is uniform in the spacing, and $N \rightarrow \infty$ at fixed spacing is not a continuum limit. The zero-selected family carries no dictionary between τ and the spacing (Section 1.1). As an illustration from a different model, the uniform dictionary $\tau = 96/g^4$ of Section 6.2 needs $g^4 \geq 9.6 \times 10^9$ for the cap and $g^4 \geq 32$ for the coefficient box, so every path with vanishing spacing and $g \rightarrow 0$ leaves both. The cover R is two coarse factors, a block of $4 \times 2 \times 2$ fine lattice spacings: fixed in lattice units, it shrinks in physical units as the spacing decreases. Both checkers reject a claim of uniformity in the spacing, and a reading of uniformity in N as a continuum statement (`not_uniform_in_a`).

Phrasing rule. Both checkers scan the reports and the exported sentences. They reject a definite-article phrase for a single AQ-type state or for a single infinite-volume limit, and any sentence that mentions uniqueness without “not”. They also require the phrase “a chosen subsequential”. AQ1’s chosen state is a chosen subsequential limit of F1, and every statement here holds for each such limit separately.

7.1.9 Controls, replays and review

Error ledger. The preregistered terms are the following.

- **state_boundary:** D per limit and $2D$ per pair (126), the AV1 forward tier (ii), uniform in N , the cutoff and the family.
- **second_order_difference:** $K'_2\tau^2 \approx 1.34175275439 \times 10^{-12}$ per limit and $2K'_2\tau^2$ per pair (137), the five items of (136) added linearly.
- **arithmetic:** no numeric cost. The headline constants are exact rationals. Directed brackets enter only for $\sqrt{10}$ (the trace norm of $\rho_R^{(1)}$), for $\sqrt{2}$ (a labelled variant) and for $e^{1/8}$, and the reverse producer’s directed 10^{-40} roundings enter only its fidelity cross-check and brackets.

A root-sum-of-squares combination and division by $\sqrt{N}$ or by 64 are rejected.

Contract controls. The contract lists 21 controls. Seven of them carry new semantics for AY1.

- **topology_named:** weak-* for states, strong continuity relabelled as norm continuity, one topology for both, and a dynamical claim for F2 limits are rejected.
- **two_families_named:** one family only, “all boundary conditions”, a third family (the literal vertex boxes), orthant boxes substituted, and F2 collapsed onto F1 (616 faces fewer on Λ_2) are rejected.
- **subsequence_versus_whole_sequence:** whole-sequence convergence and a $1/N$ rate claimed for an alternating sequence whose parity subsequences converge are rejected.
- **local_closeness_not_uniqueness:** equality inferred from closeness or from a common enclosing interval, a mandatory sentence without its negation, and a gate field set true are rejected.
- **common_clock:** F2’s normalized coefficient read in units of α (8τ), different couplings, second-order closeness across $\pm\tau$ (the first-order densities differ by $\sqrt{10}|\tau|/36$), and a normalized clock for one family are rejected.
- **padding_family_contraction_proved:** the mutations of Section 7.1.3 are rejected.
- **not_uniform_in_a:** “uniform in lattice spacing”, $g \rightarrow 0$ and $N \rightarrow \infty$ called a continuum limit are rejected.

The inherited controls keep their meaning. Among them, **tier_mixing_rejected** rejects the tier-(i) value in the closeness, the AV1 reverse 82-face refinement used as the admitted constant, a crude item inside K'_2 and $t = t_1$ without the remainder. The control **first_order_mean_charged** rejects a zero first-order density, which the AW2 lower end contradicts because it exceeds $K'_2\tau^2$. The control

vector_versus_scalar_centering rejects scalar subtraction as vector centering and the W -only K_2^+ used as a trace-norm constant. The control **reverse_premise_isolation** rejects the triage, the Jung note or the forward AY1 report in the reverse inventory.

The two named examples. Contract item 5 also names two examples, and both checkers run them.

- *Fixed vector versus moving vector.* In the forward checker the moving $|e_n\rangle\langle e_n|$ has pairwise trace distance 2, weak limit 0 and unbounded energy, and a claim of compactness is rejected. For $U(t)e_j = e^{ijt}e_j$ and $Ae_j = e_{2j}$, the fixed vector e_1 gives at most $4/n$ and the moving e_n gives 2, so strong continuity is not norm continuity. In the reverse checker the moving vectors $w_N = \Omega + \lambda e_{1+(N \bmod 2)}$, $\lambda = 1/10000$, give densities that each lie within $D_{\text{ex}} = 2\lambda/\sqrt{1+\lambda^2}$ of P , yet the sequence has two distinct subsequential limits. Closeness to a fixed vector implies neither convergence of the moving sequence nor equality of its limits.
- *A common enclosing interval is not equality.* The forward checker rejects two limits in the AW2 enclosure (74) declared equal, and two distinct states in the D -ball declared equal. The reverse checker exhibits $\tau/144 \pm K_2^+ \tau^2/2$, which both lie in the AW2-type enclosure and differ.

Checkers. The forward checker runs 42 exact checks and implements all 21 controls as damaging mutations raised through explicit exceptions, never assertions. It rejects 97 mutations in all, 72 of them inside the 21 control checks; the report’s “All 21 contract controls ... 97 in total” counts every rejection in the checker (review finding N7). The reverse checker runs 41 exact checks and rejects 94 mutations, 87 inside the control checks, and it leaves no control unimplemented. Both use only the standard library and exact rationals, and both reject float, Boolean, NaN and zero-denominator inputs. Both verify the contract snapshot’s SHA-256 before parsing and record their own SHA-256 before evaluation. A byte edit without a rehash aborts both, and so does a coherently rehashed target 1/125000: the forward checker pins the target, and the reverse checker ties it to twice the AV1 target. Both export **continuum_claim**, **uniform_wilson_claim**, **resolved_interaction_shift** and **scientific_priority_verified** as false.

Exact fixtures. Finite creation algebras, labelled with **transfers_to_aq** false, audit the algebra of Sections 7.1.5 and 7.1.6. The forward fixture has two sites of R of dimension 3, two outside sites and up to three decoupled spectators, with creations of every support type. It checks (135), the component bounds and the outside-excitation lemma at both outside sites. It also checks that ρ_R does not depend on the spectators, while the unnormalized weights take four different values. It rejects dropping σ , dropping the straddling creations from ξ , the unnormalized weight and dropping the two-creation term. The reverse fixture has four qubit sites, $\lambda = 1/50$, and two environments that share only the first-order creation inside R . It checks the vector overlap identity and the ledger bound in both environments. The two densities differ, yet twice their Hilbert–Schmidt distance stays below the sum of the two ledger bounds and below $\|\lambda\rho^{(1)}\|_1$. The squared Hilbert–Schmidt norm of the difference drops by a factor of about 9889 when λ is divided by 10, so the difference scales as λ^2 .

Independent pre-comparison package. The skeptic’s package holds a contract review, a derivation and a checker with 96 exact checks, implementing 26 controls (the 21 contract ids and five more) with 70 rejected mutations. It was hash-frozen and committed (55be992, 02:39:42Z) before the reverse commit (d1e6ac6, 02:49:24Z) and the forward commit (f7b9839, 02:50:55Z), and a fresh replay reproduces its frozen output byte for byte under normal and optimized Python. Its predictions matched every producer headline rational for rational. They covered D , $2D$ and $2D_i$; the counts 49, 82, 10, 72, 66, 33, 16, 6 and 27 and the seven anchors; the F2 values $J = 28|\tau|$, support 4, termination 8, reset $98|\tau|$ and $C_F = 56|\tau||F|$, the extra-face count $28N(5N+1)$ and the

distance $N - 1$; the ten-face $\rho_R^{(1)}$ with trace norm $\sqrt{10}|\tau|/72$; K'_2 with all five items, the $\sqrt{2}$ variant and the W -projected analogue; $2K'_2\tau^2$; the $\pm\tau$ separation; and the upper end of the observation. In particular, the three values that the gate binds for AY2, $2D$, K'_2 and $2K'_2\tau^2$, are the same exact rational in the forward output, the reverse output and the pre-comparison package. The producers' further labelled values, the 288-majorant, combined and admitted- ε variants, the Bures constant and the lower end of the observation, were re-derived exactly after the comparison. The skeptic records that its own triage seeded the $2D$ observation and the goal-4 controls, and that its AV1 and AW1 reviews are shared premises. The agreement is therefore correlated model-agent work, not independent discovery. The skeptic also discloses one early recursive search whose output was filtered, so that no line from the producer folders was displayed.

Post-comparison review. The review adds 55 exact checks, including 28 source-mutation runs, for 151 independent checks in all. Two unmutated copies reproduce the frozen outputs byte for byte. All 22 must-abort source edits abort at the intended check. In the forward package they include 4ϱ replaced by 2ϱ , the straddling pin 72 replaced by 66 and the density item zeroed, all caught by the second code path; the F2 rule replaced by the F1 rule; straddling faces added to F_R ; the amplitude $\tau/576$; a claim flag set true (uniqueness and continuum are not claimed); contract edits; the reverse report added as an input; and a forbidden sentence appended to the report. In the reverse package they include a halved AM2 item, the F2 rule replaced, the padding removed, a doubled Bures bound, a contract byte edit, a rehashed target that is not twice the AV1 target, the triage or the forward report added as an input, and a forbidden phrase appended. Four silent edits run to completion in the producer checkers and are caught only by the review's value validator: the forward lower end of the observation computed with $K'_2\tau^2/2$; the forward $\sqrt{2}$ rounded down; the reverse ε_R pinned to 49 faces instead of 82; and the reverse straddling pin 72 replaced by 66. The reverse has no second code path for its ledger, and the forward's second path does not cover its supplementary bounds or its variant. AY2's checker should therefore read $2D$, K'_2 and $2K'_2\tau^2$ from the AY1 gate as exact rationals and reject any recomputed or smaller value (review finding N2).

Replays and isolation. The declared-direction replay tools verify both closures file by file, 36 forward files and 34 reverse. They compare every premise snapshot with its repository source and replay each producer under normal and optimized Python into fresh external directories. All four runs reproduce the frozen outputs byte for byte, and the freezing tool reports both closures verified. The reverse inputs are exactly AGENTS.md, the contract and the 28 shared premises, 30 files. Their only skeptic files are the declared AV1 and AW1 reviews, and they hold no triage, Jung loop-2, deliberation, panel, expert, forward-AY1 or skeptic-AY1 file. The forward inputs are those 30 plus the two forward-additional premises, 32 files. Isolation between the producers rests on disclosure and content, not on mechanism (review finding N3). The Jung sentence form reached the reverse producer through its task prompt, not through a file. A final `git status` by the reverse producer, after its work was complete, showed the names of the forward checker and report. The forward report was last written at 02:49:43Z, after the reverse freeze at 02:48:44Z, and it contains no reverse-specific construct: no Bures route, vector overlap identity or 288-majorant variant. The private scratch folders were checked by names and times only, and no file in the shared scratchpad root was written after the AY1 freeze except the AX2 manuscript folder. The disclosed protocol and style reads (the tools documentation, the freezing tool, parts of earlier checkers and output conventions) carry no premise weight (N9). Isolation is verified for repository inputs only.

7.1.10 Contract wording and non-blocking findings

The gate records the contract's wording defects, in the forward producer's numbering W1–W7, as non-blocking.

- *W1.* Item 3's "(not the whole-box K_2^+)": K_2^+ is uniform in N and specific to W , and the

R -local trace-norm K'_2 is about 4.0 times larger. The phrase is read as “built from R -local pins” (Section 7.1.6).

- *W2*. The claim exclusions of the contract and of the preregistration use the definite-article phrase for a single AQ-type state that the Jung phrasing rule forbids. Both producers quote it only verbatim and negated.
- *W3*. The contract’s field `rate_claimed` and the Jung field `rate_in_N_claimed` differ in name; both are false.
- *W4*. The preregistered state provenance names only the F1 construction; F2 has its own diagonal extraction (Section 7.1.3).
- *W5*. The contract does not fix the F2 box geometry; it is read as the same centered Λ_N (Section 7.1.2).
- *W6*. Item 5 lists “freeze; byte-identical replays” as controls, but they are not among the 21 control ids. They are protocol steps, verified in Section 7.1.9.
- *W7*. Verified, no defect: item 3’s sign $-(c^{(1)}\Omega_0^* + \text{h.c.})$ gives $+\tau/144$ with the AV1 convention (Section 7.1.5).

The gate also records four findings from the review as non-blocking. The reverse route’s “derivative of the fidelity identity” holds in vector form only, because the scalar derivative vanishes (Section 7.1.5). The two producers use different valid padding conventions (Section 7.1.3). The reverse producer does not export `rate_in_N_claimed`, and the gate adopts the closeness order $[1, 2]$ (Section 7.1.8). Reverse isolation holds for repository files, and the Jung sentence wording reached the reverse producer through its task prompt, as disclosed (Section 7.1.9).

7.1.11 Limitations recorded by the gate

The AY1 gate records the following limitations verbatim.

- Only the zero-selected patterned family (selected triple (0,0,0), Haar reference), the cover $R=\{0, e_z\}$, fixed spacing and $|\tau| \leq 10^{-8}$, with the two named families on centered cubes Λ_N , $N \geq 2$; nothing transfers to literal vertex boxes (I1 section 7), periodic or orthant boxes, nonzero selected triples, the uniform route-B model (its K_2 was not transferred in AX1), weak coupling or the continuum
- Uniform local closeness on the fixed region R at the same coupling, not uniqueness, whole-sequence convergence, a rate in N , translation invariance or boundary independence of the dynamics; two limits may differ by up to $2K_2' \tau^2$ on R and without bound elsewhere; 2D alone also bounds states at different couplings (P_R is τ -independent), so the boundary-comparison content is the matching first-order term
- The first-order density is a statement about a coefficient through the uniform remainder bound, not a τ -derivative of a subsequential limit (subsequences depend on τ)
- K_2' is an upper bound dominated (99.99%) by the generic AM2 majorant 4ρ and exceeds K_2^+ ($\sim 3.9995x$); the contract’s ‘(not the whole-box K_2^+)’ is read as ‘built from R -local face pins’, not as a smaller constant; labelled variants ($\sqrt{2}$ sectors ~ 9487.945 , 288-majorant ~ 10978.18 , both ~ 7763.06 , admitted-eps ~ 13417.81) are valid, not headline
- Supplementary two-sided observation $\sqrt{10}|\tau|/72 \mp K_2' \tau^2$, i.e. $||\rho_{R-P_R}||_1$ in $[4.3786e-10, 4.4055e-10]$ for every limit of either family: a labelled observation only, not a contract target, not an admitted tier and not a resolved interaction shift; it does not replace D
- Inherited without re-proof: AM2 majorant, fixed point, excited-sector exclusion and cutoff passage (applied to F2 through their hypotheses); AV1 product split and cutoff-vector removal; AQ1 compactness, diagonal extraction and Nachtergaele-Sims dynamics (named only); AW1 $c^{(1)}$ and outside-excitation lemma; the I1 dictionary; Peter-Weyl/Haar orthogonality, Uhlmann/Fuchs-van de Graaf and Nachtergaele-Sims cited, not machine-checked
- Independence limited to proof routes, enumeration and code: the forward-only premises (triage goal-4 note, Jung loop-2 notes) and the shared contract, selection note and AV1/AW1 reports and reviews state the mechanism; the Jung sentence form

reached the reverse through its task prompt, and the reverse saw two forward file names; isolation is verified for repository inputs only; all agents are correlated model agents

- Producer checkers do not pin every exported value: the forward supplementary lower end and sqrt-2 rounding direction and the reverse 82/72 ledger pins ran silently when edited and were caught only by the skeptic validator; AY2 should read 2D, K_2' and 2K_2' tau^2 from this gate as exact rationals
- The -tau evaluation is a replay of the same |tau| formula (rho^(1)_R is odd under the AW1 flip), not a second confirmation
- Scientific priority unverified

Full derivations and reviewed implementations: [AY1 forward](#), [AY1 reverse](#), [independent pre-comparison derivation](#), [independent model-agent review](#) and the [frozen contract](#). The gate selects AY2 next. The [AY2 selection note](#) asks AY2 to state, with the AY1 gate constants, what is proved for the set of subsequential limits of both families, and to list each obligation that the present bounds do not settle, with its missing premise and a candidate route. It also asks AY2 to certify the two-sided first-order distance of Section 7.1.7 under its own contract, as a static statement and never as an interaction-shift or dynamical claim. That note is a planning record and admits nothing.

7.2 AY2: what the closeness lemma does and does not establish

Contribution HNM-C-AY2. The AY2 contract, recorded under the title “Hruday statement on state identification: what the closeness lemma does and does not establish”, was frozen after the AY1 gate and before production. It closes sub-round 4 and is investigation 8 of 10. The model, the two construction families, the cover, the clock and the constants are those of AY1 (Section 7.1). The preregistered model identifier is `AQ_patterned_zero_selected`, the coupling is $\tau = \pm 10^{-8}$ with the selected triple exactly $(0, 0, 0)$, and the states are the subsequential limits of both AY1 families, obtained through finite-volume uniform bounds. The observable is the two-sided tier for $\|\rho_R - P_R\|_1$ together with the obligations statement. Its reference is P_R on the Haar route; there is no centering and no Euclidean node. The preregistered target is the width of that tier relative to its centre, at most $1/100$. The contract’s own note puts the expected value at about 6×10^{-3} and adds that the obligations table must be complete, with six rows. The itemized error terms are `first_order_term`, the value $\sqrt{10}|\tau|/72$ with $\sqrt{10}$ enclosed by exact rationals; `second_order_remainder`, the AY1 gate’s $K'_2\tau^2$ for one state and $2K'_2\tau^2$ for two limits; and no other terms.

AY2 is a statement loop (`statement+skeptic`). One forward producer writes the statement, the obligations table and an exact checker. The second admission input is the skeptic’s independent derivation and replays (`single_direction_independent_replay`). The contract’s direction note gives the reason: both AY1 producers reached the same constants by different routes, so a second producer for the statement would add no independence. The producer received `AGENTS.md`, the contract and its 21 shared premises, 23 files in all. They include the AY1 gate, both AY1 reports and the skeptic’s AY1 review, and no current skeptic, expert, deliberation or panel file. No reverse package exists, and the contract sets `reverse_premise_isolation` to false. Both packages were developed at the same time after the contract froze at 03:09:12Z, and the commit order is the reverse of AX2’s (Section 6.13.9): the forward commit `ce6bca8` (03:39:41Z) precedes the skeptic’s pre-comparison commit `f640111` (03:40:16Z). The skeptic’s exact values were final at 03:34:06Z, before the producer’s checker (03:38:22Z) and report (03:38:56Z) were last written. Isolation therefore rests on disclosure and content, not on commit order (Section 7.2.7).

Notation follows Section 7.1. Σ_τ is the set of all subsequential limits of F1 or F2 at the coupling τ ; nothing proved so far decides whether it has one element or many, so each statement holds for a chosen subsequential limit, every one separately. For $f \in F_R$ we write $e_f = 2W_f\Omega_R$ and $v = \sum_{f \in F_R} e_f$, and $r_R = \rho_R - P_R - \rho_R^{(1)}$ as in (135). The directed bracket $s_- < \sqrt{10} < s_+$ is that of Section 7.2.5, and L and U are the certified ends of the tier.

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The gate’s decision binds the producer’s exact rationals for the tier, with its 10^{-30} bracket for $\sqrt{10}$, and records the skeptic’s 10^{-40} enclosure as the tighter independent one. It also records the non-blocking points of Section 7.2.8. The rest of this subsection derives the statement, and the limitations recorded by the gate close it (Section 7.2.9).

Every constant is read as an exact rational from the AY1 gate, as the AY1 review asked (Section 7.1.9). The producer's checker parses each rational from the hash-bound gate snapshot and types none in.

It then recomputes each one from the gate’s own itemization and requires equality: $a = |\tau|/144$, $J = 28|\tau|$, $T = 49a/(1-352J)$, $\varrho = 352JT$, $\varepsilon = 2T+T^2$, $D = 2\varepsilon(1+\varepsilon)/(1+\varepsilon^2)$, $\varepsilon_R = 82a+2\varrho+(33a+\varrho)^2$, the five items of K'_2 in (136), and $K_2^+\tau^2 = \varrho + T \cdot T + T^2 + \varepsilon^2 + a\varepsilon^2$. The pins 82, 72, 33 and 10 are re-derived from the parsed I1 table and the fine-lattice link tails. Table 27 lists the constants.

Table 27: The AY1 gate constants used by AY2, at $\tau = \pm 10^{-8}$. Each is read as an exact rational from the AY1 gate; decimals are truncated previews.

Constant	Exact value	Preview	What it bounds
D	(23)	$1.36124528702 \times 10^{-8}$	$\ \rho_R - P_R\ _1$, one limit
$2D$	(126)	$2.72249057405 \times 10^{-8}$	$\ \rho_R - \rho'_R\ _1$, two limits
$\ \rho_R^{(1)}\ _1$	$\sqrt{10} \tau /72$	$4.39205230578 \times 10^{-10}$	the common first-order density
K'_2	(137)	$1.34175275439 \times 10^4$	$\ r_R\ _1/\tau^2$, one limit
$K'_2\tau^2$	(137)	$1.34175275439 \times 10^{-12}$	one limit against $P_R + \rho_R^{(1)}$
$2K'_2\tau^2$	(137)	$2.68350550879 \times 10^{-12}$	two limits, after $\rho_R^{(1)}$ cancels
K_2^+	(63)	$3.35480322946 \times 10^3$	the single observable W

For every $\omega, \omega' \in \Sigma_\tau$, from the same family or from different families, at the same τ ,

$$\begin{aligned}
(a) \quad & \|\rho_R(\omega) - \rho_R(\omega')\|_1 \leq 2D, \\
(b) \quad & \left\| \rho_R(\omega) - P_R - \rho_R^{(1)} \right\|_1 \leq K'_2\tau^2, \\
(c) \quad & \|\rho_R(\omega) - \rho_R(\omega')\|_1 \leq 2K'_2\tau^2, \\
(d) \quad & \frac{\sqrt{10}|\tau|}{72} - K'_2\tau^2 \leq \|\rho_R(\omega) - P_R\|_1 \leq \frac{\sqrt{10}|\tau|}{72} + K'_2\tau^2.
\end{aligned} \tag{140}$$

Items (a)–(c) are AY1’s: (a) is (126), (b) is (133), and (c) follows from (b) by the triangle inequality, because $\rho_R^{(1)}$ is the same operator for every box of both families. Item (d) is the AY2 tier of Section 7.2.5. Items (a) and (c) are uniform local closeness on a fixed region at a fixed coupling. Item (b) fixes a first-order coefficient, not a state: it is not a τ -derivative of a subsequential limit, because subsequences may depend on τ (Section 7.1.5). By trace duality, (a) and (c) say that $|\omega(A) - \omega'(A)|$ is at most $2D$ and at most $2K'_2\tau^2$ for every $A \in B(\mathcal{H}_R)$ with $\|A\| \leq 1$.

The mandatory sentence. Contract item 1 requires the mandatory sentence template with the constants filled in. Unlike AY1’s, the AY2 preregistration has no template field (skeptical reading 5); the template is recoverable verbatim from the AY1 gate, a snapshotted premise. The producer’s report fills the Jung loop-2 form of the AY1 forward report, whose fixed text the checker parses from that snapshot and finds identical in the AY1 reverse snapshot, with the exact $2D$ and $2K'_2\tau^2$. The skeptic asked the gate to carry the AY1-contract template with its constants as well (review finding N2), and item (1) of the gate statement does so. It opens with the template, word for word as in Section 7.1.8:

For every pair of subsequential limits of the named construction families F1 and F2 at the same coupling, on the fixed cover R and for the frozen observable class, the reduced densities satisfy $\|\rho_R - \rho'_R\|_1 \leq 2D$ and agree to first order in τ ; this does not assert equality of the states, whole-sequence convergence, translation invariance, boundary independence of the dynamics, or a rate in N.

It then fills the slots with the exact $2D$ of (126), the first-order density $\rho_R^{(1)}$ of (129), K'_2 of (137) and $2K'_2\tau^2$, all as in Table 27, and it closes with the Jung form, in trace duality:

Equivalently, for every A in B(H_R) with $\|A\| \leq 1$, $|\omega(A) - \omega'(A)| \leq 2D$ (order τ) and $\leq 2K'_2\tau^2$ (order τ^2 after the common first-order density cancels); two limits satisfying this may differ by the stated order on R and without bound elsewhere.

Both forms speak of a chosen subsequential limit of either family, every one separately. The producer’s checker scans its report with code spans removed: it finds the phrase “a chosen subsequential”, no definite-article phrase for a single AQ-type state or for a single infinite-volume limit, and “not” in every sentence that mentions uniqueness. The skeptic’s scan finds the same; the one definite-article phrase occurs only inside the negated verbatim exclusion (defect D2).

What the statement does not say. It does not say that Σ_τ has a single element, or that two limits coincide on R or anywhere else. It does not say that any whole sequence of boxes converges, that any limit is translation invariant or that anything holds at a rate in N . It does not say that the dynamics of F1 and F2 agree on compact time windows, or that F2 has a dynamics at all, and it says nothing outside R . Each of these is an obligation of Section 7.2.3. The closeness is also not a variational statement about which boundary condition the infinite-volume theory selects (Section 7.1.8). Every constant is uniform in N at fixed spacing and fixed τ , and none is uniform in the lattice spacing.

Gate fields. The gate exports six claim fields as false, which record what is not claimed.

- not claimed, exported as false: `uniqueness_claimed`, `whole_sequence_claimed`, `rate_claimed`, `rate_in_N_claimed`, `translation_invariance_claimed` and `boundary_independence_of_dynamics_claimed`.

Item 6 requires `rate_in_N_claimed`, although the preregistered `gate_fields_required` omits it (defect D3). The descriptive fields read, verbatim:

- `states_compared`: all subsequential limits of F1 and F2 at the same tau (every pair, same or different family); for the tier, each single subsequential limit of either family against the Haar product P_R
- `region`: $R=\{0,e_z\}$ fixed before production (complete cover of the original xz Wilson loop; 48 links, 36 endpoints)
- `topology`: trace norm on $B(H_R)$; dynamics named in the norm on compact time windows (no dynamical statement)
- `closeness_order`: [1,2]
- `tier_label`: `first_order_distance_from_product`

The closeness order is 1 for $2D$ and 2 after $\rho_R^{(1)}$ cancels. Both packets also export `continuum_claim`, `uniform_wilson_claim`, `resolved_interaction_shift`, `scientific_priority_verified` and `weak_coupling_claim` as false, and the skeptic’s review adds `euclidean_node_certified` and `dynamical_claim`, also false.

7.2.3 The obligations table

Contract item 2 names six obligations. The producer’s checker reads the six names from the contract, parses the table from its report, and requires exactly these rows, in this order, each marked unproved. Table 28 gives them with the gate’s routes.

The constants of the dynamics rows. Contract item 2 asks for “a Lieb–Robinson locality argument with the AM2 constants”. AM2 supplies the finite-volume gap and the creation fixed point, which an algebraic comparison of dynamics does not use. The Lieb–Robinson constants are AQ1’s Nachtergaele–Sims constants (skeptic reading 6): $F(r) = (1+r)^{-4}$ on the ℓ^1 metric, whose shells have exactly $4r^2 + 2$ sites (checked for $r \leq 9$), so $\|F\| \leq 7$ and $C \leq 224$; stars have ℓ^1 diameter two, so $\|\Phi\|_F \leq 81 \cdot 28|\tau| = 2268|\tau|$. The F2 owner-set interaction has per-site sum $49|\tau|/3$ and gives $\|\Phi'\|_F \leq 1323|\tau|$ (Section 7.1.2). The per-site sum $J \leq 28|\tau|$ is common to AM2 and AQ1. The gate names AQ1 and records the contract’s “with the AM2 constants” as wording only; the producer’s route text for O5 repeats that wording, while its constants column names the AQ1 constants (review

Table 28: The six obligations that the AY1 bounds leave open. Every row is unproved; each route is a candidate, named and not executed.

Obligation	Status	Missing premise	Candidate route
O1: uniqueness of the limit	unproved	Not available: a two-state estimate, or a classification of every state in the class, that forces two subsequential limits to agree on R ; every admitted bound is a one-state ball, and uniqueness is not implied by closeness.	HTW-type local stability with evaluated constants (the Yarotsky constants $c_1(S)$, $c_2(S)$ of I1 §5, and $c_{\text{HTW}}(1,1)$), or a Dobrushin-type condition at the cap, not formulated for this model. None is evaluated.
O2: whole-sequence convergence	unproved	A Cauchy estimate in N for the reduced densities of each family. The admitted bounds hold with one constant in every box, and compactness yields subsequences only.	A Cauchy estimate in N , for instance from a boundary-influence refinement of the AM2 fixed point; or within-family identification of all subsequential limits plus AQ1 compactness, which gives no rate.
O3: translation invariance	unproved	Covariance of the construction together with O1 or O2. The model is covariant under coarse translations, but the centered boxes are not, and a translated box sequence is a different sequence.	Covariance of the construction plus identification of the limits over every exhausting sequence, translated boxes included.
O4: a rate in N	unproved	Any N -dependent estimate. D , K'_2 and the reset budget $98 \tau $ are uniform in N , and the subsequence construction has no rate.	A quantitative Cauchy estimate, with a per-shell factor $\kappa < 1$ and a prefactor; summing its tail would give a rate.
O5: boundary independence of dynamics on compact time windows	unproved	A comparison of the F1 and F2 finite-volume dynamics of local observables near R , uniform on compact windows. AY1 names this topology but constructs no F2 dynamics.	A Duhamel and Lieb–Robinson comparison over the $28N(5N+1)$ extra F2 boundary faces at distance $N-1$, with the AQ1 Nachtergaele–Sims constants $F(r) = (1+r)^{-4}$, $C \leq 224$, $\ \Phi\ _F \leq 2268 \tau $ and $\ \Phi'\ _F \leq 1323 \tau $. Unwritten.
O6: dynamics of the padded family itself	unproved	An infinite-volume dynamics for F2. AQ1 places F1 in the Nachtergaele–Sims theorem; for F2 no such application, stationarity of its limits, GNS continuity or generator positivity is proved.	Nachtergaele–Sims with the owner-set interaction ($\ \Phi'\ _F \leq 1323 \tau $, $C \leq 224$), whose restrictions are the F2 box Hamiltonians, followed by AQ1 §§4–5. Unwritten.

finding N3). Every constant the two dynamics rows need is available, and the proofs are unwritten. In the skeptic's sketch of O5 the finite-volume Hamiltonians differ by the $28N(5N + 1)$ shell faces, each of norm $|\tau|/3$, and since $F \leq N^{-4}$ on the shell the difference is expected to be of order N^{-2} ; that is not proved. A comparison at the level of GNS generators of two possibly different states would also need O1.

Links and ranking. O1, O2 and O3 share one missing input: an estimate that compares two boxes, two families or two subsequences directly, instead of comparing each with a fixed centre. O5 and O6 are linked: a Lieb–Robinson comparison that goes to zero with N , uniformly on compact windows, would also produce the F2 limiting dynamics as the same norm limit. That is a candidate, not a result. The skeptic ranks the rows by the constants they need. O6 and O5 have all their constants and need only writing, O2 reduces to O1, and O3, O1 and O4 need constants that nobody has evaluated. The HTW constants c_1 , c_2 and $c_{\text{HTW}}(1, 1)$ are existential, and the AM2 gate states that AM2 does not evaluate them. The skeptic's pre-comparison adds two candidates for O1: a complex- τ analyticity route native to AM2, whose inequalities as written hold for real $|\tau| < 1/37888$ (self-map) and $|\tau| < 1/19712$ (exclusion), route constants and not admitted extensions; and a quasi-adiabatic route using the AM2 gap $\frac{1}{2}$. Both are candidates only. None of the six rows is proved, and none may be called closed by a route.

7.2.4 The falsifying scenario

Two subsequential limits at the same τ , for instance one of F1 and one of F2, may have densities on R that agree to first order, since both carry $\rho_R^{(1)}$, that differ at second order by up to $2K'_2\tau^2 \approx 2.68350550879 \times 10^{-12}$, and that differ without bound outside R . Nothing admitted excludes this. The scenario is not asserted to occur: the limits may coincide, and the present bounds are silent on the question.

Why the present bounds cannot exclude it. Every constraint that the admitted material places on a subsequential limit ω is a one-state condition, proved box by box and passed to limits through a closed set:

- $\|\rho_R - P_R\|_1 \leq D$ (AV1 tier (ii));
- $\|\rho_R - P_R - \rho_R^{(1)}\|_1 \leq K'_2\tau^2$ (AY1);
- $|\omega(W) - \tau/144| \leq K_2^+\tau^2$ (AW1, as bound in the AY1 gate);
- the reset energy $\omega(h_R) \leq 98|\tau|$ in normalized units (125);
- gauge invariance, positivity and unit trace.

None of them compares two boxes, two families or two subsequences. A set cut out by one-state conditions around a common centre contains distinct points, and the triangle inequality controls only its diameter.

The exact witness. Let $e_{yz0}, \dots, e_{yz3}$ be the vectors e_f of the four yz faces of F_R , with fine phases $r = 0, \dots, 3$ and $q = 0$. On the 11-dimensional span of Ω_R and the ten e_f put

$$\psi_{\pm} = \Omega_R + \frac{\tau}{144} v \pm \nu_w w', \quad w' = e_{yz0} - e_{yz1} + e_{yz2} - e_{yz3}, \quad \nu_w = \frac{K'_2\tau^2(1 - 10^{-6})}{4}, \quad (141)$$

and $\rho_{\pm} = |\psi_{\pm}\rangle\langle\psi_{\pm}|/\|\psi_{\pm}\|^2$. These are density matrices on $\mathcal{H}_R$. The first two terms of $\psi_{\pm}$ form $x = \Omega_R + (\tau/144)v$, and $w' \perp x$. The producer's checker verifies each constraint in exact arithmetic.

- $\|\rho_{\pm} - P_R - \rho_R^{(1)}\|_1 \leq K'_2\tau^2$, by an exact decomposition of $\|\psi_{\pm}\|^2(\rho_{\pm} - P_R - \rho_R^{(1)})$ into six pieces of rank one or two, with directed square roots for their trace norms; the upper bound is $1.34175150939 \times 10^{-12}$.

[illegible]

Step 1: the rank-two structure of $\rho_R^{(1)}$. The ten faces of F_R are anchored at 0: the xz classes with $r = 0, 1, 2$ and $q = 0, 1$, and the yz classes with $r = 0, \dots, 3$ and $q = 0$. The checker derives them from the parsed I1 table and the fine-lattice link tails. The Haar moments $\mathbb{E}[W^n]$ agree by two routes, the Clebsch–Gordan invariant count and Weyl–Wallis integration; in particular $\mathbb{E}[W] = 0$ and $\mathbb{E}[W^2] = \frac{1}{4}$. For $f \neq g$ a link of f that is not in g occurs once in $W_f W_g$, and the centre grading $U_e \mapsto -U_e$ kills the Haar integral. So the e_f are orthonormal and orthogonal to Ω_R , and

Each face piece $(\tau/72)(|W_f\Omega_R\rangle\langle\Omega_R| + \text{h.c.})$ is a rank-one operator plus its adjoint, so it has rank two, with eigenvalues $\pm|\tau|/144$. All ten pieces share Ω_R , so their sum again has rank two, with eigenvalues $\pm\sqrt{10}\tau/144$, each simple, and 0 on the orthogonal complement. Its trace norm is the sum of their absolute values, $\sqrt{10}|\tau|/72$. The checker verifies $\hat{\rho}^3 = (10/144^2)\hat{\rho}$, $\text{Tr } \hat{\rho} = 0$ and $\text{Tr } \hat{\rho}^2 = 20/144^2$ for $\hat{\rho} = \rho_R^{(1)}/\tau$; for a real symmetric operator these put the spectrum in $\{0, \pm c\}$ with exactly one pair $\pm c$. It also verifies $\text{Tr}(\rho_R^{(1)}W) = +\tau/144$. The factor $\sqrt{10}$ is $\|v\|$ for ten exactly orthonormal vectors, not a statistical root- N . The skeptic reaches the same values by explicit Haar integration: the 11×11 Gram matrix of $\{\Omega_R, e_f\}$ is exactly the identity, and the characteristic polynomial per unit τ is $\lambda^9(\lambda^2 - 10/20736)$. The producer corrects a relayed instruction that described “ten rank-one pieces” with eigenvalues $\pm\sqrt{10}\tau/72$ (R1), and the skeptic applies the same correction to its own pre-comparison wording (S1). No value changes.

Step 3: the reverse triangle inequality. $\|\rho_R - P_R\|_1 - \|\rho_R^{(1)}\|_1 \leq \|r_R\|_1 \leq K'_2 \tau^2$, which is (143).

$$\begin{aligned} s_- &= \frac{197642353760523708249930846527}{625000000000000000000000000000}, \\ s_+ &= \frac{3162277660168379331998893544433}{10^{30}}, \quad s_-^2 < 10 < s_+^2. \end{aligned} \quad (146)$$
$$\begin{aligned} \frac{U-L}{s_-|\tau|/72} &\leq \frac{966771578474926086618624139607815612133987277450873561227041}{158230021383939647982640996438347350340375381295486435500687360} \\ &\approx 6.10991245541 \times 10^{-3} \leq \frac{1}{100}, \end{aligned} \quad (147)$$

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The tier is the labelled observation of Section 7.1.7, now certified under the AY2 contract. L and U match the AY1 gate’s recorded interval $[4.3786 \times 10^{-10}, 4.4055 \times 10^{-10}]$ to its printed digits and lie within 10^{-22} of the exact ends verified in the skeptic’s AY1 review; the upper end is about 30.9 times below D . It is not a new mathematical result: AY1’s forward, reverse and skeptic packages already wrote it down. The AV1 and AX1 gates list “no lower bound on $\|\rho_R - P_R\|_1$ ” among their limitations (Section 2.10), and the tier supplies a certified lower end for the zero-selected family.

7.2.6 K'_2 versus K_2^+

Contract item 5 asks for this comparison, recorded correctly, with the AY1 wording corrected and the variants given as previews only.

- *Duality fixes the direction.* Since $\|W\| \leq 1$, $\text{Tr}(P_R W) = 0$ and $\text{Tr}(\rho_R^{(1)} W) = \tau/144$,

$$|\omega(W) - \tau/144| = |\text{Tr}(r_R W)| \leq \|r_R\|_1 = \sup_{A \in B(\mathcal{H}_R), \|A\| \leq 1} |\text{Tr}(r_R A)|. \quad (148)$$

Every valid trace-norm constant is therefore a valid W constant, and the smallest valid trace-norm constant is at least the smallest valid W constant. A trace-norm constant over all of $B(\mathcal{H}_R)$ can never be sharper than the best single-observable constant.

- *The AM2 remainder dominates both.* $\varrho/\tau^2 \approx 3354.108$ is 99.979% of K_2^+ , and $4\varrho/\tau^2 \approx 13416.43$ is 99.992% of K'_2 . With the admitted remainder $\varrho = 352JT$, every trace-norm constant assembled from the AW1 items is at least $2\varrho/\tau^2 = 47162500000/7030557 \approx 6708.2167$, which already exceeds K_2^+ . Face restriction changes only the count-dependent items (straddling 0.340, two-creation 0.105, density 0.649; about 1.09 in all), not the dominant one. So face restriction cannot make K'_2 smaller than K_2^+ , and $K'_2/K_2^+ \approx 3.99950$ (138).
- *Scope of that statement.* It concerns constants built from the admitted items; the skeptic notes that it is not an impossibility theorem about every trace-norm constant. The W -projected R -local analogue, about 3354.4994, lies below K_2^+ , but it is a constant for W , not a density bound (Section 7.1.6).
- *Corrected wording.* The AY1 contract’s “not the whole-box K_2^+ ” is corrected (defect D5): K_2^+ is volume-uniform and specific to W , not a whole-box constant. What distinguishes K'_2 is that it is built from the R -local face pins 82, 72, 33 and 10 and bounds the full trace norm on $B(\mathcal{H}_R)$. It is larger, not smaller.
- *Variants, previews only.* The $\sqrt{2}$ -sectors value ≈ 9487.945 , the 288-majorant value ≈ 10978.18 , both combined ≈ 7763.06 and the admitted- ε value ≈ 13417.81 (Table 25) are valid upper bounds, all above K_2^+ . None is admitted, and none enters the tier. The skeptic’s checklist records that K_2^+ in the tier would give $[4.38870 \times 10^{-10}, 4.39541 \times 10^{-10}]$, which is not a trace-norm bound.

7.2.7 Controls, replays and review

Error ledger. The preregistered terms are the following.

- **first_order_term:** $\sqrt{10}|\tau|/72$, about $4.39205230578 \times 10^{-10}$, enclosed in $[s_-|\tau|/72, s_+|\tau|/72]$; the lower end of the tier uses s_- and the upper end s_+ .
- **second_order_remainder:** $K'_2\tau^2 \approx 1.34175275439 \times 10^{-12}$ for one state and $2K'_2\tau^2 \approx 2.68350550879 \times 10^{-12}$ for two limits, the AY1 headline with its items added linearly. The preregistration’s “single state versus product” is read as one state against $P_R + \rho_R^{(1)}$ (skeptic reading 9).
- **No other terms:** **not_applicable**, with the stated reason. This is a statement loop, the only new certificate is the tier, and all arithmetic is exact except the $\sqrt{10}$ bracket, whose 10^{-30} width is charged in L and U .

Contract controls. The contract lists 21 controls, and its preregistration mirror lists the same 21. Two ids are new.

- `lower_bound_is_static_not_dynamic`: the tier relabelled as dynamical, as an interaction shift or as a Euclidean node, the lower bound read as an identification, and `resolved_interaction_shift` set true are rejected.
- `obligations_table_complete`: the padded-family dynamics row removed, O1 marked proved, the Cauchy route emptied, a seventh row added and rows reordered or renamed are rejected.

Several inherited ids carry AY2 semantics.

- `first_order_mean_charged` rejects a zero first-order density, which $\omega(W) \geq \tau/144 - K_2^+ \tau^2 \approx 6.91089641214 \times 10^{-11}$ contradicts because it exceeds $K_2' \tau^2$, and a zero lower end.
- `common_clock` rejects opposite signs, separated by at least $2L$, different couplings, and mixed units or a normalized clock for one family.
- `tier_mixing_rejected` rejects K_2^+ as the trace-norm remainder, a labelled variant as the headline, the pair constant as the single-state remainder, D as a second-order remainder, and a lower end from K_2^+ with an upper end from K_2' .
- `root_n_misuse` rejects a root sum of squares of two single-state remainders, division by the square root of the face count, $\sqrt{10}$ read as a statistical root- N , and the linear face sum $10|\tau|/72$ as the trace norm.
- `vector_versus_scalar_centering` rejects vector or scalar centering imposed on the uncentered density, and the scalar Wilson mean $\tau/144$ as the centre of the tier.
- `changed_model_relabelled` rejects τ above the cap, a nonzero triple, the selected-strip reference, route B, the literal vertex boxes as a family and the witness relabelled as AQ data.
- `coherent_evidence_tampering` rejects eleven edits with the packet hash rebound, among them a raised lower end, a quartered K_2' , a halved relative width, a removed obligation row, a replaced AY1 gate hash and a doubled witness separation.

Checker. The forward checker runs 42 exact checks. It implements all 21 controls as damaging mutations, with none deferred, and records 85 rejections inside the control checks and 114 in the whole checker. It verifies the contract snapshot’s SHA-256 before any evaluation, reads τ , the target, the reference, the control list and the six obligation names from the contract, records its own SHA-256 before evaluation, and hashes all 23 input snapshots. It uses only the standard library and exact rationals, and it rejects float, Boolean, NaN and zero-denominator inputs. Its outputs under `-B` and `-B -0` are byte-identical. It records the AY1 gate hash but does not pin it (review finding N4). The producer also disclosed a private source-edit audit of sixteen damaging edits, all aborting, which is not evidence. It disclosed as well that its map test at first accepted any backticked control id, so that removing a map row ran silently; the test now requires one row per control and per item.

Independent pre-comparison package. The skeptic’s package holds a contract review, a derivation and a checker with 82 exact checks and 29 controls, the 21 contract ids and eight more. It rejects 88 mutations, 73 of them inside the contract controls. Its routes differ from the producer’s: explicit $SU(2)$ Haar polynomial integration on S^3 for the Gram matrix and for the vanishing R -marginal of the 72 straddling faces, where the producer uses the centre-grading parity rule; a Faddeev–LeVerrier characteristic polynomial, where the producer uses the cube identity; a 10^{-40} bracket; its own witness in the χ_1 direction; and its own obligations table with route constants. A fresh replay reproduces its frozen output byte for byte under normal and optimized Python, and seventeen must-abort source edits, run in private scratch, abort at the intended check. The contract, the selection note, the AY1 gate and the skeptic’s own AY1 review state every headline value, so the producer could have copied the headline. The replay is independent in route and code only, and

the skeptic records that its triage, its AY1 review and its AY1 pre-comparison package seeded the statement AY2 makes.

Post-comparison review. The review adds 27 exact checks, 109 independent checks with the 82, and 46 source-mutation runs. One unmutated copy reproduces both output files byte for byte. All 42 must-abort edits abort with the intended message: 21 weakenings of the validator of one contract control each, each ending in “damaging mutation accepted”; nine value edits, among them half the remainder at the lower end, the upper $\sqrt{10}$ bracket at the lower end, a halved AM2 item and a doubled witness amplitude; five input edits, among them a contract target of $1/1000$ with its hash rebound, which aborts because the checker requires the preregistered $1/100$, and an edited digit of K'_2 in the AY1 gate snapshot; and seven report edits, among them O1 marked proved, O6 removed, the negation removed from the sentence, a forbidden phrase appended and the exact lower end removed. Three edits run silently and are accounted for (N4, N5). An edit to an unparsed field of the AY1 gate snapshot is caught by the snapshot comparison and by the review’s validator, which pins the gate hash. Moving W to another face of F_R changes nothing, because $\text{Tr}(\rho_R^{(1)} W_g) = +\tau/144$ for every $g \in F_R$. Removing the witness’s reset-energy test changes nothing, because that constraint holds with a margin of about 10^{12} ; the review re-checked it. The review’s own validator rejects 14 damaged packets. The eleven report previews it checked are truncations of the exact values.

Replays and isolation. The existing replay tools do not apply unchanged to a statement loop: one always replays two directions and stops on the absent reverse freeze, and the other accepts a single producer only with direction **single+skeptic**. Both are bound by earlier reviews and gates, so neither was edited (N9). The review runs the same steps with the unchanged freeze verification. The closure of 27 listed files plus **freeze.json** is verified file by file. The inputs are exactly **AGENTS.md**, the contract and the 21 shared premises, each byte-identical to its repository source, and the only skeptic files among them are the declared AY1 and AM2 reviews. No **reverse/ay2** package exists. Replays under normal and optimized Python into fresh external directories reproduce the frozen outputs byte for byte, the freezing tool reports the closure verified, and the checker SHA-256 recorded before evaluation equals the frozen file. On isolation (N1), the skeptic’s frozen results differ from its 03:34:06Z results only in the checker hash and the incidental-exposure line; its later edits, at 03:39:05Z and 03:39:43Z, only extended the disclosure after **git status** showed the producer’s file names. Each side saw only the other’s file names and opened none. The producer disclosed reads outside its inputs for protocol and style only: the tools documentation, the freezing tool, parts of the AY1 forward checker, the first 60 lines of the AX2 forward report, and file names and line counts; none carries premise weight. Of its premises it read some in full, some in part and three method references not at all, and all 23 are hash-verified. The private scratch folders were checked by name and time only, and after the AY2 freeze the shared scratchpad root held only manuscript files. Isolation is verified for repository inputs only; all agents are correlated model agents.

7.2.8 Contract wording and non-blocking findings

The gate records the contract’s wording defects, in the producer’s numbering D1–D6, as non-blocking; the skeptic verified each against the frozen text.

- *D1.* **selected_after** was left as the placeholder **<preceding gate>** at freeze time; the preceding gate is AY1.
- *D2.* The preregistration’s claim exclusions use a definite-article phrase for a single AQ-type state, which item 1 forbids; both packets quote it only verbatim and negated.
- *D3.* **gate_fields_required** lacks **rate_in_N_claimed**, which item 6 requires; both rate fields are exported false.
- *D4.* Item 3’s “both lie within 2D of the product” should read “each lies within D ”; its “at least $8.7\text{e-}10$ ” is a preview of $\sqrt{10}|\tau|/36 - 2K'_2\tau^2 \approx 8.757 \times 10^{-10}$ (Section 7.2.4).

- *D5*. The K_2^+ wording is corrected (Section 7.2.6).
- *D6*. The selection reason’s “certified” two-sided distance anticipated AY2: the AY1 gate recorded it as a labelled observation only.

The relayed-instruction correction R1 and the skeptic’s self-correction S1 concern the rank of the face pieces (Section 7.2.5). The gate also records the review’s findings as non-blocking. The mandatory sentence in the producer report is the Jung loop-2 form, and the gate carries the AY1-contract template with its constants (N2). The route text of O5 repeats “AM2 constants” while its constants column names AQ1 (N3). The checker records but does not pin the AY1 gate hash (N4). Two edits are silent by design (N5). The skeptic’s pre-comparison list of one-state constraints omitted the reset energy, which its fixture satisfies retroactively (N7). The two relative-width definitions differ but agree (N8), and the replay tools do not accept a statement loop unchanged (N9). The commit order is recorded in N1.

7.2.9 Limitations recorded by the gate

The AY2 gate records the following limitations verbatim.

- Only the zero-selected patterned family (selected triple (0,0,0), Haar reference), the cover $R=\{0,e_z\}$, fixed spacing and $|\tau|\leq 10^{-8}$, with the two named families on centered cubes Λ_N , $N\geq 2$; nothing transfers to literal vertex boxes, periodic or orthant boxes, nonzero selected triples, the uniform route-B model, weak coupling or the continuum; every constant is uniform in N at fixed spacing, never in a
- Uniform local closeness on the fixed region R at the same coupling plus a static two-sided distance tier; not uniqueness, whole-sequence convergence, translation invariance, a rate in N , boundary independence of the dynamics or dynamics of the padded family (all six obligations unproved); two limits may differ by up to $2K_2'\tau^2$ on R and without bound elsewhere
- The tier `first_order_distance_from_product` is static: not an interaction shift, Euclidean node or dynamical statement; its positive lower end excludes the Haar product on R but does not identify any limit; it is the AY1 labelled observation certified under the AY2 contract, not a new mathematical result
- The falsifying witness is a pair of density matrices on H_R that satisfies every admitted one-state constraint; it shows what the bounds leave open and is not a constructed limit (`transfers_to_aq` false)
- The candidate routes are named, not executed: HTW/Yarotsky constants $c_1, c_2, c_{HTW(1,1)}$ are unevaluated (AM2 gate); a Dobrushin-type condition is not formulated for this model; the Lieb-Robinson constants for the dynamics rows are AQ1’s Nachtergaele-Sims constants ($F(r)=(1+r)^{-4}$, $C\leq 224$, $||\Phi||_F\leq 2268|\tau|$) and the named $||\Phi||_F\leq 1323|\tau|$, not AM2 constants; the contract’s ‘with the AM2 constants’ is wording only
- Inherited without re-proof: AM2 majorant, fixed point, exclusion and cutoff passage; AV1 split and cutoff-vector removal (F22); AY1 F13/F14 and the passage to every limit through closed trace-norm balls; AQ1 compactness and extraction; the I1 dictionary; SU(2) Haar orthogonality and the spectral theorem are standard
- Constants are upper bounds except the certified lower end; K_2' is 99.99% the generic AM2 majorant 4ρ
- The mandatory sentence in the producer report is the Jung loop-2 form (AY1 forward section 6); the AY1-contract template (verbatim in the AY1 gate) is carried here in the supported statement with the constants filled; the AY2 preregistration has no template field
- Single producer, correlated model agents: the contract, selection note, AY1 gate and the skeptic’s AY1 review state the headline values; independence is limited to route and code; isolation is shown by disclosure, file-name-only exposure and content, not by commit order (the skeptic pre-comparison commit f640111 follows the producer commit ce6bca8; the skeptic’s exact values were final at 03:34:06Z, before the producer’s check.py and report were last written)
- The `-tau` evaluation is a replay of the same $|\tau|$ formula, not a second confirmation
- Scientific priority unverified

Full derivation and reviewed implementation: [AY2 forward](#), [contract review](#), [independent pre-](#)

comparison derivation, independent model-agent review and the frozen contract. The advisor’s findings record that AY2 closes the fourth Round31 roadmap goal within its stated scope: a quantitative boundary comparison with explicit constants and an explicit list of what remains unproved. It identifies no state. Sub-round 4 is complete, and the gate names panel update 4 as the next step.

8 Sub-round 5: the continuum trajectory and a finite graph

The fifth Round31 roadmap goal, “Establish a continuum matching trajectory”, asks for a matched lattice family connected to a nontrivial four-dimensional continuum theory with the required axioms and a positive physical gap. Its recorded missing premise reads “A regulator/coupling trajectory, physical scale control, continuum limits and full reconstruction hypotheses remain unproved” (Round31 roadmap). Its proposed first test is to “State the regulator and coupling dictionary, then identify one necessary uniform estimate that the present fixed-lattice family actually supplies or fails to supply”. Sub-round 5 carries out that first test and nothing more. AZ1 is a statement loop: it names a trajectory as a hypothesis, applies the AL1 dictionary along it, cites the one necessary estimate that the uniform Kogut–Susskind model of Section 6 supplies, and shows exactly where that estimate stops. AZ2 computes exact first- and second-order Wilson coefficients on the Round11 two-plaquette graph, a different finite model, with sign-certified enclosures that transfer nothing to $\mathbb{Z}^3$ (Section 8.2). Neither investigation constructs a continuum theory, and no number in this section is a fraction of the continuum problem.

8.1 AZ1: what the fixed-spacing family supplies along a trajectory

Contribution HNM-C-AZ1. The AZ1 contract, recorded under the title “Hruday continuum-trajectory statement: the one uniform estimate the fixed-spacing family supplies and the one it fails to supply”, was frozen after the AY2 gate and panel update 4, and before production. It opens sub-round 5 and is investigation 9 of 10. The model is the uniform Kogut–Susskind SU(2) family of AX1 (Section 6): every elementary face carries $\nu = \alpha\tau/24$, handled by route B with the Haar reference, centered whole-star boxes and the cover $R = \{0, e_z\}$. It is followed along a named trajectory (a_n, g_n) , with $E_\star$ fixed and no plateau fit. The preregistered model identifier is `AQ_uniform_routeB` along `(a_n, g_n)`, the coupling is the cap $\tau = 1/100000000$ with both signs evaluated, and the state provenance is AQ1’s centered whole-star subsequences through a finite-volume bound, read as volume-uniform at fixed a (defect D9). The observable is the statement itself, with no centering, no reference value and no Euclidean node. The target, “dictionary identities verified exactly and the toy-trajectory crossover indices computed”, at least 1, is declared a feasibility and format check on admitted quantities, not a blind discovery threshold; the statement is the deliverable. The skeptic reads it as a Boolean that is met by construction. It catches arithmetic slips but cannot test the failure statement, the scope of the word “uniform”, the requirement table or the verdict, which rest on the controls and on the review. The error ledger has one entry, `not_applicable`, with its reason (Section 8.1.8).

AZ1 is a statement loop (`statement+skeptic`), as planned. One forward producer writes the statement, the requirement table and an exact checker. The second admission input is the skeptic’s independent derivation and replays (`single_direction_independent_replay`). The contract’s direction note records that the historical, Jung and modern lenses all kept AZ1 unchanged after sub-round 4; the skeptic notes that the panel kept the goal but adopted new contract requirements, a clerical point (reading 16). The producer received `AGENTS.md`, the contract and its 27 shared premises, 29 files in all: the AL1 report and gate; the AX1 gate, both AX1 reports and the skeptic’s AX1 review; the AM2 gate, both AM2 reports and the skeptic’s AM2 review; the AQ1 and AQ2 reports; the I1 and AT4 reports; the method snapshots; the selection note; and the modern and historical lens notes of sub-round 4. No reverse package exists, and `reverse_premise_isolation`

is false.

Both packages were developed at the same time after the contract froze at 04:29:40Z. This time the skeptic committed first, the reverse of AY2 (Section 7.2). Its pre-comparison checker was final and replayed at 04:54:11Z, and the package was committed as 649db92 at 04:54:45Z. The producer’s report was last written at 04:58:01Z and its checker at 04:58:33Z; the producer froze at 04:58:58Z and committed 8359876 at 04:59:51Z. Isolation therefore rests on disclosure, name-only exposure and content, and this time also on commit order (Section 8.1.8).

Notation: g is the bare coupling and a the lattice spacing; α , λ , $r = \lambda/\alpha$, $\delta = \alpha/8$ and τ are the scales of Section 6.2. Δ is the physical gap of the finite-volume and GNS Hamiltonians, $E_\star > 0$ the fixed comparison energy, and $n^\star$ a crossover index (Section 8.1.5). The dictionary is written with $\hbar = c = 1$; restoring units, $\alpha = \hbar c g^2/(2a)$ and the gap bound is $\hbar c g^2/(32a)$. The word “uniform” is used in two senses that are kept apart, as Section 8.1.7 explains.

8.1.1 Reviewed scope

The [AZ1 gate](#) records the verdict `accepted_within_scope` with no sub-label: its `sub_label` field is null. Of the five preregistered sub-labels, three name certificates (`reference_unresolved`, `sign_certified_finite_graph` and `sign_certified_below_cap`) and two name state or observable statements (`uniform_local_closeness_not_uniqueness` and `static_not_dynamic`). AZ1 is a dictionary and trajectory statement, so none fits, and the producer proposed none. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this subsection. It is the contract’s mandatory sentence template with the reviewed constants inserted (Section 8.1.7). In it, `a` is the lattice spacing and `a_n` the spacing along the trajectory; `Delta` and `Delta_n` are the physical gap Δ ; `E_star` is $E_\star$; `R` is the cover; `n*` is a crossover index; and `s` in `a->a/s` is a rescaling factor, not the Euclidean clock.

```
Along the named trajectory (a_n,g_n) -- a hypothesis of asymptotic-freedom form with
g_n->0 as a_n->0, not derived -- the fixed-spacing family (uniform Kogut-Susskind SU(2)
at fixed spacing, strong bare coupling, as admitted by AX1: every face alpha*tau/24,
route B, Haar reference, centered whole-star boxes, cover R={0,e_z}; AL1 dictionary
alpha=g^2/(2a), lambda=2/(g^2 a), tau=24 lambda/alpha=96/g^4, so alpha*lambda*a^2=1;
E_star and hbar fixed) supplies exactly one uniform estimate, the volume-uniform gap
alpha/16=g^2/(32a) for g^4>=9.6x10^9 (0<tau<=10^-8) at fixed a (every finite
complete-factor route-B volume and every AQ subsequential state, AX1 gate items (2)-(3),
cited; in lattice units a*Delta>=g^2/32>=1250 sqrt 6, directed lower end
19595917942265424785578272597647131/64000000000000000000000000000000, about 3061.86),
and fails to supply any estimate uniform along a_n->0 on a trajectory with g_n->0 (the
admitted regime is left at a finite index and the AL1 bridge g^4>=32, i.e. tau<=3,
fails from a finite index on; on the toy trajectory g_n=g_0/n the first indices below
9.6x10^9 and below 32 are n*=2 and n*=132 for g_0^4=9.6x10^9 and n*=4 and n*=421 for
g_0=1000; common rescaling a->a/s at fixed g moves no admission ratio; a finite
physical gap along a_n->0 needs a_n*Delta_n->0, impossible inside the admitted regime;
this is failure of a sufficient certificate, not a statement that the gap closes); this
is not a statement that the continuum limit exists, that any estimate is uniform in a,
that the coupling is weak, or that any fraction of the continuum problem is thereby
resolved.
```

The gate’s decision admits four items. (1) The three contract identities and eight derived ones hold exactly as rational-function identities and on a 35-point grid, and the cap is $g^4 = 9600000000$ at $\tau = 10^{-8}$. (2) The one necessary estimate that the family supplies is the volume-uniform gap at fixed a , cited from AX1 and not re-derived. Every trajectory with $g_n \rightarrow 0$ leaves the admitted regime after finitely many steps, the AL1 bridge fails from a finite index on, and the lattice-units floor shows that the admitted regime cannot host a finite-mass limit. (3) The toy crossovers are exact, and the Faizal–Shabir construction is disclosed as an unaudited lead only. (4) The bridge obstruction and the failure of common rescaling are retained, and the additional requirements of a continuum construction are listed with the missing premise for each. The gate binds the producer’s

exact values, which the skeptic’s review verified exactly; the review records its own 10^{-40} bracket of $\sqrt{6}$ as the tighter enclosure of the floor (Section 8.1.3). The gate records the non-blocking findings of Section 8.1.9, and it admits nothing about the continuum, weak coupling or an estimate uniform in a . The rest of this subsection derives the statement, and the limitations recorded by the gate close it (Section 8.1.10).

8.1.2 Trajectory, dictionary and admitted regime

The named trajectory. Contract item 1 asks for a trajectory (a_n, g_n) with $g_n \rightarrow 0$ as $a_n \rightarrow 0$, named as a hypothesis of asymptotic-freedom form and not derived. The producer states it as

$$H_{AF} : (a_n, g_n), n \geq 1, \quad a_n \rightarrow 0, \quad g_n \rightarrow 0, \quad g(a)^2 \sim \frac{1}{2b_0 \log(1/(a\Lambda))} \quad (a \rightarrow 0), \quad (149)$$

for some $b_0 > 0$ and scale Λ . No beta function is computed, no numerical b_0 is used or source-audited, and no sequence is claimed to be the physical renormalization trajectory. Every conclusion below uses only $g_n \rightarrow 0$. The checker exports the trajectory with the fields `derived` and `claimed_physical_trajectory` false. The control `trajectory_named` rejects the trajectory marked derived, an unnamed a_n , a constant g_n and a claimed physical renormalization trajectory.

The dictionary. AL1 fixes the bulk dictionary of the uniform model in the Bauer et al. convention, with the raw magnetic term $\lambda \sum_p (1 - W_p)$ and $W_p = \frac{1}{2} \text{Tr } U_p$ (HNM-AL1.1–2 of the [AL1 forward report](#); (80)):

$$\alpha = \frac{g^2}{2a}, \quad \lambda = \frac{2}{g^2 a}, \quad r = \frac{\lambda}{\alpha} = \frac{4}{g^4}, \quad \tau = \frac{24\lambda}{\alpha} = \frac{96}{g^4}, \quad \delta = \frac{\alpha}{8}. \quad (150)$$

Contract item 5 names three identities: $\alpha/16 = g^2/(32a)$, $\tau = 96/g^4$ and the cross-identity $\alpha\lambda a^2 = 1$. The producer adds eight more from the AL1, AX1 and I1 premises and verifies all eleven as literal identities (Table 29), by two routes. The first writes each side as a quotient of integer-coefficient polynomials in (g, a, c) and compares by cross-multiplication. The second uses direct exact rational arithmetic on a grid of 35 points: seven values of g^2 ($1/7, 1, 3/2, 97979, 100000, 1000000, 1048576/3$) times five of a ($1/1000, 1/10, 1, 7/3, 1000$), with the rescaling factors $1/3$ and 5 . The cross-identity says that the product of the electric and magnetic scales is $1/a^2$ whatever the coupling. It catches the commonest slip: $\alpha = g^2/a$ makes $\alpha\lambda a^2 = 2$ identically.

Table 29: The eleven dictionary identities, verified exactly by the producer and re-proved to the zero polynomial by the skeptic. Rows 1–3 are named by the contract; rows 4–11 are added from the AL1, AX1 and I1 premises.

	Identity	Reading
1	$\alpha/16 = g^2/(32a)$	the supplied gap (AX1 gate; AX1 reverse report)
2	$\tau = 96/g^4$	the AL1 coupling (HNM-AL1.2; AX1 gate)
3	$\alpha\lambda a^2 = 1$	the cross-identity (contract item 5)
4	$r = 4/g^4$	the ratio λ/α (HNM-AL1.1)
5	$\alpha\tau/24 = \lambda$	every face coefficient ν equals λ (HNM-AL1.2)
6	$(\alpha/8)(\tau/3) = \lambda$	the normalized face coefficient $\tau/3$ in units of δ
7	$7\tau = 672/g^4$	the AL1 local norm $7 \tau $
8	$\tau/144 = 2/(3g^4)$	the first-order Wilson mean (AX1 gate item 7)
9	$(\alpha/8)/2 = g^2/(32a)$	the normalized gap $\frac{1}{2}$ in units of δ
10	$a\alpha/16 = g^2/32$	the supplied gap in lattice units
11	$\tau g^4 = 96$	the cap equivalence

The skeptic’s pre-comparison proves nine identities of rational functions in (g^2, a) by cross-multiplication to the zero polynomial. Eight of them are among the producer’s eleven, and the

ninth is the inverse map $g^4 = 4\alpha/\lambda$, which with $\alpha\lambda a^2 = 1$ makes $(a, g^2) \mapsto (\alpha, \lambda)$ a bijection of the open positive quadrant; τ depends on the ratio λ/α alone. The post-comparison review maps each of the producer’s eleven to an explicit pair of rational functions and checks it to the zero polynomial, so the union of twelve is verified (review finding N7).

The admitted regime. For real g , $\tau = 96/g^4 > 0$, and for every $T > 0$, $\tau \leq T$ if and only if $g^4 \geq 96/T$. At the preregistered cap $T = 10^{-8}$ of AM2 and AX1,

$$|\tau| \leq 10^{-8} \iff g^4 \geq \frac{96}{10^{-8}} = 9600000000 = 9.6 \times 10^9, \quad (151)$$

that is, $g^2 \geq 40000\sqrt{6} \approx 9.79795897113 \times 10^4$, at fixed a . This is strong bare coupling, and the model label is “uniform Kogut–Susskind SU(2) at fixed spacing, strong bare coupling”. The value 9600000000 is read from the AX1 gate, the skeptic’s AX1 table and four places in the contract, and it is recomputed as $96/\tau$ from the preregistered $\tau = 1/100000000$. Since g^2 at the cap is irrational, every numerical value at the cap carries a directed bracket of $\sqrt{9.6 \times 10^9}$. A negative τ has no real g . In the uniform model it is the U_E mirror of $+|\tau|$ (Section 6.9), not a point of any trajectory (defect D8).

8.1.3 The one estimate supplied

The estimate, cited. Contract item 2 asks for exactly one necessary estimate, cited and not re-derived. It is the AX1 gap of Proposition 1. For both signs of $|\tau| \leq 10^{-8}$, AX1 gate item (2) admits that every finite complete-factor route-B volume has a unique gauge-invariant ground and a full-space gap at least $\frac{1}{2}$ normalized, $\alpha/16$ physical, with a nonzero physical excited sector. Item (3) re-applies the gap $\alpha/16$ to every AQ subsequential state. In the dictionary (150),

$$\Delta \geq \frac{\alpha}{16} = \frac{\alpha}{8} \cdot \frac{1}{2} = \frac{g^2}{32a} \quad \text{for } g^4 \geq 9.6 \times 10^9, \text{ at fixed } a. \quad (152)$$

This is one lower bound, the same for every finite complete-factor volume, so it is volume-uniform, at one fixed spacing, and it passes to the GNS representation of every AQ subsequential state. Its proof is the AM2/AX1 creation-operator contraction with the re-frozen $J'_0 = 29/10^8$ and the per-site sum $J' = 29|\tau|$, incoming stars included: $J'_0 G(R_a) < 1073/175000000 < 1/64$ and $2J'_0 G'(R_a) < 319/1562500 < 1$ (87). The producer’s checker re-reads these constants from the hash-bound AX1 gate and re-proves nothing. Without the proof citation the contract’s verdict would have been limited.

Why this estimate. A continuum mass gap needs, at the least, a lower bound on the gap in physical units that survives along the trajectory, and (152) is the only estimate of that kind the family supplies. AX1 also admits other volume-uniform constants at fixed a : the state bound D'_{ii} and the reset budget $102|\tau|$, and, the skeptic adds, $\varepsilon_R \leq 17|\tau|$ and the Nachtergaele–Sims constants. The template’s “exactly one uniform estimate” is therefore read as exactly one necessary estimate, not as the only volume-uniform bound (defect D2, skeptic reading 6). The control `one_uniform_estimate_identified` rejects no estimate, a second estimate, the patterned-model gate cited instead of AX1, the estimate read along a_n , and the estimate re-derived instead of cited.

Units and clock. The gap is read in units of the fixed $E_\star$, with $\hbar$ fixed. In units of α it is $\frac{1}{16}$ at every (a, g) , and in units of δ it is $\frac{1}{2}$; these numbers are constant only because the unit α moves with (a, g) . The skeptic’s non-unit fixture, $\alpha = 5$ and $\hbar = 7$, gives the physical frequency $\alpha/(16\hbar) = 5/112$ against the normalized misread $1/14$. The control `wrong_delta_alpha_hbar_clock` rejects the normalized gap $\frac{1}{2}$ read in units of α , the gap-one route $(\alpha/8)$, the Haar on-site gap six read as the interacting gap, a normalized clock and the mixed $g^2/(16a)$. The control `e_star_fixed_no_plateau` rejects $E_\star = 0$, an $E_\star$ that varies as $1/a_n$, a plateau fit and post-hoc node selection.

$$(a\Delta)^2 \geq \frac{g^4}{1024} \geq \frac{9600000000}{1024} = 9375000 = (1250\sqrt{6})^2, \quad a\Delta \geq \frac{g^2}{32} \geq 1250\sqrt{6} \approx 3061.86. \quad (153)$$
$$a\Delta \geq \frac{19595917942265424785578272597647131}{6400000000000000000000000000} \approx 3.06186217847 \times 10^3, \quad (154)$$
$$\begin{aligned} 1250\sqrt{6} &\geq \frac{24494897427831780981972840747058913919659}{8000000000000000000000000000000}, \\ 1250\sqrt{6} &\leq \frac{1224744871391589049098642037352945695983}{4000000000000000000000000000000}. \end{aligned} \quad (155)$$

The admitted regime cannot host a finite-mass limit. The certified gap is more than three thousand times the inverse spacing, so it is a cutoff-scale quantity. A trajectory with $a_n \rightarrow 0$ that stayed inside the admitted regime would have a certified $\Delta_n \geq 3061/a_n$, which diverges in units of the fixed $E_\star$. At $a_n = 1/10, 1/100, 1/1000$ and $1/10000$ fm the certified lower ends are about $3.0618621 \times 10^4, 3.0618621 \times 10^5, 3.0618621 \times 10^6$ and 3.0618621×10^7 fm $^{-1}$. A finite physical gap along $a_n \rightarrow 0$ requires $a_n \Delta_n \rightarrow 0$, which is impossible inside the admitted regime, where $a_n \Delta_n > 3061$ for every n . Equivalently, $(\alpha a)^2 = g^4/4 \geq 2.4 \times 10^9$, so the electric scale is at least $20000\sqrt{6}/a$. This is the physical-scale face of the same failure point (row R3 of Table 32). The skeptic adds a conditional remark, which is not a premise and was not required. If $\Delta_n \rightarrow m < \infty$ and $g_n^2/a_n \rightarrow \infty$, as on any trajectory of asymptotic-freedom form, then $\Delta_n \geq g_n^2/(32a_n)$ must fail for all large n . The form of the estimate, and not only its proof, is then incompatible with the hypothesized limit.

8.1.4 The failure point

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The scope $g_n \rightarrow 0$ is essential. At fixed g in the admitted regime the estimate survives any change of spacing. With $g^2 = 10^5$ and $a_n = 1/n$, the bound $g^2/(32a_n) = 3125n$ holds at every n and increases. That sequence is common rescaling: $a_n\Delta \geq 3125$ stays fixed, and in units of $E_\star$ the bound diverges like $1/a_n$, so it gives no finite-mass limit either. A negated statement about estimates along $a_n \rightarrow 0$ is therefore true only on a trajectory with $g_n \rightarrow 0$. Every such statement in this subsection is scoped that way, as the gate statement is (“on a trajectory with $g_n \rightarrow 0$ ”; skeptic reading 9, review finding N3).

The AL1 frozen path. On AL1’s explicit test path $a_n = a_0/n$, $g_n^2 = 1/n$, the bridge fails at every $n \geq 1$: $g_n^4 = 1/n^2 \leq 1 < 32$ and $\tau_n = 96n^2$, while $\alpha_n a_0 = 1/2$ stays fixed. AL1 records it as “an explicit algebraic test path, not a claim that this is the physically correct renormalization trajectory”. The checker re-verifies it for $n = 1, \dots, 1000$. The contract’s “the AL1 bridge condition $g^4 \geq 32$ also fails” therefore means “fails from a finite index on” on a trajectory with $g_n \rightarrow 0$, and “fails at every n ” only on the AL1 path (defect D5).

The cap is not a sharp threshold. 10^{-8} is the preregistered cap at which the AM2 and AX1 certificates are admitted. The AM2 reverse report states that it does not establish a sharp threshold, and the contract’s “the AM2 contraction needs $\tau \leq 10^{-8}$ ” is read as “is admitted only for” (defect D3, skeptic reading 7). The contraction inequalities alone would allow the route radii $|\tau| \leq 7/274688$ for the self-map, about 2548.3 times the cap ($g^4 \geq 26370048/7 \approx 3.767 \times 10^6$), and $|\tau| < 1/20416$ for the exclusion ($g^4 > 1959936$); see Section 6.4. No gate froze or reviewed them. They are not admitted regimes, because the rest of the AX1 chain was evaluated only at the cap. Nor do they change the conclusion. The contraction’s smallness parameter is $J' = 29|\tau| = 2784/g^4$, so any finite contraction radius bounds g^4 from below. The declared toy trajectory leaves the two route radii at $n^* = 8$ and $n^* = 9$ (Table 31).

Failure of a certificate, not of the gap. Nothing here says that the gap closes where the certificate stops. The AL1 gate records its obstruction as “failure of a specified sufficient certificate, not a no-gap or continuum-impossibility theorem”, and the AM2 report “does not interpret rejection beyond the chosen cap as actual gap failure”. The producer states “This is failure of a sufficient certificate, not absence of a gap”, and along the toy trajectory “this is not a statement that the gap vanishes” (Section 8.1.5). The control `insufficient_verdict_retained` rejects the failure point relabelled resolved.

8.1.5 The toy trajectory

Contract item 3 asks for the failure on at least one explicit, clearly labelled toy trajectory. The producer declares

$$\begin{aligned} \text{T1: } g_n &= \frac{g_0}{n}, \quad a_n = \frac{a_0}{n}, \quad g_0^4 = 9.6 \times 10^9; \\ \text{T2: the same law with } g_0 &= 1000, \text{ so that } g_n^4 = \frac{10^{12}}{n^4}, \end{aligned} \tag{156}$$

with $n \geq 1$ and $E_\star$ fixed. T1 is the contract’s declared example and the producer’s primary one. T2 is the panel rehearsal of the modern lens’s update 4, labelled a second example. The contract names g_n and $E_\star$ but not a_n ; the producer declares $a_n = a_0/n$ (defect D7). Because τ does not depend on a , the crossover indices do not depend on that choice. The toy law is algebraic, a power law in n that decays faster than the logarithmic form (149). It is not of asymptotic-freedom form and is not a physical trajectory.

A crossover index n^* is the first $n \geq 1$ with g_n^4 strictly below the threshold. The cap is inclusive: $g_n^4 = 9.6 \times 10^9$, that is $|\tau| = 10^{-8}$, lies inside the admitted regime. So $n = 1$ of T1, which sits exactly on the cap, is not a crossover. Table 30 lists the four indices. The producer finds each by

two routes. The first is the exact integer fourth root, $n^* = \text{iroot}_4(F) + 1$ with $F = \lfloor g_0^4/T \rfloor$ and $m^4 \leq F < (m+1)^4$ checked; the second is an exact rational scan $n = 1, 2, \dots$. For T1 and the bridge, $F = 300000000$ and $131^4 = 294499921 \leq F < 132^4 = 303595776$. For T2 and the bridge, $420^4 = 31116960000$ and $421^4 = 31414372081$. The skeptic computes each index by an integer root with a monotone correction and a brute-force cross-check, and finds the same four.

Table 30: Exact crossover indices on the toy trajectory $g_n = g_0/n$ for the cap $g^4 \geq 9.6 \times 10^9$ and the bridge $g^4 \geq 32$, with g_n^4 at the last index inside the threshold and at the crossover, and $\tau = 96/g^4$ at the crossover. Decimals are truncated previews of the exact rationals.

Toy	Threshold	n^*	g^4 at $n^* - 1$ (inside)	g^4 at n^* (outside)	τ at n^*
T1	cap	2	9600000000 ($n = 1$, on the cap)	600000000	$1/6250000$ (1.6×10^{-7})
T1	bridge	132	$9600000000/294499921$ (≈ 32.5976318343)	$12500000/395307$ (≈ 31.6209933039)	$1185921/390625$ (≈ 3.03595776)
T2	cap	4	$10^{12}/81$ ($\approx 1.23456790123 \times 10^{10}$)	3906250000	$6/244140625$ ($\approx 2.4576 \times 10^{-8}$)
T2	bridge	421	$6250000/194481$ (≈ 32.1368154215)	$10^{12}/31414372081$ (≈ 31.8325636884)	$94243116243/31250000000$ (≈ 3.01577971977)

So the declared toy keeps the admitted regime only at $n = 1$, and it keeps the much weaker bridge condition only through $n = 131$. The rehearsal keeps them through $n = 3$ and $n = 420$. The control `toy_trajectory_crossover_exact` rejects the boundary $n = 1$ of T1 counted as a crossover, the T1 bridge indices 131 and 133, the T2 indices 3 and 420, the threshold $32/3$ and the cap decade 9.6×10^8 . The control `root_n_misuse` rejects a square-root crossover index (11 for T2) and g_n^4 read as g_0^4/n^2 . The skeptic’s controls also reject $g_n = g_0/n$ read as $g_n^2 = g_0^2/n$, which gives the bridge indices 17321 and 176777, and g_n^2 compared with 32, which gives 177 at $g_0 = 1000$.

The panel rehearsal and its arithmetic slip. The producer reproduces every index and power that the modern lens’s rehearsal states: $n^* = 4$, $n^* = 421$, $g_4^4 = 3906250000$, $420^4 = 31116960000$ and $421^4 = 31414372081$. It finds one arithmetic slip in the lens text, which writes $g_3^4 = 10^{12}/81 = 12345679012 + 34/81$. Since $81 \times 12345679012 = 99999999972$, the remainder is 28, and $10^{12}/81 = 12345679012 + 28/81$ exactly (premise note P1). No index is affected, because g_3^4 is above the cap either way. The skeptic’s pre-comparison reproduced the indices but did not check the mixed number, and its review records P1 as new and valid (review finding N10).

What the family supplies along the toy. Along T2 with $a_0 = 1/10$ fm and $E_* = \hbar c/(1 \text{ fm})$, the supplied estimate gives $\Delta_n/E_* \geq g_0^2/(32a_0nE_*) = 312500/n$ for $n = 1, 2, 3$, that is 312500, 156250 and 312500/3. From $n = 4$ on the family supplies no certificate; this is not a statement that the gap vanishes. Along T1 with the same a_0 and E_* only $n = 1$ is covered, with Δ_1/E_* at least about $3.06186217847 \times 10^4$.

Labelled extras. Table 31 lists further crossovers, none of them an admitted regime. A slower toy, $g_n^4 = g_0^4/n$ with $g_0 = 1000$, crosses the cap at $n^* = 105$ and the bridge at $n^* = 31250000001$. Every one of these indices is finite, as the lemma of Section 8.1.4 requires.

The plain statement and the one recorded lead. No rigorous construction of a genuine asymptotic-freedom trajectory satisfying either bound uniformly was found in this or any prior sub-round’s source search. The statement rests on the searches recorded in the snapshotted lens notes. The modern memo lists “any estimate uniform along $a \rightarrow 0$ with $g(a) \rightarrow 0$ ” under what the family fails to supply. The historical and modern updates of sub-round 4 found no rigorous result of that kind to import, and the producer ran no search of its own. The skeptic notes that the

Table 31: Labelled extra crossovers on the toys T1 and T2 of (156): the first index at which each condition fails. None is an admitted regime.

Condition	T1	T2	Computed by
self-map radius $ \tau \leq 7/274688$, i.e. $g^4 \geq 26370048/7$	8	23	producer and skeptic
exclusion radius $ \tau < 1/20416$, i.e. $g^4 > 1959936$	9	27	producer; verified in review
AL1 ends $g^4 \geq 8$	187	595	skeptic
$ \tau/144 = 2/(3g^4) \leq 1$, i.e. $g^4 \geq 2/3$	347	1107	skeptic

sentence is also true by the shape of the regime, since no sequence $g_n \rightarrow 0$ satisfies $g_n^4 \geq c$ for all n and any $c > 0$. That content is carried by the lemma of Section 8.1.4. The substantive statement is that no premise supplies an estimate valid at weak bare coupling, that is, a replacement for the contraction (skeptic reading 10).

The same searches found one claimed construction, and it is disclosed next to that sentence. Faizal and Shabir [23] (arXiv:2606.19362; Fortschr. Phys. 74 (2026) e70097) claim a reflection-positive $SU(N)$ construction in which the strong-coupling and asymptotically-free continuum limits coincide. The modern lens read it at abstract depth only, and neither the producer nor the skeptic read it. It is unaudited and not imported: no premise, number or conclusion of AZ1 depends on it, and no priority or correctness statement about it is made. The skeptic gives the reasons it changes nothing here.

- AZ1 states what this fixed-spacing Kogut–Susskind family supplies. A different construction, a Euclidean reflection-positive $SU(N)$ lattice, cannot supply an estimate for this family without an identification theorem.
- Its central claim, if correct, concerns a continuum theory. It is not an estimate for this family at $g^4 \geq 9.6 \times 10^9$.
- Any route from strong coupling to a continuum with finite mass must cross the whole coupling range, because $a\Delta$ is bounded below in the strong-coupling regime. What would have to be audited is exactly the estimate that row R1 of Table 32 lists as missing.

The producer’s checker requires “unaudited” and “not imported” in the prose, and removing the paragraph aborts it. The lead is recorded for a future round’s skeptical diff. That diff should establish which of its estimates hold along $a_n \rightarrow 0$, with what constants and coupling range, whether they apply to the Hamiltonian limit of the Wilson action, and an independent replay.

8.1.6 Bridge obstruction, common rescaling and continuum requirements

The AL1 bridge obstruction. Contract item 4 asks for it to be retained. In AL1 the selected bridge needs $r = 4/g^4 \leq 1/8$, that is $g^4 \geq 32$ or $\tau \leq 3$, and the ends need $r \leq 1/2$, that is $g^4 \geq 8$ (HNM-AL1.3). In the uniform model this is the condition $|\tau|/24 \leq 1/8$ on the uniform selected triple (Section 6.2). The AL1 gate records that its frozen path violates the bridge for every $n \geq 1$, and that every path with $g \rightarrow 0$ eventually violates it. The R18-B2 bound $g^4 \geq 32/3$ concerned a different finite-graph coefficient box. Its substitution is a rejected mutation, and dropping the bridge is rejected by `bridge_obstruction_retained`.

Common rescaling fails. The coupling $\tau = 24\lambda/\alpha$ depends on g alone. Exactly, for every $c > 0$,

$$\tau(ca) = \tau(a), \quad \alpha(ca) = \frac{\alpha(a)}{c}, \quad \lambda(ca) = \frac{\lambda(a)}{c}, \quad \frac{g^2}{32ca} = \frac{1}{c} \cdot \frac{g^2}{32a}, \quad r(ca, c\lambda) = r. \quad (157)$$

In the skeptic’s homogeneity degrees in (g^2, a) , α has degree $(1, -1)$, λ has $(-1, -1)$, τ and r have $(-2, 0)$, the gap has $(1, -1)$ and the lattice gap has $(1, 0)$. The producer’s exponents in g are twice the degrees in g^2 . A common rescaling $(\alpha, \lambda) \rightarrow (s\alpha, s\lambda)$ is exactly $a \rightarrow a/s$ at fixed g . Changing

the spacing at fixed g , changing the energy unit or rescaling the whole Hamiltonian therefore moves the gap $g^2/(32a)$ but never moves τ or r . No admission ratio moves, and only g decides whether a point lies in the admitted regime or satisfies the bridge, so common rescaling can neither repair nor break the cap or the bridge. The AL1 gate selected a later loop to test common scale changes, but that result is not a shared premise of AZ1, and the invariance is derived here from the AL1 dictionary alone (defect D4). An independent magnetic multiplier $\lambda \rightarrow s\lambda$ at fixed α is a change of model. It changes g^4 to g^4/s , and the bridge then needs $s \leq g^4/32$. On the declared toy $s_{\max} < 1$ from $n = 132$ on, about 0.98816 at $n = 132$, and $s_{\max} \rightarrow 0$ as $g \rightarrow 0$ (skeptic).

What a continuum construction would additionally require. Contract item 4 names three requirements. Table 32 gives each with what the fixed-spacing family supplies and its missing premise. The control `coherent_evidence_tampering` rejects a requirement marked supplied, and the report scan rejects the removal of a row.

Table 32: What a continuum construction would additionally require. Every row is missing; the text is the producer’s, as bound by the gate.

Requirement	Supplied by the fixed-spacing family	Missing premise	Status
R1: uniform-in- a estimates	Not along $a_n \rightarrow 0$: at each fixed a with $g^4 \geq 9.6 \times 10^9$ only, the volume-uniform gap $\alpha/16$ and the AX1 state bounds, and nothing once $g_n^4 < 9.6 \times 10^9$.	A lower bound $\Delta(a_n, g_n) \geq m E_\star$ with a fixed $m > 0$ for every n along a trajectory with $a_n \rightarrow 0$ and $g_n \rightarrow 0$, with the matching control of local states and correlations. It requires a certificate valid for $g^4 < 9.6 \times 10^9$, outside strong bare coupling, and no admitted or source-audited result supplies one.	missing
R2: reconstruction hypotheses	Fixed- a lattice objects only, no limit object: AQ subsequential states and their GNS Hamiltonians with the gap $\alpha/16$, on a lattice with no rotation or Lorentz covariance.	Osterwalder–Schrader axioms for limiting Euclidean Schwinger functions (reflection positivity passed to the limit, restored Euclidean invariance, regularity, clustering), or the Hamiltonian analogue: a limit of the Hilbert spaces and local algebras along $a_n \rightarrow 0$ carrying a self-adjoint nonnegative Hamiltonian with a unique vacuum, local fields and Poincaré covariance. No such limit is constructed in the admitted material.	missing
R3: physical scale control	Not a physical scale: in lattice units $a\Delta \geq g^2/32 > 3061$ throughout the admitted regime, so the certified gap is a cutoff-scale quantity.	A relation between a_n and g_n that holds a physical quantity fixed in units of $E_\star$ (the renormalization-group trajectory, here only the hypothesis H_{AF}), with a proof that $a_n\Delta_n \rightarrow 0$ while $\Delta_n/E_\star$ stays bounded above and below. Inside the admitted regime $a_n\Delta_n > 3061$ for every n , so the certified $\Delta_n/E_\star$ diverges as $a_n \rightarrow 0$.	missing

The skeptic’s pre-comparison adds detail to the three rows and three rows of its own, none of them a contract requirement.

- *R1.* The contraction parameter $J' = 2784/g^4$ is unbounded as $g \rightarrow 0$, so the AM2/AX1 fixed point and gap exclusion cannot be continued. The skeptic names multiscale renormalization-group control of Balaban type as a candidate route, not read here and not a premise, and notes that shared notation does not transfer a gap between models.
- *R2.* Strong-resolvent convergence alone does not carry a gap to a limit: the vacuum eigenvalue can disappear or change multiplicity. On different lattice Hilbert spaces a gap bound that holds along the whole sequence passes to the limit only with identification maps, convergent renormalized ground energies and a surviving vacuum vector. Reflection positivity at each

spacing survives limits; everything else needs bounds along $a_n \rightarrow 0$, and none is available.

- *State identification at fixed a .* The six AY2 obligations of Table 28 are unproved even at fixed spacing.
- *Nontriviality.* Any non-Gaussian estimate that survives $a_n \rightarrow 0$ is missing. The only rigorous non-Abelian scaling limit in more than two dimensions in the modern lens’s SOTA table, for SU(2) Yang–Mills–Higgs, is Gaussian.
- *Observables.* Which operators converge, with operator renormalization and an a -dependent normalization, is open.

The gate’s limitations name the AY2 state obligations and nontriviality as missing as well.

No fraction. No number in this subsection is a fraction of the continuum problem. AZ1 is investigation 9 of 10 of Round32. That is an index of executed investigations, not a measure of progress on the continuum problem, and no percentage or completion estimate is given. The control `loop_count_not_fraction` rejects the loop-count flag set true, an investigation index exported as a fraction field and a fraction sentence appended to the report.

8.1.7 The mandatory sentence, gate fields and the two senses

The template. Contract item 5 requires the preregistered mandatory sentence template in the report. The producer quotes it verbatim, exactly once:

Along the named trajectory (a_n,g_n), the fixed-spacing family supplies exactly one uniform estimate, the volume-uniform gap $\alpha/16=g^2/(32a)$ for $g^4 \geq 9.6 \times 10^{-9}$, and fails to supply any estimate uniform along $a_n \rightarrow 0$; this is not a statement that the continuum limit exists, that any estimate is uniform in a , that the coupling is weak, or that any fraction of the continuum problem is thereby resolved.

The gate statement of Section 8.1.1 is this template with four insertions and no other change. The first insertion is set off by dashes and takes in the template’s comma after (a_n,g_n). Table 33 lists the four. The negated closing clause is unchanged.

Table 33: The insertions with which the gate fills the mandatory sentence template.

After the template’s words	The gate inserts
“Along the named trajectory (a_n,g_n)”	the hypothesis of asymptotic-freedom form with $g_n \rightarrow 0$ as $a_n \rightarrow 0$, not derived
“the fixed-spacing family”	the AX1 model (every face $\alpha\tau/24$, route B, Haar reference, centered whole-star boxes, cover $R = \{0, e_z\}$), the AL1 dictionary with $\alpha\lambda a^2 = 1$, and $E_\star$ and $\hbar$ fixed
“for $g^4 \geq 9.6 \times 10^{-9}$ ”	$0 < \tau \leq 10^{-8}$ and “at fixed a ”, then every finite complete-factor route-B volume and every AQ subsequential state, AX1 gate items (2)–(3), cited, and the lattice-units floor $1250\sqrt{6}$ with its directed lower end
“fails to supply any estimate uniform along $a_n \rightarrow 0$ ”	“on a trajectory with $g_n \rightarrow 0$ ”, then the exits from the regime and the bridge, the four toy crossovers, common rescaling, the finite-mass obstruction and the certificate-not-gap reading

Three readings go with the template. It omits “at fixed a ”, which the scope string, the producer’s text and the gate statement supply next to it (skeptic reading 5). Its “exactly one uniform estimate” is exactly one necessary estimate (defect D2). It quotes the first forbidden phrase inside a negation (defect D1), so the producer’s scan removes the template’s one verbatim occurrence before scanning; the review finds that the scan also removes every inline code span and fenced block (finding N1). In this addendum the first forbidden phrase occurs only inside the two quoted, negated sentences above, and the second nowhere.

Two senses of one word. Two senses of “uniform” occur, and they are kept apart. The N-sense is the one positive claim: volume-uniform, that is, one bound for every finite complete-factor volume at one fixed spacing. The a-sense, uniform in a or along $a_n \rightarrow 0$ on a trajectory with $g_n \rightarrow 0$, is never claimed. A third, unrelated use is the model name, uniform Kogut–Susskind SU(2), with the same coefficient on every face. The producer’s checker enforces three rules on its report. Every occurrence of the word is qualified in place by one of these senses, every a-sense sentence carries a negation or a requirement word, and the two senses are never conjoined; the template is checked separately and is qualified by apposition. The skeptic’s validator rejects an a-sense occurrence that shares a sentence with an unqualified occurrence and no N-qualifier, and an a-sense occurrence that no negator precedes, including the “not only ... but also” construction. The control `uniform_in_N_not_in_a` rejects four appended sentences: one with the unqualified word, one with the a-sense, one with both senses together and one asserting validity along the whole trajectory. The control `uniform_label_strong_coupling` rejects a weak-coupling or continuum label and a label without “fixed spacing” or “strong bare coupling”. This subsection follows the same rules.

Gate fields. The preregistration requires four gate fields, and contract item 5 adds a fifth with its scope string. The gate exports them as follows.

- not claimed, exported as false: `continuum_claim`, `uniform_in_a_claimed`, `weak_coupling_claim` and `loop_count_fraction_claimed`;
- claimed, exported as true with a scope string: `uniform_in_N_claimed`.

The scope field reads, verbatim:

- `uniform_in_N_scope`: volume-uniform at fixed a , strong bare coupling

The `sub_label` field is null. The packets export further flags as false.

- both packets: `uniform_wilson_claim`, `resolved_interaction_shift` and `scientific_priority_verified`;
- the skeptic’s review in addition: `dynamical_claim` and `euclidean_node_certified`.

The AX1 gate’s `uniform_wilson_claim` true meant only the model label, which AZ1 carries in its own label, so AZ1 exports the field as false, as its task instruction asked (relayed note R1); AZ1 makes no Wilson-mean claim. The preregistered `gate_fields_required` lists only the four false fields (skeptic reading 2). The control `no_priority_or_continuum_claim` rejects a continuum, priority, weak-coupling, Wilson-mean or shift flag set true, and an a-sense flag set true.

8.1.8 Controls, replays and review

Error ledger. The preregistration itemizes one entry, `not_applicable`, with the stated reason: a statement loop with exact integer and rational dictionary arithmetic and exact crossover indices only, and no enclosure. The only irrational quantity, $g^2 = \sqrt{9.6 \times 10^9}$ at the cap, is bracketed by directed integer square roots of width 10^{-30} and enters only as a lower bound. The skeptic notes that the entry holds only because every printed numerical gap at the cap carries its directed bracket, as it does here (reading 15). All twelve report previews that the review checked are truncations of the exact values toward zero.

Contract controls. The contract lists 21 controls, and its preregistration mirror lists the same 21. Eight ids have AZ1-specific semantics.

- new: `uniform_in_N_not_in_a`, `dictionary_arithmetic_exact` and `toy_trajectory_crossover_exact`;

- from the skeptic’s prospective continuum group: `trajectory_named`, `bridge_obstruction_retained`, `one_uniform_estimate_identified`, `e_star_fixed_no_plateau` and `loop_count_not_fraction`.

The producer implements all 21 as damaging mutations raised through explicit exceptions, with none deferred (Table 34). The inherited ids concern R -local constants that AZ1 only cites. They guard the cited quantities and discriminate little for a dictionary statement (skeptic reading 17).

Checker. The forward checker runs 42 exact checks. It implements all 21 controls as damaging mutations, with none deferred, and records 107 rejections inside the control checks and 121 in the whole checker. It verifies the contract snapshot’s SHA-256 before any evaluation, reads the cap, the dictionary, the identities, the target and the control list from the contract, records its own SHA-256 before evaluation, and hashes all 29 input snapshots. Of these, 22 are byte-identical to what the AX1 gate binds, and the AL1 report is what the AL1 gate binds. The checker uses only the standard library and exact rationals, and its outputs under `-B` and `-B -O` are byte-identical. It records the AX1 gate hash but does not pin it (review finding N4). The producer also disclosed a private source-edit audit of thirteen damaging edits, all aborting, plus one unmutated copy; it is not evidence.

Independent pre-comparison package. The skeptic’s package holds a contract review with 18 readings, a derivation and a checker with 82 exact checks and 28 controls, the 21 contract ids and seven more. The checker rejects 122 mutations, 101 of them inside the contract controls. In 28 source-edit runs in private scratch, 24 must-abort edits all aborted at the intended check and three silent edits were accounted for. The package was written from the contract without opening any file under `forward/az1/`; the skeptic listed only the names of the 29 input snapshots and hashed them against their sources. Its routes differ from the producer’s. It proves the identities as rational functions in (g^2, a) , computes crossovers by an integer root with a monotone correction and a brute-force scan, and uses a directed 10^{-40} bracket of $\sqrt{6}$, a negation-aware phrase and two-senses validator, and the fixed- g counterexample of Section 8.1.4. Its predictions matched the producer’s exact values: the crossovers 2, 132, 4 and 421, the cap g^4 , the self-map crossovers 8 and 23 and the gap formula. The skeptic records its correlated ancestry. Five of the 21 control ids come from its prospective controls, and its AX1 review is a shared premise. The selection note and the modern update 4, both premises, already print the statement, the three contract identities and the rehearsal indices 4 and 421. The agreement is therefore a replay of elementary algebra, independent in route and code only.

Post-comparison review. The review adds 28 exact checks, for 110 independent checks with the 82, and 46 source-mutation runs. One unmutated copy reproduces the frozen output byte for byte. All 41 must-abort edits abort with the intended message.

- Twenty-one weaken the validator of one contract control each; each run then aborts on the producer’s own damaging mutation, which shows that every control is live.
- Seven are value and logic edits, among them $\tau g^4 = 95$, a crossover root shifted by two, $a_0 = 1/5$ fm and `continuum_claim` set true.
- Six are input edits, among them a contract byte edit, the AX1 gate’s physical gap changed to $\alpha/8$, the AL1 gate’s bridge changed to $g^4 \geq 31$ (caught because the AX1 gate binds the AL1 gate) and an undeclared skeptic input.
- Seven are report edits: row R3 removed, the template’s negation removed, a forbidden phrase appended, an asserted a-sense sentence appended, $n^* = 132$ changed to 131, a map row removed and the Faizal–Shabir paragraph removed.

Four edits run silently in the producer’s checker, and all are accounted for. Two appended sentences that state a completion percentage or fraction pass the producer’s literal scan and are caught by the

Table 34: The 21 contract controls and the damaging mutations the forward checker rejects under each.

Control	Damaging mutations rejected
missing_incoming_stars	outgoing stars only (two anchors, per-site sum $8 \tau $); single-factor groups dropped ($28 \tau $); a single star at 0; the incidence (7 stars, 2 groups, 153 faces, 88 meeting R , $82 + 6$, 16 inside) is recomputed from the I1 owner map
full_original_wilson_cover	a single-factor cover $\{0\}$; the four drawn links as the cover (48 links and 36 endpoints required)
wrong_delta_alpha_hbar_clock	$\frac{1}{2}$ read as $\alpha/2$; the gap-one route $\alpha/8$; the Haar gap six; a normalized clock; $g^2/(16a)$
vector_versus_scalar_centering	vector or scalar centering imposed on a statement; a zero reference energy for the gap
first_order_mean_charged	the mean set to zero; $-\tau/72$; $\tau/72$; the mean is $\tau/144 = 2/(3g^4)$, $1/14400000000$ at the cap
tau_scaling_exponent	τ as $96/g^2$; τ depending on a ; the lattice gap $a\alpha/16$ labelled physical
changed_model_relabelled	the patterned zero-selected model; the toy point $n = 2$ ($\tau = 16 \times 10^{-8}$) or an AL1-path point labelled admitted; $\tau < 0$ as a real- g coupling; a finite graph; the selected-strip reference; the trajectory points relabelled as one model
coherent_evidence_tampering	sixteen edits with the packet hash rebound, among them a flipped control, the T1 indices 131 and 1, the cap decade, the scope string, the AX1 gate hash, the mandatory sentence, a second estimate, a derived trajectory, a supplied requirement and the verdict
insufficient_verdict_retained	a missing citation, an overclaim or inexact arithmetic relabelled; the AL1 path failure dropped; the failure point relabelled resolved
exact_arithmetic_admission	float, Boolean, NaN and zero-denominator inputs; a decimal preview as an admission value; a float coupling in a crossover; a non-integer polynomial coefficient
root_n_misuse	a square-root crossover index (11 for T2); g_n^4 as g_0^4/n^2 ; the gap divided by a root of the volume
no_priority_or_continuum_claim	a continuum, priority, weak-coupling, Wilson-mean, shift or a-sense flag set true
trajectory_named	the trajectory claimed derived; a_n unnamed; g_n constant; a claimed physical renormalization trajectory
bridge_obstruction_retained	the bridge dropped; $32/3$ substituted; common rescaling claimed to reach the cap; an a -dependent τ
one_uniform_estimate_identified	no estimate; a second estimate; the patterned-model gate cited; the estimate read along a_n ; the estimate re-derived
e_star_fixed_no_plateau	$E_\star = 0$; $E_\star$ varying as $1/a_n$; a plateau fit; post-hoc node selection
loop_count_not_fraction	the loop-count flag set true; an investigation index as a fraction field; a fraction sentence appended
uniform_label_strong_coupling	a weak-coupling or continuum label; “fixed spacing” or “strong bare coupling” missing
uniform_in_N_not_in_a	four appended sentences: the unqualified word, the a-sense, both senses together, validity along the whole trajectory
dictionary_arithmetic_exact	$\alpha = g^2/a$; $\lambda = 1/(g^2a)$; $\tau = 12\lambda/\alpha$; the gap $g^2/(16a)$; $\alpha\lambda a^2 = 2$; a grid with α denominator 1; a float grid coupling; a shifted cap decade
toy_trajectory_crossover_exact	$n = 1$ of T1 as a crossover; T1 bridge 131 or 133; T2 indices 3 and 420; $32/3$; the cap decade 9.6×10^8

review’s pattern scan (N2). An appended negated sentence about estimates along $a_n \rightarrow 0$ without the $g_n \rightarrow 0$ scope is wording level (N3). An edit to an unparsed field of the AX1 gate snapshot is caught by the snapshot comparison and by the review’s validator, which pins the gate hash (N4). The review’s own validator accepts the frozen packet and rejects 18 damaged packets.

Replays and isolation. The existing replay tools do not accept a statement loop unchanged. `replay_loop.py` verifies and replays the forward closure without a mismatch and then stops on the absent reverse freeze, and `replay_declared.py` accepts a single producer only with direction `single+skeptic`. Both are bound by earlier reviews and gates, so neither was edited (N12, as AY2’s N9). The review runs the same steps: the unchanged freeze verification before and after, two fresh external replay directories, a byte comparison with the frozen output and the freezing tool’s verification. Both replays reproduce `results.json` and `source-manifest.json`. The closure lists 33 files plus `freeze.json`, and every one verifies. The inputs are exactly `AGENTS.md`, the contract and the 27 shared premises, each byte-identical to its repository source. The only skeptic files among them are the declared AX1 and AM2 reviews, and no `reverse/az1` package exists. The producer’s private scratch folder matches its disclosure by names and times. The producer disclosed reads outside its inputs for protocol and code style only: the tools documentation, the freezing tool, and parts of the AY2 forward checker and report. The project’s instruction file was shown to it as session context. None of these carries premise weight. It read some premises in full, some in part and three method references not at all, and all 29 are hash-verified. Two name exposures are disclosed (N11). After its freeze, a `git status` run showed the producer two file names; the exposure post-dates the frozen closure and carries no premise weight. During the review, a whole-tree `git status` and a scratch listing showed the skeptic the names of four untracked AZ2 files and two AZ2 scratch folders, none of which it opened. Isolation is verified for repository inputs only, and all agents are correlated model agents.

8.1.9 Contract wording and non-blocking findings

The gate records the producer’s contract wording defects D1–D9 as non-blocking. The skeptic verified each against the frozen text; seven coincide with its own readings, D6 is read differently and D9 is new.

- *D1.* The template contains the first forbidden phrase, negated; the producer’s scan removes the template before scanning.
- *D2.* “supplies exactly one uniform estimate” is read as exactly one necessary estimate, because AX1 also admits other volume-uniform constants at fixed a , D'_{ii} and the reset $102|\tau|$.
- *D3.* “the AM2 contraction needs $\tau \leq 10^{-8}$ ”: 10^{-8} is the admitted cap, not a sharp threshold; the route radii are not admitted, and the failure point holds for every fixed cap.
- *D4.* “the failure of common rescaling” refers to work after AL1 that is not a shared premise; the invariance is derived here from the dictionary.
- *D5.* The bridge fails from a finite index on along a trajectory with $g_n \rightarrow 0$, and at every n only on the AL1 frozen path.
- *D6.* The preregistered selected triple `["0","0","0"]` is read by the producer as the route-B Haar reference. The skeptic reads it as a defect copied from the patterned-family block and asks the gate to record it as a vocabulary defect (N5). Both bind the uniform coefficient $\tau/24$.
- *D7.* The toy trajectory names no a_n ; $a_n = a_0/n$ is declared, and the crossovers do not depend on it.
- *D8.* A real- g trajectory has $\tau > 0$; $\tau < 0$ is the U_E mirror, not a trajectory point.
- *D9.* The state provenance’s “finite_volume uniform bound” carries an unqualified occurrence of the word; it is read as volume-uniform at fixed a .

The premise note P1 is the panel’s arithmetic slip of Section 8.1.5: in the modern lens’s update 4, $10^{12}/81$ has remainder 28, not 34, and no index is affected. The relayed-instruction notes record that

`uniform_wilson_claim` is exported false as instructed (R1), and that the relayed freeze command omitted the `--verdict` argument that the freezing tool requires, so the proposed forward verdict text was passed (R2).

The gate also records the review’s findings as non-blocking.

- *N1*. The producer’s scan removes the template and also every inline code span and fenced block, which is broader than D1 states. In this report only the verbatim claim-exclusion list sits in code spans, so nothing is hidden, but a future report could hide an assertion there.
- *N2*. The producer’s forbidden-phrase scan is literal, so a percentage or a differently worded fraction passes it. The review’s pattern scan finds none in the frozen report.
- *N3*. Four negated sentences about estimates along $a_n \rightarrow 0$ in the producer’s report carry no $g_n \rightarrow 0$ scope in the sentence. One is scoped by the preceding sentence and one defines the a-sense. The other two deny the a-sense for the other AX1 constants, which is false as a free-standing sentence at fixed g , because D'_{ii} depends on τ alone. The gate scopes the a-sense to trajectories with $g_n \rightarrow 0$; this is wording, not a claim.
- *N4*. The AX1 gate hash is not pinned in the checker, as in AY2’s N4; no file binds the AX1 gate itself.

The remaining findings concern the preregistered zero triple (N5), the two declared gap examples (N6), the identity sets (N7), the labelled extras (N8), isolation and independence (N9), the skeptic’s own miss of P1 (N10), the name exposures (N11) and the replay tools (N12). The gate’s decision also records the commit order: the skeptic’s pre-comparison 649db92 before the producer’s 8359876.

8.1.10 Limitations recorded by the gate

The AZ1 gate records the following limitations verbatim.

- Only the uniform Kogut-Susskind SU(2) model at fixed spacing and strong bare coupling ($g^4 \geq 9.6 \times 10^9$, $|\tau| \leq 10^{-8}$) as admitted by AX1, with the AL1 dictionary; the supplied gap is volume-uniform at fixed a and never uniform in a along a trajectory with $g_n \rightarrow 0$
- The trajectory is a hypothesis of asymptotic-freedom form, not derived; no beta function is computed or used; the toy trajectory $g_n = g_0/n$ ($a_n = a_0/n$ declared) is algebraic and not of asymptotic-freedom form; physical-units numbers are illustrations without scale setting
- Failure of a sufficient certificate only: nothing is said about the actual gap where the certificate stops; the route radii 7/274688 and 1/20416 are not admitted regimes
- The continuum requirements (uniform-in- a estimates, reconstruction hypotheses, physical scale control) are missing, as are the AY2 state obligations and nontriviality; no loop count is a fraction of the continuum problem
- Inherited without re-proof: the AX1 gap (AM2 contraction, cutoff removal, AQ passage), the AL1 dictionary and bridge, the I1 geometry
- The Faizal-Shabir construction (arXiv:2606.19362) is an unaudited lead read at abstract depth by the modern lens; neither producer nor skeptic read it; nothing depends on it
- Single producer, correlated model agents; the premises print the headline; the agreement is a replay of elementary algebra
- Scientific priority unverified

Full derivation and reviewed implementation: [AZ1 forward](#), [contract review](#), [independent pre-comparison derivation](#), [independent model-agent review](#) and the [frozen contract](#). The advisor’s findings record that AZ1 closes the fifth Round31 roadmap goal in its honest form: the dictionary along a trajectory, and an explicit identification of the single volume-uniform estimate supplied and of the estimates missing. The roadmap’s continuum goal now names the missing premise exactly. The lattice-units floor $a\Delta \geq g^2/32$ in the admitted regime is the quantitative reason why the strong-coupling family cannot approach a finite-mass continuum by itself. The gate selects AZ2 next; its contract was already frozen and in production.

8.2 AZ2: finite-graph first- and second-order Wilson coefficients

Contribution HNM-C-AZ2. The AZ2 contract, recorded under the title “Hruday exact finite-graph first- and second-order Wilson coefficients on the two-plaquette graph (a different finite model)”, was frozen at 04:29:40Z, 0.03 s after AZ1, after panel update 4 and before production. It closes sub-round 5 and is investigation 10 of 10. Its model is not a $\mathbb{Z}^3$ model at all. It is the Round11 two-plaquette graph [7]: two adjacent squares sharing one link, seven links, six Gauss constraints, the gauge-invariant sector, a polynomial-degree cutoff $D \in \{6, 8\}$ with a certified tail, a coupling τ_{FG} on a declared grid and the I1.5 sign convention. The preregistered model identifier is `FG(two-plaquette, D in {6,8}, tau_FG grid, I1.5, gauge-invariant)`, the selected triple is `n/a (finite graph)`, and the coupling is the grid $\tau_{\text{FG}} \in \{1/1000, 1/100, 1/10\}$ at both signs, declared a model change against the previous loop. The observable is $\langle W \rangle_{\text{FG}}$ of one square with its coefficients, without centering; the reference is the graph’s own free value on the route `own_finite_graph`, and there is no Euclidean node. The three itemized error terms are `truncation_jmax`, `eigenvector_residual` and `arithmetic`. The target, “certified tail and enclosure widths reported at every grid point and both cutoffs; first-order coefficient enclosure contains 1/6”, has the value 1 with the comparator $\geq$ and is declared a feasibility and format check on a rehearsed finite-model computation, not a blind discovery threshold. The acceptance clause reads “Enclosures with certified tails at every grid point; first-order coefficient consistent with the dictionary; labels present”; a tail certified only at $D = 6$, or an unitemized product channel, would have been `limited`. The contract’s stop field reads “Investigation 10 of 10; scientific production stops after this review.”

AZ2 runs in one direction, as planned, because it instantiates certificates on a computable finite model; its contract declares `single+skeptic`. There is one forward producer. The second admission input is the skeptic’s pre-comparison derivation and replay, written from the contract alone (`single_direction_independent_replay`), followed by the post-comparison review. The producer received `AGENTS.md`, the contract and its 26 shared premises, 28 files in all, and read all of them in full: the selection note; the Round11 README, advisor record, solver README and solver source; the modern lens’s loop-2 response, loop-3 sign-off and update 4; the AW1 gate and AW1 forward report; the complete-residual reference note; the AT4, I1, AM2 (forward and reverse), AQ1 and AQ2 reports; the AM2 gate and the skeptic’s AM2 review; and the four method snapshots with their three reference notes. No reverse package exists, and `reverse_premise_isolation` is false.

Both packages were developed at the same time after the contract froze. The skeptic’s first scratch file dates from 04:41:37Z and the producer’s from 04:46:16Z. The skeptic’s checker was last written at 05:17:07Z, and its final replays followed; the producer’s checker and report were last written at 05:27:23Z and 05:27:39Z and its output at 05:28:27Z. The skeptic froze at 05:29:35Z and the producer at 05:29:49Z. As for AZ1, the skeptic committed first: `4bae265` at 05:30:10Z, before the producer’s `dc817d8` at 05:33:11Z. The two packages also chose different models for the same contract text (Section 8.2.2), which is itself evidence that neither read the other. Isolation rests on disclosure, name-only exposure, content and commit order (Section 8.2.9).

Notation: $x = W_1$, $y = W_2$ and z are the three trace coordinates of Section 8.2.2; K is the kinetic operator and H_{FG} the Hamiltonian in units of α ; P_D is the span of the monomials of total degree at most D and $Q_D = I - P_D$. For a Ritz vector $v \in P_D$ with unit vector $\hat{v}$, μ is its Rayleigh quotient, $r = (H_{\text{FG}} - \mu)\hat{v}$ its complete residual in the full graph space and $\rho = \|r\|$. $b = 3 - 2|\tau_{\text{FG}}|$ is the certified separation, $t_\ell = \text{tail_lower}$ the first omitted-shell threshold, and s_A and s_B are the Davis–Kahan and Eckart angle bounds. Decimals are previews of exact rationals. The word “interaction” refers only to this graph.

8.2.1 Reviewed scope

The [AZ2 gate](#), completed at 06:06:38Z, records the verdict `accepted_within_scope` with two sub-labels.

- The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this subsection. In it, **tau_FG** is τ_{FG} ; **H_FG**, **K**, **W_1**, **W_2** and **z** are as in Section 8.2.2; **psi_0** is the ground state and **psi_1** the first-order vector; $\langle \mathbf{W_1} \rangle$ is $\langle W_1 \rangle = \langle \psi_0, W_1 \psi_0 \rangle$; **E_0** is the ground energy; **h1**, ..., **vM** are link names; **rho** is Round11’s shared-link weight, written ρ_{11} below; and **tail_lower** is t_ℓ . The sentence beginning “The finite-graph derivative” is the contract’s mandatory sentence (Section 8.2.8).

The gate’s decision admits the statement for the two-face model it names and records how that model was fixed: the contract fixed neither the one-face nor the two-face coupling, the producer disclosed its choice in its verdict, report and results, and the skeptic’s one-face pre-comparison reading remains admissible, with its values recorded as the other model. It admits four items. (1) The derivative is exactly $[1/6, 1/6]$ at both cutoffs with tail term exactly 0, the free reference is exactly 0 in the same code path, and $1/6 \leftrightarrow 1/144$ under $\tau_{\text{FG}} = \tau/24$ is a consistency check, never a transfer. (2) The twelve exact enclosures, with the recorded half-widths, are certified by the exact full-space residual on the first omitted shell, the certified gap, Temple’s bound with the Round11 tail comparison and an Eckart angle bound. (3) The coefficients are exact and cutoff-independent, and the one-face model differs certifiably at third order. (4) The five-part ledger is complete, and the python-flint Arb preview is labelled preview only. The gate binds the producer’s exact rationals. The skeptic’s independent two-face enclosures lie inside them, and its one-face enclosures are disjoint from them, because they belong to a different model. The rest of this subsection derives the statement, and the

limitations recorded by the gate close it (Section 8.2.11).

8.2.2 The graph, the Hamiltonian and the two admissible readings

The graph. The Round11 graph (G1–G2 of the [Round11 advisor record](#)) has the vertices TL, TM, TR, BL, BM, BR and seven oriented links, each with the normalized Haar measure of SU(2):

link	h1	h2	h3	h4	vL	vM	vR
orientation	TL→TM	TM→TR	BL→BM	BM→BR	TL→BL	TM→BM	TR→BR

Independent SU(2) gauge transformations act at all six vertices; these are the six Gauss constraints. With both loops based at TM,

$$\begin{aligned} U &= v_M h_3^{-1} v_L^{-1} h_1, & V &= h_2 v_R h_4^{-1} v_M^{-1}, \\ x &= W_1 = \frac{1}{2} \text{Tr } U, & y &= W_2 = \frac{1}{2} \text{Tr } V, & z &= \frac{1}{2} \text{Tr}(UV). \end{aligned} \quad (158)$$

Square 1 is U ; the shared link v_M lies in both squares and cancels in the outer loop z . Round11 reduces the gauge-invariant sector to $L^2(\Omega, (2/\pi^2) dx dy dz)$ (its Q1–Q3). The ground state of every Hamiltonian below is unique (Round11 Theorem F, and $E_0 \leq \mu < b \leq E_1$ at every grid point).

The admitted Hamiltonian. In units of α ,

$$\begin{aligned} H_{\text{FG}} &= K - \tau_{\text{FG}} (W_1 + W_2), & K &= \sum_e C_e, \\ H_{\text{FG}} &= H_{\text{R11}}(\alpha = 1, \lambda_1 = \lambda_2 = \tau_{\text{FG}}, \rho_{11} = 1) - 2\tau_{\text{FG}}, \end{aligned} \quad (159)$$

where K is the sum of the seven link Casimirs $j(j+1)$, both faces carry the same coupling, and $\rho_{11} = 1$ is Round11’s shared-link weight, written ρ_{11} here to keep it apart from the residual norm ρ . The constant $-2\tau_{\text{FG}}$ moves no state. For $\tau_{\text{FG}} < 0$ the Round11 solver’s interface, which requires $\lambda \geq 0$, does not apply; every bound below uses only $\|W_1 + W_2\| \leq 2$ and needs no sign. The sign follows I1.5: the star operator is $\phi_b = -(\tau/3) \sum W_f$ in units of δ , that is $-(\tau/24)W_f$ in units of α , so the first-order mean is positive for positive coupling. A static ground-state mean does not depend on α , and the clock of (6) is named only. The normalization dictionary of the contract is

$$\tau_{\text{FG}} = \frac{\tau}{24}, \quad (160)$$

because the $\mathbb{Z}^3$ face coefficient in units of α is $\nu/\alpha = \tau/24$ (I1) and the graph’s coupling multiplies one face each.

The kinetic operator. In quotient coordinates $K = 3C_U + 3C_V + S$: C_U acts on (x, z) , C_V on (y, z) , and the shared-link term S , with derivative $L_U - R_V$, on (x, y) . The checker reproduces the Round11 K5 identities exactly:

$$K1 = 0, \quad Kx = 3x, \quad Ky = 3y, \quad Kz = \frac{9}{2}z, \quad K(xy) = \frac{13}{2}xy - \frac{1}{2}z. \quad (161)$$

The last identity is the shared-link cross term. The span of x and y is the exact eigenspace of K at its free gap 3.

Which faces carry the coupling: a contract wording defect. The skeptic’s pre-comparison package read the contract as $H_{\text{FG}} = K - \tau_{\text{FG}}W_1$, with the second face uncoupled; the producer computed (159). The review finds that the contract fixes neither. Its model string names one coupling on a declared grid but not which faces carry it, and λ_2 appears nowhere. `parameters.observable`

names the observable, “ $\langle W \rangle$ of one square”, not the coupling. The only face wording is the dictionary’s parenthetical “the two-plaquette graph’s own coupling multiplies one face”, which explains the per-face normalization and reads equally well as “multiplies one face each”. The Round11 defaults $\lambda_1 = \lambda_2 = 0$ decide nothing. The shared premise of the modern lens’s loop-2 response specifies $H/\alpha = K + \kappa_1(1 - x) + \kappa_2(1 - y)$, with both faces coupled, and the uniform $\mathbb{Z}^3$ model that the dictionary quotes couples every face. So the two-face reading is at least as well supported as the one-face reading, and the one-face reading remains admissible. The producer named its Hamiltonian in its verdict, report and results and in its model control, and computed the one-face $E_2 = -1/12$ as a labelled control value, so there is no undisclosed model change. The review records the point as contract wording defect W-A, not a blocking issue, and the gate names the two-face model exactly.

The exact coefficients identify the model. The review recomputes the untruncated Rayleigh–Schrödinger series of both models in Round11 monomials. With $\langle W_1 \rangle = \sum_n a_n \tau_{\text{FG}}^n$ and $E_0 = \sum_n e_n \tau_{\text{FG}}^n$:

Model	a_1	a_2	a_3	a_5	e_2	e_4
two faces (admitted)	1/6	0	$-187/33696$	$767713/2523156480$	$-1/6$	$187/67392$
one face (the skeptic’s reading)	1/6	0	$-5/864$	$289/829440$	$-1/12$	$5/3456$

Both models share the derivative $1/6$, the zero second-order coefficient of $\langle W \rangle$ (a centre flip on v_M in one model, on a link of U in the other), the dictionary comparison $1/6 \leftrightarrow 1/144$, the own free reference 0, and the dimensions, tails and channel thresholds of Section 8.2.3. They first differ at third order. The skeptic’s pre-comparison package had recorded the two-face values $-187/33696$ and $-1/6$ as side values, and they equal the producer’s exactly.

The spectator identity holds only in the one-face model. With only U coupled, the spin carried by h_2 , v_R and h_4 is conserved, because x acts only on the four links of U and K is diagonal in spin-network labels. The ground state lies in the sector of class functions of U , where $K = 4C$ and, in the orthonormal characters $U_k(x)$, H is the Jacobi matrix $\text{diag}(k(k+2)) - (\tau_{\text{FG}}/2)(\text{shift up} + \text{shift down})$. Under (160), $8H_{\text{FG}} = 32C - (\tau/3)x$, which is exactly AW1’s one-plaquette fixture (60), and indeed $-\frac{5}{864} \cdot 24^{-3} = -\frac{5}{11943936}$. In that model the second square is a spectator. In the admitted two-face model it is active, and the model differs from the AW1 fixture at third order: $-187/(33696 \cdot 24^3)$ against $-5/11943936$. The two-face model is therefore a genuinely different finite model beyond first order, and the spectator identity is not claimed for it.

The second-square shift is certified. Exactly,

$$\langle W_1 \rangle_{\text{two faces}} - \langle W_1 \rangle_{\text{one face}} = \frac{\tau_{\text{FG}}^3}{4212} + O(\tau_{\text{FG}}^5), \quad -\frac{187}{33696} + \frac{5}{864} = \frac{1}{4212}. \quad (162)$$

The review certifies it at all twelve points from its own two-face enclosures and its frozen one-face enclosures, with the sign of τ_{FG} at every point; at $D = 8$ the midpoints are about $2.3697588999 \times 10^{-7}$, $2.3741248805 \times 10^{-10}$ and $2.3741685992 \times 10^{-13}$ at $\tau_{\text{FG}} = +1/10$, $+1/100$ and $+1/1000$. The producer’s twelve enclosures and the skeptic’s twelve one-face pre-comparison enclosures are therefore disjoint at every point. The one-face values are recorded in the gate as the skeptic’s alternative reading, not admitted, and they are not used below.

8.2.3 Basis, cutoff and the first omitted shell

Why the complete residual is finite. P_D is spanned by the monomials $x^a y^b z^c$ with $a+b+c \leq D$, in Round11 order, of dimension $\binom{D+3}{3}$: 84 at $D = 6$ and 165 at $D = 8$. K is triangular in degree,

with diagonal

$$\varepsilon(a, b, c) = 3j(j+1) + 3k(k+1) + \ell(\ell+1), \quad j = \frac{a+c}{2}, \quad k = \frac{b+c}{2}, \quad \ell = \frac{a+b}{2} \quad (163)$$

(Round11 E2). K preserves every P_d , and multiplication by x or y raises the degree by at most one. Hence $H_{\text{FG}}v \in P_{D+1}$ for every $v \in P_D$: the complete full-space residual is a finite polynomial, computed exactly, and nothing lies beyond shell $D+1$. D is the total polynomial degree. No per-link j_{max} is well-posed for this basis, as the contract states.

Exactly orthogonal sectors. For a monomial m_s of degree $D+1$, $Q_D m_s = m_s - P_D m_s$ is a single spin-network sector (j, k, ℓ) of shell $D+1$ by Peter–Weyl on each link: m_s has link spins at most (j, k, ℓ) coordinatewise, and the monomials of a down-closed label set span exactly the spin networks with those labels. The checker verifies orthogonality exactly: the Schur complement $G_{D+1, D+1} - B^\top G_D^{-1} B$ has zero off-diagonal entries, for 36 sectors and 630 pairs at $D=6$ and 55 sectors and 1485 pairs at $D=8$; the top retained shells, with 28 and 45 sectors, pass the same test. The Gram matrix is block diagonal in the four link-flip parity classes (blocks 24/20/20/20 at $D=6$ and 45/40/40/40 at $D=8$), and its exact $LDL^\top$ has all pivots positive. Every omitted norm is therefore an exact sum of sector weights.

Per-link spins and the product channel. The degree-two vertices force one spin along each three-link path: j on h_1, v_L and h_3 ; k on h_2, v_R and h_4 ; and ℓ on the shared link v_M . A shell- $(D+1)$ sector exceeds the retained per-link maximum $D/2$ on the j links if and only if $b=0$, on the k links if and only if $a=0$, and on v_M if and only if $c=0$. The *product channel* is the set of sectors with $a, b, c \geq 1$: every link spin is within $D/2$, yet the joint degree is omitted, so these states are not implied by any per-link item. The exact thresholds are minima of ε over all omitted shells. The enumeration stops once the Round11 E3 shell minimum $m_d = \frac{5}{8}d^2 + 2d + \frac{3}{8}(d \bmod 2)$, which increases with d , exceeds every class minimum (Table 35). Both joint thresholds equal the contract’s 45 and 69 and the E3 formula, $m_7 = 45$ and $m_9 = 69$. The lowest omitted state excites the shared link. The skeptic reaches the same floors independently: ε increases strictly in each exponent, so no later shell undercuts shell $D+1$ and $QKQ \geq m_{D+1}Q$ on the whole omitted space.

Table 35: Exact first-omitted-shell thresholds, in units of α , with the sector (a, b, c) that attains each. The joint threshold is `tail_lower`.

D	joint $t_\ell = m_{D+1}$	v_M per link	h_1, v_L, h_3 (and h_2, v_R, h_4)	product channel
6	45 at (3, 4, 0)	45 at (3, 4, 0)	123/2 at (6, 0, 1)	48 at (3, 3, 1)
8	69 at (4, 5, 0)	69 at (4, 5, 0)	96 at (7, 0, 2)	145/2 at (4, 4, 1)

8.2.4 The five-part ledger and the enclosure theorem

The complete residual. Take a rational Ritz vector $v \in P_D$, its unit vector $\hat{v}$, its Rayleigh quotient μ and its complete residual $r = (H_{\text{FG}} - \mu)\hat{v} \in P_{D+1}$. Then

$$\rho^2 = \|H_{\text{FG}}\hat{v}\|^2 - \mu^2 = \rho_P^2 + \rho_Q^2, \quad \rho_Q^2 = \tau_{\text{FG}}^2 \|Q_D(x+y)\hat{v}\|^2, \quad (164)$$

exactly, where ρ_P is the retained part and ρ_Q the omitted part. Contract item 1 asks for five separately itemized parts, and a single nonzero omitted matrix element, it adds, proves leakage but does not bound it. The producer itemizes all five at each of the 14 points (the free reference and the twelve grid points, at both cutoffs):

- (a) *Per-link representation tail*, one row for each of the seven links: the retained maximum spin $D/2$, the exact threshold of the states whose spin on that link exceeds $D/2$, the exact weight of

- r in those sectors, the link's boundary layer in $\hat{v}$ (the shell- D sectors with spin exactly $D/2$) and the certified per-link tail of the true ground state (item 7 below).
- (b) *Joint product channel across all seven links:* ρ_Q^2 as the exact sum over all shell- $(D+1)$ sectors, the product-channel weight and its threshold, the corner overlaps between per-link classes, and the x - and y -face channels with their interference $2XY$. That interference is positive, so the two face channels are not orthogonal and may not be added in quadrature. The checker verifies exactly that the per-link classes plus the product channel minus the corners equal ρ_Q^2 , and it reports the certified joint tail T_{joint} of the true ground state.
 - (c) *Gauge-invariant projection:* exactly 0. Every basis element is a polynomial in traces of closed holonomies, invariant under all six vertex actions, and H_{FG} commutes with the gauge action. An exact rational-quaternion fixture checks the invariance: seven rational unit quaternions give $(x, y, z) = (13/18, 1/6, -4/27)$, unchanged under four sets of six vertex transformations, while the open three-link path $h_3^{-1}v_L^{-1}h_1$ changes; z does not depend on v_M , and the v_M flip maps (x, y, z) to $(-x, -y, z)$.
 - (d) *Ritz and eigenvector residual:* ρ_P^2 , which is below 5×10^{-199} everywhere because the Ritz proposal is rounded at 10^{-100} ; ρ^2 ; the certified gap; and the angle bounds s_A and s_B .
 - (e) *Arithmetic:* exact **Fraction** arithmetic throughout. The only non-rational steps are square roots, rounded upward by integer square roots with width at most 10^{-55} . Exported ends are rounded outward to denominators 10^{45} , which contributes at most about 2×10^{-45} of each width, included. The 110-digit decimal inverse iteration only proposes v ; its rounding enters, measured exactly, in item (d).

The preregistered error terms map onto the ledger as `truncation_jmax` $\rightarrow$ (a)+(b)+(c), the D -cutoff truncation (producer defect D1); `eigenvector_residual` $\rightarrow$ (d); and `arithmetic` $\rightarrow$ (e). The skeptic's contract review read the map the same way, with (c) reported as not applicable with its reason; the producer reports it as exactly 0 with the same reason.

The enclosure theorem. Write $b = 3 - 2|\tau_{\text{FG}}|$. The certificate chains seven items; Round11 T1–T5 and E3–E4 are inherited, and Kato, Rayleigh–Ritz, Weyl, Temple, Eckart and Davis–Kahan are cited [28].

1. *Complete residual.* For $v \in P_D$, $r \in P_{D+1}$ and (164) is exact.
2. *Certified gap (Weyl).* $E_1(H_{\text{FG}}) \geq E_1(K) - \|\tau_{\text{FG}}(W_1 + W_2)\| \geq 3 - 2|\tau_{\text{FG}}| = b$, where $E_1(K) = 3$ comes from the complete free spectrum (Round11 E3, with $\text{span}\{x, y\}$ at 3), and $E_0 \leq \mu$ by Rayleigh–Ritz. At every point $\mu < b$, so the ground state is simple and $E_1 - E_0 \geq b - \mu$, which ranges from 2.80 to 3.00.
3. *Davis–Kahan and Temple.* The angle θ between $\hat{v}$ and the ground state satisfies $\sin \theta \leq \rho/(b - \mu) =: s_A$, and $E_0 \geq \mu - \rho^2/(b - \mu)$.
4. *Round11 tail comparison.* Put $t = t_\ell - 2|\tau_{\text{FG}}|$, so that $Q_D H_{\text{FG}} Q_D \geq t Q_D$ (E4), and take $R = b < t$. Young's inequality gives $H_{\text{FG}} \geq B \oplus R Q_D$ with

$$B = A - \frac{CC^*}{t - R}, \quad A = P_D H_{\text{FG}} P_D, \quad CC^* = P_D V Q_D V P_D, \quad (165)$$

so $E_0 \geq \lambda_0(B)$ because $\lambda_0(B) \leq \mu < R$. With $\beta_0 = \langle \hat{v}, B \hat{v} \rangle = \mu - \rho_Q^2/(t - R)$ and the exact $\eta = \|(B - \beta_0)\hat{v}\|$, finite Temple gives $\lambda_0(B) \geq \beta_0 - \eta^2/(b_B - \beta_0)$ with $b_B = b - 4\tau_{\text{FG}}^2/(t - R) \leq \lambda_1(B)$. This energy bound is 14 to 24 times sharper than Temple's, because the omitted leakage is divided by $t - R$, about 42 at $D = 6$ and 66 at $D = 8$, instead of by $b - \mu \approx 2.8$.

5. *Eckart.* $\sin^2 \theta \leq (\mu - E_0)/(E_1 - E_0) \leq (\mu - E_0^{\text{low}})/(b - \mu) =: s_B^2$. At every grid point $s_B < s_A$, by a factor of 3.7 to 4.9 (the review records 3.74 to 4.85 exactly). The headline enclosure therefore uses the tail comparison together with the certified gap.
6. *Observable.* With $\|W_1\| \leq 1$, vector centering and $s = \min(s_A, s_B)$,

$$|\langle \hat{v}, W_1 \hat{v} \rangle - \langle \psi_0, W_1 \psi_0 \rangle| \leq 2s\sigma + 2s^2 \leq 2s + 2s^2, \quad \sigma = \|(W_1 - \langle W_1 \rangle)\psi_0\| \leq 1. \quad (166)$$

7. *Certified tails of the true ground state.* From $Q_D(H_{\text{FG}} - E_0)\psi_0 = 0$ and $Q_D H_{\text{FG}} Q_D \geq t Q_D$, $\|Q_D \psi_0\| \leq (\rho_Q + 2|\tau_{\text{FG}}|s)/(t - \mu) =: T_{\text{joint}}$. For a link class e let Π_e project onto spin above $D/2$; it commutes with K , and its range lies in Q_D by the triangle inequality. With c_e the number of faces containing the link, t_e its threshold and Λ_e its spin-exactly- $D/2$ layer, $\|\Pi_e \psi_0\| \leq c_e |\tau_{\text{FG}}|(\|\Lambda_e \hat{v}\| + s)/(t_e - 2|\tau_{\text{FG}}| - \mu)$. The smaller of the per-link and joint bounds is reported: for v_M the joint bound, for the other six links the per-link bound, smaller by factors of about 50 to 140.

At every grid point and cutoff, then, $\langle W_1 \rangle$ lies in $[w_R - (2s + 2s^2), w_R + (2s + 2s^2)]$ rounded outward, where $w_R = \langle \hat{v}, W_1 \hat{v} \rangle$; E_0 lies between the larger of the Temple and tail-comparison bounds and μ ; and the ground state's omitted weight is at most T_{joint} .

8.2.5 The derivative at zero and the own free reference

The free reference, in the same code path. The certificate routine used at every grid point, called at $\tau_{\text{FG}} = 0$, returns the constant as its Ritz vector (the rounded proposal is exactly 10^{100} times the constant monomial), with $\mu = 0$, $\rho = 0$ and the enclosure $[0, 0]$ at both cutoffs. The value is exactly $\mathbb{E}[W] = \mathbb{E}[x] = 0$ by Haar symmetry: the central flip of v_M maps x to $-x$ and preserves the normalized Haar measure. The control `own_free_reference` rejects the $\mathbb{Z}^3$ value $1/144$ imported as the reference, $\mathbb{E}[W^2] = 1/4$ as the reference, and a reference computed in another code path.

The exact derivative. $H_{\text{FG}}(\tau_{\text{FG}}) = K + \tau_{\text{FG}}V$ with $V = -(W_1 + W_2)$ bounded, and the free ground state $E_0(0) = 0$ is simple with gap 3, so Kato's theory makes $\langle W_1 \rangle(\tau_{\text{FG}})$ real-analytic near 0 with Rayleigh–Schrödinger coefficients. The first-order vector is

$$\psi_1 = K^{-1}Q_0(W_1 + W_2)\Omega = \frac{W_1 + W_2}{3} \quad (167)$$

exactly in the full graph space, because $\text{span}\{x, y\}$ is the exact eigenspace of K at 3 (161). The finite-basis resolvent on P_D therefore coincides with the full resolvent on this vector, the tail term is exactly 0, and

$$\left. \frac{d\langle W_1 \rangle}{d\tau_{\text{FG}}} \right|_{\tau_{\text{FG}}=0} = 2\langle x, \psi_1 \rangle = \frac{2(\mathbb{E}[x^2] + \mathbb{E}[xy])}{3} = \frac{2 \cdot \frac{1}{4}}{3} = \frac{1}{6} \quad (168)$$

at $D = 6$ and $D = 8$ alike, and on the untruncated graph. The skeptic checks the same value in the producer's basis without the producer's code: the Galerkin identity $K_{\text{form}}(e_x/3) = M_X e_1$ holds entrywise, and the exact inertia of the kinetic form is $(0, 1, n - 1)$, so its kernel is the constants and the solution is unique.

A second route: certified finite differences. Independently of the Rayleigh–Schrödinger vectors, for complex $|\tau_{\text{FG}}| \leq 3/8$ the Riesz projection $P(\tau_{\text{FG}})$ around $|z| = 3/2$ satisfies $\|(H(\tau_{\text{FG}}) - z)^{-1}\| \leq (2/3)/(1 - 4|\tau_{\text{FG}}|/3) \leq 4/3$, hence $\|P(\tau_{\text{FG}})\| \leq 2$ and $|\text{Tr}(HP)| \leq 3$. Cauchy's estimate then gives $|a_n| \leq 2(8/3)^n$ and $|e_n| \leq 3(8/3)^n$, so the bias of a finite difference is fully bounded. The odd part of the certified $\pm 1/1000$ enclosures encloses a_1 in $[166628734921/10^{12}, 166704587313/10^{12}] \approx [0.1666287, 0.1667046]$ at both cutoffs, which contains $1/6$. The interval is about 10^{-4} wide, valid, and retained as a weaker tier (review finding N4).

- *Channels.* The omitted residual is dominated by the shared-link channel, the sectors $x^a y^b$ with $a + b = D + 1$, where v_M carries spin $(D + 1)/2$; the v_M class carries essentially all of ρ_Q^2 . The corner z^{D+1} has exactly zero weight, since shell D cannot reach it in one step. The product channel is small but nonzero, and it is itemized.
- *Scaling.* ρ_Q^2 scales as $\tau_{\text{FG}}^{2(D+1)}$; the certified ratio between 1/100 and 1/1000 at $D = 8$ is 9.99983×10^{17} . The product channel scales as τ_{FG}^{2D+4} : the review finds it zero at leading order.
- *Tail sharpening.* T_{joint} is four to five times smaller than $\sin \theta$, which shows how the tail threshold sharpens the omitted weight.

Table 38: Omitted-residual channels of the Ritz vector (upward-rounded previews): the complete omitted weight ρ_Q^2 , the j -class weight (equal to the k -class weight), the product-channel weight and the positive face interference $2XY$. The retained part ρ_P^2 is below 5×10^{-199} at every point.

D	τ_{FG}	ρ_Q^2	j class	product channel	$2XY$
6	1/1000	3.81105×10^{-57}	2.897×10^{-61}	4.98938×10^{-66}	1.68597×10^{-57}
6	1/100	3.81098×10^{-43}	2.897×10^{-47}	4.98929×10^{-50}	1.68594×10^{-43}
6	1/10	3.80477×10^{-29}	2.894×10^{-33}	4.98003×10^{-34}	1.68320×10^{-29}
8	1/1000	3.91821×10^{-76}	7.128×10^{-82}	3.31098×10^{-85}	1.78252×10^{-76}
8	1/100	3.91814×10^{-58}	7.128×10^{-64}	3.31092×10^{-65}	1.78249×10^{-58}
8	1/10	3.91131×10^{-40}	7.122×10^{-46}	3.30449×10^{-45}	1.77939×10^{-40}

Table 39: Angle bounds and certified tails of the true ground state (previews): the Davis–Kahan and Eckart bounds on $\sin \theta$, the joint tail T_{joint} , and the per-link tails for the h_1, v_L, h_3 class (equal to the h_2, v_R, h_4 class) and for v_M .

D	τ_{FG}	s_A (Davis–Kahan)	s_B (Eckart)	T_{joint}	T_j	T_{v_M}
6	1/1000	2.059×10^{-29}	5.502×10^{-30}	1.372×10^{-30}	1.759×10^{-32}	1.372×10^{-30}
6	1/100	2.072×10^{-22}	5.518×10^{-23}	1.375×10^{-23}	1.840×10^{-25}	1.374×10^{-23}
6	1/10	2.202×10^{-15}	5.687×10^{-16}	1.402×10^{-16}	2.682×10^{-18}	1.402×10^{-16}
8	1/1000	6.603×10^{-39}	1.407×10^{-39}	2.869×10^{-40}	5.708×10^{-43}	2.869×10^{-40}
8	1/100	6.642×10^{-30}	1.411×10^{-30}	2.874×10^{-31}	7.033×10^{-34}	2.873×10^{-31}
8	1/10	7.059×10^{-21}	1.454×10^{-21}	2.917×10^{-22}	2.075×10^{-24}	2.916×10^{-22}

8.2.7 Second- and higher-order coefficients as finite-model values

The exact series. The producer computes the Rayleigh–Schrödinger series about the free ground state exactly, by polynomial back-substitution, since K is triangular in degree. Every ψ_n lies in P_n , and the full-space equation $K\psi_n = -V\psi_{n-1} + \sum_k E^{(k)}\psi_{n-k}$ holds as a polynomial identity for $n \leq 5$. The coefficients are therefore identical at $D = 6$ and $D = 8$, and their truncation error is exactly zero, not merely bounded. The first two vectors are

$$\psi_1 = \frac{x+y}{3}, \quad \psi_2 = \frac{x^2}{24} + \frac{4xy}{39} + \frac{y^2}{24} + \frac{4z}{351} - \frac{1}{48}, \quad (170)$$

and Table 40 lists the coefficients. In closed form,

$$\langle W_1 \rangle = \frac{\tau_{\text{FG}}}{6} - \frac{187}{33696} \tau_{\text{FG}}^3 + \frac{767713}{2523156480} \tau_{\text{FG}}^5 + \cdots, \quad E_0 = -\frac{\tau_{\text{FG}}^2}{6} + \frac{187}{67392} \tau_{\text{FG}}^4 + \cdots. \quad (171)$$

Checks on the table.

- *Hellmann–Feynman.* $a_n = -(n+1)e_{n+1}/2$ holds exactly for $n \leq 4$; for instance $a_1 = -e_2 = 1/6$ and $a_3 = -2e_4 = -187/33696$.

Table 40: Exact Rayleigh–Schrödinger coefficients of the admitted two-face model, identical at $D = 6$ and $D = 8$ with truncation error exactly 0. These are finite-model values, never fitted.

Quantity	order 0	order 1	order 2	order 3	order 4	order 5
$\langle W_1 \rangle$	0	1/6	0	$-187/33696$	0	$767713/2523156480$
E_0	0	0	$-1/6$	0	$187/67392$	0
$\langle W_1^2 \rangle$	1/4	0	7/576	0	$-787717/1261578240$	—
$\langle W_1 W_2 \rangle$	0	0	79/2808	0	$-20159/10513152$	—
$\langle z \rangle$	0	0	7/216	0	$-349/151632$	—

- *Why the second-order coefficient of $\langle W_1 \rangle$ vanishes.* The shared-link flip U_E with $E = \{v_M\}$ meets both faces once, sends both W_f to $-W_f$ and commutes with every Casimir and with the gauge action. So $\langle W_1 \rangle(-\tau_{FG}) = -\langle W_1 \rangle(\tau_{FG})$ and every even coefficient is zero. This is the finite-graph analogue of the AW1 flip lemma (Section 4.4), recorded as a consistency observation only; it is a finite-graph centre-flip fact and says nothing about K_2 or the AQ remainder.
- *A second route for the second-order values.* The even parts of the certified $\pm 1/1000$ enclosures give $a_2 \in [-1.0114 \times 10^{-4}, 1.0114 \times 10^{-4}]$ and $e_2 \in [-0.16681837, -0.16651496]$ at both cutoffs, which contain 0 and $-1/6$.
- *Grid remainder, a labelled comparison.* $(\langle W_1 \rangle - \tau_{FG}/6)/\tau_{FG}^3$ is about $-5.54961982868 \times 10^{-3}$ at $1/1000$ and $-5.54657925470 \times 10^{-3}$ at $1/10$, against $-187/33696 \approx -5.5496201 \times 10^{-3}$.

The control `fg_coefficients_not_fitted` rejects the least-squares slope through the grid, about 0.1666117, and the secant; a coefficient read off one enclosure is not the coefficient. These are finite-model values only. No remainder constant of any other model is bounded or estimated from them.

8.2.8 The dictionary, the mandatory sentence and the gate fields

The comparison, consistency only. Contract item 3 asks for a comparison of the first-order coefficient with the matched $\mathbb{Z}^3$ value under (160). The derivative $1/6$ corresponds to $(1/6)/24 = 1/144$, the first-order coefficient of the static Wilson mean admitted by AW1 under I1.5 (59); the Cauchy interval divided by 24, $[6.9428640 \times 10^{-3}, 6.9460245 \times 10^{-3}]$, also contains $1/144$. Two facts make the agreement expected rather than informative. Both models share the per-link Casimir normalization, so $W\Omega$ has energy 3 in units of α in both; and in both the first-order mean is $2 \times \text{coefficient} \times \mathbb{E}[W^2]/3$. The skeptic adds that the slope is $2/(\alpha(9 + 3\rho_{11}))$ in general, so it equals $1/6$ only at $\alpha = \rho_{11} = 1$ (its reading 2, defect W-B). The finite graph is therefore consistent with the matched coefficient and nothing more. Nothing about the AQ model’s shift, its remainder constant or roadmap goal 2 is inferred.

The mandatory sentence. The preregistered template, which the producer’s report and the gate quote verbatim, reads:

The finite-graph derivative of $\langle W \rangle_{FG}$ at $\tau_{FG}=0$, under the stated normalization dictionary, is consistent with the matched $\mathbb{Z}^3$ first-order coefficient $1/144$; this is a consistency check on a different, finite model, not a prediction, not a confirmation of the $\mathbb{Z}^3$ value, and it does not transfer to any subsequential limit of the AQ construction or resolve roadmap goal 2.

The gate’s accepted statement carries it unchanged and adds one sentence: the finite graph is consistent with the matched coefficient and nothing more, and its one-face variant is a different finite model that differs at third order. Contract item 6 forbids two phrasings, a verb of prediction and a phrase of confirmation of the $\mathbb{Z}^3$ value. The producer’s checker scans its report for both as lowercase substrings with code spans removed; the skeptic’s whole-word scan of the prose finds

neither. The only two occurrences sit in code spans of the producer’s limitations section, where the contract’s claim-exclusion list is quoted. Appending the forbidden verb to the report aborts the producer’s checker. Neither phrasing occurs in this addendum.

Gate fields and flags. The preregistration requires three gate fields, and the gate exports them as required.

- gate fields: `transfers_to_aq` false, `model_is_finite_graph` true and `fg_coefficients_fitted` false;
- further flags exported false by the producer’s packet: `continuum_claim`, `weak_coupling_claim`, `uniform_wilson_claim`, `roadmap_goal_2_resolved`, `aq_K2_statement_from_graph`, `scientific_priority_verified` and `resolved_interaction_shift`, which refers to the AQ and $\mathbb{Z}^3$ shift;
- added by the review: `dynamical_claim` false.

The preregistered exclusions all apply: no interaction claim from a reference inside an enclosure, no uniqueness of any subsequential limit, no whole-sequence convergence or rate in N , no continuum or weak coupling, no transfer from a finite graph, no static-to-dynamic relabelling and no priority.

The Arb preview. The optional `arb_preview.py` is labelled PREVIEW ONLY. With python-flint 0.9.0 [26] it rebuilds the Haar moments (Round11 Q4) and the kinetic operator directly from the Round11 K4 differential formula as exact FLINT rational matrices, not from the checker’s Casimir decomposition, and runs Arb-ball inverse iteration at 400 bits, recording balls for the Ritz $\langle W_1 \rangle$ and E_0 at all 14 points in about four minutes. The checker reads its output only for a comparison record: all 14 Wilson balls, of width at most 6×10^{-60} , lie inside the exact enclosures. The checker never imports the preview (the review inspects its imports), and a source-edit run that deletes the preview output completes with every headline and point value unchanged (review finding N6). A finite-matrix value is not a full-space certificate, and the preview decides no admission.

8.2.9 Controls, replays and review

Contract controls. The contract lists 20 controls, and its preregistration mirror lists the same 20.

- named in contract item 5 and specific to the finite graph: `finite_graph_model_id`, `complete_residual_all_channels`, `complete_residual_ledger_itemized`, `certified_representation_tail`, `own_free_reference`, `no_transfer_to_aq`, `fg_coefficients_not_fitted` and `sign_convention_fixture`;
- inherited: the other twelve, which name $\mathbb{Z}^3$ or AQ objects; the producer gives each an explicit finite-graph analogue (defect D3).

All 20 reject damaging mutations through explicit exceptions, with none deferred (Table 41).

Checker. The forward checker runs 35 exact checks, implements all 20 controls as damaging mutations and records 80 rejections inside the control checks and 90 in the whole checker. It verifies the contract snapshot’s SHA-256 and reads from the contract the grid, the cutoffs, the dimensions, the tails, the dictionary, the target, the gate fields, the mandatory sentence and the forbidden phrasings; it records its own SHA-256 before evaluation and binds the Round11 solver’s exact source strings (basis, moments, kinetic coefficients, tail formula) to the snapshot. It uses only the standard library and never imports numpy, scipy, python-flint, the Round11 solver or the preview. The target, read as a 0/1 feasibility indicator (defect D5), evaluates to 1. A replay takes about 41 s, dominated by the 14 certificates and a 110-digit Ritz proposal of about 4 s each at $D = 8$, and outputs under `-B` and `-B -O` are byte-identical. The producer also disclosed a private source-edit audit in scratch,

Table 41: The 20 contract controls, their finite-graph reading in the forward checker and the damaging mutations rejected.

Control	Finite-graph reading and mutations rejected
missing_incoming_stars	both faces coupled (W_2 touches v_M of square 1) and all seven Casimirs; rejects W_2 omitted, $\rho_{11} = 0$ and the independent rotor
full_original_wilson_cover	W_1 uses all four links of square 1 in the seven-link space; rejects the open three-link path, a basis without z and the outer perimeter
wrong_delta_alpha_hbar_clock	units of α ($\tau/24$, energy 3) and of δ ($\tau/3$, energy 24) both give $1/144$; rejects $\tau/18$, $\tau/1152$ and the u -clock; non-unit fixture $\alpha = 5$, $\hbar = 7$
vector_versus_scalar_centering	the AT4 control is reproduced and item 6 uses vector centering; rejects scalar subtraction as a centering bound
first_order_mean_charged	$1/6$ is charged and the enclosures exclude 0; rejects a zero first order (parity misapplied) and a first order absorbed into the remainder
tau_scaling_exponent	certified ratios: linear mean, cubic remainder, quadratic energy, leakage $\tau_{FG}^{2(D+1)}$; rejects the remainder labelled second order and other mislabels
changed_model_relabelled	rejects $\rho_{11} = 2$, one plaquette, $D = 4$, an off-grid coupling, unequal faces, the AQ identifier and a flipped sign
coherent_evidence_tampering	twelve coherent tamperings with the packet hash rebound, rejected by exact recomputation
insufficient_verdict_retained	the acceptance rule; weaker tiers retained; rejects a tail certified only at $D = 6$ relabelled accepted
exact_arithmetic_admission	rejects float, Boolean, NaN and zero-denominator inputs and the decimal proposal as a value
root_n_misuse	the face channels interfere ($2XY > 0$); rejects a root-sum-square of face channels without interference, division by 7 and a root-mean-square over link classes
no_priority_or_continuum_claim	flags false; rejects continuum, priority, uniform-Wilson, weak-coupling and goal-2 flags set true
finite_graph_model_id	finite-graph identifier, graph name and labels; rejects the flag false, the AQ identifier and a missing graph name
complete_residual_all_channels	complete residual = retained + omitted = exact sector sum; rejects a dropped y channel, dropped product sectors, one sector as the norm, the finite-matrix residual and one face only
certified_representation_tail	$t_\ell = 45/69$ exact, with T_{joint} and per-link tails; rejects m_D , an added diagonal, an unsubtracted potential, the product or a per-link threshold used as the joint one, and a sampled sector
own_free_reference	the same code path gives $[0, 0] = \mathbb{E}[W]$; rejects $1/144$, $1/4$ and another path
no_transfer_to_aq	transfers_to_aq false; rejects a K_2 or goal-2 statement from the graph
fg_coefficients_not_fitted	exact Rayleigh–Schrödinger values; rejects the least-squares slope and the secant
sign_convention_fixture	a two-state fixture gives $+1/6$ under I1.5 and $-1/6$ when flipped, with a positive exact enclosure at $+1/10$; rejects the flipped reading
complete_residual_ledger_itemized	five items at 14 points; rejects a missing (b) or (e), a zero entry without a reason and a missing v_M row

which is not evidence: each listed edit aborts, among them the y face dropped from Hv , $\|W_1 + W_2\|$ set to 1 or 3, the shared-link Casimir weight set to 2, the tail read from the last retained shell, the free gap set to 4 and a changed Haar normalization. The free-gap edit was at first caught only through the report’s bound values; the checker now ties the free gap to $m_1 = 3$ and the potential norm to the identity configuration.

Independent pre-comparison package. The skeptic’s package holds a contract review with 15 readings, a derivation and a checker with 81 exact checks and 23 controls, the 20 contract ids and three extra ones: `ground_sector_spectator_disclosed`, `minus_tau_mirror_replay` and `truncated_bracket_not_full_graph`. It rejects 75 mutations, 71 of them inside the contract controls. In 14 private source-edit runs, 12 edits abort for the intended reason. Two do not, and both are accounted for: flipping the sign of V in the Jacobi series is harmless because E_0 is even, and replacing the sum of ledger items by their maximum is invisible below the 10^{-100} square-root rounding, which leaves a proof-text item. The package was written from the contract without opening any file under `forward/az2/`: the skeptic listed the 28 input names and hashed them against their repository sources. Its routes differ from the producer’s. It reduces the one-face model exactly to the Jacobi matrix of the x sector; certifies by Davis–Kahan with the one-component leak and the separation $3 - |\tau_{FG}|$ (route C); cross-checks by Hellmann–Feynman concavity with the Round11 Feshbach tail (route F), which is where the tails 45 and 69 enter; places the Ritz energy of the producer’s basis inside its own bracket by exact inertia of the Round11 pencils; and computes the Haar moments by a Beta-function route, cross-checked on 1330 triples. It uses no numpy, no python-flint and no import of the Round11 solver, and a run takes about 1 min 50 s. Its model-independent predictions all held: $1/6$ at both cutoffs, the free reference 0, $a_2 = 0$, the dimensions, the tails and the channel thresholds. Its one-face predictions belong to the other model (finding N8).

Correlated ancestry. The skeptic’s loop-2 response set the conditions “the Round11 two-plaquette graph, or smaller” and “coefficients computed algebraically, never fitted”, and its triage wrote the finite-graph labelling. The contract, the selection note and the modern update 4, all premises, already print $1/6$, the dimensions and the tails, so the single producer could copy the headline. The agreement is replay and re-derivation, independent in route and code only.

Post-comparison review. The review adds 28 exact checks, for 109 independent checks with the 81. It recomputes the two-face series of $\langle W_1 \rangle$, E_0 , $\langle W_1^2 \rangle$, $\langle W_1 W_2 \rangle$ and $\langle z \rangle$. It builds its own two-face enclosures from a degree-12 Rayleigh–Schrödinger vector with the exact complete residual and the Weyl separation, bounded by Davis–Kahan, with the negative points from the exact $(-1)^n$ parity of ψ_n ; these certify the full-graph value, are not cutoff- D certificates, and lie inside every producer enclosure. By its own exact Gram projections on the parity blocks it recomputes the leading-order omitted residual, $\tau_{FG}^{2D+2} \|Q_D(x+y)\psi_D\|^2$, which agrees with the producer’s ρ_Q^2 to relative 1.6×10^{-7} ($D = 6$) and 1.8×10^{-7} ($D = 8$) at $1/1000$; the j -class weight agrees to relative 9×10^{-8} , and the product channel vanishes at leading order. The review runs 27 source-edit runs on copies of the producer closure outside the checkout, always under `python -B`; a run takes about 4.5 minutes.

- One unmutated copy reproduces the frozen output byte for byte.
- Twenty weaken the validator of one contract control each; each copy then accepts the producer’s own damaging mutation and aborts at that control, which shows that every control is live.
- Six are semantic edits. Five abort: the one-face default in the series, the one-face certificate Hamiltonian, the free gap 4, the omitted residual dropped and the forbidden verb appended to the report. The sixth, deleting the preview output, completes by design.

The review’s own validator accepts the frozen packet and refuses 18 damaged packets, among them the Hamiltonian relabelled one-face while the numbers stay two-face, the one-face E_2 inserted into

the two-face series, an unnamed model, an enclosure replaced by the one-face interval, the derivative $1/144$, the free reference $1/4$, a nonzero a_2 , a tail of 56, the product threshold 45, the v_M row dropped and a dropped sub-label.

Widths. Table 42 compares the half-widths. The one-face pre-comparison widths are narrower, by about 30 times at $D = 6$ and 160 times at $D = 8$, and they belong to a different model. In the one-face model the omitted residual of the ground vector is a single spin-network component, $|(D+1)/2, 0, (D+1)/2\rangle$ with amplitude $\tau_{FG}v_D/2$. In the two-face model $(x+y)$ applied to the top shell leaks into the whole v_M channel $x^a y^b$ with $a+b = D+1$, with binomially grown amplitudes: about 115 times the one-face leak at $D = 6$ (6.2×10^{-15} against 5.4×10^{-17} at $1/10$). The producer’s Eckart angle recovers part of that gap. Within one model the producer’s cutoff- D certificates are the tightest of their kind, and the degree-12 vector is tighter only because it leaves P_8 . All intervals are valid (finding N3).

Table 42: Half-widths of the certified enclosures of $\langle W_1 \rangle$ at the positive grid points.

Certificate	1/1000	1/100	1/10
producer, two faces, $D = 6$ (Eckart with tail comparison)	1.10×10^{-29}	1.10×10^{-22}	1.14×10^{-15}
producer, two faces, $D = 8$	2.81×10^{-39}	2.82×10^{-30}	2.91×10^{-21}
skeptic pre-comparison, one face, $D = 6$ (Davis–Kahan)	3.59×10^{-31}	3.60×10^{-24}	3.71×10^{-17}
skeptic pre-comparison, one face, $D = 8$	1.78×10^{-41}	1.79×10^{-32}	1.84×10^{-23}
review, two faces, degree-12 vector (not a cutoff certificate)	7.27×10^{-48}	7.31×10^{-35}	7.77×10^{-22}

Replays and isolation. Unlike the statement loop AZ1, a single-direction loop is accepted unchanged by `replay_declared.py`, which reuses `replay_loop.py`; neither was edited. Two producer replays, under `-B` and `-B -0`, into fresh external directories reproduce `results.json` and `source-manifest.json` byte for byte, in about 41 s each, and `freeze.py verify` reports verified, inside the tool and in a separate run. The closure lists 34 files plus `freeze.json`, and every one verifies. The inputs are exactly `AGENTS.md`, the contract and the 26 shared premises, each byte-identical to its repository source; the only skeptic file among them is the Round29 AM2 review, and no `reverse/az2` package exists. The producer’s private scratch folder matches its disclosure by names and modification times. The producer read, outside its inputs and for protocol and code style only, the tools documentation, the freezing tool, most of the AY2 forward checker and the header of the AX2 report; the project’s instruction file was in its session context. None carries premise weight (finding N7). The skeptic saw, in a `git status` run after its final pre-comparison replays, only the names of the producer’s untracked files, and opened none of them before its freeze. Isolation is verified for repository inputs only, and all agents are correlated model agents.

8.2.10 Contract wording and non-blocking findings

The gate records the following as non-blocking.

- *W-A.* The contract does not fix which faces carry τ_{FG} . The two-face and one-face readings are both admissible and differ at third order; the gate names the two-face model (Section 8.2.2). Later contracts that reuse a finite graph should name every coupled face and its coefficient (finding N1).
- *W-B.* α , ρ_{11} and λ_2 are pinned only implicitly; the producer pins $\alpha = \rho_{11} = 1$ and $\lambda_1 = \lambda_2$.
- *D1.* `truncation_jmax` is preregistered although no per-link $j_{\max}$ is well-posed for the basis; it is mapped to ledger items (a)+(b)+(c). The skeptic adds that in the one-face ground sector the degree cutoff is a per-link cutoff.
- *D2.* `state_provenance` is the AQ-family string copied from the $\mathbb{Z}^3$ loops; the state is the unique ground state of H_{FG} , and no subsequence is involved.

- *D3*. Twelve control ids name AQ or $\mathbb{Z}^3$ objects; each is given a finite-graph analogue.
- *D4*. Item 4 asks for a tail residual bounding the truncation error of the coefficients; that error is exactly zero, and the second-order coefficient of $\langle W \rangle$ is exactly zero.
- *D5*. The target, the value 1 with $\geq$ on a conjunction of conditions, is read as a feasibility indicator.
- *D6*. `selected_after` names the AZ1 gate, which is not a premise and did not exist when AZ2 was frozen.
- *D7*. “must reproduce” in `parameters.observable` is read only as the consistency comparison of item 3.
- *D8*. The clock’s forbidden “exponent 24” coexists with the dictionary’s denominator 24, which is a coupling ratio, not an exponent.

Two review findings concern the controls. *N2*: the producer’s `missing_incoming_stars` control rejects “ W_2 omitted” as a damaging mutation, but relative to the frozen text that is the alternative admissible model; the control encodes the producer’s model choice and is not cited as refuting the one-face reading. *N5*: the one-face certificate-Hamiltonian edit aborts through an internal square-root guard, not a named check; a named consistency check between the proposal and certificate Hamiltonians would be clearer. The remaining findings concern the weaker Cauchy tier (*N4*), the preview (*N6*), the producer’s reads outside its inputs (*N7*) and the skeptic’s own one-face predictions (*N8*).

The retained correction of update 4. The modern lens’s update 4, a shared premise, describes a “certified-tail method” for the assistant-4 rehearsal. The skeptic’s contract review finds it wrong on two points, and the gate records the correction.

- Update 4 says each exact bracket of the rehearsal “is already a certified two-sided enclosure” with the tail threshold as its floor. But the solver’s `exact_bracket` brackets an eigenvalue of the *truncated* matrix and never reads the tail threshold; only Round11’s `certificate` does. The rehearsal’s Ritz values for $\langle W \rangle$ therefore enclose the truncated $\langle W \rangle_D$, not the graph’s $\langle W \rangle$.
- Its heuristic error estimate $\tau_{\text{FG}}^2/(t_\ell - 3)$ has exponent 2. The certified truncation effect in $\langle W \rangle$ has order τ_{FG}^{2D+1} and the leak order τ_{FG}^{D+1} , so the heuristic overstates the error by 13 to 33 orders of magnitude, for example 1.5×10^{-4} against a certified 1.8×10^{-23} at $D = 8$, $\tau_{\text{FG}} = 1/10$.

A packet following either reading would have been retained as `insufficient`; the skeptic’s extra control `truncated_bracket_not_full_graph` rejects it. The producer used neither: it certifies full-graph enclosures with the complete residual and the Round11 tail comparison.

8.2.11 Limitations recorded by the gate

The AZ2 gate records the following limitations verbatim.

- Only the Round11 two-square patch and the named two-face model (both faces coupled equally, $\alpha=\rho=1$), D in $\{6,8\}$ and the declared grid; a static ground-state mean with no dynamics, correlator, clock or mass-gap reading
- No statement about the AQ model, its subsequential limits, K_2 or the $\mathbb{Z}^3$ shift; roadmap goal 2 is not resolved; no continuum or weak-coupling claim; the zero second-order coefficient is a finite-graph centre-flip fact
- The dictionary $1/6 \leftrightarrow 1/144$ is a consistency check expected from the shared normalization (energy 3, $E[W^2]=1/4$); not a prediction and not a confirmation of the $\mathbb{Z}^3$ value
- The one-face model is not admitted; its values are the skeptic alternative reading, with the certified second-square shift $\tau^3/4212 + O(\tau^5)$ between the two
- Inherited without re-proof: the Round11 reduction (Q1-Q3), kinetic operator, spectrum, tail threshold and Theorem F; Kato, Weyl, Temple, Eckart and Davis-Kahan are cited; the checkers re-verify fixtures, not the reduction
- Model-agent review with correlated ancestry; not human peer review or formal verification; scientific priority unverified

Full derivation and reviewed implementation: [AZ2 forward](#), [contract review](#), [independent pre-comparison derivation](#), [independent model-agent review](#) and the [frozen contract](#). The advisor’s findings record AZ2 as the finite-model comparison attached to roadmap goal 2, in the only form the contract allows: exact coefficients of a computable, different finite model, which show what first- and second-order Wilson coefficients are on a truncated graph and are never transferred to $\mathbb{Z}^3$. The goal itself was met in its re-targeted static form by AW2 (Section 5), and AZ2 neither resolves nor extends it. The exact zero second-order coefficient on the graph mirrors the AW1 flip lemma as a consistency observation only; K_2 and the AQ model are untouched. The gate closes the cycle: investigation 10 of 10, after which scientific production stops.

9 Process history of the Round32 cycle

This section records how the cycle was run, using the repository files named in each paragraph. It adds no mathematical claim, and none of the planning numbers below is an admitted value. All times are UTC. The freezing tool wrote the `frozen_at` stamps and the gate tool wrote the `completed_at` stamps, both from the machine clock. Like commit dates, they are self-reported and are not trusted timestamps. The roles are those of the status section: an advisor who owns contracts, gates and publication; a skeptic; three methodological lenses; one research assistant per lens after each sub-round; and forward and reverse producers. All of them are model agents with correlated ancestry, working under the project’s workflow contract [1].

9.1 Authorization and brief

A user instruction of 23 September 2026 authorized Round32 as an adaptive cycle. It named an expert panel of Newton/Tesla, Jung/Pauli and Penrose/Feynman lenses, with occult and mystical source reading, together with an advisor, a skeptic and subagents for coding and simulation. The structure was three panel deliberation loops, then five sub-rounds of two investigations each. After every sub-round each lens would receive a research assistant, and the panel could change goals, directions, skills and roadmap. Publication was by merge to the main branch. The panel was also allowed to produce statements, derivations and experiment-setup images where it judged them more useful than a numerical certificate.

The advisor’s brief of the same day [8] restated the admitted Round29 and Round31 results with their quantifiers [3, 4, 5, 6]. It listed the five ranked Round31 roadmap goals and five technical observations for the panel to test, none of them admitted. Before reading any panel memo, the advisor optimized the AT4 Poisson certificate over its cutoff. The result was a dynamics-plus-tail floor of about 1.27×10^{-6} at $s = 1$, whatever the state term. An Arb/mpmath cross-check tool [26, 27] confirmed that independent enclosures lie inside the recorded AT5 intervals. Both were planning observations; AV2 later proved the floor as a retained failure (Section 3.7).

9.2 Three deliberation loops

Loop 1: triage, memos and plan v1. The skeptic wrote first. Its prospective triage was written before any Round32 expert memo, contract or producer work existed, and it read nothing under `experts/` [9]. It inventories every admitted result with its quantifiers, sets a complete error ledger for each roadmap goal, assesses the brief’s five observations, takes a position on direction and fixes a minimum acceptable deliverable for each sub-round. The companion catalogue `prospective-controls.json` names 60 control ids in seven groups: universal 12, cap 15, shift 9, finite graph 5, uniform 9, state 5 and continuum 5. Each id names a damaging mutation that every producer checker must reject by explicit exceptions that stay active under `-0`. The catalogue’s status is `prospective_skeptic_requirements_not_executed`.

Each lens then wrote a memo, a source ledger and a structured recommendation [11], and the Jung/Pauli lens also wrote the occult reading ledger [12]. The advisor’s synthesis records eight points of agreement:

- ordering the creation product removes the square root;
- the Poisson floor is an artifact of the kernel;
- the shift of the centered correlation is second order, so the only first-order Wilson effect is the uncentered mean $\tau/144$ and roadmap goal 2 was re-targeted to it;
- route B for the uniform model;
- finite-graph certificates are a different model;
- the direction rules;
- the Jung pre-registration rules;
- no occult or historical source became a premise.

Three disagreements were left open: a second-order Duhamel comparison against the window kernel, preview estimates of K_2 spanning 1.9×10^4 to 1.6×10^8 , and the rank of the finite-graph loop. Plan v1 already had the five sub-rounds of Table 3. The modern memo arrived after the synthesis draft and was merged into the same note. It added the Gauvin lead, the J_0 problem for the uniform model and a concrete design on the Round11 two-plaquette graph [7].

Loop 2: critique of plan v1. The inputs were the skeptic’s loop-2 response and the three lens responses. Three loop-2 sessions were interrupted by an account spend limit and relaunched on a smaller model; their written responses are complete. The skeptic reported two facts that changed numbers in every memo, including its own. A bulk factor lies in 49 omitted-face owner sets, not the bound 84 [2]. The second-order multiplier of the Wilson mean is 1, not 4. The response also listed eight vetoes.

The advisor adopted ten corrections:

- the count of 49 faces per factor;
- the multiplier of 1, with the resulting range for K_2 ;
- the AV1 target $4/10^7$, set by AV2 feasibility;
- reverse premise isolation, enforced by the tools;
- separate forward and reverse routes;
- confirmation of the window kernel;
- the J_0 re-freeze for the uniform model;
- the finite graph after sub-round 2, never as the resolution of goal 2;
- an interim Gauvin sentence;
- the Jung pre-registration block with the skeptic’s four amendments.

Five vetoes were recorded (Section 1.3). The modern lens confirmed the window constants with sympy and Arb and withdrew its own taper. It also proposed the link-flip set E , which the advisor checked in all three orientations on a 7^3 block, and Eckart’s inequality as a candidate route for the cutoff-vector step. Each lens accepted the errors the skeptic itemized in its memo, in its loop-2 response or in loop 3.

Loop 3: objection pass and freeze. Each lens and the skeptic read the draft freeze note, `plan.json` and the AV1 and AV2 draft contracts, and wrote a sign-off. Nobody vetoed the plan. The skeptic signed off the plan and both contract texts, but vetoed freezing AV1 until its selection note stopped contradicting the contract in three places: the note gave a 10^{-6} target where the contract has $4/10^7$; it asked producers to find unaided a normalization trap that the contract spells out; and it said “independently” where “separately” is correct. The skeptic’s loop-3 note also fixed the conditions under which the link-flip antisymmetry holds. These are the identity $H(\tau, \kappa) \mapsto H(-\tau, -\kappa)$, antisymmetry in τ only at the zero triple, a whole-set statement for AQ limits, and no $O(\tau^3)$ remainder from oddness alone, and they entered AW1 as obligations. The

advisor applied every condition, recorded the change from plan v1 to v2, adopted the attribution sentence fixed by the skeptic’s Gauvin diff record (Section 9.7) and froze plan v2.

Rules added to the method snapshots at this point [13]:

- *Newton*: compute the target’s first-order coefficient exactly before charging a first-order error budget;
- *Tesla*: name source, load, clock and transfer element with their exact constants, and use the $\pm\tau$ mirror as a control;
- *panel guide*: the Jung pre-registration rules, with two cautionary cases, Jung’s 1952 astrology statistics and Besant and Leadbeater’s occult chemistry [37, 42].

The plan record. `plan.json` carries the status `frozen_after_loop3`. Its `history` field held six dated entries at the AZ2 gate: the change from plan v1 to v2 (23 September), the four panel updates, and one entry for a vocabulary extension. Its `preregistration_vocabulary_extensions` field holds two entries, both dated 24 September. The first followed the Jung assistant-2 audit. It allows exact τ -dependent selected triples, the `statement+skeptic` direction, and `n/a` reference routes and grid couplings for statement and finite-graph loops. AX1 was already frozen, so it is recorded as grandfathered with its contract unamended. The second followed the Jung assistant-4 audit and closes five further gaps:

- a state provenance may name more than one family;
- the required gate fields are topic-specific and include `rate_in_N_claimed` where a rate is at stake;
- a `mandatory_sentence_template` field is required wherever a template is cited;
- the phrase “uniqueness of the AQ state” becomes “uniqueness of any subsequential limit”;
- a certification loop declares its target as a feasibility check.

AY1 and AY2 are grandfathered in the same way.

9.3 Contracts, gates and chronology

Table 43 lists the ten contracts in freeze order. Every contract was stamped by the freezing tool. The tool checks the required fields and the declared premises and refuses to re-freeze. Since sub-round 1 it also requires the pre-registration control list to equal the contract’s control list [14]. Five contracts are paired (AV1, AV2, AW1, AX1, AY1), three are single-direction with an independent skeptic replay (AW2, AX2, AZ2) and two are statement loops with a skeptic review (AY2, AZ1).

The ten gated investigations. They ran from 21:23 on 23 September to 06:06 on 24 September, and every verdict is `accepted_within_scope`. Nine of them took between 38 and 62 minutes from contract freeze to gate. AZ2, frozen together with AZ1 and gated after it, took 97 minutes. For each of them the skeptic wrote a contract review from the frozen text as part of its pre-comparison package. Each successor contract names its predecessor’s gate in `selected_after`, with three exceptions:

- AV1 names the loop-3 freeze note;
- AY2 carries the literal placeholder `<preceding gate>`, which its gate records as contract defect D1;
- AZ2 names the AZ1 gate, which did not yet exist when AZ2 was frozen.

Only AV1 carries a recorded amendment. Before any producer read it, `plan.json` was removed from its shared premises, because the plan is a living record and premise snapshots must stay byte-identical to their sources.

Table 43: Round32 chronology from the contract and gate records (UTC, 2026). Dir. is the contract’s direction: P = paired, S = single+skeptic, T = statement+skeptic.

Loop	Dir.	Contract frozen	Gate completed	Verdict	Sub-label
AV1	P	09-23 21:23:47	09-23 22:10:05	accepted	UL [†]
AV2	P	09-23 22:10:41	09-23 22:59:05	accepted	RU
AW1	P	09-23 22:59:30	09-24 00:01:23	accepted	SD [†]
AW2	S	09-24 00:01:24	09-24 00:40:12	accepted	SD
AX1	P	09-24 00:40:52	09-24 01:30:55	accepted	UL [†]
AX2	S	09-24 01:30:56	09-24 02:11:13	accepted	RU
AY1	P	09-24 02:11:14	09-24 03:06:48	accepted	UL
AY2	T	09-24 03:09:12	09-24 04:00:01	accepted	UL; SD
AZ1	T	09-24 04:29:40	09-24 05:18:19	accepted	none
AZ2	S	09-24 04:29:40	09-24 06:06:38	accepted	SF; SD

“accepted” is the verdict `accepted_within_scope`. Sub-labels: RU = `reference_unresolved`; SD = `static_not_dynamic`; UL = `uniform_local_closeness_not_uniqueness`; SF = `sign_certified_finite_graph`. For AY2, UL is the primary sub-label and SD the secondary one; for AZ2, SF is the primary sub-label and SD the secondary one. [†]The label is named by the skeptic’s review and the producer packets; the gate states the same restriction in words but names no label.

AZ1 and AZ2. Both contracts were frozen after panel update 4, 0.03 s apart, and both packages were developed in parallel. In both loops the skeptic’s pre-comparison package was committed before the producer’s: 649db92 before 8359876 for AZ1, and 4bae265 before dc817d8 for AZ2. The AZ1 gate was completed at 05:18:19 with the verdict `accepted_within_scope` and no sub-label (Section 8.1). The AZ2 gate was completed at 06:06:38 with the verdict `accepted_within_scope`, the sub-label `sign_certified_finite_graph` and the secondary sub-label `static_not_dynamic` (Section 8.2). The gate resolved a contract wording defect, the unfixed face coupling of the finite graph, by naming the two-face model that the producer had computed and disclosed.

Closing statement. The Round32 cycle consists of exactly ten gated investigations, AV1 to AZ2, in five sub-rounds of two, and all ten are accepted within their stated model scope. Five are paired, three are single-direction with an independent skeptic replay and two are statement loops with a skeptic review. As its contract and gate state, scientific production stopped after AZ2. Nothing after the AZ2 gate, including this manuscript, the closing panel update, the site data and the release checks, adds a research loop or changes an admission.

9.4 Sub-round updates, assistants and lens reviews

After each sub-round, each lens reviewed the gates and wrote an update, one assistant per lens ran the tests that lens had requested, and the advisor wrote a panel update [8, 11]. The assistant packages count zero research loops, and nothing in them decides an admission. Each of the first four updates kept the goals of the remaining sub-rounds.

Panel update 1 (23 September, after AV1 and AV2). The Newton/Tesla assistant reproduced the exact Haar moments by two routes, rebuilt the 49/15/82 face enumeration from the I1 table without the producers’ code, reproduced the forward AV1 constant bit for bit, and reproduced the window budget of about 1.832×10^{-7} . The Jung/Pauli assistant ran the $\tau = 0$ null replay and the τ -scaling exponents, 1.0000 for the linear tiers against exactly 0.5 for the square-root bound. Its audit of the AV2 pre-registration found that the frozen contract lists 24 controls while its mirror lists 23. Both producers had implemented the missing id, so the gate records a non-blocking clerical defect, and the freezing tool now rejects such a mismatch. The modern assistant built a python-flint harness and checked the flip parity exhaustively on 6^3 and 9^3 blocks. The AW1 contract gained

the free-link cross-check, a full flip enumeration, pinned K_2 terms and a positive demonstration at nonzero κ . The AZ2 cutoff was reread as a polynomial degree.

Panel update 2 (24 September, after AW1 and AW2). Roadmap goal 2 was recorded as met in its re-targeted, static form. The AX1 and AX2 contracts gained three requirements: a proof that route B ties every face coefficient to τ ; an itemized incidence table with a no-double-count control; and binding of the admitted AX1 bound, never the target. The update also records that the gate-bound K_2^+ is the skeptic’s conservative itemization. The scratch-folder rule of Section 9.5 was adopted here.

Panel update 3 (24 September, after AX1 and AX2). The historical lens preferred a single producer plus skeptic for AY1, and its dissent was recorded; AY1 stayed paired because its padding-family verification, first-order density and R-local K_2' are new derivations. The update also settled the content of AZ1 and the recipe for AZ2. AX1’s frozen contract carried a mis-rounded prose threshold. Both producers and the skeptic caught it, and the gate binds exact rationals only.

Panel update 4 (24 September, after AY1 and AY2). The Jung assistant’s audit recorded nine defects in the sub-round-4 record, none of which decides admission. They led to the second vocabulary extension and to a new final-integration audit item on shared field names. The modern lens withdrew its sub-round-3 expectation that K_2' would be smaller than K_2^+ : it is about four times larger, and the panel update records the expectation as a category error. The historical lens withdrew its AY1 dissent. The AZ1 and AZ2 contracts gained the requirements that the panel update lists.

Panel update 5 (24 September, after AZ1 and AZ2). After the AZ2 gate the cycle closed with a fifth panel update. The closing lens updates and the assistant-5 packages, one per lens, are recorded in [panel update 5](#), which the advisor writes after this manuscript pass. As for the earlier updates, the assistant work counts zero investigations and decides no admission; no further investigation follows it in Round32.

Web checks. The historical and modern lens updates record dated web searches after every sub-round, and the Jung lens searched in sub-round 4. None found a rigorous result matching the question at hand. The leads were read at abstract depth:

- a quantum Dobrushin condition at high temperature [21], a named lead for the uniqueness obligation of AY2;
- Lieb–Robinson bounds with exponential-in-volume tails [22], a candidate for its boundary-dynamics obligation, with hypotheses unchecked;
- Nair’s outline for a three-dimensional gap [24], a proposal and not a four-dimensional theorem;
- a review of gauge-theory quantum simulation [25]. The Jung ledger records its abstract as read, while the historical lens saw it only as a search summary.

The modern lens recorded Faizal and Shabir’s claimed reflection-positive $SU(N)$ construction [23] as a lead for a future skeptical diff only, with no priority, correctness or import statement. Its claim that the strong- and weak-coupling limits coincide is exactly the kind of estimate that AZ1 is contracted to state this family does not supply.

9.5 Process rules added during the cycle

Four rules were added as repairs by record. None of them edits a frozen file.

Scratch disclosure. The skeptic’s AW1 review, finding N1, found that both AW1 producers had used the shared scratchpad root between 23:14Z and 23:20Z without disclosing any reads. The forward producer’s `k2.py` had overwritten the skeptic’s own pre-comparison scratch file at the same path. Whether any agent read another’s file cannot be determined from the files. Isolation was therefore verified for repository inputs only, and scratch-channel independence is unrefuted but unaudited. From sub-round 3 on, producers keep private scratch folders and disclose any scratchpad read in their reports. The skeptic keeps its own work in private folders as well, and checks the disclosed folders by name and time.

Control mirror. After the AV2 clerical defect, the freezing tool refuses any contract whose pre-registration control list differs from its control list.

Prose never admits. A threshold written in prose never enters the admission path. Checkers read exact targets from the hash-checked contract snapshot.

Vocabulary. The vocabulary extensions are recorded once, in the plan, and later contracts write the extended vocabulary directly.

Two earlier rules held throughout. Same-author passes are never called independent review, and agreement between producers is described as “separate”, not “independent”, when the producers share a premise.

9.6 Tools

The shared tools compute no scientific result [14]:

- `freeze.py` snapshots the declared premises into a producer’s inputs. It replays the checker under normal and `-O` Python into fresh external directories, requires byte-identical outputs, writes and verifies `freeze.json`, and rejects a reverse inventory that differs from `AGENTS.md`, the contract and its shared premises.
- `freeze_contract.py` validates and stamps a contract.
- `record_gate.py` writes a gate from the skeptic’s reviewed statement and limitations, with complete evidence bindings, and can never widen the reviewed scope.
- `arb_crosscheck.py` compares exported rational enclosures with python-flint Arb balls and mpmath intervals [26, 27]. It is never the admission arithmetic.

Every producer checker writes a non-empty `checks` list and Boolean claim flags. All admission arithmetic uses exact `fractions.Fraction`.

9.7 Sources, attribution and disclosure

Gauvin. The modern memo identified Gauvin’s arXiv:2503.15539 [15] as the closest external analogue of AM2 and AQ, and asked for a version diff before any attribution. The Round29 source ledger already recorded the v3 supplement as the AM template, so AM2 is not claimed to be independent in any case. The skeptic downloaded v1, v2, v3 and both supplements, recorded their hashes and diffed them [10]. Every item of the fixed-cutoff theorem, its lemmas and supplement A.6–A.11 first appears in v2, submitted 11 September 2026; none appears in v1. The v3 files are byte-identical to those bound in Round29. Gauvin credits the creation-operator method to Bravyi, DiVincenzo and Loss [16]. The record proposes this sentence, quoted with its section signs and dashes typeset:

“The AM2 creation-expansion gap and the AQ construction follow strategies that first appear publicly in Gauvin arXiv:2503.15539v2 (2026-09-11; §II.C–D and supplement A.6–A.11). The

project read them in v3 (2026-09-14) and records them as its template. Gauvin credits the creation-operator method to Bravyi–DiVincenzo–Loss (CMP 284, 2008, §§2.1–2.4). v1 (2025-03-06) contains none of these items. The SU(2) selected-reference application and its explicit constants are this project’s record; scientific priority is not claimed.”

The loop-3 freeze adopted the shorter form printed in the status section, which also says that no independence is claimed. Under both wordings `priority_statement_allowed` stays false. No inference about the direction of influence is drawn.

Historical, occult and methodological reading. The lenses read selected, located passages, and their ledgers record each passage’s edition, reading depth and access failures:

- *Newton*: the General Scholium’s “I frame no hypothesis” paragraph and three Latin sentences of his “Hermes” commentary [30, 31];
- *Tesla*: two patent specifications, one page of the Colorado Springs notes, an essay and a lecture [32, 33, 34, 35, 36];
- *Jung/Pauli*: Jung’s astrology experiment, Pauli’s Kepler essay, Kepler’s recorded null on harmonic proportions in the periodic times, Dee, Agrippa, Besant and Leadbeater, and the 1995 AIR remote-viewing evaluation [37, 38, 39, 40, 41, 42, 43].

The Pauli–Jung letters, a CERN letter record and the ETH Pauli archive remained leads or were inaccessible. Two CIA locators redirected to a generic page. In sub-round 4 the Jung lens read two pre-registration sources [44, 45] as methodological precedent only.

These readings motivated methodological rules only: pre-registration, complete accounting, the $\pm\tau$ mirror as a control, and the two cautionary cases. The one government record read concludes against the programme it evaluates, and government custody is treated as provenance only. *No occult, mystical or historical source changed any premise.* The occult ledger states that no occult proposition became a physical premise, in loop 1 and again in sub-round 4 [12]. Each of the four panel updates records that no such source changed a premise in that sub-round.

Sources named in the text. The addendum uses the following sources without re-deriving them:

- the Kogut–Susskind Hamiltonian form [17];
- the Nachtergaele–Sims dynamics used by AQ1 [18];
- Yarotsky’s and Henheik–Teufel–Wessel’s stability theorems, whose constants are not evaluated for this model [19, 20];
- the standard results collected in [28].

Each is cited, not machine-checked. The continuum problem these fixed-spacing certificates leave open is the Clay problem statement [29]. Investigation counts are never a fraction of it.

AI assistance. The advisor, the skeptic, the lenses, the assistants and the producers are AI model agents directed by the human author. The AY1 reverse report, for example, identifies its author as Claude, an AI model. The text of this addendum was also drafted with AI assistance. The agreement of these roles is correlated model-agent work, not external human peer review.

10 What the Round32 record does and does not establish

All ten Round32 gates record the verdict `accepted_within_scope`, and each acceptance holds only inside the model its gate names. The record uses three models, and they are kept apart:

- the zero-selected, nonuniform AQ family at fixed spacing (AV1, AV2, AW1, AW2, AY1 and AY2), with the selected triple exactly (0,0,0), the Haar reference and the cover $R = \{0, e_z\}$;

- the uniform Kogut–Susskind SU(2) model at fixed spacing and strong bare coupling, $g^4 \geq 9.6 \times 10^9$ at the cap, handled by route B (AX1, AX2 and AZ1);
- the Round11 two-plaquette graph with both faces coupled (AZ2), a different finite model.

A result proved in one model is not a claim about another. AX1 re-derived the admitted chain for the uniform model; it did not import it. AZ2 transfers nothing to either lattice model. Table 44 lists, for each gate, its model, its sub-labels and the claims its frozen contract excludes. Every contract also carries the seven preregistered exclusions: no interaction claim when the free reference lies inside an enclosure; no uniqueness of the AQ state, or of any subsequential limit in the wording of AZ1 and AZ2; no whole-sequence convergence or rate in N ; no continuum or weak coupling; no transfer from a finite graph; no static shift relabelled as dynamical; and no scientific priority.

Table 44: The ten gated statements: model, sub-labels and the claim exclusions of each frozen contract, in addition to the seven preregistered ones. Sub-labels are abbreviated as in Table 43 (RU, SD, UL and SF). “In words” means that the gate states the restriction without naming the label.

Loop	Model	Sub-labels	Excluded by the contract
AV1	zero-selected AQ family, both signs of $ \tau \leq 10^{-8}$, every subsequential limit	none (UL in words)	a Euclidean certificate; an interaction shift or its sign; uniform Wilson theory; uniqueness, whole-sequence convergence or a rate in N ; continuum; Yarotsky/HTW constants; priority; the value or sign of $\omega(W)$ beyond $ \omega(W) \leq D$; the coefficient 1/144; the vanishing of first-order $C(s)$ or $\omega(W^2)$
AV2	the same, $\tau = +10^{-8}$ (-10^{-8} a replay), node $s = 1$	RU	a resolved shift (free value inside); the full $[0, 128]$ grid; uniform Wilson theory; uniqueness or a rate; continuum; priority; the second-order Dyson refinement; any shift of $C(s)$
AW1	the same, both signs	none (SD in words)	an admitted enclosure of $\omega(W)$; a dynamical or mass-gap correction; any third-order remainder from oddness alone; uniform Wilson theory; uniqueness or a rate; continuum; priority
AW2	the same, $\tau = \pm 10^{-8}$ fixed by the frozen AW1 rule	SD	a dynamical correction or mass-gap statement; a correlation shift; uniform Wilson theory; uniqueness; continuum; priority
AX1	uniform Kogut–Susskind model, route B, both signs of $ \tau \leq 10^{-8}$	none (UL in words)	weak coupling or continuum; a uniform-model Euclidean datum; a uniform Wilson-mean sign certificate; uniqueness or a rate; priority
AX2	the same, $\tau = +10^{-8}$ (the U_E mirror a replay), $s = 1$	RU	a resolved shift; a uniform Wilson-mean sign certificate; weak coupling or continuum; uniqueness; priority
AY1	zero-selected AQ family, families F1 and F2, both signs	UL	uniqueness; whole-sequence convergence or a rate in N ; translation invariance; continuum; priority

Loop	Model	Sub-labels	Excluded by the contract
AY2	the same	UL; SD	uniqueness; whole-sequence convergence or a rate in N ; translation invariance; boundary independence of dynamics; an interaction shift or dynamical claim from the static tier; continuum; priority
AZ1	uniform model along a named trajectory (a_n, g_n) , a hypothesis	none	any continuum construction; any estimate uniform in a ; weak coupling; priority; the two forbidden phrasings of its template
AZ2	Round11 two-plaquette graph, two-face model, $D \in \{6, 8\}$, six grid points	SF; SD	any statement about the AQ model's shift or K_2 from the graph; resolution of roadmap goal 2; continuum; priority; the two forbidden phrasings of item 6

No shift of the centered correlation. AV2 and AX2 certify the centered Euclidean Wilson correlation at the single node $s = 1$, with radii about 1.832×10^{-7} and 1.912×10^{-7} against the target 10^{-6} . In both, the free value $e^{-3}/4$ lies inside the interval, so no shift, sign or coefficient of $C(s)$ is resolved, and AW1 proves that its first-order shift vanishes. The only interaction effect certified in the $\mathbb{Z}^3$ models is the sign of the uncentered, static Wilson mean at $\tau = \pm 10^{-8}$ in the zero-selected family (AW2). It is an equal-time property of the states, not a dynamical correction, mass shift, susceptibility or correlation shift, and it is not certified in the uniform model: AX1 transfers only the first-order coefficient $+\tau/144$, not a remainder constant. The AY2 two-sided tier, which puts $\|\rho_R - P_R\|_1$ between about 4.379×10^{-10} and 4.405×10^{-10} for every limit, is also static.

No uniqueness. AY1 proves that any two subsequential limits of the two named construction families, at the same coupling, lie within $2D \approx 2.72 \times 10^{-8}$ of each other in trace norm on R , share one explicit first-order density, and differ by at most $2K'_2\tau^2 \approx 2.7 \times 10^{-12}$ beyond it, with $K'_2 \approx 13418$. That is closeness on one fixed region at one coupling. It is not uniqueness, whole-sequence convergence, translation invariance, a rate in N or boundary independence of the dynamics. AY2 lists what remains unproved, and Table 45 summarizes its obligations table (Table 28 gives the full rows). AY2 also records a falsifying scenario that the present bounds cannot exclude: two admissible densities that agree to first order and differ at second.

Nothing uniform in a . Every lattice result holds at one fixed spacing. AZ1 names the one necessary estimate that the fixed-spacing family supplies along a trajectory: the volume-uniform gap $\alpha/16 = g^2/(32a)$ at fixed a , for $g^4 \geq 9.6 \times 10^9$. Every trajectory with $g_n \rightarrow 0$ leaves that regime after finitely many steps, so the family supplies no estimate uniform along $a_n \rightarrow 0$ on such a trajectory. Inside the regime the lattice-unit floor $a\Delta \geq g^2/32 \geq 1250\sqrt{6} \approx 3061.86$ holds, so the certified gap is a cutoff-scale quantity, and a finite physical mass cannot be approached inside the admitted regime. That is the failure of a sufficient certificate, not a statement that the gap closes. The requirements of a continuum construction are all missing (Table 32): estimates uniform in a , reconstruction hypotheses and physical scale control, together with the AY2 obligations and nontriviality. A rate in N would not be a rate in a .

Nothing transferred from the finite graph. AZ2's exact coefficients belong to the two-face model on the Round11 graph. The derivative $1/6$ corresponds to $1/144$ under $\tau_{\text{FG}} = \tau/24$, which is a consistency check expected from the shared normalization and nothing more. The zero second-order

Table 45: The six AY2 obligations, all unproved, in summary. Each route is a named candidate, not an executed proof.

Obligation	Missing premise	Candidate route
O1: uniqueness of the limit	a two-state estimate forcing two limits to agree on R	HTW- or Dobrushin-type stability with evaluated constants
O2: whole-sequence convergence	a Cauchy estimate in N	a boundary-influence refinement of the AM2 fixed point
O3: translation invariance	covariance together with O1 or O2	identification over every exhausting sequence
O4: a rate in N	any N -dependent estimate	a quantitative Cauchy estimate with a per-shell factor below one
O5: boundary independence of dynamics	a comparison of the F1 and F2 dynamics on compact windows	Duhamel and Lieb–Robinson with the AQ1 Nachtergaele–Sims constants
O6: dynamics of the padded family	an infinite-volume dynamics for F2	Nachtergaele–Sims for the owner-set interaction

coefficient is a finite-graph centre-flip fact. Nothing about the AQ model, its subsequential limits, K_2 , K'_2 or a $\mathbb{Z}^3$ shift follows from the graph, and roadmap goal 2 is not resolved by it. The one-face variant, which the contract text also admits, is a different finite model; it is not admitted, and it differs at third order.

Standing of the evidence. Every admission uses exact rational arithmetic with directed rounding; the decimals in this addendum are previews, and the Arb and mpmath comparisons never admit a value. The producers, the skeptic, the advisor, the lenses and the assistants are model agents with correlated ancestry. Their agreement is separate model-agent work, never external human peer review or formal verification, and in several loops the premises already printed the headline values. Contract wording defects were recorded as non-blocking by the gates and did not change any frozen text. Inherited premises (I1, AL1, AM2, AQ1, AQ2, AT4 and the Round11 graph) are cited, not re-proved, and the standard theorems of [28] are cited, not machine-checked.

Attribution and priority. No scientific priority is claimed or verified for any Round32 result. The attribution sentence fixed by the skeptic’s Gauvin version-diff record and adopted in the plan freeze stands unchanged: the AM2 creation-expansion gap and the AQ construction follow strategies present in Gauvin arXiv:2503.15539 [15] from v2 (2026-09-11; absent from v1 of 2025-03), which itself credits the creation-operator method to Bravyi, DiVincenzo and Loss (2008) [16]; the SU(2) selected-reference application and its explicit constants are this project’s record; scientific priority is unverified and no independence is claimed. The claimed reflection-positive SU(N) construction of Faizal and Shabir [23] remains an unaudited lead, read at abstract depth only; nothing in Round32 depends on it.

What comes next. The advisor’s [Round32 roadmap](#) records the next goals as planned and unexecuted, each with a named missing premise. Any of them needs a fresh prospective contract. The number of investigations, pages or named contributions is not a measure of progress on the continuum problem. The four-dimensional continuum Yang–Mills problem remains open [29].

A Contribution and evidence ledger

The names below are internal attribution aliases for this project, taken from the advisor’s [findings record](#). Established methods keep their existing authorship: Peter–Weyl and Weyl integration,

Kato perturbation theory, the Duhamel formula, the Temple, Eckart, Weyl and Davis–Kahan inequalities, Nachtergaele–Sims dynamics and the creation-operator method credited by Gauvin to Bravyi, DiVincenzo and Loss. No alias establishes scientific priority. The dependency identifiers are those of the findings record: **r32-** marks a Round32 investigation, **r29-**, **r31-** and **r21-** earlier admitted premises, and **r11-** the Round11 two-plaquette operator and its gap.

Table 46: The ten Round32 contributions. Each summary is scoped by its gate’s accepted statement; decimals are previews of exact rationals.

Alias	Investigation and scoped contribution	Depends on	Gate
HNM-C-AV1	<i>First-order local state lemma for the zero-selected AQ subfamily.</i> A trace-distance bound on the Wilson cover, linear in τ and the same in every volume: about 1.3612×10^{-8} at the cap for every AQ1 subsequential limit; the crude tier fails and is retained.	r29-am2, r29-aq1, r29-aq2, r31-at4	av1-gate.json
HNM-C-AV2	<i>Window-kernel certificate for the centered Wilson correlation at the original cap.</i> A C^2 window replaces the Poisson cutoff; $C(1)$ lies within about 1.832×10^{-7} of an exact datum at $\tau = 10^{-8}$, with the free reference inside.	r32-av1, r31-at4, r31-at5, r29-aq1	av2-gate.json
HNM-C-AW1	<i>Parity theorem, link-flip antisymmetry and the first-order Wilson mean with an itemized second-order remainder.</i> No first-order shift of the centered correlation; the Wilson mean is odd in τ , equal to $+\tau/144$ up to $K_2^+ \tau^2$ with $K_2^+ \approx 3354.80$; the AW2 coupling rule is frozen in advance.	r32-av1, r29-am2, r29-aq1, r29-aq2	aw1-gate.json
HNM-C-AW2	<i>Sign-certified enclosure of the Wilson mean at the AW1-frozen coupling.</i> The static Wilson mean lies in about $[6.911, 6.978] \times 10^{-11}$ at $\tau = 10^{-8}$, mirrored at -10^{-8} ; zero is excluded with margin about 206.	r32-aw1, r32-av1	aw2-gate.json
HNM-C-AX1	<i>Route-B re-derivation for the uniform Kogut–Susskind $SU(2)$ Hamiltonian at fixed spacing.</i> Re-frozen contraction constant $J'_0 = 29/10^8$, a unique ground state and gap $\alpha/16$, the 88-face incidence on the cover, a state bound of about 1.4446×10^{-8} , and the flip identity and $+\tau/144$ transferred by proof.	r32-av1, r32-aw1, r29-am2, r29-al1	ax1-gate.json
HNM-C-AX2	<i>Uniform-model window certificate for $C(1)$ at the original cap.</i> In the uniform model at strong bare coupling, radius about 1.912×10^{-7} at $s = 1$, with the free reference inside.	r32-ax1, r32-av2	ax2-gate.json
HNM-C-AY1	<i>Uniform local closeness of all AQ-type subsequential states and the boundary-independent first-order density.</i> Any two limits of the two named families lie within $2D \approx 2.72 \times 10^{-8}$ on R , share one first-order density and differ by at most $2K'_2 \tau^2 \approx 2.7 \times 10^{-12}$ beyond it, $K'_2 \approx 13418$; not uniqueness.	r32-av1, r32-aw1, r29-am2, r29-aq1, r21-i1	ay1-gate.json

Alias	Investigation and scoped contribution	Depends on	Gate
HNM-C-AY2	<i>Statement on state identification.</i> What the closeness lemma does and does not establish: six unproved obligations with their missing premises, a falsifying scenario, and a certified static two-sided distance from the Haar product on R of about 4.38 to 4.41×10^{-10} .	r32-ay1, r32-av1, r32-aw1, r29-aq1	ay2-gate.json
HNM-C-AZ1	<i>Continuum-trajectory statement.</i> Along a trajectory with $g_n \rightarrow 0$ the fixed-spacing family supplies one necessary estimate, the volume-uniform gap $g^2/(32a)$ at fixed a for $g^4 \geq 9.6 \times 10^9$, and none uniform along $a_n \rightarrow 0$; the lattice-unit floor $a\Delta \geq 1250\sqrt{6}$ and the missing continuum requirements are stated.	r32-ax1, r29-a11, r29-am2	az1-gate.json
HNM-C-AZ2	<i>Exact finite-graph Wilson coefficients on the two-plaquette graph, a different finite model.</i> With both faces coupled: derivative exactly $1/6$, second-order coefficient 0 , third-order $-187/33696$, and sign-certified enclosures at six grid points and two cutoffs; $1/6 \leftrightarrow 1/144$ is a consistency check only, and nothing transfers.	r32-aw1, r11-operator, r11-gap	az2-gate.json

Process contributions. The cycle also produced four process records. They are methods of working, not scientific results, and they count as zero investigations.

- *The pre-registration block* (`hnm-r32-prereg-v1`), frozen and hashed in every contract before any outcome: the model identifier, selected triple, coupling and signs, state provenance, observable with its centering and reference route, nodes, itemized error terms, target, outcome types, allowed sub-labels, claim exclusions and required control ids, with a control list that must equal the contract’s (Section 1.3).
- *The sentence template.* From sub-round 4 on, a contract may fix a `mandatory_sentence_template`, the gate fields it requires and, for AZ1 and AZ2, forbidden phrasings; the producer quotes the template verbatim and the gate fills it with reviewed constants (Sections 8.1.7 and 8.2.8).
- *The two vocabulary extensions* recorded in the plan after the Jung assistant-2 and assistant-4 audits (Section 9.2), with AX1, AY1 and AY2 grandfathered and their contracts unamended.
- *The tools* of Section 9.6: the freezing tool, the contract freezer, the gate recorder and the Arb cross-check, together with the skeptic’s replay tools `replay_loop.py` and `replay_declared.py`. None of them computes a scientific result.

Evidence records. The immutable decision records are the ten gates listed above. Each binds its contract, the frozen producer reports, checkers, input snapshots and outputs, and the skeptic’s review evidence. Fresh normal and optimized replays reproduce every producer output byte for byte, and gate-specific method snapshots preserve the instructions actually used, so later guidance cannot rewrite an earlier admission. This addendum’s builder refuses to run until all ten gates exist and verifies every gate binding. Its build receipt binds the scientific and authoring sources to the PDF, and a separate QA record binds the page geometry, the reference resolution and the review of every rendered page to the PDF’s digest. From the repository root:

```
python -B papers/round32-addendum/build.py --render
python -B papers/round32-addendum/build.py --check
```

A hash or a passing program does not replace the stated mathematical hypotheses, and the release

checks are distinct from the page review.

Exactly ten investigations are closed. The deliberation loops, the assistant packages, the lens updates, repairs, replays, this manuscript and publication add no research loop. Draft03, the Round31 addendum and every earlier scientific verdict are preserved.

References

- [1] Hruday N M (BUNZEEY), *Yang–Mills workbench research-workflow contract*, [AGENTS.md](#). It is snapshotted into every Round32 producer closure and bound by every Round32 gate.
- [2] Hruday N M (BUNZEEY), with model-agent assistance, *Round21 I1 dictionary report*, [I1 forward report](#). This is a shared premise of all ten Round32 contracts: the face-class table, the I1.5 star convention, the padded family of §6 and the Yarotsky constants of §5.
- [3] Hruday N M (BUNZEEY), with model-agent assistance, *Round29 AM2: finite-volume gap by a commuting creation expansion*, [AM2 gate](#); [forward report](#); [reverse report](#); [skeptical review](#). These are shared premises of all ten contracts and are bound by all ten Round32 gates.
- [4] Hruday N M (BUNZEEY), with model-agent assistance, *Round29 AQ1 and AQ2: subsequential thermodynamic states, dynamics and physical gap*, [AQ1 forward report](#); [AQ2 forward report](#). These are shared premises of all ten contracts and are bound by all ten Round32 gates.
- [5] Hruday N M (BUNZEEY), with model-agent assistance, *Round29 AL1: uniform dictionary $\tau = 96/g^4$* , [AL1 gate](#); [AL1 forward report](#). These are premises of AX1 and AZ1 and are bound by the AX1 gate.
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